

Solitons and their applications

*in Supersymmetry, Holography, Brane dynamics and Condensed Matter**

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Chapter 1

Introduction and Definitions

1.1 Introduction

The uses of solitons in the context of particle physics started some 40 years ago when exact solutions to the classical equations of motion of certain non-linear field theories were employed, within a semiclassical approach, to study non-perturbative aspects of quantum field theory (QFT)(see [1]-[2] for a list of references).

As we shall see below, solitons for particle physics can be defined as finite energy localized static solutions that could be taken as the starting point of a quantum field theory WKB-like approximation [3]. This approach is particularly appropriate to study the strong coupling regime in models used to describe the fundamental interactions. In certain cases solitons could even be connected with string theory and also have relevance in cosmology.

In the same context as solitons, instantons [2], which are time-dependent finite action solutions to the equations of motion of non-linear field theories in Euclidean space-time were proposed to implement in a concrete way Dirac-Feynmann ideas on the role of the action S as the phase of the quantum wave function. Within the path integral framework, instantons were considered the dominant part of the path integral variables which were then written as

$$|\text{FIELD}\rangle = |\text{instanton}\rangle + \sqrt{\hbar}|\text{field}\rangle$$

Here $|\text{FIELD}\rangle$ represent the original path-integral field variables while $|\text{field}\rangle$ can be considered as quantum fluctuation around the classical dominant instanton configuration.

In this way, the “partition function” Z defining the Euclidean quantum theory can be calculated as

$$Z = \exp\left(-\frac{1}{\hbar}S[\text{instanton}]\right) (1 + \hbar K[\text{field}] + \dots)$$

where $S[\text{instanton}]$ is the classical action calculated at the instanton solution and K and the following terms in the expansion can be explicitly computed term by term in powers of $\hbar$.

In particle physics, the study of solitons arising in Abelian gauge theories coupled to charged scalars (vortices) [8] and their non-Abelian extensions (monopoles) [5]-[6] together with instantons arising as solution of Yang-Mills theory [2] concentrated a big deal of efforts in the study of non-perturbative aspects of high energy physics and cosmology.

More or less simultaneously solitons were also employed, following a similar approach, to describe condensed matter phenomena as for example high-Tc superconductivity and Hall effect. In this case skyrmion solitons attracted much attention, particularly in the last years in which they are an important ingredient in the study of topological insulators.

The study of soliton solutions to classical equations of motion of bosonic field theories has also made contact with Supersymmetry. Indeed it was known since 1976 [1] that under certain circumstances, the second order equations of motion leading to soliton solutions could be reduced to simpler first order equations also known as self-dual equations, BPS equations (for Bogomolny and Prasad-Sommerfield papers [9]-[10]), and that this reduction had to do with supersymmetry. More specifically, BPS solitons can emerge in those supersymmetric theories in which superalgebras are centrally extended and in many instances the corresponding central charges are seen at the classical level. The connection was clarified after the work of Olive and Witten [11] and then it was found that in the so-called BPS sectors of supersymmetric theories like super Yang-Mills some data can be computed at strong coupling even when the full theory is not solvable [12]. An extensive review on the uses of solitons in supersymmetric theories can be found in [13].

If one searches in the Stanford University High Energy Physics database Inspire-HEP for papers having the word “soliton” in their title one finds that the first record corresponds to a 1976 publication and that at the end of August 2014 the number has grown to more than 4000, a bit less than the number of titles with the word “monopole”. Now, there is a word for a younger object which is closely connected to solitons, namely the word “brane” first introduced in 1987 [14] in the context of string theory, that has collected since then about 8000 citations.

It should be stressed that branes have come to mean many things to many people. Broadly speaking it refers to certain objects in field theory and string theory. One of its uses is as synonym of soliton solutions of (super)gravity theory and (super) string theories. The word brane comes from “membrane”, which can be seen as a two-dimensional brane if one defines a p -brane as sweeping out a $p + 1$ volume as it propagates through space-time. Such p -brane can then be identified with a soliton in the $p + 1$ dimensional world-volume.

Another kind of branes are “D branes”, defined as a hypersurface on which a string may end. In field theory both the brane and the string arise as BPS solitons, the brane as a domain wall and the string as a flux tube. Supersymmetric gauge theories can be considered, in a

sense, dual to string/D-brane theory.

A brief review on the subject of branes and solitons will be presented in this series of lectures.

Finally, in the last 5 years a new and surprising field connecting gravity and condensed matter physics. This is a result of a remarkably gauge/gravity duality (See [15] and references therein) in turn originated in the well-honored AdS/CFT correspondence (See [16] and references therein). A key ingredient in this context is the so called black-hole hair that can exist when General Relativity is coupled to scalar fields, gauge fields, etcætera. That a black hole is hairy means that the equations of motion of the Einstein-Hilbert-field theory have classical solutions which can be thought as the curved space-time extensions of solitons like vortices, monopoles, etc.

Once these solutions are constructed, the gauge/gravity duality enters into the game: with anti-de Sitter space boundary conditions, string theory (which includes gravity) is completely equivalent to a (non-gravitational) gauge theory living on the boundary at infinity.

In this way an appropriate choice of gravity theory and matter and gauge fields in the AdS bulk (for example general relativity with a negative cosmological constant coupled to a Maxwell field and a charged scalar in $3 + 1$ space-time dimensions can reproduce through the gauge/gravity duality some basic thermodynamical properties of planar superconductors described by the dual theory in the boundary (the temperature on the boundary theory being provided by the black-hole temperature of the classical solution found in the bulk). The last part of these lectures will describe in some detail these issues.

1.2 Definitions

Let us start with the definition of a soliton. Originally the names solitary waves and solitons referred to certain solutions of non-linear wave equations with two main characteristics

1. They are localized solution depending on spacespace-time (wave-packets) travelling with constant velocity and no distortion in shape.
2. After collision of two localized packets they retain their original shapes and velocities.

Such kind of waves were described for the first time in 1834 by a civil engineer and naval architect, John Scott Russell (1808-1882), who observed a solitary wave in the Union Canal in Scotland. He reproduced the phenomenon in a wave tank and named it the “Wave of Translation”¹.

¹ Scott Russell describes his observation in a report to the the British Association for the Advancement of Science as follows: *I was observing the motion of a boat which was rapidly drawn along a narrow channel*



Figure 1.1

Scott Russell's observation took place on the Union Canal at Hermiston, very close to the Riccarton campus of Heriot-Watt University, Edinburgh. The photography was taken on July 1995, when an international gathering of scientists witnessed a re-creation of the solitary wave on the Union Canal near Edinburgh. They were attending a conference on nonlinear waves in physics and biology at Heriot-Watt University.

131 years after Scott Russell observation, Zabusky and Kruskal introduced the name soliton in a classic paper [18] and some years after particle physicists proposed to quantize semiclassically certain classical solutions of non-linear equations of motion of various fields theories in order to describe elementary particles. Soliton-like solution were natural

by a pair of horses, when the boat suddenly stopped-not so the mass of water in the channel which it had put in motion; it accumulated round the prow of the vessel in a state of violent agitation, then suddenly leaving it behind, rolled forward with great velocity, assuming the form of a large solitary elevation, a rounded, smooth and well-defined heap of water, which continued its course along the channel apparently without change of form or diminution of speed. I followed it on horseback, and overtook it still rolling on at a rate of some eight or nine miles an hour (14 km/h), preserving its original figure some thirty feet (9 m) long and a foot to a foot and a half (300 – 450 mm) in height. Its height gradually diminished, and after a chase of one or two miles (2 – 3 km) I lost it in the windings of the channel. Such, in the month of August 1834, was my first chance interview with that singular and beautiful phenomenon which I have called the Wave of Translation.[17]

candidates in view of the properties enumerated above.

A definition in broad sense of the solitons as understood in particle physics slightly differs from the one above. We shall adopt this definition.

- **Solitons**

Solitons are static solutions of (non-linear) equations of motion which are stable, localized in space and have finite energy.

Note that one can consider time-dependent solitons (see [2] for a detailed discussion) but we shall not cover this subject in the present lecture.

- **Topological solitons [19]**

In some cases soliton solutions can be associated to a topological quantity² (a non-trivial “topological charge”, usually an integer number) which cannot change by smooth deformations of the solution. The topological properties of the solitons have to do in this case with the boundary configuration of classical solutions of eqs. of motion at spatial infinity and they ensure stability of the solution. These properties are characterized by quantities called “topological charges” that are preserved but this property is not related to the conservation of Noether charges related to symmetries of the theory.

The boundary conditions at infinity for a topological soliton state should necessarily be topologically different from those of the trivial vacuum state. In turn, this requires degenerate vacuum states.

In the context of relativistic classical field theory, we see that topological solitons look very much like classical particles. Their energy density is localized at a finite region, their total energy is finite, and they correspond to stable solutions. Moreover, because the field equations are Lorentz invariant, once one has found a solution $\phi_{sol}(x)$, one also has boosted solutions $\phi_{sol}(x - vt/\sqrt{1 - v^2})$ for arbitrary v , $|v| < 1$ ($c = 1$): solitons can move in space.

In a famous series of Erice lectures on solitons, Coleman [3] uses the word **lump**³, and gives for “soliton” a more restrictive definition (solitary waves of permanent form which can interact with other solitons and emerge from the collision unchanged except for a phase shift).

²The word topology comes from the greek *topos* and *logos* (τόπος and λόγος) meaning “place” and “study”. Literally, topology is the “study of the place”. Its task is to define a place (also called “space”) and its properties (particularly its shape). Today it is understood as the branch of mathematics studying the properties that are preserved under continuous deformations of spaces.

³**Lump**: a compact mass of a substance; In Spanish: grumo.

- **Non-topological solitons [21]**

Non-topological solitons are solutions that are rendered stable by the existence of a conserved Noether charge carried by fields confined to a finite region of space. The Noether charge is associated to an unbroken continuous symmetry. The minimum charge of these solitons depends upon ratios of coupling constants.

The boundary conditions at infinity for a non-topological soliton are the same as those for the trivial vacuum state. Thus there is no need for degenerate vacuum states. The necessary condition for the existence of non-topological solitons is that there should be an additive conservation law.

In these series of lectures I will solely discuss topological solitons.

Solitons could exist in different number of spatial dimensions:

- **1 + 1 dimensional solitons: kinks**

A kink is a soliton solution of the classical equations of motion of a field theory in 1 + 1 space-time dimensions. The Oxford Dictionary defines the word *kink* as a sharp twist or curve in something that is otherwise straight. The twist can be caused by a doubling or winding of something such as a rope, cable or hair upon itself. This winding has an associated integer number related to the topological charge.

In order to clarify kink taxonomy let us indicate that the kink solution that interpolates between the different boundary conditions related by discrete $\mathbf{Z}_2$ transformations is called a “ $\mathbf{Z}_2$ kink”.

- **$d > 1$ dimensional solitons**

Magnetic vortices and monopoles with magnetic flux connected to the topology resulting from the spontaneous breaking of a gauge symmetry gather the features required by solitons by an appropriate choice of spatial dimensions.

Also skyrmions solution are solitons associated to equations of motion of non-linear models (like the nonlinear sigma model) in different number of dimensions

- **Domain walls in different space-time dimensions**

When kink solutions are placed in more than one spatial dimension, they become extended planar structures called “domain walls”. They arise when a discrete symmetry as the above mentioned $\mathbf{Z}_2$ symmetry is spontaneously broken.

Unstable domain walls can also appear if spontaneously broken discrete symmetry is approximate and there is the metastable vacuum.

In $3 + 1$ -dimensional space time, domain walls are two-dimensional extended objects. In $2 + 1$ dimensions they are called “domain lines” and in $1 + 1$ dimensions they reduce to kinks.

- **Instantons**

Instantons are localized finite action solutions to the Euclidean classical equations of motion of a given theory (i.e., the equations of motion extremizing a classical field theory action in imaginary time).

They cannot then be directly associated to particles as for the previously mentioned solitons but they can be considered as “static” solitons if one takes the Euclidean space-time coordinates as the spatial coordinates in a space-time in one more dimension.

The word “instanton” was coined by ’t Hooft in a seminar given at École Normale Supérieure in January 1976 but they were originally introduced by Belavin, Polyakov, Schwartz and Tyupkin [22] under the name of *pseudoparticles*.

Bibliography

- [1] C. Rebbi and G. Soliani, *Solitons and Particles*, World. Sci., Singapore, 1984.
- [2] R. Rajaraman, *Soliton and Instantons*, Elsevier, Amsterdam, 1996.
- [3] R. F. Dashen, B. Hasslacher and A. Neveu, “The Particle Spectrum in Model Field Theories from Semiclassical Functional Integral Techniques,” *Phys. Rev. D* **11** (1975) 3424.
- [4] H. B. Nielsen and P. Olesen, *Nucl. Phys. B* **61** (1973) 45.
- [5] G. ’t Hooft, “Magnetic Monopoles in Unified Gauge Theories,” *Nucl. Phys. B* **79** (1974) 276.
- [6] A. M. Polyakov, “Particle Spectrum in the Quantum Field Theory,” *JETP Lett.* **20** (1974) 194 [*Pisma Zh. Eksp. Teor. Fiz.* **20** (1974) 430].
- [7] A. A. Belavin, A. M. Polyakov, A. S. Schwartz and Y. S. Tyupkin, *Phys. Lett. B* **59** (1975) 85.
- [8] H. J. de Vega and F. A. Schaposnik, “A Classical Vortex Solution of the Abelian Higgs Model,” *Phys. Rev. D* **14** (1976) 1100.
- [9] M. K. Prasad and C. M. Sommerfield, “An Exact Classical Solution for the ’t Hooft Monopole and the Julia-Zee Dyon,” *Phys. Rev. Lett.* **35** (1975) 760.
- [10] E. B. Bogomolny, “Stability of Classical Solutions,” *Sov. J. Nucl. Phys.* **24** (1976) 449 [*Yad. Fiz.* **24** (1976) 861].
- [11] E. Witten and D. I. Olive, “Supersymmetry Algebras That Include Topological Charges,” *Phys. Lett. B* **78** (1978) 97.

- [12] L. Alvarez-Gaume and S. F. Hassan, “Introduction to S duality in N=2 supersymmetric gauge theories: A Pedagogical review of the work of Seiberg and Witten,” *Fortsch. Phys.* **45** (1997) 159
- [13] M. Shifman and A. Yung, *Supersymmetric Solitons* Cambridge U. Press, Cambridge, UK, 2009.
- [14] A. Achucarro, J. M. Evans, P. K. Townsend and D. L. Wiltshire, “Super p-Branes,” *Phys. Lett. B* **198** (1987) 441.
- [15] G. T. Horowitz, “Using general relativity to study superconductivity,” *J. Phys. Conf. Ser.* **484** (2014) 012002.
- [16] O. Aharony, S. S. Gubser, J. M. Maldacena, H. Ooguri and Y. Oz, “Large N field theories, string theory and gravity,” *Phys. Rept.* **323** (2000) 183 [hep-th/9905111].
- [17] John Scott Russel, *Report on Waves* a report to the Fourteenth meeting of the British Association for the Advancement of Science, Published in the BA reports VI, 417-468, plus plates 1-8. 1837; <http://www.macs.hw.ac.uk/~chris/Scott-Russell/SR37.pdf>
- [18] N. J. Zabuski and M. D. Kruskal is *Interaction of “Solitons” in a Collisionless Plasma and the Recurrence of Initial States*, *Phys. Rev. Lett.* **15** (1965)240.
- [19] R. Jackiw and C. Rebbi, “Topological Solitons And Instantons,” *Comments Nucl. Part. Phys.* **8** (1978) 129.
- [20] S. Coleman, “Classical Lumps and their quantum descendants” in “Aspects of symmetry”, Cambridge University Press, Cambridge, UK, 1985.
- [21] T. D. Lee and Y. Pang, “Nontopological solitons,” *Phys. Rept.* **221** (1992) 251.
- [22] A. A. Belavin, A. M. Polyakov, A. S. Schwartz, Yu. S. Tyupkin, “Pseudoparticle solutions of the Yang-Mills equations”. *Phys. Lett. B* **59** (1975) 85.

Chapter 2

Kinks

2.1 A simple scalar model: the anharmonic oscillator in $1 + 1$ dimensions

As stated in Lecture 1, the Oxford Dictionary defines a **kink** as a sharp twist or curve in something that is otherwise straight. It can be caused by a doubling or winding of something such as a rope, cable or hair upon itself.

The word was introduced in Physics by Dashen et al [1] in their study of non-perturbative methods applied to the analysis of extended hadrons. They called kink the solutions of the equations of motion of a real scalar field theory in $1+1$ dimensional space-time with dynamics governed by the Lagrangian

$$L = \frac{1}{2}(\dot{\phi})^2 - \frac{1}{2}(\phi')^2 - V[\phi] \quad (2.1.1)$$

where henceforth a dot (a prime) represents differentiation with respect to time t (to space variable x). The potential $V[\phi]$ can be any positive semi-definite functional of ϕ reaching a value of zero for some value or values of ϕ . For simplicity, we shall consider the particular (and paradigmatic) quartic potential

$$V[\phi] = \frac{1}{4}\lambda(\phi^2 - m^2/\lambda)^2 \quad (2.1.2)$$

the extension to field theory of the well-known anharmonic potential in particle mechanics.

The energy of any field configuration is given by

$$E = T + V = \int dx \left(\frac{1}{2}(\dot{\phi})^2 + \frac{1}{2}(\phi')^2 + V[\phi] \right) \quad (2.1.3)$$

As mentioned in Lecture 1, starting from a static soliton solution of a Lorentz invariant theory, we can construct a boosted soliton moving in space, a solitary wave that is also called a kink. We start with the static case in which we can apply the Hamilton principle minimizing

Given the energy (2.1.3) we can find static solutions to the equations of motion by minimizing the energy (2.1.3), which coincides with a Lagrangian if one takes x as time and a potential which is $-V$. Then, the equations of motion of this “Lagrangian” will be

$$\phi'' = \frac{dV}{d\phi} \quad (2.1.4)$$

or

$$\phi'' = \lambda\phi^3 - m^2\phi \quad (2.1.5)$$

The solution can be found by quadratures. Indeed, multiplying (2.1.5) by ϕ' and integrating one gets

$$\int dx \phi' \phi'' = \int dx \phi' \frac{dV}{d\phi} \quad (2.1.6)$$

or

$$\frac{1}{2}(\phi')^2 = V[\phi] \quad (2.1.7)$$

from which we find

$$x - x^0 = \pm \int_{\phi(x_0)}^{\phi(x)} \frac{d\bar{\phi}}{(2V[\bar{\phi}])^{1/2}} = \pm \int_{\phi(x_0)}^{\phi(x)} \frac{d\bar{\phi}}{\sqrt{\lambda/2}(\bar{\phi}^2 - m^2/\lambda)} \quad (2.1.8)$$

Choosing $\phi(x_0) = 0$, integrating over $\bar{\phi}$ and inverting the result one ends with

$$\phi(x) = \pm(m/\sqrt{\lambda}) \tanh\left((m/\sqrt{2})(x - x_0)\right) \quad (2.1.9)$$

From the shape of the solution, shown in Fig.2.1, one understands why the solution is called a kink (sign +). When the $-$ sign is chosen the solution is called an anti-kink.

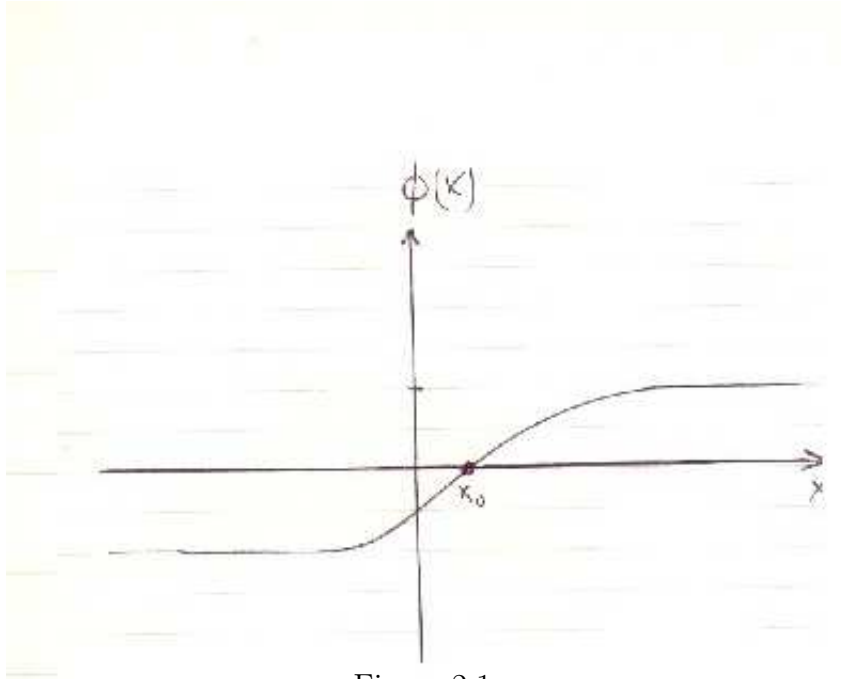


Figure 2.1

The energy density of the kink solution

$$\epsilon(x) = \frac{1}{2}(\phi')^2 + V[\phi] = \frac{m^2}{2\lambda} \operatorname{sech}^2 \left(m(x - x_0)/\sqrt{2} \right) \quad (2.1.10)$$

is plotted in Fig.2.2. where one can clearly see its localization around $x = x_0$.

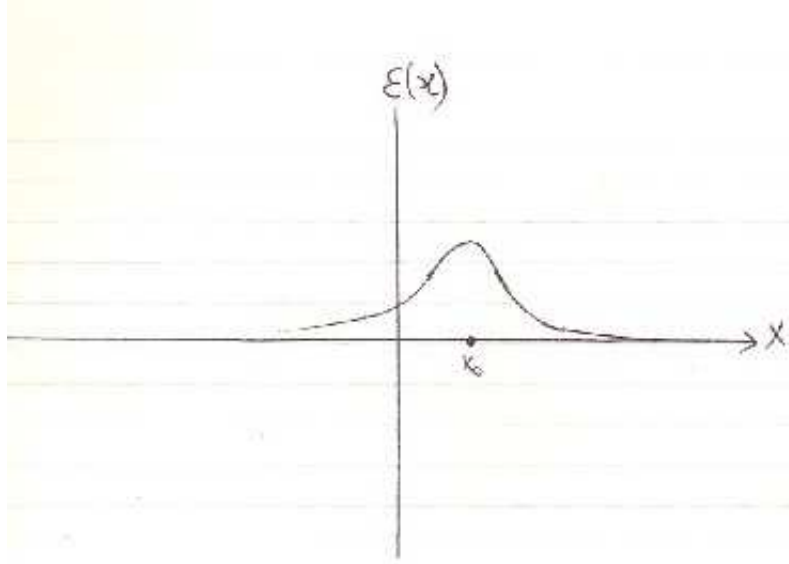


Figure 2.2

The total kink energy is given by

$$E = \int_{-\infty}^{\infty} dx \epsilon(x) \quad (2.1.11)$$

In a strict sense, a kink is not a soliton if one adopts the “narrow” definition including the condition that solitons have to survive collision without any change. In contrast, it can be interpreted as an instanton in one less dimension. Indeed, if one identifies the spatial variable x as imaginary time x^0 then the energy density (2.1.10) coincides with the Euclidean Lagrangian while the Euclidean action can be identified with a total energy (2.1.11)

$$S_E[\phi] = \int_{-\infty}^{\infty} dx^0 \left(\frac{1}{2} \left(\frac{d\phi}{dx^0} \right)^2 + V[\phi] \right) \quad (2.1.12)$$

In a quantum mechanical context one can identify $\phi(x^0)$ with the coordinate $q(x^0)$ describing the trajectory in Euclidean time of a particle of mass m subject to an anharmonic potential (i.e. quartic powers of $q(x^0)$ apart from the harmonic quadratic ones).

When kink solutions are placed in more than one spatial dimension, they become extended planar structures called **domain walls**. The field configuration associated to the model we are studying for a $\mathbf{Z}_2$ domain wall in the yz -plane in three spatial dimensions is just given by eq.(11.3.41) by adding the yz variables in the l.h.s.,

$$\phi(x, y, z) = \pm(m/\sqrt{\lambda}) \tanh \left((m/\sqrt{2})(x - x_0) \right) \quad (2.1.13)$$

The energy density of the wall is concentrated over all the yz -plane and is given by eq.(2.1.10). The new aspects of domain walls are that they can be curved and deformations can propagate along them. Another feature of the planar domain wall is that it is invariant under boosts in the plane parallel to the wall. This is simply because the solution is independent of t , y and z and any transformations of these coordinates do not affect the solution.

2.2 The sine-Gordon model

Another example of a $1 + 1$ dimensional theory having kink solutions is the sine-Gordon model introduced by the french engineer Edmond Bour in 1862 in his studies of surfaces of constant negative curvature and their deformations [2]. Instead of the potential (2.1.2), one considers the case

$$V_{SG}[\phi] = \frac{\alpha}{\beta^2}(1 - \cos \beta\phi) \quad (2.2.14)$$

with α and β real parameters which can be taken as positive without loss of generality¹. The ground states are those for which

$$\phi = \frac{2\pi}{\beta}n \quad (2.2.15)$$

If one considers small oscillations about one of these ground states, say $\phi = 0$ the potential takes the form

$$V_{SG} = \frac{\alpha}{2}\phi^2 - \frac{\alpha\beta^2}{4!}\phi^4 + \dots \quad (2.2.16)$$

so that α can be seen as the mass parameter and $\alpha\beta^2$ as the coupling constant of the anharmonic oscillator considered previously.

Note that formula (2.1.8) is valid for any potential $V[\phi]$,

$$x - x^0 = \pm \int_{\phi(x_0)}^{\phi(x)} \frac{d\bar{\phi}}{(2V[\bar{\phi}])^{1/2}} \quad (2.2.17)$$

so that given a potential, one can find the kink solutions (if it exists) just by quadratures. However, most of these solutions are not of finite energy. To begin with, ϕ should go to one a zeros of $V[\phi]$ as x goes to plus or minus infinity. The following conditions should be tested in order to decide if finite energy solutions exists

¹The name of the model is inspired by the presence of the sine term in the equation of motion, compared with the Klein-Gordon equation arising in the free field case. In a letter that Finkelstein addressed to Coleman [3] he states that the nomenclature was originated in a private joke between Julio Rubinstein and himself which became public in a paper. However, in his paper Rubinstein states that the name is due to Professor Martin Kruskal [4].

1. If V has only one zero (i.e., the ground state is unique) there are no non-trivial time independent solutions of finite energy.
2. If V has more than one zero (i.e. the theory has more than one ground state) there always exist non trivial time-independent solutions of finite energy. For such solutions ϕ moves from one zero of V at $x = -\infty$ to an adjacent zero of V at $+\infty$.

Note that because of translation invariance there is an arbitrary parameter x_0 in eq.(2.2.17). Then, given a particular solution $\phi_{sol}(x)$ the general solution is $\phi_{sol}(x - x_0)$ centered at the origin

The sine-Gordon equations has the following 1-soliton solution

$$\phi^{(1)}(x, t) = \frac{4}{\beta} \tan^{-1} \exp(\sqrt{\alpha} x) \quad (2.2.18)$$

The choice of the inverse tangent branch determines which two consecutive zeros of the potential are considered. Multisoliton solutions can be obtained by applying the so-called Bäcklund transformations to the 1-soliton solution [5].

2.3 More spatial dimensions: Derrick's theorem [6]

Derrick theorem is a no go theorem for the search of kink solutions of scalar theories in more than one spatial dimensions.

Theorem. *Let $\phi = (\phi^1, \phi^2, \dots)$ be a set of scalar fields in one time dimension and d space dimension with dynamics defined by*

$$L = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V[\phi] \quad (2.3.19)$$

where $V[\phi]$ is nonnegative and equal to zero for the ground state(s) of the theory. Then, for $d \geq 2$, the only non-singular time independent finite energy solutions of the equations of motion are the ground states.

Proof. Define

$$E_1 = \frac{1}{2} \int d^d x (\nabla \phi)^2 \quad (2.3.20)$$

$$E_2 = \int d^d x V[\phi] \quad (2.3.21)$$

E_1 and E_2 are both non negative and are simultaneously equal to zero only for the ground states (which are identified with constant solutions). Let $\phi(x)$ denote a time-independent solution and consider a one parameter family of field configurations defined by

$$\phi(x; \lambda) = \phi(\lambda x) \quad (2.3.22)$$

with λ a positive number. Inserting this expression in eqs.(2.3.21) and making the change of variables $x = y/\lambda$ one gets

$$E(\lambda) = \lambda^{(2-d)} E_1 + \lambda^{-d} E_2 \quad (2.3.23)$$

Consider the cases $d=2$ and $d=3$:

$$E(\lambda) = E_1 + \frac{1}{\lambda^2} E_2, \quad d = 2 \quad (2.3.24)$$

$$E(\lambda) = \frac{1}{\lambda} E_1 + \frac{1}{\lambda^3} E_2, \quad d = 3 \quad (2.3.25)$$

In both cases $E(\lambda)$ decreases monotonically as λ increases so there is no stationary point and no nontrivial solution of the equations of motion exists. The same happens for $d = 4, 5, \dots$

Now, if $d = 1$ $E(\lambda)$ takes the form

$$E(\lambda) = \lambda E_1 + \frac{1}{\lambda} E_2, \quad d = 1 \quad (2.3.26)$$

which is stationary at $\mu = E_2/E_1$ so in this case solutions are not ruled out.

Note that in the particular case $d = 2$ if there is no potential $E(\lambda) = E_1$ which is independent of λ and so the proof above does not hold anymore. But in this case we can then we can apply Hamilton principle to V_1 alone and then one trivially see that V_1 must vanish.

there are several ways to evade the Derrick no-go theorem. As we shall discuss in future lectures the non-linear sigma model in $d = 2$ spatial dimensions has non-trivial soliton solutions due to the fact that instead of a positive definite potential one has a Lagrange multiplier term with no definite sign. Another way to evade Derrick theorem is to include in the Lagrangian higher powers of the derivatives of the scalar fields, as it happens in the case of skyrmion and baby skyrmion models.

2.4 The kink and a topological conservation law

Whenever there exist degenerate vacua, as for example in the case of the potential (2.1.2) which has two degenerate vacua²

$$q_{vac} = \pm q_0 \quad (2.4.27)$$

with $q_0 = m/\sqrt{\lambda}$, there exist topological conservation laws which are not derived *à la* Noether from any symmetry of the Lagrangian, but rather from the topology of the manifold of solutions (or field manifold). In general, soliton solutions interpolate degenerate vacua at spatial infinity.

Consider the case of the anharmonic oscillator defined by Lagrangian (2.1.1) with potential (2.1.2). There is a “trivial” (in the sense that it is easily found) conservation law with an associated conserved current given by

$$j^\mu(x) = \epsilon^{\mu\nu} \partial_\nu \phi(x, t) \quad \epsilon^{01} = -\epsilon^{10} = 1 \quad (2.4.28)$$

Evidently

$$\partial_\mu j^\mu = 0 \quad (2.4.29)$$

independently of any field equation. This current is not associated with any symmetry of the Lagrangian and it is related to the existence of degenerate vacua. To see this last property let us construct the charge Q associated to the topological current,

$$Q \equiv \int_{-\infty}^{+\infty} dx j^0 = \int_{-\infty}^{+\infty} dx \frac{\partial \phi}{\partial x} = \phi(+\infty, t) - \phi(-\infty, t) \quad (2.4.30)$$

In the case of the kink we have been discussed for the anharmonic potential we then have

$$Q = 2\phi_0 \quad (2.4.31)$$

In the case of the sine-Gordon model, we have (see eq.(2.2.15))

$$Q = \frac{2\pi}{\beta} N \quad (2.4.32)$$

with N integer.

As Coleman remarks [3] these conservation laws, like the conservation of energy, states that a certain quantity constructed from the fields is independent of time. However, unlike

²We call vacua the possible values that take the field at spatial infinity. These values must correspond to a minimum of the potential and they are usually associated to the classical approximation of the quantum vacuum states.

the conservation of energy, it is not a consequence of a symmetry of the theory. It is instead a consequence of a topological property of the space of non-singular finite energy solutions of the equations of motion of the field theory, the fact that such space is not connected. We shall come back to these topological or “homotopic” [8] conservation laws in the next lectures.

Let us end this section by comparing the nature of topological conserved current versus Noether conserved currents

Topological conserved current

1. It is conserved independently of the equations of motion.
2. j^0 contains only canonical coordinates and no momenta. Therefore its Poisson brackets with coordinates vanish and it generates no symmetry.
3. j^0 is a spatial divergence. Its spatial integral is non-zero only if the field satisfies the boundary conditions $\phi(\pm\infty)$ in a (topologically) non-trivial way (for example $\phi(\pm\infty) \in \{2N\pi/\beta : N = 0, +1, +2, \dots\}$ in the sine-Gordon model).

Noether conserved current

1. Its conservation requires use of the equations of motion.
2. j^0 contains canonical momenta.

In summary we can conclude that when there are degenerate vacua there can exist a new sort of conserved current, a topological current, transporting a conserved quantity which, unlike a Noether current or charge, is not associated with any manifest symmetry of the Lagrangian.

2.5 The kink as an instanton

In order to see how working with Euclidean time one can use instantons to compute the ground state energy of a quantum anharmonic oscillator let us consider the transition amplitude for a quantum anharmonic oscillator with potential (2.1.2) defined in terms of Feynman’s path integrals. a

$$Z_{i \rightarrow f} = \langle q_f | \exp \left(-\frac{HT}{\hbar} \right) | q_i \rangle = \mathcal{N}^{-1} \int \mathcal{D}\phi \exp \left(-\frac{1}{\hbar} S[\phi] \right) \quad (2.5.33)$$

with, $T = x_f^0 - x_i^0$ and $q_i = q(x_i^0)$ and $q_f = q(x_f^0)$ (The time integration interval defining the action is taken as (x_f^0, x_i^0) and eventually it will be $(-\infty, \infty)$).

To see how physical quantities can be calculated from $Z_{i \rightarrow f}$ using instantons, let us observe that the ground state energy can be derived in quantum mechanics as follows: one considers

$$H|n\rangle = E_n|n\rangle \quad (2.5.34)$$

and then, one can write

$$\sum_n \langle q_f | n \rangle \langle n | q_i \rangle \exp\left(-\frac{E_n T}{\hbar}\right) = Z_{i \rightarrow f} \quad (2.5.35)$$

Now, in the $T \rightarrow \infty$ limit, it is the ground state the one which makes the most important contribution to the sum

$$\lim_{T \rightarrow \infty} Z_{i \rightarrow f} \sim \langle \phi_f | 0 \rangle \langle 0 | \phi_i \rangle \exp\left(-\frac{E_0 T}{\hbar}\right) \quad (2.5.36)$$

Then, from the exponential damping of $Z_{i \rightarrow f}$ when $T \rightarrow \infty$ we can read the ground state energy

$$\lim_{T \rightarrow \infty} Z_{i \rightarrow f} = \lim_{T \rightarrow \infty} \int \mathcal{D}\phi \exp\left(-\frac{1}{\hbar} S[\phi]\right) \sim \lim_{T \rightarrow \infty} \exp\left(-\frac{E_0 T}{\hbar}\right) \quad (2.5.37)$$

At this point one can understand the reason why the path integral approach requires working in Euclidean time by comparing the anharmonic potential in real and imaginary time. In Dirac-Feynmann spirit, classical trajectories (extrema of the action) dominate the quantum process so that it is natural, starting from the path-integral (2.5.37), to proceed to the change of variables

$$q(x^0) = q_{cl}(x^0) + \sqrt{\hbar} y(x^0) \quad (2.5.38)$$

where $y(x^0)$ can be seen as quantum fluctuations around the classical trajectory $q_{cl}(x^0)$.

Now, in Minkowski space, given the form of potential $V(q)$, represented in figure 2.3, there are just two trivial classical solutions of the equations of motion namely $q_{cl} = \pm q_0$ with $q_0 = m/\sqrt{\lambda}$.

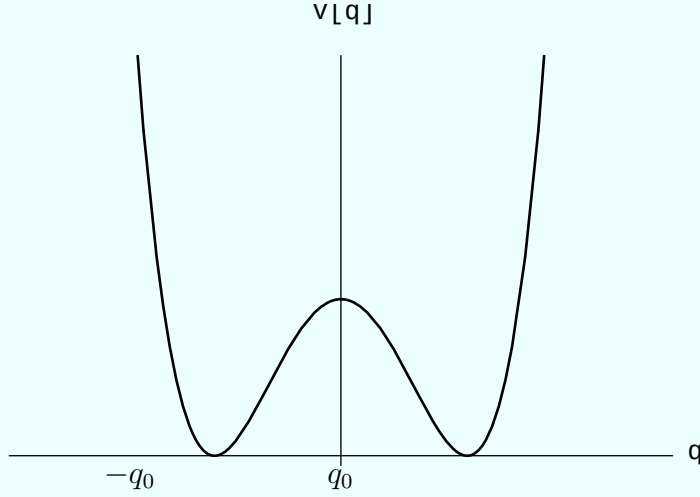


Figure 2.3

There are in Minkowski space two natural candidates for the path-integral change of variables,

$$q(t) = q_0 + \sqrt{\hbar}y \quad (2.5.39)$$

or

$$q(t) = -q_0 + \sqrt{\hbar}y \quad (2.5.40)$$

Both changes of variables are useless for describing the well-known tunnel effect that takes place in this problem. Indeed, one learns in any quantum mechanics course that the ground state $\psi_0(q)$ of a particle in such a potential is in fact a superposition of two gaussian wave-packets $\psi_{\pm}(q)$ centered at each minima of the potential,

$$\psi_0(q) = \alpha\psi_+(q - q_0) + \beta\psi_-(q + q_0) \quad (2.5.41)$$

Being the problem parity invariant, the parity operator $\mathcal{P}$,

$$\mathcal{P}\psi(q) = \psi(-q) \quad (2.5.42)$$

commutes with the Hamiltonian, $[H, \mathcal{P}] = 0$ and then, necessarily, the ground state has to be an eigenfunction of $\mathcal{P}$,

$$\mathcal{P}\psi_0 = \pm\psi_0 \quad (2.5.43)$$

But then

$$\mathcal{P}\psi_0(q) = \alpha\psi_+(q + q_0) + \beta\psi_-(q - q_0) = \pm\psi_0(q) \quad (2.5.44)$$

(remember that the wave-packets are gaussians). One then has

$$\alpha = \pm\beta = \pm\frac{1}{\sqrt{2}} \quad (2.5.45)$$

Now, the ground state has to be symmetric and then,

$$\psi_0 = \frac{1}{\sqrt{2}}(\psi_+(q - q_0) + \psi_-(q + q_0)) \quad (2.5.46)$$

Concerning the antisymmetric combination, it corresponds to the first excited state.

$$\psi_1 = \frac{1}{\sqrt{2}}(\psi_+(q - q_0) - \psi_-(q + q_0)) \quad (2.5.47)$$

The form of these wave functions makes evident the role of tunneling.

What about a path-integral approach to this problem? On the one hand, we know that if we make a smart choice for q_{cl} for the shift of path-integral variables, this q_{cl} will dominate the quantum result and we will be able to control quantum fluctuations non perturbatively. To achieve this, such classical solution should have, in itself, some information of the tunneling but this in turn becomes an oxymoron: tunneling is a phenomenon that does not exist classically. That is, it should interpolate between both minima. But, on the other hand, there is no classical solution in Minkowski space to take into account this. And, if one insists in performing a change of variables like (4.2.89), one will just study fluctuations around the minimum at q_0 . If, instead, one makes the change of variables (2.5.40), one will describe fluctuations around the minimum at $-q_0$. Notice that using q_0 or $-q_0$ to center the fluctuation analysis implies breaking the original symmetry under parity ($q \rightarrow -q$) of the problem. In order to describe the true ground state one needs a classical trajectory joining both minima. But there is no such trajectory in Minkowski space.

Now, looking to the Euclidean action (2.1.12) we see that the potential reverse its sign, so that it looks as in the following figure,

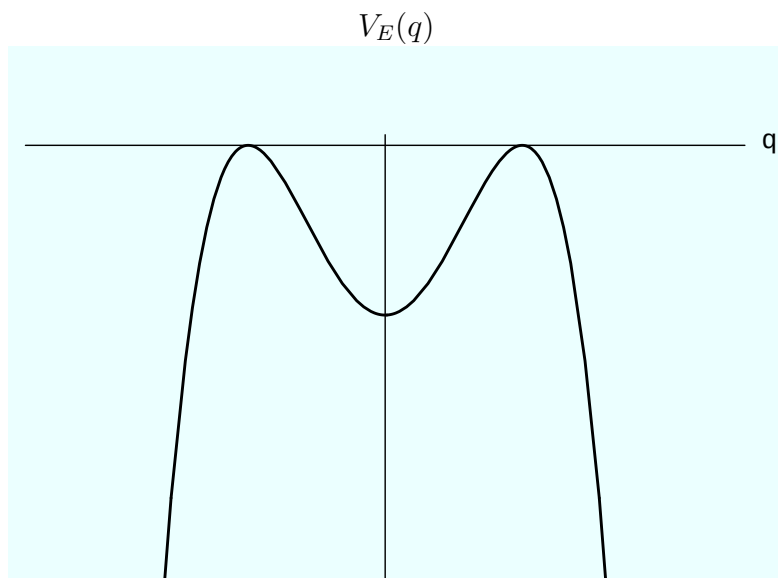


Figure 2.4

Then, for such an Euclidean potential, one should expect to have classical solutions interpolating between the two extrema and in fact such solutions are nothing but the kink we have been discussing in this lecture. Within this framework, the kink “restored” the symmetry under parity transformations. A complete analysis of the path-integral approach to the tunneling effect using kinks as instantons can be found in [7]

Bibliography

- [1] R. F. Dashen, B. Hasslacher and A. Neveu, “Nonperturbative Methods and Extended Hadron Models in Field Theory. 2. Two-Dimensional Models and Extended Hadrons,” *Phys. Rev. D* **10** (1974) 4130.
- [2] E. Bour, *Théorie de la déformation des surfaces*, J. École Impériale Polytech. **19** (1862) 1.
- [3] S. Coleman, *Classical lumps and their quantum descendants* in “Aspects of Symmetry: Selected Erice Lectures”, Univ. of Cambridge, Cambridge, USA, 1985.
- [4] J. Rubinstein, *Journal of Math. Phys.* **11** (1970) 258.
- [5] C. Rogers, W. K. Schief *Bäcklund and Darboux Transformations: Geometry and Modern Applications in Soliton Theory*. Cambridge Texts in Applied Mathematics. New York: Cambridge University Press, 2002.
- [6] G. H. Derrick, “Comments on nonlinear wave equations as models for elementary particles,” *J. Math. Phys.* **5** (1964) 1252.
- [7] F. A. Schaposnik, “24 Lectures on Path Integrals”,
<https://sites.google.com/site/schaposnik/>
- [8] C. v. Westenholz, *Ann. Inst. Henri Poincaré* **30** (1979) 353

Chapter 3

Vortices

3.1 The Abelian Brout-Englert-Higgs (ABEH) model

The ABEH model was proposed in 1964 as a theory in which gauge vector mesons could acquire a mass without losing gauge invariance and, a fortiori, renormalizability [1]-[12]. In their seminal paper Englert and Brout [1] motivated their proposal *in the possibility that strong-interaction physics originates from massive gauge fields related to a system of conserved currents*. [1]. Concerning Higgs works, he argues in ref. [3] that the symmetry breaking mechanism could be of relevance to describe weak-interactions: *It may be expected that when a further mechanism (presumably related to the weak interactions) is introduced in order to break Y^1 conservation, one of these gauge fields will acquire mass, leaving the photon as the only massless vector particle*. [3].

The gauge fields acquire a mass because the choice of the particular quartic potential for the complex scalar field implies the existence of a degenerate vacuum. As it is at present well-known, this mechanisms (usually dubbed the Higgs mechanism) is the tenor in the Standard Model opera. In fact the scalar mass arising from the quadratic part of the potential is the only dimension-full parameter in the Standard Model so that all other massive particles get their mass due to its presence.

We shall here describe the Higgs mechanism in the simplest case of a $U(1)$ gauge symmetry. We start with the ABEH Lagrangian in the way Higgs wrote it

$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}(D_\mu\phi)^*(D^\mu\phi) - V(|\phi|) \quad (3.1.1)$$

¹Y: weak hypercharge (sometimes called Y_H), the generator of the $U(1)$ component of the electroweak gauge group, $SU(2) \times U(1)$.

Here $F_{\mu\nu}$ is the field strength,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu , \quad (3.1.2)$$

ϕ is a complex charged scalar,

$$\phi = \phi^1 + i\phi^2 , \quad (3.1.3)$$

the covariant derivative is defined as

$$D_\mu = \partial_\mu + ieA_\mu , \quad (3.1.4)$$

and the potential taken in the form

$$V(|\phi|) = \lambda (|\phi|^2 - \phi_0^2) \quad (3.1.5)$$

We are working in $(3+1)$ -dimensional space-time ($\mu = 0, 1, 2, 3$) so that the real positive constant ϕ_0 has [mass] dimensions m^1 .

Lagrangian (3.1.1) is invariant under local $U(1)$ gauge transformations

$$\phi(x) \rightarrow \phi'(x) = \exp(i\alpha(x))\phi(x) , \quad A_\mu \rightarrow A'_\mu = A_\mu - \frac{1}{e}\partial_\mu\alpha(x) \quad (3.1.6)$$

The form of the potential ensures that its minimum occurs at

$$\langle\phi\rangle = \phi_0 \quad (3.1.7)$$

or at any other value related by the $U(1)$ symmetry (3.1.6). This set of minima $\{\exp(i\alpha(x))\phi_0\}$ are identified with the degenerate vacua of the quantum theory and that is the reason why usually this parameter of the classical potential is called the Higgs vacuum value and why the notation $\langle \rangle$ is used.

Let us expand the Lagrangian (3.1.1) about the vacuum (3.1.7). To this end, we write

$$\phi(x) = \phi_0 + \varphi \quad (3.1.8)$$

where the complex field φ will represent, at the quantum level, the field fluctuation about the vacuum. This expansion induces, through the Lagrangian's covariant derivative term squared, a mass term for the gauge field

$$L_{m_A} = \frac{1}{2}m_A^2 A_\mu A^\mu \quad (3.1.9)$$

with

$$m_A^2 = e^2 \phi_0^2 \quad (3.1.10)$$

Expanding the potential one finds also that the real part of the complex φ field has a mass

$$m_{\varphi_r}^2 = 4\lambda\phi_0^2 \quad (3.1.11)$$

The massless imaginary scalar part is a Goldstone boson which plays no role as an independent physical particle. One can see this by working in the so-called unitary gauge in which it completely decouples. This corresponds to a gauge function $\alpha(x)$ in (3.1.6) such that $\phi(x)$ becomes real so that the Goldstone boson is removed from the theory (“eaten” by the gauge field). Note that if the following relation between the gauge coupling constant e and the scalar self-interaction coupling constant λ holds,

$$\kappa^2 \equiv \frac{4\lambda}{e^2} = 1 \quad (3.1.12)$$

the two masses in the theory coincide. As we shall see, this condition has important consequences both for the classical equations of motion and for the possibility of a ($\mathcal{N} = 2$) supersymmetric extension of the model (which requires the scalar and the gauge field to have the same mass [5]-[7]). Parameter κ plays an important role in the phenomenological theory of superconductivity where it is called the “Landau parameter”.

Some 20 years after BEH proposal, Nielsen and Olesen [8] put forward the idea of constructing field theories for the dual string. As an example they show that the Lagrangian (3.1.1) allows for vortex-like solutions, in analogy of the vortex lines in a type-two superconductor [15]. These vortex lines can approximately be identified with the Nambu-Goto string [16]-[12] but most of the more than 2000 citations of their paper do not address to that connection but to the role of these solutions as solitons in particle physics or cosmic strings in cosmology. To see how these solutions arise let us write the equations of motion for the Abelian Higgs model defined by Lagrangian (3.1.1) take the form

$$\begin{aligned} \partial^\mu F_{\mu\nu} &= j_\nu \\ D^\mu D_\mu \phi &= -\frac{\delta V}{\delta \phi^*} \end{aligned} \quad (3.1.13)$$

with the Higgs field current given by

$$j_\nu = \frac{ie}{2}(\phi\partial_\nu\phi^* - \phi^*\partial_\nu\phi) + e^2|\phi|^2 A_\nu \quad (3.1.14)$$

3.2 Nielsen-Olesen ansatz

The Nielsen-Olesen strategy to construct nontrivial regular finite energy (per unit length) solutions to the equations of starts from a trivial (constant) solution and implies the following steps:

1. Trivial zero-energy solution $\phi = \phi_0 \in \mathbf{R}$, $A_i = 0$
2. non-trivial but singular solution (“fluxon” with N units of magnetic flux $4\pi/e$) which in polar coordinates reads

$$\phi = \phi_0 \exp(-iN\varphi) , \quad A_i = \frac{N}{e} \partial_i \varphi \Rightarrow \varepsilon_{ij} F_{ij} = N \left(\frac{4\pi}{e} \right) \delta^{(2)}(\vec{x}) \quad (3.2.15)$$

One can easily prove this result by writing $z = x^1 + ix^2 = r \exp(i\varphi)$ and using the Cauchy-Riemann conditions for $\log z$ leading to $\varepsilon_{ij} \partial_i \partial_j \varphi = \nabla^2 \log r = 2\pi \delta^2(\vec{x})$. From this the magnetic flux takes the form

$$\Phi = \int \varepsilon_{ij} F_{ij} d^2x = N \left(\frac{4\pi}{e} \right) , \quad N \in \mathbb{Z} \quad (3.2.16)$$

with $\varepsilon_{ij} F_{ij}$ related to the z -component of the magnetic field

Configuration (3.2.15) is a gauge transformed of the trivial one if one accepts singular gauge transformations. Indeed, the appropriate gauge group element to consider is

$$g_N(\varphi) = \exp(-iN\varphi) \quad (3.2.17)$$

which is ill-defined at the origin.

3. Regular Nielsen-Olesen vortex ansatz

$$\phi = f(r) g_N(\varphi) , \quad A_i = a(r) \frac{1}{i} g_N(\varphi) \partial_i g_N^{-1}(\varphi) , \quad A_0 = 0 \quad (3.2.18)$$

with $f(r)$ and $a(r)$ real functions satisfying the following conditions at the origin and at infinity

$$f(0) = a(0) = 0 , \quad f(\infty) = \phi_0 , \quad a(\infty) = 1 \quad (3.2.19)$$

This ansatz leads to the same magnetic flux as that in (3.2.16) but there is no singularity in view of the boundary conditions that one imposes at the origin on f and a (note that at infinity the scalar field is at its vacuum value and the gauge field is a pure gauge (flat connection, no curvature)).

Being axially symmetric, the ansatz (3.2.18) can be interpreted as a magnetic tube with quantized flux (a vortex). Being z -independent, in 3 spatial dimensions such a solution has

infinite energy. However, if we define an energy E per unit length (along z), conditions (3.2.19) ensure its finiteness,

$$E = \int d^2x \left(\frac{1}{4} F_{ij}^2 + \frac{1}{2} |D_i \phi|^2 + \lambda (|\phi|^2 - \phi_0^2)^2 \right) \quad (3.2.20)$$

Being z independent, such a static vortex solution can be also interpreted as a soliton solution in $2 + 1$ dimensions. Because it depends only on x, y and $A_0 = 0$, it can also be thought as an instanton solution in $d=2$ Euclidean dimensions. In any two of this cases, space at infinity can be seen as a circle S_∞^1 .

Finite energy condition forces each term in the integrand of the r.h.s. of eq.(3.2.20) to vanish at infinity. That is

$$F_{ij}|_\infty = 0 \quad (3.2.21)$$

$$D_i \phi|_\infty = 0 \quad (3.2.22)$$

$$|\phi|_\infty^2 = \phi_0^2 \quad (3.2.23)$$

and this is guaranteed by eqs.(3.2.18)-(3.2.19)

The gauge transformation introduced in eq.(3.2.17) is a representative of the homotopy class N . Any other element of this class can be obtained from g_N by continuous deformations. In contrast, one can not deform an element in a given class into another of a different class. Given an element g in the class N , one can compute its “topological charge” or “winding number” T from the formula

$$T = \frac{i}{2\pi} \oint_C g_N \partial_i g_N^{-1} dx^i = N \quad (3.2.24)$$

since all elements in the class have the same topological charge. We then see the connection between the vortex magnetic flux (3.2.16) and the topological charge (3.2.24). This connection can be made more manifest through the following arguments. Condition (3.2.22) implies that

$$(\partial_i \phi + ie A_i \phi)|_\infty = 0 \quad (3.2.25)$$

or

$$-i\phi * \partial_i \phi|_\infty = e A_i|_\infty \quad (3.2.26)$$

Integrating both sides over the circle at infinity one then has

$$T = \frac{e}{2\pi} \oint_{C_\infty} A_i dx^i = \frac{e}{4\pi} \int \epsilon_{ij} F_{ij} d^2x \quad (3.2.27)$$

3.3 Topological conservation laws and homotopy classes

Before discussing the solutions of equations of motion (3.1.13) we shall describe here the topological aspects related to the flux results more in detail. In particular, we shall learn to divide the space of the initial value data associated to the equations of motion into disconnected components.

The analysis will be useful not just for the Abelian Brout-Englert-Higgs model. Instead of discussing the specific Abelian case, we shall consider that Higgs fields are n -vectors and the gauge group is a more general compact connected Lie group G with an n -dimensional unitary representation of G , $D(g)$. Given any function of space-time g which we indicate as $g(x)$ we define gauge transformation of a field ϕ which is

$$g(x) : \phi(x) \rightarrow D(g(x))\phi(x) \quad (3.3.28)$$

To simplify notation from here on we shall write the precedent equation in the form

$$g(x) : \phi(x) \rightarrow g(x)\phi(x) \quad (3.3.29)$$

The appropriate generators of the Lie algebra $\mathcal{G}$ associated to G will be $n \times n$ Hermitian matrices t^a with $a = 1, 2, \dots, \dim \mathcal{G}$ and commutation relations

$$[t^a, t^b] = if^{abc}t^c \quad (3.3.30)$$

The gauge field associated to these generators are a set of real vectors $A_\mu^a(x)$. Taken as elements of the Lie algebra of G they will be written as $A_\mu(x) = A_\mu^a(x)t^a$. Under a gauge transformation they change according to

$$A_\mu = g^{-1}A_\mu(x)g(x) - ie^{-1}g^{-1}(x)\partial_\mu g(x) \quad (3.3.31)$$

where e is a real constant called the gauge coupling.

From this objects one defines the covariant derivative of $\phi(x)$

$$D_\mu \phi = \partial_\mu \phi + ieA_\mu \phi \quad (3.3.32)$$

and the field strengths

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \quad (3.3.33)$$

which in components takes the form

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - ef^{abc}A_\mu^b A_\nu^c \quad (3.3.34)$$

Concerning the Lagrangian, it is the obvious generalization of the $U(1)$ one (eq.(3.1.1))

$$L = -\frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} + \frac{1}{2}(D_\mu\phi)^\dagger(D^\mu\phi) - V(|\phi|) \quad (3.3.35)$$

where V is now some invariant semi-positive function of scalar fields and equals zero at its minima. A ground state of the theory will be identified with a field configuration for which the gauge field vanishes and ϕ is a constant and a zero of V . For such ground state $\phi = \phi_0$ we define H as the subgroup of G by saying that h is an element of H if and only if $h\phi_0 = \phi_0$. For this reason H is called the unbroken group (or little group in mathematical literature).

If we study small oscillations about this ground state, the gauge fields associated with the generators of H remain massless, while the other gauge field components acquire a mass. Note that since V is gauge invariant, given the zero of V that we call ϕ_0 , $g\phi_0$ is also a zero of V . In what follows we shall make the following assumption:

- All the zeros of V are of the form $g\phi_0$ for some $g \in G$.

This assumption is equivalent to say that the set of zeros of V can be identified with the coset space G/H (The coset space is then composed of the zeros that are not identical after a gauge transformation).

One more ingredient that we need to topologically characterize a field configuration is the specification of space at infinity (i.e., $r \rightarrow \infty$), where the field goes to the minimum of the potential. Physically, this implies that all points at infinity are identified so that space is just a sphere S_∞^n with n the required number of angular variables to locate a point for a given number of spatial dimensions (the ∞ subindex indicates that one is dealing with the spatial n^{th} sphere).

Coming back to the ABEH model in which the gauge group is $U(1)$, since the vortex ansatz involves just 2 spatial coordinates (r, φ) so that space at infinity is just the circle S_∞^1 . Concerning the coset $G/H = U(1)/I = U(1)$. Now, the elements of $U(1)$ are complex numbers with modulus 1 so that they can be written as $g(x) = \exp(i\alpha(\varphi))$.

$$\phi_0 g(x) = \phi_0 \exp(i\alpha(\varphi)) \quad (3.3.36)$$

so that the $g(x)$ manifold can be identified with a circle which we denote $S_{U(1)}^1$ or simply S^1 . Elements $g(x)$ calculated on the circle S_∞^1 then establish a map between the circle at infinity and the group manifold

$$g(x)|_\infty : S_\infty^1 \rightarrow S_{U(1)}^1 \quad (3.3.37)$$

Such maps are classified by what are known as homotopy classes Π_1 according to the formula

$$\Pi_1(S^1) = \mathbb{Z} \quad (3.3.38)$$

(One usually represent homotopy classes with the letter Π , in this case with a subindex indicating the dimension of the sphere, the circle S_∞^1 . The argument indicates the symmetry group G/H , in this case $U(1) \sim S^1$).

Equation (3.3.38) means that there are different families of maps $g_1, g_2, \dots, g_n$ which correspond to equivalent classes. A representative of the family g_1 is $\exp(i\varphi)$, of the family g_2 is $\exp(2i\varphi)$, and so on. Any member of the n equivalence class can be continuously deformed to the above defined representative but not to one in the $m \neq n$ class.

A more detailed treatment of topology and homotopy classes can be found in the Appendix

3.4 The vortex solution

Unlike monopoles or instantons, no analytic solution to the equations of motion is known. However, numerical solutions can be easily constructed and there are also proofs that solutions exist (see [14] and references therein).

If one inserts the ansatz in the equations of motion one gets the following coupled non-linear second order radial equations:

$$a''(r) - \frac{1}{r}a'(r) - e(n + eA)f^2 = 0 \quad (3.4.39)$$

$$f''(r) + \frac{1}{r}f'(r) - \frac{1}{r^2}(n + ea(r))^2 f = 2\lambda f(f^2 - \phi_o^2) \quad (3.4.40)$$

which has to be solved with the conditions

$$f(0) = a(0) = 0 \quad (3.4.41)$$

$$\lim_{r \rightarrow \infty} f(r) = \phi_o \quad (3.4.42)$$

$$\lim_{r \rightarrow \infty} a(r) = -\frac{n}{e} \quad (3.4.43)$$

This equations has to be solved numerically. One can however find analytically the asymptotic behavior of the solutions,

$$\begin{aligned} a(r) &= -\frac{1}{e} + \sqrt{m_V r} \exp(\exp(-m_V r)) \\ \phi(r) &= \phi_0 + O\left(\frac{1}{\sqrt{r}} \exp(-m_\phi r)\right) \end{aligned} \quad (3.4.44)$$

where m_V and m_ϕ are the masses of the vector and scalar particles respectively,

$$m_V = e\phi_0 \quad (3.4.45)$$

$$m_\phi = 2\sqrt{\lambda}\phi_0 \quad (3.4.46)$$

Note that when condition (3.1.12) is satisfied both masses coincide, this being the first indication that the model can be viewed as describing the bosonic sector of an $N = 2$ supersymmetric theory with spontaneous gauge symmetry breaking. We shall come back to this connection in lecture 6. Also, it is when $m_V \sim m_\phi$ we have a well defined vortex line or a well defined string of extension $1/m_V \sim 1/m_\phi$. Inside the string one basically has magnetic field while the scalar field vanishes. Outside the string it is the magnetic field which is nearly zero while the Higgs scalar is practically in its vacuum value $|\phi| = \phi_0$.

Explicit solutions in the whole r range can be more easily found observing that, under the magic condition (3.1.12) the second order Euler-Lagrange equations can be reduced to first-order ones.

There are different ways of arriving to this result. One was discovered in [9] and starts by demanding that the space-space energy-momentum components (T_{rr} and $T_{\varphi\varphi}$) vanish. The other one establishes a bound (Bogomolny bound) for the vortex energy (per unit length) which is saturated whenever (3.1.12) holds [10]. Being configurations static, such a bound also corresponds to a bound for the action and hence they satisfy the equations of motion.

The Bogomol'nyi bound

Let us see how Bogomol'nyi found in ref.[10] a bound for the energy starting from the energy density calculated from the Lagrangian. (3.3.35). We first write the scalar field in terms of its real and imaginary components

$$\phi = (\phi^1, \phi^2) = (\phi^a), \quad a = 1, 2 \quad (3.4.47)$$

and scale fields and coordinates according to

$$\phi^a \rightarrow \phi_0 \phi^a, \quad A_i \rightarrow \phi_0 A_i, \quad x \rightarrow \frac{1}{e\phi_0} y \quad (3.4.48)$$

so that the energy reads now

$$E = \phi_0^2 \int d^2 y \left(\frac{1}{4} F_{ij}^2 + \frac{1}{2} D_i \phi^a D_i \phi^a + \kappa^2 (|\phi|^2 - 1)^2 \right) \quad (3.4.49)$$

where the covariant derivative D_i^{ab} is defined as

$$D_i^{ab} = \delta^{ab} \partial_i + \varepsilon^{ab} A_i \quad (3.4.50)$$

One easily sees that the energy can be rewritten as

$$\begin{aligned}
E = & \phi_0^2 \int d^2 y \left(\frac{1}{4} \left(F_{ij} \pm \frac{1}{2} \varepsilon_{ij} (\phi^a \phi^a - 1)^2 \right)^2 \right. \\
& + \frac{1}{4} (D_i \phi^a \mp \varepsilon^{ab} \varepsilon_{ij} D_j \phi^b)^2 + (\kappa^2 - 1) (\phi^a \phi^a - 1)^2 \\
& \left. \pm \frac{1}{4} \left(\varepsilon_{ij} F_{ij} \mp \frac{1}{2} \varepsilon^{ab} \varepsilon_{ij} \partial_i (\phi^a D_j \phi^b) \right) \right) \quad (3.4.51)
\end{aligned}$$

Here κ^2 as defined in eq.(3.1.12) relates e and λ according to $\kappa^2 = 4\lambda/e^2 = 1$ Now, the term

$$\frac{1}{2} \varepsilon^{ab} \varepsilon_{ij} \partial_i (\phi^a D_j \phi^b) \quad (3.4.52)$$

gives no contribution to the energy, as can be seen by using the Gauss-Stokes theorem and the fact that $D_i \phi^b = 0$ on S_∞^1 . Then, using (3.2.16) we have

$$\begin{aligned}
E = & \phi_0^2 \int d^2 y \left(\frac{1}{4} \left(F_{ij} \pm \frac{1}{2} \varepsilon_{ij} (\phi^a \phi^a - 1)^2 \right)^2 \right. \\
& + \frac{1}{4} (D_i \phi^a \mp \varepsilon^{ab} \varepsilon_{ij} D_j \phi^b)^2 + (\kappa^2 - 1) (\phi^a \phi^a - 1)^2 \left. \right) \pm \phi_0^2 \pi N \quad (3.4.53)
\end{aligned}$$

Were the third term absent, the energy would just be the sum of two positive definite terms plus a term proportional to the topological charge. This is precisely what happens for the Bogomol'nyi point $\kappa^2 = 1$ for which one then has

$$E \geq \phi_0^2 \pi |N| \quad (3.4.54)$$

The bound is attained when the two perfect squares vanish, and this leads to the so-called Bogomol'nyi or BPS equations for vortices,

$$\begin{aligned}
F_{ij} &= \mp \varepsilon_{ij} (\phi^2 - 1) \\
D_i \phi^a &= \pm \varepsilon^{ab} \varepsilon_{ij} D_j \phi^b \quad (3.4.55)
\end{aligned}$$

The ‘‘PS’’ refers to M. K. Prasad and C. M. Sommerfield who found exact monopole solutions for the non-Abelian version of the model for a particular (Bogomol'nyi) value of

the coupling constant [11] ($\lambda = 0$). That their solution corresponds to a Bogomol'nyi bound was first noted in [12].

As mentioned above, eqs.(3.4.55) were independently obtained by asking the stress tensor components T_{ij} to vanish. This is consistent with the eqs. of motion only if $\kappa^2 = 1$. When matter interaction exceeds that of the electromagnetic one ($\kappa^2 < 1$) vortices attract whereas in the opposite case they repel. In the former case a system of many superimposed vortices decays into separated vortices. In the case $\kappa^2 = 1$ vortices do not interact and this is consistent with the fact that the energy is proportional to the topological charge.

The solution of the Bogomolny equations has been found in [1]. The profiles of the scalar field, the gauge field and the magnetic field are shown in the next figure

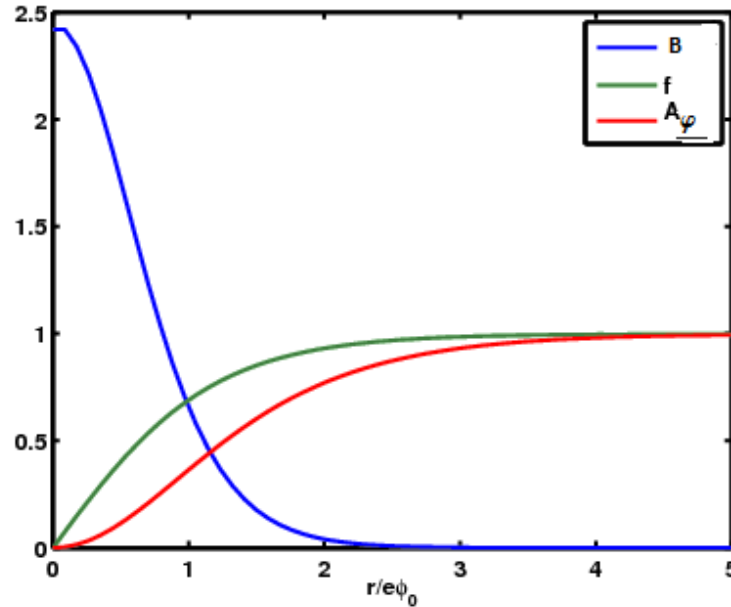


Figure 3.1

The Higgs field $f(r)$ (red) and the gauge field radial function $\alpha(r)$ (blue) as a function of r at the Bogomolny point

It should be noted that the so-called Bogomolny equations 3.4.55 were already reported in 1963 by J.L.Harpen and V. Arp in an article on “The Lower Critical Field in the Ginzburg-Landau Theory of Superconductivity” published in the Cryogenics journal [13]. Concerning

the exact (numerical) solution to the system (3.4.55) was presented in [9].

The existence theorem for vortex (and monopole) solutions can be found in A. Jaffe and C. H. Taubes book on Vortices and Monopoles [14].

3.5 The Ginzburg-Landau phenomenological theory for superconductivity

A material is called a superconductor when it shows zero electrical resistance and (a certain) expulsion of magnetic fields when cooled below a characteristic critical temperature. The phenomenon was discovered by Heike Onnes in 1911.

In 1950 Lev Landau and Vitaly Ginzburg proposed a phenomenological theory to describe superconductivity. In their argument, a Type-I superconductor had positive free energy of the superconductor-normal metal boundary. The “Type I” characterization means when an external magnetic field is applied at a given temperature, the superconductor has a single critical field, above which all superconductivity is lost. Their model was postulated without examining their microscopic origin and properties.

At that time, all accepted experimental results on superconductors corresponded to Type-I material but in the original paper Ginzburg and Landau observed the possibility of two types of superconductors depending on the energy of the interface between the normal and superconducting states. Landau considered physically acceptable only the positive energy density solution corresponding to the Type-I case. The second case with negative energy density, today called Type-II superconductivity, was considered unphysical.

There were however experiments done in 1937 on superconducting alloys Lev Shubnikov discovered anomalous behavior [19]². Because of this and of some other theoretical contradictions, Abrikosov decided to look what would happen for larger values of the GL parameter and his research led to the discovery of type-II superconductors.

In Type-II superconductivity there are two critical fields (that is the reason of the “II”) between which there is partial penetration of the magnetic field. The theory for the behavior of the Type-II superconducting state in magnetic field was developed by Alexei Alexeyevich Abrikosov and their existence in Nature was corroborated in thin films of metal alloys or complex oxide ceramics.

In Type I superconductors, superconductivity is abruptly destroyed when the strength of the applied field rises above a critical value H_c . In Type II superconductors, raising the

²These anomalies would have made him to discover the type-II superconductivity but unfortunately, he was accused of an attempt to organize an “anti-Soviet strike”, and was arrested and executed by the KGB the same year of his publication.

applied field past a critical value H_{c1} leads to a mixed state (also known as the vortex state) in which an increasing amount of magnetic flux penetrates the material, but there remains no resistance to the flow of electric current as long as the current is not too large. At a second critical field strength H_{c2} , superconductivity is destroyed.

In Ginzburg-Landau phenomenological model the equations of the theory depend only on a dimensionless “material” constant, which was later called the Ginzburg-Landau (GL) parameter. The value of this parameter could be defined from the surface energy between the normal and superconducting phases and the experimental data at that time corresponded to very small values of the GL parameter and therefore the calculations in the paper by Ginzburg and Landau were done for this limiting case, which corresponds to the domain in which type-I superconductivity takes place.

The order of the superconducting phase transition was long a matter of debate. In the 1980s it was shown theoretically with the help of a disorder field theory, in which the vortex lines of the superconductor play a major role, that the transition is of second order within the type II regime and of first order (i.e., latent heat) within the type I regime, and that the two regions are separated by a tricritical point.

3.5.1 Second-order phase transitions

Second order phase transitions, also called continuous phase transitions, are characterized by the following properties

- power-law decay of correlations
- a divergent correlation length as one approaches the critical temperature T_c
- power law behaviors for susceptibility, specific heat, ...

In the old (Ehrenfest) classification second-order phase transitions were characterized by the fact that the first derivative of the Gibbs free energy with respect to some thermodynamic variable is continuous (the order parameter, which is the first derivative of the free energy with respect to the external field, is continuous across the transition). This should be contrasted with first-order phase transitions for which the derivative of the Gibbs free energy is discontinuous (eg. in solid-liquid-gas transition, the density, which is given by derivative of the free energy with respect to the chemical potential, changes discontinuously. There is latent-heat, the property that defines this kind of transition in the modern classification).

Second order phase transitions exhibit discontinuities in higher order derivatives of the free energy. Examples of second-order phase transitions are

- ferromagnetic transition

- type-II superconductor transition
- superfluid transition

Landau phenomenological theory of superconductivity phase transitions

Generically second order phase transitions separate ordered and disordered phases. Accordingly, one introduces an order parameter Ψ such that

- ordered phase: $\Psi \neq 0$
- disordered phase: $\Psi = 0$

The order parameter can be a spin 0 field (s-wave) Ψ , a spin 1 field (p-wave) Ψ_μ , a spin 2 field (d-wave) $\Psi_{\mu\nu}$. In the case of superconductors there is a spin 0 order parameter Ψ that is related to the density of superconducting electrons

Landau assumed analyticity of the free energy $F = F(T, \Psi)$ so that it can be expanded in (even) powers of Ψ

$$F(T, \Psi) = F_0 + \alpha[T]\Psi^2 + \frac{\beta[T]}{2}\Psi^4 + \dots \quad (3.5.56)$$

Since one is interested in the behavior near $T = T_c$ the coefficients F_2 and F_4 are expanded in powers of $T - T_c$. In order to ensure stability and a change of behavior at T_c one takes, for temperatures just below T_c

$$\alpha = a(T - T_c), \quad \beta = b/2 \quad \text{with } a, b > 0 \quad (3.5.57)$$

Now, the minimum condition

$$\frac{\partial F}{\partial \Psi} = 0 \quad (3.5.58)$$

leads to the condition

$$\Psi(\alpha + 2\beta\Psi^2) = 0 \quad (3.5.59)$$

with two possible solutions

$$\Psi_1 = 0, \quad \Psi_2 = \frac{a}{2b}(T_c - T) \quad (3.5.60)$$

For $T > T_c$ the sensible solution is just the trivial one while for $T < T_c$ to describe the existence of an ordered phase the one has the non-trivial one,

$$\begin{aligned} \Psi &= 0 && \text{for } T > T_c && \text{disordered phase} \\ \Psi &\sim (T_c - T)^{1/2} && \text{for } T < T_c && \text{ordered phase} \end{aligned} \quad (3.5.61)$$

3.5.2 Type-I superconductivity

Consider a superconducting metal or alloy sample. As we have seen, in Landau theory the free energy is expanded in powers of a complex order parameter $\Psi(\vec{r})$ so that $|\Psi|^2$ corresponds to $1/2$ of the superconducting electron density. One should now include the contribution to the free energy of the magnetic field $\vec{H} = \vec{\nabla} \times \vec{A}(\vec{r})$ and also the spatial variations of the order parameter and the current flow associated to Ψ . There is no general criteria for not including higher derivatives in Ψ except by assuming that they are likely to be small when the free energy is small (In fact, in order to describe tricritical points Lifshitz developed a theory including higher derivatives [20]).

With all this, the free energy of the superconducting state takes, near a second order phase transition the form

$$\mathcal{F}_{sH_e} = \mathcal{F}_{n0} + \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4 + \frac{1}{8\pi}|\vec{\nabla} \times \vec{A}|^2 + \frac{1}{2m} \left| -i\hbar\vec{\nabla}\Psi - \frac{e^*}{c}\vec{A}\Psi \right|^2 \quad (3.5.62)$$

Here $\mathcal{F}_{sH_e}$ is the free energy of the superconducting state in an external magnetic field $\vec{H}_e$ which could exist in the absence of the specimen; $\mathcal{F}_{n0}$ is the free energy of the normal state in zero field, α and β are the phenomenological parameters already introduced in (3.5.56), m an effective mass with the same status as α and β ; e^* is the effective charge (two times that of the electron).

Note that the external field $\vec{H}_e$ does not appear explicitly in the r.h.s. of (3.5.62) but it is included in the magnetic energy since $\vec{\nabla} \wedge \vec{A}$ represents the magnetic field contribution measured with respect to H_e .

The fact that Ginzburg and Landau allow the order parameter to be a position dependent-function makes necessary to take into account the contribution to its change to the free energy through a derivative term. Gauge invariance then force to introduce a covariant derivative which is what appears in the last term of (3.5.62).

By minimizing the free energy with respect to fluctuations in the order parameter one arrives at the Ginzburg-Landau equation for Ψ together with its complex conjugate, to the Maxwell equation

$$\frac{1}{2m} \left(-i\hbar\vec{\nabla} - \frac{e^*}{c}\vec{A} \right)^2 \Psi + \alpha\Psi + \beta|\Psi|^2\Psi = 0 \quad (3.5.63)$$

$$\frac{c}{4\pi} \vec{\nabla} \wedge (\vec{\nabla} \wedge \vec{A}) = \frac{e^*\hbar}{2mi} (\Psi^*\vec{\nabla}\Psi - \Psi\vec{\nabla}\Psi^*) - \frac{(e^*)^2}{mc} |\Psi|^2 \vec{A} \quad (3.5.64)$$

Ec. (3.5.63) can be interpreted as the time independent Schrödinger equations for “Lagrangian” (3.5.62) and (3.5.64) as the Maxwell equation for $\vec{A}$ in the presence of a source

given by Schrödinger probability current. Hence, one can see Ψ as the usual Schrödinger spatial wave function. It should be stressed that Ψ is not the wave function for electrons in the material but the wave function for the correlated electron pairs or Cooper pairs³.

Also, eqs.(3.5.63)-(3.5.64) are nothing but the static equations for the Abelian Higgs model with a quartic potential, $A_0 = 0$. The Nielsen-Olesen ansatz with a non-trivial angular dependence in the Higgs field (which plays the role of the order parameter) just implies the formation of magnetic filaments (vortices) characteristic of type-II superconductivity. The Bogomolny point we discussed before just corresponds to the limiting point separating the two types of superconductivity.

For temperatures just below $T = T_c$ if the sample exhibits perfect Meissner effect no magnetic field can exist in the sample $\Psi(\vec{r})$ should take its zero-field value and $\vec{\nabla}\Psi \approx 0$. Landau and Ginzburg called Ψ_∞ the order parameter zero-field value to stress that one should be well inside the sample, far from its boundary.

Under these conditions eq.(3.5.63) reduces to

$$\alpha + \beta\Psi_\infty = 0 \quad (3.5.65)$$

One has to write a second equation for Ψ_∞ noting that in the case in which the applied magnetic field equals the critical field $\vec{H}_c$ above which superconductivity is destroyed the superconducting and normal free energies should be equal so that

$$\alpha|\Psi_\infty|^2 + \frac{\beta}{2}|\Psi_\infty|^4 = \frac{1}{8\pi}\vec{H}_c^2 \quad (3.5.66)$$

From these two equations one gets

$$\frac{1}{8\pi}\vec{H}_c^2 = \frac{\alpha^2}{2\beta} \quad (3.5.67)$$

Let us now consider the appropriate boundary condition for eq.(3.5.63) ensuring that no supercurrent leaves the sample. At the surface of the sample the order parameter should be zero so the natural boundary condition seems to be $\Psi|_{\text{bound}} = 0$. The convenient boundary condition is instead

$$\check{n} \cdot \left(i\hbar\vec{\nabla}\Psi + \frac{e^*}{c}\vec{A}\Psi \right) \Big|_{\text{bound}} = 0 \quad (3.5.68)$$

where $\check{n}$ is the surface normal.

³A Cooper pair is a composite boson (its total spin is 0 or 1) with twice the charge of the electron, i.e. $e^* = 2e$. Correspondingly the Ψ is symmetric under particle interchange. Concerning the effective mass m^* disregarding quantum corrections one can put it to the value $m^* = 2m$. The tendency for all the Cooper pairs in a body to “condense” into the same ground quantum state is responsible for the peculiar properties of superconductivity.

3.5.3 Type II superconductivity

In 2003, when he was 75 years old, Alexei Alexeievich Abrikosov received the Nobel Prize in Physics together with Vitaly Ginzburg and Antony Legget "for pioneering contributions to the theory of superconductors and superfluids".

Abrikosov contribution [15] for which he was awarded concerns type-II superconductors characterized for the formation of magnetic vortices in the interior of a bulk superconductor when a magnetic field H is applied. This should be compared with the case of type-I superconductivity in which the magnetic field is completely expelled from the material for $H < H_c$. Above H_c type-I superconductivity is lost.

There are two critical magnetic fields in type-II superconductivity: above a certain critical magnetic field H_{c1} magnetic vortices are formed. At a higher critical field H_{c2} superconductivity is completely destroyed.

We shall not discuss the Abrikosov solution of the Ginzburg-Landau equations leading to the possibility of type-II superconductivity with the formation of magnetic filaments inside the superconductor. We should stress that the equations cannot be integrated in closed form so that numerical approaches should be used. We shall just indicate the main results.

One defines the surface energy σ_{ns} in terms of Gibbs free energies $\mathcal{G}$. The Gibbs free energy is the Legendre transformed of the Helmholtz free energy $\mathcal{F}$ that we have been using and is more convenient because, as a thermodynamic function it depends on the magnetic field H and not on the magnetic induction $\vec{\nabla} \wedge \vec{A} = B$ introduced in $\mathcal{F}$, eq.(3.5.62).

The surface energy σ_{ns} is defined as the difference between the actual Gibbs free energy (per unit area) and its asymptotic value.

3.5.4 EGH model/Superconductivity correspondence

- The energy resulting from the BEH Lagrangian when static configurations are considered coincides with the free energy in the Ginzburg-Landau phenomenological model
- The parameter κ defined for the Abelian Higgs model in eq.(3.1.12) is nothing but the Ginzburg Landau parameter in their phenomenological model, with an appropriate normalization.
- In the sector where $\kappa < 1$ the range of the matter self-interaction exceeds that of the electromagnetic one, whereas for $\kappa > 1$ the opposite is true.
- Materials with $\kappa < 1$ exhibit type I superconductivity while those with $\kappa > 1$ exhibit type II superconductivity.

- The interaction between widely separated vortices has been studied and it has been found that vortices attract each other for $\kappa < 1$ and repel for $\kappa > 1$
- The stability of a rotationally symmetric configuration of many superimposed vortices has been investigated: it is found that for $\kappa > 1$ the vortex the system is unstable against decay into separated vortices.
- When $\kappa = 1$ it is possible to derive a lower bound on the energy.
- When $\kappa = 1$ vortices do not interact, $T_{ij} = 0$

3.6 Non Abelian vortices

The first reference to (local) vortex solutions in non-Abelian gauge theories can be found in the appendix of Nielsen and Olesen pioneer paper [8]. However, it was rapidly recognized that the boundary conditions imposed in their work correspond to solutions that could be deformed continuously to topologically trivial ones [22]. It was then understood that in order to have topologically stable solutions, symmetry breaking has to be complete [23]-[24] and it was then pointed that one needs $\Pi_1(G_L/H)$ to be nontrivial.

From here on we shall specialize to the case of a $SU(2)$ gauge group but the generalization to $SU(N)$ can be equally attacked [25]-[26].

To start with, let us stress that the choice of representation for the Higgs fields in order to have complete symmetry breaking is a central aspect for making the ansatz. Indeed, if we choose for the scalars the fundamental representation of $SU(2)$ or other faithful representation (i.e., isomorphic to the group, that in which which different elements g of the group are represented by distinct linear mappings $D(g)$) the fundamental group $\Pi_1(G/H)$ is trivial since H coincides with the identity one then has

$$\Pi_1(G/H) = \Pi_1(SU(2)/I) = \Pi_1(SU(2)) \quad (3.6.69)$$

Now, $SU(2)$ is simply connected, i.e. every path between two points in the group manifold can be continuously transformed, staying within the manifold, into any other such path while preserving the two endpoints in question. This can be seen recalling that $SU(2)$ is isomorphic to the sphere S^3 and spheres are simply connected⁴. Then, we have

$$\Pi_1(SU(2)) = 0 \quad (3.6.70)$$

⁴Just as an ordinary sphere (or S^2 sphere) is a two-dimensional surface that forms the boundary of a ball in three dimensions, a S^3 -sphere is an object with three dimensions that forms the boundary of a ball in four dimensions.

and there cannot exist non-trivial topological solution.

The situation changes if the scalars are in the adjoint representation, this meaning, for $SU(2)$ that they are 3-dimensional vectors ϕ^a ($a = 1, 2, 3$). Writing

$$\phi = \phi^a t^a \quad (3.6.71)$$

with t^a the generators in the adjoint, under a group transformation g they change in the form

$$\phi \rightarrow g^{-1} \phi g \quad (3.6.72)$$

Taken as a vacuum state a particular minimum of the Higgs potential, say ϕ_0 , not all elements of $SU(2)$ will change ϕ_0 to a new vacuum. Indeed, the so called center of the group which consist of the roots of the identity (the center Z_n of $SU(N)$ for $N \geq 2$) leaves ϕ_0 unchanged. For $SU(N)$ the elements of the center can be represented by the roots of the identity, which are of the form

$$g_n = \exp \left(i \frac{2\pi n}{N} \right) I \quad (3.6.73)$$

In the $SU(2)$ case the center $Z_2 = \{I, -I\}$, two elements which obviously leave invariant ϕ_0 which changes according to (3.6.72).

We conclude that the subgroup H leaving invariant the vacuum is Z_2 and then we have to consider the homotopy group $\Pi_1(SU(2)/Z_2) = Z_2$ (see appendix).

To search for vortex solutions recall the route followed by Nielsen-Olesen to find the $U(1)$ axially symmetric vortex ansatz:

$$\text{trivial : } \phi = \phi_0 \rightarrow \text{singular non-trivial : } \phi_0 \exp(in\varphi) \rightarrow \text{regular : } f(r) \exp(in\varphi)$$

The analogous to the singular nontrivial solution is now

$$\phi_{sing}(\varphi) = \phi_0 \begin{pmatrix} 0 & \exp(in\varphi) \\ \exp(-in\varphi) & 0 \end{pmatrix} \quad (3.6.74)$$

which can be written as

$$\phi_{sing}(\varphi) = \phi_0 (\sigma^1 \cos n\varphi + \sigma^2 \sin n\varphi + \sigma^3 0) \quad (3.6.75)$$

or, as a 3-vector

$$\phi_{sing}(\varphi) = \phi_0 \begin{pmatrix} \cos n\varphi \\ \sin n\varphi \\ 0 \end{pmatrix} \quad (3.6.76)$$

Now, the non-trivial regular solution is simply obtained by replacing ϕ_0 by a function $f(r)$ vanishing at the origin so as to cancel the singularity

$$\vec{\phi}(r, \varphi) = f(r) \begin{pmatrix} \cos n\varphi \\ \sin n\varphi \\ 0 \end{pmatrix}, \quad f(0) = 0, \quad f(\infty) = \phi_0 \quad (3.6.77)$$

Now, this configuration is still invariant under rotations in the third direction ($g_3 = \exp(\sigma^3 a_3)$). Since we need the symmetry to be completely broken (except for the center), we have to introduce a second Higgs scalar in the model chosen as to change under g_3 transformations. The simplest choice is just

$$\vec{\psi} = \psi_0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.6.78)$$

This ensures that the ansatz for ϕ and ψ is such that the two fields are orthogonal and hence the symmetry is completely broken. Note that proposing a constant instead of a function as in the other scalar ansatz we are ensuring orthogonality but a lower energy (no contribution either for the kinetic nor for the potential energies).

An important point concerning topologically inequivalent configurations is the following. Consider the $n = 2$ configuration taken as a vacuum state at infinity,

$$\phi^{(2)}(\varphi) = \phi_0 \begin{pmatrix} \cos 2\varphi \\ \sin 2\varphi \\ 0 \end{pmatrix}, \quad \psi^{(2)} = \psi_0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.6.79)$$

There exist a deformation $D(\varphi, \beta)$, $0 \leq \beta \leq 2\pi$ which connects this configuration with the trivial vacuum.

$$\begin{aligned} \vec{\phi}_D(\varphi, \beta) &= \begin{pmatrix} -1 + (1 + \cos \beta) \cos^2 \varphi \\ 1 + (1 + \cos \beta) \cos \varphi \sin \varphi \\ \sin \beta \cos \varphi \end{pmatrix}, \quad \vec{\psi}_D(\varphi, \beta) = \begin{pmatrix} \sin \beta \cos \varphi \\ -\sin \beta \sin \varphi \\ \cos \beta \end{pmatrix}, \quad 0 \leq \beta \leq \pi \\ \vec{\phi}_D(\varphi, \beta) &= \begin{pmatrix} \cos \beta \\ 0 \\ \sin \beta \end{pmatrix}, \quad \vec{\psi}_D(\varphi, \beta) = \begin{pmatrix} \sin \beta \\ 0 \\ \cos \beta \end{pmatrix}, \quad \pi \leq \beta \leq 2\pi \end{aligned} \quad (3.6.80)$$

Defined in this way the D deformation is continuous and interpolates between the configuration $n = 2$ and the $n = 0$ trivial configuration

$$\phi^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \psi^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.6.81)$$

The non-trivial configuration with for which one puts $n = 2k + 1$, are on their hand all equivalent to all equivalent to that with $n = 1$. Then, the topological charge of a configuration like (3.6.77) should not be associated to a topological charge n but with $n(\text{mod } 2)$, this confirming that the homotopy group is

$$\Pi_1(SU(2)/Z_2) = Z_2 \quad (3.6.82)$$

An important point concerning topologically inequivalent configurations is the following. Consider the $n = 2$ configuration taken as a vacuum state at infinity,

$$\phi^{(2)}(\varphi) = \phi_0 \begin{pmatrix} \cos 2\varphi \\ \sin 2\varphi \\ 0 \end{pmatrix}, \quad \psi^{(2)} = \psi_0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.6.83)$$

There exist a deformation $D(\varphi, \beta)$, $0 \leq \beta \leq 2\pi$ which connects this configuration with the trivial vacuum.

$$\begin{aligned} \vec{\phi}_D(\varphi, \beta) &= \begin{pmatrix} -1 + (1 + \cos \beta) \cos^2 \varphi \\ 1 + (1 + \cos \beta) \cos \varphi \sin \varphi \\ \sin \beta \cos \varphi \end{pmatrix}, \quad \vec{\psi}_D(\varphi, \beta) = \begin{pmatrix} \sin \beta \cos \varphi \\ -\sin \beta \sin \varphi \\ \cos \beta \end{pmatrix}, \quad 0 \leq \beta \leq \pi \\ \vec{\phi}_D(\varphi, \beta) &= \begin{pmatrix} \cos \beta \\ 0 \\ \sin \beta \end{pmatrix}, \quad \vec{\psi}_D(\varphi, \beta) = \begin{pmatrix} \sin \beta \\ 0 \\ \cos \beta \end{pmatrix}, \quad \pi \leq \beta \leq 2\pi \end{aligned} \quad (3.6.84)$$

Defined in this way the D deformation is continuous and interpolates between the configuration $n = 2$ and the $n = 0$ trivial configuration

$$\phi^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \psi^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.6.85)$$

The non-trivial configuration with for which one puts $n = 2k + 1$, are on their hand all equivalent to all equivalent to that with $n = 1$.

The, the topological charge of a configuration like (3.6.77) should not be associated to a topological charge n but with $n(\text{mod } 2)$, this confroming that the homotopy group is

$$\Pi_1(SU(2)/Z_2) = Z_2 \quad (3.6.86)$$

Let us now consider the dynamics of the model for which we shall search non-Abelian vortex solutions. It consist of gauge fields $A_\mu = A_\mu^a t^a$ taking values in the Lie algebra

of $SU(2)$ coupled to two scalars in the adjoint representation $\phi^a t^a$, $\psi^a t^a$ with a symmetry breaking potential. The Lagrangian governing dynamics of the model reads

$$L = -\frac{1}{4}F^{a\mu\nu}F_{\mu\nu}^a + \frac{1}{2}(D_\mu\phi)^a(D^\mu\phi)^a + \frac{1}{2}(D_\mu\psi)^a(D^\mu\psi)^a - V[\phi^a, \psi^a] \quad (3.6.87)$$

Since we are looking for static axially cylindrical solutions, instead of solving the equations of motion we can look for the minima of the energy (per unit length, along the z axis),

$$E = \int d^2x \left(\frac{1}{4}\vec{F}_{ij} \cdot \vec{F}_{ij} + \frac{1}{2}D_i\vec{\phi} \cdot D_i\vec{\phi} + \frac{1}{2}D_i\vec{\psi} \cdot D_i\vec{\psi} + V[\vec{\phi}, \vec{\psi}] \right) \quad (3.6.88)$$

where we have adopted the more convenient vector notation for object with the algebra index so that we write

$$\vec{F}_{ij} = \partial_i\vec{A}_j - \partial_j\vec{A}_i + e\vec{A}_i \wedge \vec{A}_j \quad (3.6.89)$$

$$D_i\vec{\phi} = \partial_i\vec{\phi} + e\vec{A}_i \wedge \vec{\phi} \quad (3.6.90)$$

$$D_i\vec{\psi} = \partial_i\vec{\psi} + e\vec{A}_i \wedge \vec{\psi} \quad (3.6.91)$$

Concerning the potential, we choose the form

$$V = \frac{g}{8}(\vec{\phi} \cdot \vec{\phi} - \eta^2) + \frac{g'}{8}(\vec{\psi} \cdot \vec{\psi} - \eta^2) + \frac{g''}{8}(\vec{\phi} \cdot \vec{\psi})^2 \quad (3.6.92)$$

The first two terms are the usual Higgs potentials for each one of the scalar fields. The third term has been added in order to impose the orthogonality of the potential minima (i.e., of the vacuum expectation values) in order to ensure topological non-triviality, as discussed above.

It will be convenient to proceed to the following change of variables so as to work with dimensionless fields,

$$\vec{A}_i \rightarrow \eta\vec{A}_i, \quad \vec{\phi} \rightarrow \eta\vec{\phi}, \quad \vec{\psi} \rightarrow \eta'\vec{\psi}, \quad r \rightarrow \frac{1}{e\eta}r \quad (3.6.93)$$

The energy per unit length now reads

$$\begin{aligned} E = & \eta^2 \int d^2x \left(\frac{1}{4}\vec{F}_{ij} \cdot \vec{F}_{ij} + \frac{1}{2}D_i\vec{\phi} \cdot D_i\vec{\phi} + \frac{\eta'^2}{2\eta^2}D_i\vec{\psi} \cdot D_i\vec{\psi} \right. \\ & \left. + \lambda_1(\vec{\phi} \cdot \vec{\phi} - 1)^2 + \lambda_2\frac{\eta'^2}{\eta^2}(\vec{\psi} \cdot \vec{\psi} - 1)^2 + \lambda_3\frac{\eta'^2}{\eta^2}(\vec{\phi} \cdot \vec{\psi})^2 \right) \end{aligned} \quad (3.6.94)$$

with

$$D_i \vec{\phi} = \partial_i \vec{\phi} + \vec{A}_i \wedge \vec{\phi} \quad (3.6.95)$$

and

$$\lambda_1 = \frac{g^2}{8e^2} \quad \lambda_2 = \frac{g'^2}{8e^2} \quad \lambda_3 = \frac{g''^2}{8e^2} \quad (3.6.96)$$

Finite energy condition implies the following conditions at infinity:

$$\lim_{r \rightarrow \infty} V[\vec{\phi}, \vec{\psi}] = 0 \quad (3.6.97)$$

$$\lim_{r \rightarrow \infty} D_i \vec{\phi} = 0$$

$$\lim_{r \rightarrow \infty} D_i \vec{\psi} = 0 \quad (3.6.98)$$

$$\lim_{r \rightarrow \infty} \vec{F}_{ij} = 0 \quad (3.6.99)$$

We have already proposed the ansatz for the Higgs fields, eqs.(3.6.77)-(3.6.78). Concerning the ansatz for $\vec{A}_i$, eq.(3.6.97) implies that

$$\lim_{r \rightarrow \infty} \vec{A}_i = - \lim_{r \rightarrow \infty} \check{\phi} \wedge \partial_i \check{\phi} \quad (3.6.100)$$

or

$$\lim_{r \rightarrow \infty} A_\varphi = -\frac{n}{r} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.6.101)$$

so that the appropriate ansatz is

$$\vec{A}_\varphi = -n \frac{a(r)}{r} \quad (3.6.102)$$

with

$$a(0) = 0, \quad a(\infty) = 1 \quad (3.6.103)$$

Concerning condition (3.6.98) the chosen ansatz implies that

$$D_i \psi = 0 \quad \forall r \quad (3.6.104)$$

This, and the fact that $\vec{\psi} \cdot \vec{\phi} = 0$ everywhere makes possible to write an inequality for the energy (even if the $\vec{\psi}$ field were not the constant making its symmetry breaking potential to vanish)

$$E \geq \bar{E} = \eta^2 \int d^2x \left(\frac{1}{4} \vec{F}_{ij} \cdot \vec{F}_{ij} + \frac{1}{2} D_i \vec{\phi} \cdot D_i \vec{\phi} + \lambda_1 (\vec{\phi} \cdot \vec{\phi} - \phi_0^2)^2 \right) \quad (3.6.105)$$

After some work one can see that this expression can be written à la Bogomolny as

$$\begin{aligned} \bar{E} = & \int d^2x \left(\frac{1}{4} \left(\vec{F}_{ij} + \frac{b}{(1+b^2)} \epsilon_{ij} (\vec{\phi} \cdot \vec{\phi} - 1) \vec{\psi} \cdot \vec{\psi} \right. \right. \\ & \left. \left. + \frac{1}{2} \frac{1}{(1+b^2)} (D_i \vec{\phi} + b \epsilon_{ij} \vec{\psi} \wedge D_j \vec{\psi})^2 + \left(\lambda_1 - \frac{1}{2} \frac{b^2}{1+b^2} \right) (\vec{\phi} \cdot \vec{\phi} - 1)^2 \right) \right. \end{aligned} \quad (3.6.106)$$

$$\left. - \frac{b}{1+b^2} \epsilon_{ij} \vec{F}_{ij} \cdot \vec{\psi} \right) \quad (3.6.107)$$

The last term in this equation can be related to a gauge invariant “electromagnetic” field $\mathcal{F}_{ij}$ constructed from the gauge covariant field strength $\vec{F}_{ij}$,

$$\mathcal{F}_{ij} = \vec{\psi} \cdot \vec{F}_{ij} \quad (3.6.108)$$

from which one can define a magnetic flux Φ

$$\Phi = \int d^2x \epsilon_{ij} \mathcal{F}_{ij} \quad (3.6.109)$$

It is always possible to choose a gauge in which $\epsilon_{ij} \vec{A}_i \wedge \vec{A}_j = 0$ so that one finds

$$\Phi = \int d^2x \epsilon_{ij} \vec{\psi} \cdot D_i \vec{A}_j = \int d^2x \epsilon_{ij} \partial_i (\vec{\psi} \cdot A_j) = \oint dx^i \vec{\psi} \cdot \vec{A}_i = 2\pi n \quad (3.6.110)$$

so that the last term in eq.(3.6.107) is proportional to the magnetic flux.

Now, we shall choose b so that the third term in eq.(3.6.107) is always positive or 0. We find

$$|b| = 1 \text{ if } \lambda_1 \geq \frac{1}{8} \quad (3.6.111)$$

$$\frac{b^2}{2(1+n^2)} = \lambda_1 \text{ if } \lambda_1 \leq \frac{1}{8} \quad (3.6.112)$$

and the bound is saturated when the following Bogomolny equations hold:

$$\vec{F}_{ij} + \frac{b}{1+b^2} \epsilon_{ij} (1 - \vec{\phi} \cdot \vec{\phi})^2 = 0 \quad (3.6.113)$$

$$D_i \vec{\phi} + b \epsilon_{ij} \vec{\psi} \wedge D_j \vec{\phi} = 0 \quad (3.6.114)$$

It is easy to see that these two equations are compatible solely if $b = \pm 1$ which in turn implies $\lambda_1 = 1$. With this, the complete set of Bogomolny equations becomes

$$\vec{F}_{ij} \pm \frac{1}{2} \epsilon_{ij} (1 - \vec{\phi} \cdot \vec{\phi})^2 = 0 \quad (3.6.115)$$

$$D_i \vec{\phi} \pm \epsilon_{ij} \vec{\psi} \wedge D_j \vec{\phi} = 0 \quad (3.6.116)$$

$$D_i \vec{\psi} = 0 \quad (3.6.117)$$

with positive and negative signs giving $n = \pm n$ so that when the energy bound is attained one has

$$E = \eta^2 \pi |n| \quad (3.6.118)$$

Recall that the topological stable solutions are those with $|n| = 1$. Concerning the solutions, they reduce to the Abelian ones already described.

When the group is $SU(N)$, one can find the appropriate ansatz inspired in the $SU(2)$ case. Indeed, note that in the $SU(2)$ case the gauge field ansatz consist in proposing the gauge field in the direction of σ^3 , which is the diagonal generator. In the case of $SU(N)$ one has $N - 1$ diagonal generators which are elements of the so-called Cartan algebra. In contrast, the $SU(2)$ ansatz for the Higgs field had components in the rest of the generators (written in the Cartan-Weyl basis). Details of the construction can be found in [26].

3.7 Vortices in the Chern-Simons-Higgs model

We learnt that gauge fields dynamics in $3 + 1$ dimensions is governed by Maxwell action

$$S_M = -\frac{1}{4} \int d^4x F_{\mu\nu} F^{\mu\nu} \quad (3.7.119)$$

The natural extension to the non-Abelian case is the Yang-Mills action that we wrote before

$$S_{YM} = -\frac{1}{2} \text{tr} \int d^4x F_{\mu\nu} F^{\mu\nu} \quad (3.7.120)$$

. These gauge invariant actions can be also written in any number of space-time dimensions. In particular, in $2 + 1$ dimensions they can be used to describe relevant planar physics (like for example in the case of high- T_c superconductivity, Hall effect, etc)

It is relatively recent that a second kind of gauge-invariant actions has been introduced in odd-dimensional space-times [27]-[29]. Let us consider the planar case with $2 + 1$ space-time

dimensions. In the Abelian case one can write the following action,

$$\begin{aligned} S_{CS}[A] &= -\mu \int d^3x \varepsilon^{\mu\nu\alpha} A_\mu \partial_\nu A_\alpha, \quad \mu, \nu, \alpha = 0, 1, 2 \\ &= -\frac{\mu}{2} \int d^3x \varepsilon^{\mu\nu\alpha} A_\mu F_{\nu\alpha} \end{aligned} \quad (3.7.121)$$

The following points should be stressed:

- In order to have a dimensionless action (working in units such that $\hbar = c = 1$) the factor μ in the action should have mass dimensions since $[A] = m^1$.
- The equations of motion deriving from $S_{CS}[A]$ are

$$\varepsilon^{\mu\nu\alpha} F_{\nu\alpha} = 0 \quad (3.7.122)$$

Then, the solution to the equations of motion are flat connections ($F_{\mu\nu} = 0$),

$$A_\mu^{sol} = \partial_\mu \Lambda(x) \quad (3.7.123)$$

- The Abelian Chern-Simons action is gauge invariant under certain conditions. Indeed, under a gauge transformation $A_\mu \rightarrow A'_\mu = A_\mu + \partial_\mu \Omega$ one has

$$\begin{aligned} S_{CS} &\rightarrow S_{CS} + \frac{\mu}{2} \int d^3x \varepsilon^{\mu\nu\alpha} (\partial_\mu \Omega) F_{\nu\alpha} = S_{CS} + \frac{\mu}{2} \int d^3x \varepsilon^{\mu\nu\alpha} \partial_\mu (\Omega F_{\nu\alpha}) \\ &= S_{CS} + \frac{\mu}{2} \int dS^\mu \varepsilon^{\mu\nu\alpha} (\Omega F_{\nu\alpha}) \end{aligned} \quad (3.7.124)$$

Hence, for field strengths going to zero at infinity as required for finite energy, the second term in the r.h.s. vanishes. Note however that when working at finite temperature T , for which the Euclidean time integral $d\tau$ goes from $\tau = 0$ to $\tau = 1/T$ there could be problems depending on the conditions one impose on gauge fields at the endpoints of the τ interval.

- Although up to now we have been working in flat space-time, also in curved space one can write an action which coincides with (3.7.126). Indeed the factor $\sqrt{|g|}$ ($g = \det g_{\mu\nu}$) that should be introduced to define the correct volume element in curved space-time is canceled out because the indices contraction is done with the Levi-Civita tensor $\varepsilon^{\mu\nu\alpha}/\sqrt{|g|}$ (and not with metric tensors, as is the case for Maxwell action).

Then, if one computes the energy-momentum tensor by making a metric variation of the action we get

$$T_{\mu\nu} = \frac{\delta S}{\delta g^{\mu\nu}} = 0 \quad (3.7.125)$$

Since this derivation holds also in flat space, the energy density given by T_{00} vanishes so that the Chern-Simons Hamiltonian is zero and the same for the space-space components: no non-trivial dynamics!

The non-Abelian extension of the Chern-Simons action (3.7.126) now with gauge fields $A_\mu^a t^a$ reads

$$\begin{aligned} S_{CS}[A] &= -\mu \operatorname{tr} \int d^3x \varepsilon^{\mu\nu\alpha} \left(A_\mu \partial_\nu A_\alpha + \frac{2}{3} A_\mu A_\nu A_\alpha \right) \\ &= -\frac{1}{2} \mu \operatorname{tr} \int d^3x \varepsilon^{\mu\nu\alpha} \left(A_\mu F_{\nu\alpha} - \frac{1}{3} A_\mu A_\nu A_\alpha \right) \end{aligned} \quad (3.7.126)$$

Originally, the physical application for the Chern-Simons theories that Deser, Jackiw and Templeton stressed was to provide a mass to Yang-Mills gauge fields without putting it by hand or by addition of Higgs fields [28]-[29]. Indeed, starting with the Lagrangian

$$L = L_{YM} + LCS \quad (3.7.127)$$

We have taken the non-Abelian gauge coupling to 1 (in some mass units). One can recover it by adding a $1/e^2$ factor in the Yang-Mills and Chern-Simons Lagrangians and a factor e in the A -cubic term.

The equations of motion for the non-Abelian Chern-Simons theory read

$$D_\mu F^{\mu\nu} + \frac{\mu}{2} \varepsilon^{\mu\nu\alpha} F_{\nu\alpha} = 0 \quad (3.7.128)$$

Defining $*F_\mu$, the curvature dual in d=3 space-time dimensions, as

$$*F_\mu = \frac{1}{2} \varepsilon_{\mu\nu\alpha} F^{\nu\alpha} \quad (3.7.129)$$

which satisfies the Bianchi identity

$$D_\mu *F_\mu = 0 \quad (3.7.130)$$

Now, the dual of (3.7.128) is

$$D_\alpha *F_\beta - D_\beta *F_\alpha - \mu F_{\alpha\beta} = 0 \quad (3.7.131)$$

where we have used that

$$*(F) = F \quad (3.7.132)$$

Taking another covariant derivative in eq.(3.7.131) and using the non-Abelian version of Ricci identity for the gauge field curvature (the field strength $F_{\mu\nu}$),

$$[D_\mu, D_\nu] = F_{\mu\nu} \quad (3.7.133)$$

eq.(3.7.128) can be written as

$$(D_\alpha D^\alpha + \mu^2)F_\mu = \varepsilon_{\mu\nu\beta}[*F^\nu, *F^\beta] \quad (3.7.134)$$

Taking the linear approximation of the equation (3.7.134) one can determine the gauge field mass $m_A^2 = \mu^2$.

When the Yang-Mills field is absent, the equation of motion for the non-Abelian Chern-Simons theory is analogous to the Abelian one

$$\varepsilon^{\mu\nu\alpha}F_{\mu\nu} = 0 \quad (3.7.135)$$

Now, in the non-Abelian case this is particularly relevant because flat connections are in one to one correspondence with equivalence classes of homomorphisms from the first homotopy group Π_1 on the space-time manifold M to the gauge group G up to conjugation,

$$\phi : \Pi_1(M) \rightarrow G \quad (3.7.136)$$

In a famous paper, Witten exploited this fact by giving a physical interpretation to Donaldson theory of Jones polynomials. Such problem was posed in a lecture at the Hermann Weyl Symposium last year [1] by Michael Atiyah proposed two problems for quantum field theorists. He also posed the problem to find an intrinsically three dimensional definition of the Jones polynomial of knot theory⁵. The two problems were solved by Witten who got the Fields medal for his achievement.

The way we solved the problems was based in a quantum field theory calculation, using path-integrals, of vacuum expectation values of Wilson loops $W_R[C]$

$$W_R[C] = \text{tr}_R P \exp \left(\int_C A_i dx^i \right), \quad i = 1, 2, 3 \quad (3.7.137)$$

⁵ As pointed by Witten “Jones polynomial and its generalizations deal with the mysteries of knots in three dimensional space. The puzzle on the mathematical side was that these objects are invariants of a three dimensional situation, but one did not have an intrinsically three dimensional definition”...“ This is analogous to studying a physical theory that is in fact relativistic but in which one does not know of a manifestly relativistic formulation - like quantum electrodynamics in the 1930’s.”

$$\langle W_R[C] \rangle = \int DA_i \exp(iS_{CS}[A]) \prod_{i=1}^r W_R[C_i] \quad (3.7.138)$$

where C_i are orientable closed curves, R an irreducible representation, P the path-order product. In an oriented three manifold M , one takes r oriented and non-intersecting knots C_i , $i = 1 \dots r$, whose union is what knot theorists would call a "link" L .

In contrast with the Yang-Mills action, the Chern-Simons action is not gauge invariant. Indeed, under the gauge field transformation

$$A_\mu \rightarrow g^{-1} A_\mu g + g^{-1} \partial_\mu g \quad (3.7.139)$$

the Chern-Simons Lagrangian transforms according to

$$L_{CS} \rightarrow L_{CS} + \mu \varepsilon^{\alpha\beta\gamma} \partial_\alpha \text{tr} (\partial_\beta g g^{-1} A_\gamma) + 8\pi^2 \mu^2 \omega[g] \quad (3.7.140)$$

with

$$\omega[g] = \frac{1}{24\pi^2} \varepsilon^{\alpha\beta\gamma} \text{tr} ((g^{-1} \partial_\alpha g)(g^{-1} \partial_\beta g)(g^{-1} \partial_\gamma g)) \quad (3.7.141)$$

Using this expression to determine the behavior of the action under gauge transformation we see that even if one imposes appropriate boundary conditions so as to eliminate the total derivative term in the r.h.s., there is still the $\omega[g]$ term

$$\Omega[g] = \int d^3x \omega(g) = \int d^3x \frac{1}{24\pi^2} \varepsilon^{\alpha\beta\gamma} \text{tr} ((g^{-1} \partial_\alpha g)(g^{-1} \partial_\beta g)(g^{-1} \partial_\gamma g)) \quad (3.7.142)$$

It can be recognized that $\Omega[g]$ is the *winding number* of the gauge element g satisfying the usual boundary condition (in Euclidean space-time)

$$\lim_{r \rightarrow \infty} g = I \quad (3.7.143)$$

This can be interpreted as the analogous of the $U(1)$ formula

$$-\frac{i}{2\pi} \oint g^{-1} \partial_i g dx^i = n \in \mathbb{Z} \quad (3.7.144)$$

with g so that at the origin behaves as $\exp(in\varphi)$ (i.e. singular at the origin).

Condition (3.7.143) compactifies Euclidean space time R^3 to S^3 so that g is defining an application

$$g : S^3 \rightarrow G \quad (3.7.145)$$

with G the gauge group. As in the case we discussed for the Abelian vortex, this applications are classified by the homotopy group $\Pi_3(G)$ which, for the usual $SU(N)$ gauge groups we are interested in can be determined to be (see appendix)

$$\Pi_3(SU(N)) = \mathbb{Z} \quad (3.7.146)$$

and the way to compute the winding number is precisely to calculate $\Omega[g]$,

$$\Omega[g] = n \in \mathbb{Z} \quad (3.7.147)$$

Then, under a gauge transformation with group element g the action transforms as

$$S_{CS}[A^g] = S[A] + 8\pi^2 \mu \Omega[g] \quad (3.7.148)$$

Now, in the context of quantization of gauge theories within the path integral framework the action enters solely through the exponential $\exp(iS[A])$ (Note that the i factor is present both in Minkowski and Euclidean space-time due to the metric independence of the Chern-Simons action). Then only an appropriate choice of μ can lead to a gauge invariant exponential

$$4\pi \frac{\mu}{e^2} = m \quad (3.7.149)$$

so that the change in the action is $2\pi mn$ and $\exp(iS)$ remains invariant. Here we have recovered the $1/e^2$ coupling constant factor in the Chern-Simons action (note that in $2+1$ dimensions, $[e] = m^{1/2}$).

We shall now look for vortex solutions coupling to $SU(2)$ non-Abelian gauge fields with dynamics given by a Chern-Simons action instead of a Yang-Mills action and, as before, two Higgs scalars in the adjoint representation. The Lagrangian reads

$$L = \frac{\mu}{4} \epsilon^{\mu\nu\alpha} \left(\vec{F}_{\mu\nu} \vec{A}_\alpha - \frac{1}{3} \vec{A}_\alpha (\vec{A}_\mu \wedge \vec{A}_\nu) \right) + \frac{1}{2} D_\mu \vec{\phi} D^\mu \vec{\phi} + \frac{1}{2} D_\mu \vec{\psi} D^\mu \vec{\psi} - V[\vec{\phi}, \vec{\psi}] \quad (3.7.150)$$

Note that we have not included a Yang-Mills Lagrangian so that the gauge field dynamics is governed just by the Chern-Simons term. Concerning the potential we shall write it in the form

$$V[\vec{\phi}, \vec{\psi}] = V_1[\vec{\phi}] + V_2[\vec{\psi}] + g(\vec{\phi} \cdot \vec{\psi})^2 \quad (3.7.151)$$

For the moment we do not fix the potentials V_1 and V_2 . The third term ensures that as in the previously discussed Yang-Mills-Higgs system the two scalars are orthogonal.

The Euler Lagrange equations associated to Lagrangian are

$$\frac{\mu}{2}\epsilon^{\nu\alpha\beta}\vec{F}_{\alpha\beta} = \vec{J}^\nu = \vec{\phi} \wedge D^\nu \vec{\phi} + \vec{\psi} \wedge D^\nu \vec{\psi} \quad (3.7.152)$$

$$\begin{aligned} D_\mu D^\mu \vec{\phi} &= -\frac{\delta V}{\delta \vec{\phi}} \\ D_\mu D^\mu \vec{\psi} &= -\frac{\delta V}{\delta \vec{\psi}} \end{aligned} \quad (3.7.153)$$

If one considers the case $\nu = 0$ in eq.(3.7.152) one gets

$$\frac{\mu}{2}\epsilon^{ij}\vec{F}_{ij} = \vec{J}^0 \quad (3.7.154)$$

or, multiplying by $\check{\psi}$ to the left

$$\frac{\mu}{2}\epsilon^{ij}\mathcal{F}_{ij} = \sigma \quad (3.7.155)$$

where

$$\mathcal{F}_{ij} = \check{\psi} \cdot \vec{F}_{ij} \quad (3.7.156)$$

$$\sigma = \check{\psi} \cdot (\vec{\phi} \wedge \vec{D}^0 \vec{\phi}) \quad (3.7.157)$$

Eq (3.7.155) can be rewritten as

$$\mu B = \sigma \quad (3.7.158)$$

where we have introduced the magnetic field defined as

$$B = \frac{1}{2}\epsilon^{ij}\mathcal{F}_{ij} \quad (3.7.159)$$

Now, integrating this equation on spatial coordinates we get

$$\mu\Phi = Q \quad (3.7.160)$$

where the the magnetic flux Φ and the electric charge Q are given by

$$\Phi = \int d^2x B, \quad Q = \int d^2x \sigma \quad (3.7.161)$$

We see that in the presence of a CS term if magnetic vortices exist as solutions of the Chern-Simons-Higgs system, they should carry both magnetic flux and electric charge. We have

now to determine whether such configurations have finite energy we should compute the energy-momentum tensor

$$T_{\mu\nu} = \frac{\delta S}{\delta g^{\mu\nu}} = \frac{2}{\sqrt{|g|}} \frac{\delta L}{\delta g^{\mu\nu}} \quad (3.7.162)$$

As stressed above, since the Chern-Simons action does not depend on the metric, it does not contribute to $T_{\mu\nu}$. Hence, for static configurations as the one we are looking for the energy is given by

$$E = \int d^2x T_{00} = \int d^2x \left(\frac{e^2}{2} (\vec{A}_0 \wedge \vec{\phi})^2 + \frac{e'^2}{2} (\vec{A}_0 \wedge \vec{\psi})^2 + \frac{1}{2} (D_i \vec{\phi})^2 + \frac{1}{2} (D_i \vec{\psi})^2 + V[\vec{\phi}, \vec{\psi}] \right) \quad (3.7.163)$$

Even if the Chern-Simons term does not explicitly appear in the energy, we shall see that the temporal component of eq.(3.7.152) (the Gauss law)

$$e(\vec{\Phi} \wedge D^0 \vec{\phi}) = \frac{\mu}{2} \varepsilon^{ij} \vec{F}_{ij} \quad (3.7.164)$$

does introduce a dependence on its presence.

The boundary conditions are the same that we imposed to the $SU(2)$ Yang Mills-Higgs vortices. Concerning the scalars finite energy requires

$$\lim_{r \rightarrow \infty} D_\mu \vec{\phi} = 0 \quad (3.7.165)$$

$$\lim_{r \rightarrow \infty} D_\mu \vec{\psi} = 0 \quad (3.7.166)$$

$$\lim_{r \rightarrow \infty} V[\vec{\phi}, \vec{\psi}] = 0 \quad (3.7.167)$$

$$\lim_{r \rightarrow \infty} \vec{F}_{\mu\nu} = 0 \quad (3.7.168)$$

Condition (3.7.168) can be inferred in this case from eqs.(3.7.165)- (3.7.166) and the equations of motion.

We shall look for static solutions and the complete ansatz will be the same as for the Yang-Mills-Higgs vortices

$$\vec{\phi}(r, \varphi) = \eta f(r) \begin{pmatrix} \cos n\varphi \\ \sin n\varphi \\ 0 \end{pmatrix}, \quad \vec{\psi} = \psi_0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \vec{A}_\varphi = \frac{a(r) - n}{er} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.7.169)$$

We see that this ansatz implies that $D_\mu \vec{\psi} = 0$, $\vec{\psi} \perp \vec{\phi}$, $V_1[\vec{\psi}] = 0$.

Using the Gauss law (i.e. the O^{th} component of the gauge field equation (3.7.152)) we have

$$e(\vec{\Phi} \wedge D^0 \vec{\phi}) = \frac{\mu}{2} \varepsilon^{ij} \vec{F}_{ij} \equiv \mu \vec{B} \quad (3.7.170)$$

this implying

$$\vec{\phi} \cdot \vec{B} = 0 \quad (3.7.171)$$

Moreover, multiplying by $\vec{\phi}$ using the wedge product both sides of eq.(3.7.170) one gets

$$\vec{A}_0 \wedge \vec{\phi} = \frac{\mu}{e^2 |\vec{\phi}|^2} \vec{B} \wedge \vec{\phi} \quad (3.7.172)$$

which can be inserted in the expression for the energy in eq. (3.7.163)

$$E = \int d^2x T_{00} = \int d^2x \left(\frac{\mu}{2e^2 |\vec{\phi}|^2} |\vec{B}|^2 + \frac{1}{2} (D_i \vec{\phi})^2 + V_1[\vec{\phi}] \right) \quad (3.7.173)$$

At this point we try to write the energy à la Bogomolnyi, i.e, as a sum of squares,

$$\begin{aligned} E = \int d^2x & \left(\left(\frac{1}{4} D_i \vec{\phi} \pm \varepsilon_{ij} \vec{\psi} \wedge D_j \vec{\phi} \right)^2 + \frac{\mu}{2e^2 |\vec{\phi}|^2} \left(\vec{B} \mp \frac{e^2}{2\mu^2} |\vec{\phi}|^2 (|\vec{\phi}|^2 - \eta^2) \vec{\psi} \right)^2 \right. \\ & \left. + V_1[\vec{\phi}] - \frac{e^2}{8\mu^2} |\vec{\phi}|^2 (|\vec{\phi}|^2 - \eta^2)^2 \mp \frac{e}{2} \eta^2 (\vec{\psi} \cdot \vec{B}) \right) \end{aligned} \quad (3.7.174)$$

Then, if we choose for the not yet specified $\vec{\phi}$ potential the form

$$V_1[\vec{\phi}] = \frac{e^2}{8\mu^2} |\vec{\phi}|^2 (|\vec{\phi}|^2 - \eta^2)^2 \quad (3.7.175)$$

we get a bound for the energy

$$E \geq \mp \frac{e}{2} \eta^2 \int d^2x (\vec{\psi} \cdot \vec{B}) \quad (3.7.176)$$

We recognize in the integral the magnetic flux defined in eq.(3.7.161) which for our ansatz is given by

$$\Phi = \int d^2x \vec{\psi} \cdot \vec{B} = \frac{2\pi n}{e} \quad (3.7.177)$$

so that the bound can be written as

$$E \geq \eta^2 \pi n \quad (3.7.178)$$

The bound is saturated when the following Bogomolny equations hold

$$\begin{aligned}\frac{1}{4}D_i\vec{\phi} \pm \varepsilon_{ij}\vec{\psi} \wedge D_j\vec{\phi} &= 0 \\ \vec{B} \mp \frac{e^2}{2\mu^2}|\vec{\phi}|^2(|\vec{\phi}|^2 - \eta^2)\vec{\psi} &= 0 \\ D_i\vec{\psi} &= 0\end{aligned}\tag{3.7.179}$$

In $d = 2 + 1$ space-time a sixth order potential as that in eq.(3.7.175) is acceptable since it leads to a renormalizable term in the Lagrangian. Note that the coupling constant in the potential is not arbitrary but related to the gauge coupling constant. As it happened for the Abelian Bogomolny equations, here also in the non Abelian case the present relation is required and again coincides with the one needed for the $\mathcal{N} = 2$ supersymmetric extension of the model. Also, the masses of the vector and the Higgs bosons coincide due to such relation.

Concerning the explicit solutions for the radial fields $a(r)$ and $f(r)$ one obtains the following first order equations

$$\begin{aligned}f' &= \pm \frac{af}{r} \\ \frac{a'}{r} &= \pm \frac{e^2\eta^4}{2\mu^2}f^2(f^2 - 1)\end{aligned}\tag{3.7.180}$$

The numerical solution of these equations show that the magnetic field vanishes at the origin so that the vortex is no more like a flux tube but it has an annular shape.

From the relation (3.7.160) between magnetic flux and electric charge one gets

$$Q = \frac{1}{2}mne\tag{3.7.181}$$

so that the vortex charge is also quantized. The same happens for the angular momentum,

$$J = \int d^2x \varepsilon^{ij} x_i T_{0j} = \int d^2x D_0\vec{\phi} \cdot D_j\vec{\phi} = \frac{mn^2}{4}\tag{3.7.182}$$

where we have used the quantization of the CS coefficient and the explicit form of the scalar and gauge field ansatz.

As an example, for $m = n = 1$ the vortex has electric charge $e/2$, magnetic flux $\Phi = 2\pi/e$ and angular momentum $J = 1/4$ so that it satisfies a fractional statistics. To see this, recall that for spin s particles two states in which the particles are interchanged are related according to

$$|12\rangle = \exp(i\Delta\Theta)|21\rangle\tag{3.7.183}$$

with the statistical parameter $\Delta\Theta = 2\pi s$ (with s the spin) giving a $+$ ($-$) factor for bosons (fermions). Now in the present case we have $J = 1/4$ so that

$$\Delta\Theta = 2\pi J = \pi/2 \tag{3.7.184}$$

so the vortex obeys a semi-integer Fermi statistic⁶, a property welcome in condensed matter systems like high- T_c superconductors and the Hall effect.

⁶Some authors call objects with this statistics “semions”, see [30]

Bibliography

- [1] F. Englert and R. Brout, “Broken Symmetry and the Mass of Gauge Vector Mesons,” Phys. Rev. Lett. **13** (1964) 321.
- [2] P. W. Higgs, “Broken symmetries, massless particles and gauge fields,” Phys. Lett. **12** (1964) 132.
- [3] P. W. Higgs, “Broken Symmetries and the Masses of Gauge Bosons,” Phys. Rev. Lett. **13** (1964) 508.
- [4] G. S. Guralnik, C. R. Hagen and T. W. B. Kibble, Phys. Rev. Lett. **13** (1964) 585.
- [5] A. Salam and J. Strathdee, “Supersymmetry, Parity And Fermion - Number Conservation,” Nucl. Phys. **B97**(1975)293.
- [6] P. Fayet, Il Nuovo Cimento, “Higgs Model And Supersymmetry,” **A31** (1976)626.
- [7] P. Di Vecchia and S. Ferrara, “Classical Solutions in Two-Dimensional Supersymmetric Field Theories,” Nucl. Phys. B **130** (1977) 93.
- [8] H. Nielsen and P. Olesen, “Vortex-line models for dual strings,” Nucl. Phys. **B61** 45 (1973) [Reprinted in *Solitons and Particles*, Eds. C. Rebbi and G. Soliani (World Scientific, Singapore, 1984), p. 365].
- [9] H. J. de Vega and F. A. Schaposnik, “A Classical Vortex Solution of the Abelian Higgs Model,” Phys. Rev. D **14** (1976) 1100.
- [10] E. B. Bogomolny, “Stability of Classical Solutions,” Sov. J. Nucl. Phys. **24** (1976) 449 [Yad. Fiz. **24** (1976) 861].
- [11] M. K. Prasad and C. M. Sommerfield, “An Exact Classical Solution for the ’t Hooft Monopole and the Julia-Zee Dyon,” Phys. Rev. Lett. **35** (1975) 760.

- [12] S. R. Coleman, S. J. Parke, A. Neveu and C. M. Sommerfield, “Can One Dent a Dyon?,” *Phys. Rev. D* **15** (1977) 544.
- [13] [1] J. L. Harden and V. Arp, “The Lower Critical Field in the Ginzburg-Landau Theory of Superconductivity”, *Cryogenics* **3** (1963) 105
- [14] J. Jaffe and C. H. Taubes, “Vortices and monopoles: structure of static gauge theories”, Birkhäuser, Boston, 1980.
- [15] A. A. Abrikosov, *Doklady Akademii Nauk. SSSR* **86** (1952) 489.
- [16] Y. Nambu, Lectures at the Copenhagen Summer Symposium (1970).
- [17] T. Goto, *Relativistic Quantum Mechanics of One-Dimensional Mechanical Continuum and Subsidiary Condition of Dual Resonance Model*, *Prog. Theor. Phys.* **46** (1971) 1560.
- [18] V. L. Ginzburg and L. D. Landau, *Zh. Eksp. Teor. Fiz.* **20** (1950) 1064.
- [19] L. V. Shubnikov et al., *Zh. Eksp. Teor. Fiz.* **7** (1937) 221.
- [20] E. M. Lifshitz, “On the theory of second order phase transitions I”, *Zh. Eksp. Teor. Fiz.* **11** (1941) 255; On the theory of second-order phase transitions II, *Zh. Eksp. Teor. Fiz.* **11** (1941) 269.
- [21] J.L Harden, V Arp, “The lower critical field in the Ginzburg-Landau theory of superconductivity”, *Cryogenics* **3** (1963) 105.
- [22] S. Mandelstam, “Vortices and Quark Confinement in Nonabelian Gauge Theories,” *Phys. Lett. B* **53**, 476 (1975).
- [23] H. J. de Vega, “Fermions And Vortex Solutions In Abelian And Nonabelian Gauge Theories,” *Phys. Rev. D* **18** (1978) 2932.
- [24] A. S. Schwarz and Y. S. Tyupkin, “Vortices In Unified Theories Of Weak And Electromagnetic Interactions,” *Phys. Lett. B* **90** (1980) 135.
- [25] H. J. de Vega and F. A. Schaposnik, “Electrically Charged Vortices in Nonabelian gauge theories with Cher-Simons Term,” *Phys. Rev. Lett.* **56**, 2564 (1986).
- [26] H. J. de Vega and F. A. Schaposnik, “Vortices and electrically charged vortices in non-Abelian gauge theories,” *Phys. Rev. D* **34**, 3206 (1986).

- [27] A. Schwarz, “The partition function of degenerate quadratic functional and Ray-Singer invariants”. *Lett. Math. Phys.* **2** (1978) 247.
- [28] S. Deser, R. Jackiw and S. Templeton, “Three-Dimensional Massive Gauge Theories,” *Phys. Rev. Lett.* **48** (1982) 975.
- [29] S. Deser, R. Jackiw and S. Templeton, *Annals Phys.* **140** (1982) 372 [Erratum-ibid. **185** (1988) 406] [*Annals Phys.* **185** (1988) 406] [*Annals Phys.* **281** (2000) 409].
- [30] A. Zee, “Semionics, a theory of High Temperature Superconductivity”, in *High temperature superconductivity*, Ed. K. Bedell et al, Addison Wesley, Boston, 1990.

Chapter 4

Monopoles

4.1 The Dirac monopole

4.1.1 Dirac modifies Maxwell equations

In his quest of beauty and symmetry Dirac produced in 1931 a beautiful article [1] in which he modified the Maxwell equations making them more symmetric and finding

"a reason for the existence of a smallest electric charge, given approximately by

$$\hbar c/e^2 = 137. \quad (4.1.1)$$

The theory

"while it looks at first as though it will give a theoretical value of e , is found when we worked out to give a connection between the smallest electric charge and the smallest magnetic pole. It shows in fact a symmetry between electricity and magnetism quite foreign to current views".

Maxwell equations can be written as

$$\vec{\nabla} \cdot \vec{E} = \rho \quad \vec{\nabla} \wedge \vec{B} - \dot{\vec{E}} = \vec{j} \quad (4.1.2)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \vec{\nabla} \cdot \dot{\vec{E}} + \dot{\vec{B}} = 0 \quad (4.1.3)$$

or, more compactly,

$$\partial_\mu F^{\mu\nu} = j^\nu \quad (4.1.4)$$

$$\partial_\mu^* F^{\mu\nu} = 0 \quad (4.1.5)$$

where

$$F^{0i} = -E^i, \quad F^{ij} = \varepsilon^{ijk} B^k \quad (4.1.6)$$

where j^μ is the electric 4-current and $*F^{\mu\nu}$ is the dual tensor of $F^{\mu\nu}$,

$$*F^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta} \quad (4.1.7)$$

In vacua the electric 4-current vanishes and the Maxwell equations are symmetric under the “duality” transformation

$$F^{\mu\nu} \rightarrow *F^{\mu\nu}, \quad *F^{\mu\nu} \rightarrow -F^{\mu\nu} \quad (4.1.8)$$

or, equivalently, $\vec{E} \rightarrow \vec{B}$ and $\vec{B} \rightarrow -\vec{E}$, which interchanges electricity with magnetism. If we wish this symmetry to be valid in the presence of matter we would have to introduce a magnetic current $(k_\rho) = (\sigma, \vec{k})$, on the right-hand side of equation (4.1.3) and consequently in (4.1.5), so that the new field equations take the form

$$\partial_\mu F^{\mu\nu} = j^\nu \quad \partial_\mu *F^{\mu\nu} = k^\nu \quad (4.1.9)$$

Equations (4.1.9) are now symmetric under the duality transformation of equations (4.1.8) augmented by:

$$j^\mu \rightarrow k^\mu \quad k^\mu \rightarrow -j^\mu \quad (4.1.10)$$

Now, the variables appearing in the Maxwell Lagrangian are not $F_{\mu\nu}$ but the components of the four-vector potential $(A^\mu) = (\rho, \vec{j})$, related to the electromagnetic fields through

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (4.1.11)$$

and this identity implies, through the Bianchi identity, the vanishing of $\partial_\nu *F^{\mu\nu}$ and consequently the vanishing of the magnetic current k^μ .

Dirac found the way to circumvent this difficulty, showing that a dually symmetric electromagnetic theory could be quantized, provided that for any electric charge q and magnetic charge g in theory, the condition:

$$qg = 2\pi n\hbar \quad n \text{ an integer} \quad (4.1.12)$$

was satisfied. This is the celebrated *Dirac quantization condition*. The occurrence of the Planck constant $\hbar$ indicates that, in Dirac’s approach, it is quantum mechanical in origin.

If a magnetic monopole with charge g exists, the flux of its magnetic field should take the form

$$\Phi = \int_S d^3x \vec{B}_{mon} \cdot d\vec{S} = g \quad (4.1.13)$$

for any closed surface S .

Now, if $\vec{B} = \vec{\nabla} \wedge \vec{A}$ then the integral would have to vanish. Thus $\vec{A}$ cannot exist everywhere on S , even if $\vec{\nabla} \cdot \vec{B}$ is only non-zero at the origin, and the best we can do is to find an $\vec{A}$ defined everywhere except on a line joining the origin to infinity, such that $\vec{B} = \vec{\nabla} \wedge \vec{A}$. To see that this is possible consider the field due to an infinitely long and thin solenoid placed along the negative z axis with its positive pole which has strength g at the origin. Its magnetic field would be:

$$\vec{B}_{sol} = \frac{g}{4\pi r^2} \vec{r} + g\theta(-z)\delta(x)\delta(y)\vec{z} \quad (4.1.14)$$

The line occupied by the solenoid is called in this context the “Dirac string”. This magnetic field differs from the magnetic field $\vec{B}_{mon}$ of a monopole only by the singular magnetic flux along the solenoid but it is clearly source-free since $\vec{\nabla} \cdot \vec{B}_{sol}$ vanishes, even at the origin by a vector potential $\vec{A}_{sol}$ well-defined everywhere. Then, we may write:

$$\vec{B}_{mon} = \frac{g}{4\pi r^2} \vec{r} = \vec{\nabla} \wedge \vec{A}_{sol} - g\theta(-z)\delta(x)\delta(y)\vec{z} \quad (4.1.15)$$

The effect of equation (4.1.15) is graphically represented in figure 1. We should think of the field $\vec{B}_{mon}$ as being represented not just by a gauge field but by $\vec{A}_{sol}$ together with a string $\mathcal{L}$ on which it is singular

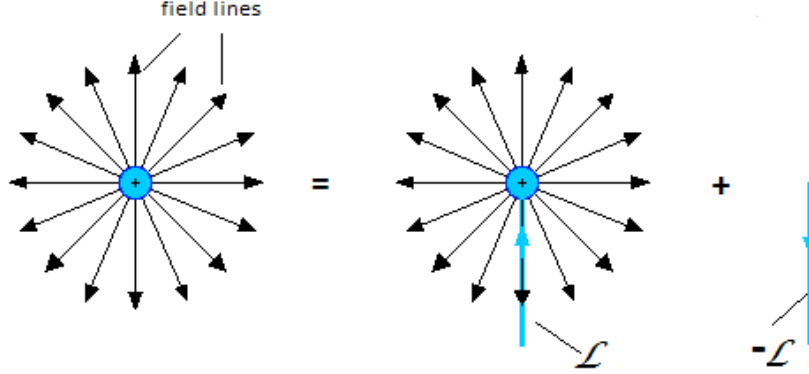


Figure 4.1

The monopole magnetic field as produced by $\vec{A}_{sol}$ together with a string $\mathcal{L}$.

4.1.2 Dirac condition

The quantum Hamiltonian describing an electron in the presence of a magnetic field is

$$\hat{H} = \frac{1}{2m} \left(\hat{\vec{p}} - \frac{e}{c} \vec{A} \right)^2 \quad (4.1.16)$$

where

$$\hat{\vec{p}} = \frac{\hbar}{i} \vec{\nabla} \quad (4.1.17)$$

and of course $\vec{B} = \vec{\nabla} \wedge \vec{A}$

Suppose there were no Dirac string so that everywhere in space (except at the origin) $\vec{B} = 0$. Then the vector potential $\vec{A} = 0$ would be a pure gauge (except at the origin)

$$\vec{A}_{ns} = \vec{\nabla} \Lambda \quad (4.1.18)$$

where the subindex ns indicates that there is no string. Then, the Hamiltonian (4.1.16) takes the form

$$\hat{H} = \frac{1}{2m} \left(\frac{\hbar}{i} \vec{\nabla} - \frac{e}{c} \vec{\nabla} \Lambda \right)^2 \quad (4.1.19)$$

and it is easy to see that the Schrödinger equation

$$\hat{H}\psi = \frac{\hbar}{i} \frac{\partial \psi}{\partial t} \quad (4.1.20)$$

has the solution

$$\Psi_{\Lambda} = \Psi_0 \exp \left(\frac{ie}{\hbar c} \Lambda(x) \right) \quad (4.1.21)$$

with Ψ_0 a free particle solution.

Now, since $\vec{A}_{ns} = \vec{\nabla} \Lambda$, we can write

$$\Psi_{\Lambda} = \Psi_0 \exp \left(\frac{ie}{\hbar c} \int_{C_{x_0}^x} \vec{\nabla} \Lambda(x') \cdot d\vec{x}' \right) = \Psi_0 \exp \left(\frac{ie}{\hbar c} \int_{C_{x_0}^x} \vec{A}_{sc}(x') \cdot d\vec{x}' \right) \quad (4.1.22)$$

where $C_{x_0}^x$ going from $\vec{x}_0$ to $\vec{x}$. Note that $\Psi_{\Lambda} = \Psi_{\Lambda}[C_{x_0}^x]$, i.e., the solution is path-dependent.

It is easy to extend this analysis to the case where there is a nontrivial gauge field. One just writes

$$\Psi[C_{x_0}^x] = \Psi_0 \exp \left(\frac{i}{\hbar} S[C_{x_0}^x] \right) \quad (4.1.23)$$

where the phase S due to the magnetic field $\vec{B}$ is simply given by

$$S[C_{x_0}^x] = \frac{e}{c} \int_{C_{x_0}^x} \vec{A} \cdot d\vec{x} \quad (4.1.24)$$

Suppose we want to perform an experiment à la Aharonov-Bohm in order not to detect the Dirac string when there is a Dirac monopole. The difference in phases of electrons passing on both sides of the string along paths $C_{1x_0}^x$ and $C_{2x_0}^x$ will be

$$\begin{aligned} \frac{(S[C_{1x_0}^x] - S[C_{2x_0}^x])}{\hbar} &= \frac{e}{\hbar c} \int_{C_{1x_0}^x} \vec{A}.d\vec{x} - \frac{e}{\hbar c} \int_{C_{2x_0}^x} \vec{A}.d\vec{x} = -\frac{e}{\hbar c} \oint A.dx \\ &= -\frac{e}{\hbar c} \int_S \vec{B}.d\vec{S} \end{aligned} \quad (4.1.25)$$

If the magnetic field is that of a monopole the final result is

$$\frac{(S^1 - S^2)}{\hbar} = -\frac{eg}{\hbar c} \quad (4.1.26)$$

Then there will be no interference and the electron will not detect the Dirac string whenever

$$\frac{eg}{\hbar c} = 2n\pi \quad (4.1.27)$$

which is Dirac quantization condition.

Dirac invented an undetectable Dirac string in the first half of the 20th century. Ignoring this, condensed matter physicist have already announced twice in this century that they measured the Dirac string. A very nice example of an oxymoron to teach in a college course...

4.2 The non-Abelian monopole

In order to find monopole solutions in electrodynamics (an Abelian $U(1)$ gauge theory) Dirac had to modify Maxwell equations developing a subtle mechanism in order to avoid a contradiction between the existence of magnetic charges and one of Maxwell's equation, $\vec{\nabla}.\vec{B} = 0$. Remarkably, when one considers certain non-Abelian gauge theories coupled to scalar fields magnetic monopoles exist as soliton solutions of the equations of motion.

The simplest model having monopole solutions to its classical equations of motion is an $SO(3)$ gauge theory broken to $U(1)$ by the vacuum expectation value of a triplet of scalar fields $\vec{\phi} = (\phi^a)$, $a = 1, 2, 3$. Gauge fields taking values in the Lie algebra of $SO(3)$ will be indicated as $\vec{A} = (A^a)$. Dynamics of the model is governed by the Yang-Mills-Higgs Lagrangian

$$L = -\frac{1}{4} \vec{F}^{\mu\nu}.\vec{F}_{\mu\nu} + \frac{1}{2} \overrightarrow{D^\mu \phi}.\overrightarrow{D_\mu \phi} - V[|\vec{\phi}|] \quad (4.2.28)$$

with

$$\begin{aligned}\vec{F}_{\mu\nu} &= \partial_\mu \vec{A}_\nu - \partial_\nu \vec{A}_\mu + e \vec{A}_\mu \wedge \vec{A}_\nu \\ \overrightarrow{D_\mu \phi} &= \partial_\mu \vec{\phi} + e \vec{A} \wedge \vec{\phi}\end{aligned}\quad (4.2.29)$$

and the symmetry breaking Higgs potential

$$V[|\vec{\phi}|] = \frac{\lambda}{4} \left(|\vec{\phi}|^2 - v^2 \right)^2 \quad (4.2.30)$$

with its minimum given by

$$|\vec{\phi}_{vac}| = v \quad (4.2.31)$$

where “*vac*” indicates that although we are considering the classical theory the value of the scalar field at the minimum of the potential will be associated to the Higgs “vacuum expectation value” when analyzing the theory at the quantum level.

Lagrangian (4.2.28) leads to the equations of motion

$$(D_\mu F^{\mu\nu})^a = \left(\vec{\phi} \wedge \overrightarrow{D^\nu \phi} \right)^a \quad (4.2.32)$$

$$(D^\mu D_\mu \phi)^a = -\lambda(|\vec{\phi}|^2 - v^2)\phi^a \quad (4.2.33)$$

Non-Abelian electric and magnetic fields

$$E^{ai} = F^{ai0}, \quad B^{ai} = \frac{1}{2} \epsilon^{ijk} F_{jk}^a \quad (4.2.34)$$

can be defined but they are not gauge invariant but gauge covariant, so they are not measurable quantities.

The covariant energy-momentum tensor is given by

$$T_{\mu\nu} = F_{\mu\alpha}^a F_\nu^{\alpha a} + (D_\mu \phi)^a (D_\nu \phi)^a - g_{\mu\nu} L \quad (4.2.35)$$

from which the energy density can be written in the form

$$T_{00} = \frac{1}{2} \left((E^{ai})^2 + (B^{ai})^2 + (D_i \phi)^a (D_i \phi)^a + (D_0 \phi)^a (D_0 \phi)^a \right) + V[|\phi|] \quad (4.2.36)$$

Note that $T_{00} \geq 0$ and it only vanishes whenever

$$\vec{F}_{\mu\nu} = 0, \quad \overrightarrow{D^\mu \phi} = 0, \quad V[|\phi|] = 0 \quad (4.2.37)$$

The first identity means that in the “vacuum” A_μ^a is a pure gauge, the other two define the Higgs vacuum.

If we choose for the scalar field vacuum

$$\vec{\phi}_{vac} = \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix} \quad (4.2.38)$$

the unbroken $U(1)$ gauge theory will correspond to rotations around the third axis. The spectrum of elementary particles in the quantized theory will include a massless gauge field component A_μ^3 which can be identified, using an electromagnetism language, with a photon. Concerning the other two components will correspond to vector bosons which can be accommodated as complex vectors

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(A_\mu^1 \pm ieA_\mu^2) \quad (4.2.39)$$

with mass $m_W = ev$ and charges $\pm e$. Finally, there is one neutral Higgs scalar

$$\phi \equiv \phi_3 \quad (4.2.40)$$

with mass $m_\phi = \sqrt{2\lambda}v$.

In view of (4.2.36) finite energy static solutions with $A_0 = 0$ will require as asymptotic behavior the same as the trivial vacuum solution (4.2.37)

$$\vec{F}_{ij} = 0 \quad (4.2.41)$$

$$\vec{D_j \phi} = 0, \quad r \rightarrow \infty \quad (4.2.42)$$

$$V[|\vec{\phi}|] = 0 \quad (4.2.43)$$

Concerning the scalar field covariant derivative, it should go to zero faster than $r^{3/2}$ this implying

$$\frac{1}{8\pi} \epsilon^{ijk} \int_{S_\infty^2} dS^i \check{\phi} \cdot (D_j \check{\phi} \wedge D_k \check{\phi}) = 0 \quad (4.2.44)$$

After writing explicitly the covariant derivatives, using properties of the scalar and vector products and integrating by parts one ends with

$$\frac{1}{8\pi} \epsilon^{ijk} \int_{S_\infty^2} dS_i \left(\check{\phi} \cdot (\partial_j \check{\phi} \wedge \partial_k \check{\phi}) - e \check{\phi} \cdot \vec{F}_{jk} \right) = 0 \quad (4.2.45)$$

where S_∞^2 is the sphere at spatial infinity. It is natural to identify the second term of the integrand in the r.h.s. of eq.(4.2.45)) with the magnetic field $\vec{B}$ associated to the U(1) massless gauge field

$$B^i|_\infty = \frac{1}{2}\epsilon^{ijk}\check{\phi}.\vec{F}_{jk}|_\infty \quad (4.2.46)$$

so that

$$\frac{e}{4\pi}\int_{S_\infty^2} dS^i\check{\phi}.B_i = N[\phi] \quad (4.2.47)$$

where

$$N[\phi] = \frac{1}{8\pi}\epsilon^{ijk}\int_{S^2} dS_i\check{\phi}.\partial_j\check{\phi}\wedge\partial_k\check{\phi} \quad (4.2.48)$$

It turns out that $N[\phi]$ has the geometrical interpretation of being the number of times the Higgs field covers the 2-sphere that it defines at infinity (which we shall call $\mathcal{M}_{\phi_{vac}}$)

$$\mathcal{M}_{\phi_{vac}} \equiv \phi_\infty^1{}^2 + \phi_\infty^2{}^2 + \phi_\infty^3{}^2 = \phi_0^2 \quad (4.2.49)$$

as r covers the space sphere at infinity S_∞^2 . Since we call vacuum the value of the Higgs field at spatial infinity (i.e. $\vec{\phi}_{vac} = \vec{\Omega}(\theta, \varphi)$) we can state that the Higgs vacuum defines a map

$$|\vec{\phi}_{vac}| : S_\infty^2 \rightarrow \mathcal{M}_{\phi_{vac}} \equiv S^2 \quad (4.2.50)$$

This map is characterized by its homotopy group $\pi_2(S^2)$ which is classified by integer numbers,

$$\pi_2(S^2) = \mathbb{Z} \quad (4.2.51)$$

Given a map $\vec{\phi}_{vac}$ the integer associated to the class to which it belongs is precisely (4.2.147) so that we have $N[\phi_{vac}] = n$ or, using (4.2.46),

$$\int dS^i\check{\phi}.B_i = \frac{4\pi}{e}n \quad (4.2.52)$$

The integer n is called the winding number associated to $\vec{\phi}_{vac}$ which, from this result, determines the monopole charge g

$$g = \frac{4\pi}{e}n \quad (4.2.53)$$

Note that the above formula differs from Dirac monopole quantization condition (4.1.12) in a factor of 2. This is because the smallest electric charge which could exists in our model is $e/2$ for a spinorial representation of $SU(2)$, the universal covering group of $SO(3)$ (while the charge of the scalar is e).

The integer n is called the Kronecker index [4] and it plays a central role in topology. It clearly shows that magnetic charge does not originate in dynamics but rather follows from the topological structure of three scalar fields in a three-dimensional space. Using the Gauss-Stokes theorem the surface integral (4.2.147) can be written as the volume integral

$$N[\phi] = \frac{1}{8\pi} \epsilon^{ijk} \int d^3x \partial_i (\check{\phi} \cdot \partial_j \check{\phi} \wedge \partial_k \check{\phi}) = \frac{1}{8\pi} \epsilon^{ijk} \int d^3x \partial_i \check{\phi} \cdot (\partial_j \check{\phi} \wedge \partial_k \check{\phi}) \quad (4.2.54)$$

suggesting to define the divergenless magnetic 4-current can be defined

$$k^\mu \equiv \frac{1}{8\pi e} \epsilon^{\mu\nu\rho\sigma} \partial_\nu \check{\phi} \cdot (\partial_\rho \check{\phi} \wedge \partial_\sigma \check{\phi}) , \quad \partial_\mu k^\mu = 0 \quad (4.2.55)$$

so that the the magnetic charge can be written as

$$g = \frac{4\pi}{e} \int k^0 d^3x \quad (4.2.56)$$

and obeys

$$\frac{dg}{dt} = 0 \quad (4.2.57)$$

A remarkable feature is that the magnetic current is completely specified in terms of the scalar triplet of Higgs fields. It is independent of the Yang-Mills fields A_μ^a . It holds true no matter what action principle determines the dynamics of ϕ^a . It is not a combination of field equations. Noether's theorem (more precisely the reciprocal of what is usually referred to as Noether's theorem) does not apply and we cannot derive a symmetry. In fact M , as can be seen from Eqs. (4.2.54) and (4.2.56), contains only the fields ϕ^a and their space derivatives and hence commutes with all dynamical variables; thus it generates no symmetry transformations. The conservation of magnetic charge is due to its topological character. The continuous time evolution of the scalar fields and their gradients can change the magnetic charge-as defined by the integral (4.2.56)only continuously. This is compatible with the integer spectrum of M only if M is time independent. By using Feynman's path integral quantization this statement is readily carried over to the quantized theory.

Finally, we note here that the charge density k^0 given by eq. (4.2.55) vanishes everywhere except at the zeros of the Higgs field. Indeed

$$\check{\phi} \cdot \check{\phi} = 1 \rightarrow \check{\phi} \cdot \partial_i \check{\phi} = 0 \quad (4.2.58)$$

so the three vectors appearing in k^0 are ortogonal to $\check{\phi}$ and hence the triple product defining k^0 vanishes except for zeros of the Higgs field. This zeros can be seen as δ -like singularities

that yield the result (8.2.11) upon integration. The zeros of the Higgs field are thus obtained by imposing three conditions $\phi^a(x_0) = 0$, $a = 1, 2, 3$, in a three-dimensional manifold (x^1, x^2, x^3) leaving thus in regular cases a zero-dimensional manifold, i. e., an isolated zero.

In eq.(4.2.46) we have identified the magnetic components of the field strength as

$$F_{ij} = \check{\phi} \cdot \vec{F}_{ij} \quad (4.2.59)$$

This was the way in which Polyakov [3] defined the $U(1)$ electromagnetic field strength. 't Hooft definition was instead

$$\mathcal{F}_{ij} = \check{\phi} \cdot \vec{F}_{ij} - \frac{1}{e} \epsilon_{ijk} \check{\phi} \cdot (D_j \check{\phi} \wedge D_k \check{\phi}) \quad (4.2.60)$$

If one is to include also electric charge, the natural “electromagnetic” $U(1)$ field strength should read

$$F_{\mu\nu} = \check{\phi} \cdot \vec{F}_{\mu\nu} , \quad (\text{Polyakov field strength}) \quad (4.2.61)$$

or

$$\mathcal{F}_{\mu\nu} = \check{\phi} \cdot \vec{F}_{\mu\nu} - \frac{1}{e} \epsilon_{\mu\nu\alpha\beta} \check{\phi} \cdot (D_\alpha \check{\phi} \wedge D_\beta \check{\phi}) , \quad ('t \text{ Hooft field strength}) \quad (4.2.62)$$

Since at infinity covariant derivatives should vanish to ensure finite energy, both definitions lead to the same magnetic charge, which is given in terms of a surface integral. 't Hooft idea was to define an expression that after gauge rotation to $\phi^a(x) = |\vec{\phi}(x)|(0, 0, 1)$ give

$$\mathcal{F}_{\mu\nu} = \partial_\mu A_\nu^3 - \partial_\nu A_\mu^3 = 0 \quad (4.2.63)$$

which is verified by his definition everywhere.

4.2.1 Noether Theorem and θ -term

Zwanziger [7] and Schwinger [6] generalized Dirac condition (4.1.12) to allow for the possibility of particles (dyons) that carry both electric and magnetic charge. A quantum mechanical theory can have two particles of electric and magnetic charges (e_1, g_1) and (e_2, g_2) only if

$$e_1 g_2 - e_2 g_1 = 2\pi n \quad (4.2.64)$$

where we have taken $\hbar = 1$. This formula may be heuristically derived [7] by considering the classical formula for the angular momentum of the electromagnetic field. The angular momentum in the field of the two particle system can be calculated readily. It has magnitude $(e_1 g_2 - e_2 g_1)/4\pi$. This has an integer or half-integer value, as expected in quantum mechanics, only if $(e_1 g_2 - e_2 g_1)/2\pi n$.

Since in Nature there are electrons of charges $(e, 0)$, the quantization condition, applied to a hypothetical magnetic monopole of charges (q, g) , requires $eg = 2\pi n$. Notice that, because the electron has no magnetic charge, the electric charge of the monopole does not contribute to $e_1 g_2 - e_2 g_1$. Therefore, the quantization condition, by itself, says nothing about the electric charge that a magnetic monopole should be expected to have.

The quantization condition does say something about the difference between the electric charges of two magnetic monopoles. Given, for instance, two monopoles of minimum allowed magnetic charge $g = 2\pi/e$ and of electric charges q and q' , one finds

$$e_1 g_2 - e_2 g_1 = 2n(q - q')/e \quad (4.2.65)$$

so that the Dirac-Schwinger-Zwanziger condition gives

$$q - q' = ne \quad (4.2.66)$$

Thus, the difference $q - q'$ must be an integral multiple of e . But, as Zwanziger and Schwinger noted, there is no restriction on q and q' separately.

If, however, the Dirac quantization condition is supplemented by CP conservation, the allowed values of the electric charge of a magnetic monopole are quantized. In fact, although the electric charge is odd under CP, the magnetic charge is even. (This is so because electric and magnetic fields transform oppositely under parity.) Applied to a monopole of charges $(q, 2\pi/e)$, a CP transformation gives a monopole of charges $(-q, 2\pi/e)$. For these two particles, $e_1 g_2 - e_2 g_1 = 4nq/e$, and is a multiple of 2π only if

$$q = ne, \text{ or } q = \left(n + \frac{1}{2}\right)e \quad (4.2.67)$$

Thus, the monopoles must have integer or half-integer charges. Moreover, in view of eq. (4.2.66), if monopoles of integer charge exist, then monopoles of half-integer charge do not, and vice-versa.

Apart from CP conservation, there is no general reason to expect magnetic monopoles to have integral (or even rational) electric charges. In nature, CP is violated, but only weakly. One may therefore suspect that monopoles, if they exist, have charges that are almost, but not quite, integers. The deviation of the monopole from integral charge would be proportional to the strength of CP violation.

Let us now consider the $SO(3)$ non-Abelian gauge theory that we have been considering and study the combined effect of introducing CP violation and considering monopole and dyon solutions.

CP violation is introduced by adding to Lagrangian (4.2.28) the so-called theta term

$$\Delta L = \theta \left(\frac{e^2}{32\pi^2} \right) \vec{F}_{\mu\nu} \cdot {}^* \vec{F}_{\mu\nu} \quad (4.2.68)$$

CP symmetry is slightly violated during certain types of weak decay while electromagnetic and strong interaction forces the θ parameter to be very small (mod. 2π), $\theta \approx 10^{-9}$.

Although the θ term can be written as a total derivative

$$\varepsilon^{\mu\nu\alpha\beta} \vec{F}_{\mu\nu} \cdot \vec{F}_{\alpha\beta} = \partial_\mu K^\mu \quad (4.2.69)$$

with

$$K^\mu = 2\varepsilon^{\mu\nu\alpha\beta} \left(\vec{A}_\nu \cdot \partial_\alpha \vec{A}_\beta + \frac{2}{3} \vec{A}_\nu \cdot (\vec{A}_\alpha \wedge \vec{A}_\beta) \right) \quad (4.2.70)$$

it modifies physics, as we shall see below. Let us mention that for static gauge field configurations K^0 is nothing but the $SO(3)$ Chern-Simons Lagrangian (in Euclidean $d = 3$ space).

As explained above, the Higgs mechanism makes that the original $SO(3)$ symmetry breaks down to $U(1)$. Let us call $\hat{N}$ the operator that generates gauge transformations leaving invariant the v.e.v (4.2.31). One can write $\hat{N}$ for an infinitesimal transformation in the form the

$$\hat{N}_\epsilon = \exp(i\epsilon \check{\phi}_{vac} \cdot \vec{\sigma}) \approx 1 + i \epsilon \check{\phi}_{vac} \cdot \vec{\sigma} \quad (4.2.71)$$

which represents a $U(1)$ transformation around the direction $\check{\phi}_{vac}$. Now this transformation is implemented on an arbitrary isovectors $\chi = \chi^a \sigma^a$ as

$$\chi^\epsilon = \hat{N}_\epsilon^{-1} \chi \hat{N}_\epsilon \quad (4.2.72)$$

which in components reads, in the infinitesimal case,

$$\vec{\chi}^\epsilon = \vec{\chi} + \check{\phi}_{vac} \wedge \vec{\chi} \quad (4.2.73)$$

or

$$\delta^\epsilon \chi = \check{\phi}_{vac} \wedge \vec{\chi} \quad (4.2.74)$$

As expected one has

$$\delta^\epsilon \check{\phi}_{vac} = 0 \quad (4.2.75)$$

so that the vacuum is invariant under the action of $\hat{N}$. This $U(1)$ invariance is associated to the electromagnetic gauge invariance and then the eigenvalues of the operator will correspond to electric charges.

Concerning gauge fields, they transform according to

$$\delta \vec{A}_\mu = \frac{1}{e} \epsilon D_\mu \check{\phi}_{vac} = \epsilon \left(\frac{1}{e} \partial_\mu \check{\phi}_{vac} + \vec{A}_\mu \wedge \check{\phi}_{vac} \right) \quad (4.2.76)$$

Now, if instead of an infinitesimal $U(1)$ transformation one considers a 2π one,

$$\hat{N}_{2\pi} = \exp(i2\pi \check{\phi}_{vac} \cdot \vec{\sigma}) \quad (4.2.77)$$

such a rotation around $\check{\phi}_{vac}$ should generate no transformation and hence

$$\hat{N}_{2\pi} = I \quad (4.2.78)$$

this meaning that the eigenvalues of the operator $\hat{N}$ are integers, a statement that corresponds to the quantization of electric charge.

Let us now compute operator $\hat{N}$ in the realization given by fields. We use Noether's formula for computing the charge density j^0 ,

$$j^0 = \frac{\delta L}{\delta \partial_0 \vec{A}_\mu} \cdot \delta \vec{A}_\mu + \frac{\delta L}{\delta \partial_0 \vec{\phi}} \delta \vec{\phi} \quad (4.2.79)$$

Using (4.2.75)-(4.2.76) and remembering to include the theta term (4.2.68) one finds

$$N = \frac{1}{e} \int d^3x D_i \check{\phi} \cdot \vec{F}_{0i} + \frac{\theta e}{8\pi^2} \int d^3x D_i \check{\phi} \cdot \left(\frac{1}{2} \varepsilon_{ijk} \vec{F}_{jk} \right) \quad (4.2.80)$$

Now, the conventional definition of electric and magnetic charge can be inferred from the expresion of the Polyakov's electromagnetic field strength (4.2.61)

$$\hat{Q} = \int d^3x D_i \check{\phi} \cdot \vec{F}_{0i} = \int d^3x \partial_i (\check{\phi} \cdot \vec{F}_{0i}) \quad (4.2.81)$$

$$\hat{M} = \int d^3x D_i \check{\phi} \cdot (\varepsilon_{ijk} \vec{F}_{jk}) = \int d^3x \partial_i (\check{\phi} \varepsilon_{ijk} \vec{F}_{jk}) \quad (4.2.82)$$

Here we have made use of the eqs. of motion to relate middle and last terms of each equation.

We conclude from equation (4.2.82) that

$$\hat{N} = \frac{1}{e} \hat{Q} + \frac{\theta e}{8\pi^2} \hat{M} \quad (4.2.83)$$

Now since

$$\exp(2\pi i \hat{N}) = I \quad (4.2.84)$$

then $\hat{N}$'s eigenvalues, which we shall call q for the electric and g for the magnetic charges should then be integers

$$\frac{1}{e}q + \frac{\theta e}{8\pi^2}g = n \in \mathbb{Z} \quad (4.2.85)$$

This means that if a 't Hooft-Polyakov monopole with magnetic charge $g = 4\pi/e$ is present then the allowed eigenvalues q of the electric charge satisfy

$$q = ne - \frac{e\theta}{2\pi} \quad (4.2.86)$$

This result is exact since it derives from a symmetry statement given by eq. (4.2.84).

In view of the fact that θ is a very small number ($\theta < 10^{-10}$) the electric charge of a 't Hooft-Polyakov magnetic monopole will not ordinarily be integral, or even rational in units of the fundamental charge e . Although the arguments presented here were developed using a simplified model, they apply also to “realistic” theories such as the SU(5).

4.2.2 Finding monopole and dyon solutions

The equations of motion (4.2.33) can be written in the form

$$D^\mu \vec{F}_{\mu i} + e\vec{\phi} \wedge D_i \vec{\phi} = 0 \quad (4.2.87)$$

$$D^\mu D_\mu \vec{\phi} - v^2 \lambda \vec{\phi} + \lambda |\vec{\phi}|^2 \vec{\phi} = 0 \quad (4.2.88)$$

together with the constraint

$$\vec{A}_0 \wedge F_{0i} + e\vec{\phi} \wedge (\vec{A}_0 \wedge \vec{\phi}) = 0 \quad (4.2.89)$$

Following [8] we shall look for solutions of the equations of motion satisfying:

- Time-independence
- Separability into radial (r) and angular (θ, φ) coordinates, i.e.:

$$\phi^a = F(r)\Omega^a(\theta, \varphi) \quad (4.2.90)$$

$$A_i^a = \frac{1}{e} \frac{K(r)}{r} B_i^a(\theta, \varphi) \quad (4.2.91)$$

$$A_0^a = J(r)C^a(\theta, \varphi) \quad (4.2.92)$$

where all these functions are a priori unknown (we do not assume the specific ansatz of Wu-Yang 't Hooft-Polyakov-Julia-Zee [9],[2]-[3],[10], but we shall derive it). We assume analyticity in the variables r, θ, φ and single valuedness.

- The solution has finite energy.

The energy of the solution is given by $\int d^3x T_{00}$ with the covariant energy momentum tensor is given by eq.(4.2.35) which, as we have seen is the sum of positive definite quantities. Therefore, each of them must be integrable and go to zero fast enough at infinity for the energy to be finite.

First the potential must vanish at infinity. Therefore if

$$\lim_{r \rightarrow \infty} \phi^a = f_0 \Omega^a(\theta, \varphi) \quad (4.2.93)$$

then

$$v^2 = f_0^2 \Omega^a \Omega^a \quad (4.2.94)$$

So $\Omega^a \Omega^a$ should be a constant which can be normalized to one,

$$\Omega^a \Omega^a = 1 \quad (4.2.95)$$

Next $(D_0 \phi^a)^2$ should be integrable so one needs that $D_0 \phi^a$ to go to zero faster than r^{-1} at infinity. Now, A_0 has mass dimensions so the asymptotic dependence of J on r could at most be $\sim 1/r$. Now,

$$D_0 \vec{\phi} = e \vec{A}_0 \wedge \phi = e F(r) J(r) \vec{C}(\theta, \varphi) \wedge \vec{\Omega} \quad (4.2.96)$$

so that $\vec{C}$ and $\vec{\Omega}$ should be parallel and we normalize $\vec{C}$ so that $\lim_{r \rightarrow \infty} r J(r) = 1$,

$$\vec{A}_0 = J(r) \vec{\Omega}(\theta, \varphi) \quad (4.2.97)$$

Finally the $(D_i \vec{\phi})^2$ term must be integrable. So $(D_i \phi)^a$ must vanish at infinity. This, together with the factorizability condition is seen to imply:

$$\vec{A}_i = \frac{1}{e} K(r) \left(\vec{\Omega} \wedge \partial_i \vec{\Omega} + (B_i(\theta, \varphi)/r) \vec{\Omega} \right) \quad (4.2.98)$$

and

$$\lim_{r \rightarrow \infty} K(r) = -1 \quad (4.2.99)$$

Now we write down the equation of motion of the Higgs fields. It depends only upon the unknown functions $F(r)$, $K(r)$, $\vec{\Omega}(\theta, \varphi)$, B_i/r and $\Omega(\theta, \varphi)$.

$$\hat{L}^2 \vec{\Omega} + K(r) \vec{\Omega} (\vec{\Omega} \cdot \hat{L}^2 \vec{\Omega}) - F(r) \Omega + r q(r) (\vec{\Omega} \wedge \vec{\partial}_i \Omega) B_i = 0 \quad (4.2.100)$$

where

$$q(r) = v^{-1} (1 + K(r))^{-1} \left((r^2 F')' + \lambda v^2 r^2 F(r) - \lambda r^2 F(r)^3 \right) \quad (4.2.101)$$

and $\hat{L}^2$ is the angular momentum operator,

$$\hat{L}^2 = -\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) - \frac{1}{\sin^2 \theta} \partial_\varphi^2 \quad (4.2.102)$$

Taking the scalar product of this equation by Ω yields:

$$\vec{\Omega} \cdot \hat{L}^2 \vec{\Omega} = \frac{q(r)}{1 + K(r)} = c \quad (4.2.103)$$

so that we obtain the equation

$$\hat{L}^2 \vec{\Omega} - c \vec{\Omega} + r K(r) (\vec{\Omega} \wedge \partial_i \Omega) B_i = 0 \quad (4.2.104)$$

If $B_i \neq 0$ one necessarily has $K(r) = -1$ everywhere since the first two terms depend only on angles. But if $K = -1$ then F_{ij} behaves as r^{-4} at the origin leading to a divergent energy. So $B_i = 0$ this in turn implying that $\vec{\Omega}$ is an eigenfunction of $\hat{L}^2$

According to the hypothesis of analyticity and single valuedness on a sphere, c must be of the form $l(l+1)$, where l is an integer. Therefore the equations are now separated into radial and orbital parts

$$\hat{L}^2 \vec{\Omega} = c \vec{\Omega} \quad (4.2.105)$$

$$q(r) = l(l+1)(1 + K(r)) \quad (4.2.106)$$

One can now show that $l = 0, 1$ are the only allowed cases as a result of the following theorem:

Th.: Let be N real analytic, single valued functions on the sphere Ω^a ($a = 1, 2, \dots, N$) satisfying

a) $\Omega^a \Omega^a$

and

b) $\hat{L}^2 \Omega^a = l(l+1) \Omega^a$

Then the only allowed values for l are such that

$$l(l+1) \leq N. \quad (4.2.107)$$

The proof is based in the fact because of the assumptions of analyticity and single-valuedness one can write

$$\Omega^a(\theta, \varphi) = \sum_{m=-l}^l C_m^a Y_{lm}(\theta, \varphi) \quad (4.2.108)$$

and it can be found in [8] where it was taken from the book on Quantum Mechanics by Messiah [11]. In the case we are interested $N = 3$ (the Higgs fields is a triplet). and then the allowed values for l are $l = 0, 1$. The case $l = 0$ corresponds to a constant solution in (φ, θ) and $l = 1$ reproduces the original 't Hooft-Polyakov-Julia-Zee ansatz

$$\begin{aligned}\Omega^a(\theta, \varphi) &= \left(\frac{4\pi}{3}\right)^{1/2} \left((Y_{11} - Y_{1-1})/\sqrt{2}, i(Y_{11} + Y_{1-1})/\sqrt{2}, Y_{10} \right) \\ &= (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) = (\check{x}, \check{y}, \check{z})\end{aligned}\quad (4.2.109)$$

Using formula (4.2.147) one can easily from eq.(4.2.109) that the magnetic charge associated to this ansatz corresponds to $n = 1$ so that the magnetic charge of the solution is according to (4.2.52)

$$\int dS^i \check{\phi} \cdot B^i = -\frac{4\pi}{e} n = -\frac{4\pi}{e} \quad (4.2.110)$$

In fact, one can see that at infinity, the electromagnetic field strength takes the form of a Dirac magnetic monopole one,

$$\lim_{r \rightarrow \infty} \phi^a F_{ij}^a = \varepsilon_{ija} \check{x}^a \left(-\frac{1}{er^2} \right) \quad (4.2.111)$$

The usual form in which the spherically symmetric ansatz is written is, instead of (4.2.90)-(4.2.92)

$$\phi^a = \frac{h(r)}{er} \Omega^a \quad (4.2.112)$$

$$A_i^a = \varepsilon_{iab} \frac{k(r) - 1}{er} \Omega^b \quad (4.2.113)$$

$$A_0^a = \frac{j(r)}{er} \Omega^a \quad (4.2.114)$$

With this notation, the radial equations of motion read

$$r^2 j'' = 2jk^2 \quad (4.2.115)$$

$$r^2 h'' = 2hk^2 - \lambda v^2 h \left(1 - \frac{1}{e^2 v^2 r^2} h^2 \right) \quad (4.2.116)$$

$$r^2 k'' = k(k^2 - j^2 + h^2 - 1) \quad (4.2.117)$$

Apart from the magnetic charge, the solution has an electric charge Q defined in eq.(4.2.81),

$$\begin{aligned}Q &= \int d^3x \partial_i (\check{\phi} \cdot \vec{F}_{0i}) = - \int d^3x \partial_i \left(\check{x}_i \frac{d}{dr} \left(\frac{j(r)}{er} \right) \right) \\ &= \frac{8\pi}{e} \int_0^\infty dr \frac{jk^2}{r}\end{aligned}\quad (4.2.118)$$

where we have used eq.(4.2.115) and eq.(4.2.89). Thus the charge is given by an expression of the form (recovering $\hbar$ and c constants)

$$Q = \left(\frac{\hbar c}{e^2} \right) e \zeta \left(\frac{\lambda}{e^2}, \frac{M}{v} \right) \quad (4.2.119)$$

where M is the scale of A_0 (to be introduced below) and M is a mass. In contrast with the magnetic charge which being of topological origin can be computed exactly, the electric charge should be calculated numerically.

Concerning boundary conditions, at the origin one should have

$$\lim_{r \rightarrow 0} j(r) = \lim_{r \rightarrow 0} h(r) = 0, \quad \lim_{r \rightarrow 0} k(r) \rightarrow 1 \quad (4.2.120)$$

Using the eqs. of motion one finds for small r

$$j \rightarrow \text{const} \times r^2, \quad h \rightarrow \text{const} \times r^2 \quad k \rightarrow 1 + \text{const} \times r^2 \quad (4.2.121)$$

The fields are differentiable at the origin ensuring a finite energy solution.

At infinity the Higgs field approaches its “vacuum expectation value” v ,

$$h(r) \rightarrow evr \quad (4.2.122)$$

From eq.(4.2.115) we see that if k goes asymptotically to zero exponentially then

$$\lim_{r \rightarrow \infty} j(r) = Mr + b + \dots \quad (4.2.123)$$

The mass parameter M sets the scale for J while b determines the dyon charge. Indeed, from eq.(4.2.81) one has

$$Q = - \int d^3x \partial_i (\check{\phi} \cdot \vec{F}_{0i}) = \lim_{R \rightarrow \infty} \int_{S^2} R^2 \sin \theta d\theta d\varphi \frac{d}{dr} \left(\frac{J(r)}{er} \right) \Big|_{r=R} \quad (4.2.124)$$

so that using the behavior (4.2.123) one gets

$$Q = - \frac{4\pi b}{e} \quad (4.2.125)$$

From eq.(4.2.117) one has, asymptotically

$$r^2 k'' = (v^2 - M^2) r^2 k + \dots \quad (4.2.126)$$

an equation that admits a solution decreasing as

$$k(r) \sim \exp \left(-\sqrt{e^2 v^2 - M^2} r \right) \quad (4.2.127)$$

which requires $|ev| > M$.

The solution of the highly nonlinear equations of motion (4.2.115)-(4.2.117) can not be obtained analytically. It has been studied numerically in [10] where it is shown that it reduces to the purely magnetic 't Hooft-Polyakov solution for $j(r) = 0$.

In the figure we represent schematically the behavior of functions $h(r)$, $j(r)$ and $k(r)$.

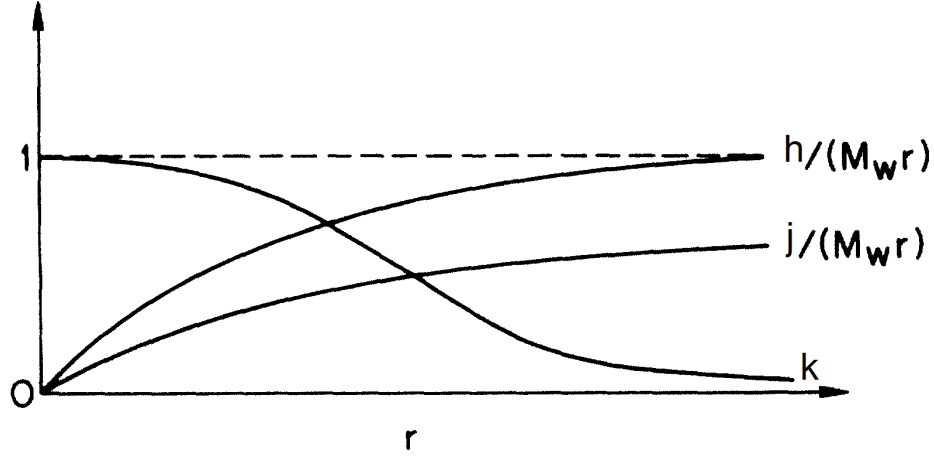


Figure 4.1

Schematic representation of the behavior of the radial functions $h(r)$ (Higgs field), $j(r)$ (gauge field component A_0) and $k(r)$ (gauge field component A_i).

Note that the the Schwinger- Zwanziger condition (4.2.66) for the case of an Abelian Dirac dyon gives

$$q = ne \quad (4.2.128)$$

or equivalently (putting $\hbar = c = 1$) in the case of the non-Abelian monopole,

$$\frac{1}{e} \zeta \left(\frac{\lambda}{e^2}, \frac{M}{v} \right) = n \quad (4.2.129)$$

Note however that the condition (4.2.128) has a quantum origine while the Julia-Zee dyon solution is just a classical solution. It is only after quantization that it could be shown to be quantized, as proved in [12].

4.2.3 Dyon charge quantization

There is an alternative way of obtaining electrically charged monopoles from 't Hooft-Polyakov solution, which turns to be more convenient concerning quantization. Indeed, working in the $\vec{A}_0 = 0$ gauge, one can make arise the dyon degree of freedom [13] by considering time-dependent transformations $U(\vec{x}, t)$, leaving the Higgs field invariant, of the form

$$U[\lambda] = \exp[i\lambda(r, t)\vec{\Omega} \cdot \vec{t}] \quad (4.2.130)$$

which act on the A'_i s so that

$$A_i^{mon} \rightarrow U[\lambda] A_i^{mon} U^{-1}[\lambda] + \frac{i}{e} U[\lambda] \partial_i U^{-1}[\lambda] \quad (4.2.131)$$

Here t^a ($a = 1, 2, 3$) are the $SO(3)$ generators, $t^a = \sigma^a/2$, and we write $A_i = \vec{A}_i \cdot \vec{t}$. Because of the underlying spherical symmetry, λ depends only on the radial coordinate r and t . It can be seen as an angular variable related to rotation associated with the surviving $U(1)$ (electromagnetic) gauge symmetry. Note however that U does not correspond to a full gauge transformation since we continue to work at fixed gauge $A_0 = 0$. Now, after change (4.2.131) there is an electric field whose radial component takes the form

$$\mathcal{F}_{0r} = \frac{1}{e} \frac{d\dot{\lambda}}{dr} \quad (4.2.132)$$

with $\dot{\lambda} = (d/dt)\lambda$, which satisfies the equation

$$\Delta \dot{\lambda} = \frac{2}{r^2} \dot{\lambda} K^2 \quad (4.2.133)$$

The following choice that satisfies eq.(4.2.133)

$$\lambda(r, t) = \alpha(t) \frac{j}{Mr} \quad (4.2.134)$$

leads to the same radial component for the electric field as that arising from the static Julia-Zee dyon, eq.(4.2.118) for $\alpha(t) = -Mt$. Now, Lagrangian (4.2.28) becomes, after transformation (4.2.131)

$$L = \frac{1}{2e^2} \partial^i (\dot{\lambda} \partial_i \dot{\lambda}) + M_{mon} \quad (4.2.135)$$

where M_{mon} is the 't Hooft-Polyakov monopole mass. At the quantum level, when working in the monopole sector, the term M_{mon} should be subtracted in order to normalize the zero-point energy so that the monopole energy is equal to zero [14].

We shall quantize the theory within the path-integral approach. We then introduce the partition function Z

$$\mathcal{Z} = \int D\vec{A}_i D\vec{\phi} \exp \left(i \int d^4x L \right), \quad (4.2.136)$$

In order to handle the collective coordinate path-integral issue we make at this point the usual path-integral change of variables separating “classical” and “quantum” parts.

$$\begin{aligned} A_i(\vec{x}, t) &= A_i^\lambda(\vec{x}, t) + a_i(\vec{x}, t) \\ \phi(\vec{x}, t) &= \phi^{mon}(\vec{x}) + \varphi(\vec{x}, t) \end{aligned} \quad (4.2.137)$$

with $A_i^\lambda(\vec{x}, t)$ the U -transformed 't Hooft-Polyakov monopole solution,

$$A_i^\lambda(\vec{x}, t) = U[\lambda] A_i^{mon}(\vec{x}) U^{-1}[\lambda] + \frac{i}{e} U[\lambda] \partial_i U^{-1}[\lambda] , \quad (4.2.138)$$

$$U[\lambda] = \exp[i\lambda(r, t) \vec{\Omega} \cdot \vec{t}] \quad (4.2.139)$$

(Collective coordinates associated to non-gauge symmetries will not be considered in what follows since they do not play any role in the charge quantization problem). One has explicitly,

$$\begin{aligned} e\vec{A}_i^\lambda &= (\partial_i \lambda) \vec{\Omega} + k \sin \lambda \partial_i \vec{\Omega} + (k \cos \lambda - 1) \vec{\Omega} \wedge \partial_i \vec{\Omega} \\ -ie\vec{F}_{0i}^\lambda &= (\partial_i \dot{\lambda}) \vec{\Omega} + \dot{\lambda} k \cos \lambda \partial_i \vec{\Omega} - \dot{\lambda} k \sin \lambda \vec{\Omega} \wedge \partial_i \vec{\Omega} \\ e\vec{F}_{ij}^\lambda &= (k^2 - 1) \partial_i \vec{\Omega} \wedge \partial_j \vec{\Omega} + \sin \lambda (\partial_i K \partial_j \vec{\Omega} - \partial_j K \partial_i \vec{\Omega}) \\ &\quad + \cos \lambda (\partial_i K \vec{\Omega} \wedge \partial_j \vec{\Omega} - \partial_j K \vec{\Omega} \wedge \partial_i \vec{\Omega}) \end{aligned} \quad (4.2.140)$$

with now $\dot{\lambda} = (d/dt)\lambda$.

In terms of the new variables the partition function reads

$$Z = \int Da_i D\varphi D\lambda \Delta \exp(i \int d^4x L[A^{mon}, \phi^{mon}, a_i, \varphi, \lambda]) , \quad (4.2.141)$$

With Δ the Jacobian associated with the change to the new integration variables. Now, using eq.(4.2.135) one has

$$Z = Z_d \times Z_f \quad (4.2.142)$$

where Z_d corresponds to the path-integral over the collective coordinate λ accounting for the dyon's degree of freedom and is obtained using eq.(4.2.140) as

$$Z_d = \int D\lambda \exp \left(\frac{2\pi i}{e^2} \int dt \int_0^\infty r^2 dr \partial^i (\dot{\lambda} \partial_i \dot{\lambda}) \right) \quad (4.2.143)$$

Concerning Z_f , it includes the contributions coming from the classical action and the integration over fluctuations a_i and φ not relevant for the analysis of the dyon charge.

Now, one can identify Z with the partition function for a one dimensional Coulomb system with a position-dependent coupling (proportional to r^2). The quantum analysis of such a system leads to the usual spectrum of dyon states [10],[12] with charge given by formula (4.2.128). Alternatively, one can relate Z with the partition function for a quantum

rotator [15] and also infer dyon charge quantization from the energy spectrum of the rotator [16]. To see this last way, let us start by noting that [13], it will be sufficient to consider the collective coordinate λ in the form (4.2.134)

$$\lambda = \frac{J(r)}{Mr} \alpha(t) \quad (4.2.144)$$

where $J(r)$ represents the (classical) electric potential of a Julia-Zee dyon and $\alpha(t)$, which describes quantum effects, is the angular variable.

Inserting (4.2.144) in Z_d given by (4.2.143) one finds

$$Z_d = \int D\alpha \exp \left(i \int dt \left(\frac{\mathcal{J}}{2} \dot{\alpha}^2 \right) \right) \quad (4.2.145)$$

where $\mathcal{J}$ can be seen as the moment of inertia of a rigid rotator and is related to the dyon charge according to

$$\mathcal{J} = -4\pi b/e^2 M = \frac{Q}{eM} \quad (4.2.146)$$

We recognize in eq.(4.2.145) the path-integral expression for the transition amplitude of a rigid rotator evolving in time from an “initial” state $\alpha' = \alpha(0)$ to a “final” state $\alpha'' = \alpha(\beta)$ modulo 2π . Now, the spectrum of such a rotator can be derived without leaving the path-integral framework (see for example [17]-[18]). One finds for the energy levels of the rotator the result

$$E_n = \frac{1}{2\mathcal{J}} n^2, \quad n \in \mathbb{Z} \quad (4.2.147)$$

We can go further in the analogy with a quantum rigid rotator and evaluate from the Lagrangian in partition function (4.2.145) the (angular) momentum p_α

$$p_\alpha \equiv \frac{\partial L}{\partial \dot{\alpha}} = \mathcal{J} \dot{\alpha} \Rightarrow p_\alpha^2 = \mathcal{J}^2 \dot{\alpha}^2 \quad (4.2.148)$$

That is, Eq.(4.2.147) shows that $p_\alpha^2 = n^2$ so that p_α is quantized according to

$$p_\alpha = \mathcal{J} \dot{\alpha} = n \quad (4.2.149)$$

On the other hand we have seen that $\alpha = Mt$ so that $\dot{\alpha} = M$ and from (4.2.146) we have

$$Q = eM\mathcal{J} = e\mathcal{J}\dot{\alpha} \quad (4.2.150)$$

so that putting all this together one ends with the dyon charge quantization condition

$$Q = en \quad (4.2.151)$$

4.2.4 BPS equations and exact solutions

The energy density of the Yang-Mills-Higgs system takes in this case the form (see (4.2.36))

$$T_{00} = \frac{1}{2} ((E^{ai})^2 + (B^{ai})^2 + (D_i \phi)^a (D_i \phi)^a + (D_0 \phi)^a (D_0 \phi)^a) \quad (4.2.152)$$

where

$$F_{i0} = E_i^a, \quad F_{ij}^a = \varepsilon_{ijk} B^{ak} \quad (4.2.153)$$

The dyon mass $\mathcal{M}$ in the center of mass frame is then given by

$$\mathcal{M} = \int d^3x \left(\frac{1}{2} (E_i^a - \sin \alpha (D_i \phi)^a)^2 + \frac{1}{2} (B_i^a - \cos \alpha (D_i \phi)^a)^2 + V[\phi] \right) + v(q \sin \alpha + g \cos \alpha) \quad (4.2.154)$$

with q and g the already defined electric and magnetic dyon charges,

$$g = \frac{1}{v} \int d^3x \vec{B}_i^a (D^i \phi)^a \quad (4.2.155)$$

$$q = \frac{1}{v} \int d^3x \vec{E}_i^a (D^i \phi)^a \quad (4.2.156)$$

To have a stable vacuum one needs $\lambda > 0$ and then we can write

$$\mathcal{M} \geq v(q \sin \alpha + g \cos \alpha) \quad (4.2.157)$$

Then, if we choose

$$\sin \alpha = \frac{q}{\sqrt{q^2 + g^2}} \quad \cos \alpha = \frac{g}{\sqrt{q^2 + g^2}} \quad (4.2.158)$$

we end with

$$\mathcal{M} \geq v \sqrt{q^2 + g^2} \quad (4.2.159)$$

For the 't Hooft-Polyakov monopole $q = 0$ so that

$$\mathcal{M} \geq v|g| \quad (4.2.160)$$

Now $|g|$ is $4\pi/e$ so

$$\mathcal{M} \geq \frac{4\pi v}{e} = \frac{4\pi e v}{e^2} = \frac{4\pi M_W}{\bar{e}^2} \quad (4.2.161)$$

where $M_W = ve$ is the mass associated to the symmetry breaking and we write $\bar{e}$ since the gauge coupling which for the $SO(3)$ group is $1/2$ while for “true” electrons is 1. We can then write

$$\mathcal{M} \geq \frac{\nu}{\alpha_f} M_W \quad (4.2.162)$$

with α_f the fine structure constant. Since α is small ($\alpha_f \sim 1/137$) the above relation implies that the monopole is much heavier than the W -boson.

From eq. (4.2.154) it is clear that the bound is not saturated unless the Higgs potential goes to zero (i.e., $\lambda \rightarrow 0$) but still imposing the condition that at infinity the Higgs field takes a non-trivial vacuum expectation value,

$$\lim_{r \rightarrow \infty} |\vec{\phi}|^2 = v^2 \quad (4.2.163)$$

so that the existence of topologically charged configurations could exist. Note that in this limit the Higgs scalar becomes massless. Now, to saturate the bound we set

$$(D_0 \phi)^a = 0 \quad (4.2.164)$$

$$E_k^a = (D_k \phi)^a \sin \alpha \quad (4.2.165)$$

$$B_k^a = (D_k \phi)^a \cos \alpha$$

with $\tan \alpha = q/g$.

The solutions of these equations automatically saturates Bogomolny bound and are referred to as the BPS states. Also note that in the BPS limit all the prerturbative excitations of the theory saturate the bound and the belong to the BPS spectrum. As we hall see bellow BPS states are very natural in the context of $\mathcal{N} = 2$ supersymmetric theories.

Finally we notice that in the acronym BPS the PS correspond to M. K. Prasad and P. M. Sommerfield who trying to solve the second order equations for the dyon using a variational method discover that hyperbolic functions optimize, for any value of λ , numerical calculations and finally realized that in the limiting $\lambda \rightarrow 0$ those hyporbolic functions were in fact exact solutions,

$$\begin{aligned} k(r) &= \frac{vr}{\sinh vr} \\ j(r) &= \sin \alpha (vr \coth(vr) - 1) \\ h(r) &= \cos \alpha (vr \coth(vr) - 1) \end{aligned} \quad (4.2.166)$$

4.2.5 Monopoles with charge $g > 1$

We have seen that by imposing spherical symmetry it is easy to find monopole solutions having necessarily magnetic charge 1. In principle it is not obvious that there exist solutions of higher charge. After many unsuccessful proposals, Ward [21] made a major breakthrough by presenting for the first time an exact analytic charge-two monopole solution. Inspired by Ward's work, Prasad an Sommerfield Exact axially symmetric multimonoopole solutions of

arbitrary topological charge were presented by Prasad and Rossi [22]. These authors used a variant of the Atiyah-Ward $\mathcal{A}_k$ -Ansatz [7] to construct an $SU(2)$ monopole of charge k .

The first rigorous proof of existence for monopoles of arbitrary charge was provided by Taubes who showed how, roughly speaking, a charge $g=k$ monopole may be constructed by gluing together k charge 1 monopoles. Furthermore, Taubes proved that the moduli space (ie: the space of solutions up to gauge equivalence) of monopoles of charge $q = k$ is a smooth manifold of dimension $4k - 1$.

Although Taubes's results resolved the issue of existence, they shed no light on how to solve the Euclidean Bogomolny equations. Progress on this front was not long coming however. Hitchin, using twistor theory, showed how a monopole corresponds to a holomorphic vector bundle on $T\mathbb{P}^1$ and using this introduced a compact algebraic curve in $T\mathbb{P}^1$, called the spectral curve, which determines the monopole up to gauge equivalence. Around the same time, Nahm, using a generalised Fourier transform (the Nahm transform), showed how the Euclidean Bogomolny equations could be reduced to the system (Nahm's equations)

$$\frac{dT_i}{ds} = \varepsilon_{ijk}[T_j, T_k] \quad (4.2.167)$$

where T_i is a $k \times k$ matrix valued function on $(0, 2)$ satisfying appropriate reality and boundary conditions. Nahm also showed how to associate an algebraic curve to a solution of these equations and Hitchin showed that this was the spectral curve of the monopole. The spectral curve thus emerged as a key feature in the theory of monopoles.

4.3 Spin from isospin

The fact that in Ansatz (4.2.112)-(4.2.114) the internal index a is coupled to the space indices i, j leads to interesting results. We have already seen this effect when we computed the angular momentum J for the non-Abelian CS vortex, eq.(??) where we observed how a fermionic object of fractional spin emerged out from boson fields through a metamorphosis of isospin into spin.

Let us note that 't Hooft-Polyakov monopole is rotationally invariant in the sense that the effects of a rotation can be completely compensated by a global gauge transformation. When written in the hedgehog gauge form of Eq. (4.2.112)-(4.2.114), it is invariant under the action of

$$\vec{J} = \vec{L} + \vec{T} \quad (4.3.168)$$

where $\vec{L}$ is related to spatial rotations and $\vec{T}$ to global $SU(2)$ internal rotations. We can see spin emerge by breaking the rotational symmetry adding a scalar field ψ that transforms under the doublet representation of the $SU(2)$. One can then prove that still a monopole

solution exists in which the solution for ψ is not invariant under $\vec{J}$ and J will take semi-integer values [26]. Following alternative routes Jackiw-Rebbi [27], Hassenfratz-'t Hooft [28] and Goldhaber [29] have shown that monopoles also exhibit this phenomenon which in fact requires more than the mere fact that the coupling of internal and spatial indices.

We refer to these papers for the derivation and discussion of these results.

Bibliography

- [1] P. A. M. Dirac, “Quantized Singularities in the Electromagnetic Field,” Proc. Roy. Soc. Lond. A **133** (1931) 60.
- [2] G. ’t Hooft, “Magnetic Monopoles in Unified Gauge Theories,” Nucl. Phys. B **79** (1974) 276.
- [3] A. M. Polyakov, “Particle Spectrum in the Quantum Field Theory,” JETP Lett. **20** (1974) 194 [Pisma Zh. Eksp. Teor. Fiz. **20** (1974) 430].
- [4] B. J. Hadamard, note in J. Tannery, *Introduction à la théorie des fonctions*, Herman, Paris, 1910), pp. 437-77.
- [5] D. Zwanziger, Phys. Rev. **176** (1968) 1480, 1489
- [6] J. Schwinger, Phys. Rev. **144** (1966) 1087; **173** (1968) 1536.
- [7] M. N. Saha, Indian J. Phys. **10** (1936) 145; M. Fierz, Helv. Phys. Acta **17** (1944) 27; H. A. Wilson, Phys. Rev. **75** (1949) 309.
- [8] E. Cremmer, F. Schaposnik and J. Scherk, Phys. Lett. B **65** (1976) 78.
- [9] T. T. Wu and C. N. Yang, in: *Properties of matter under unusual conditions*, eds. H. Mark and S. Fernbach, Interscience, N.Y. 1969.
- [10] B. Julia and A. Zee, Phys. Rev. D **11** (1975) 2227.
- [11] A. Messiah, *Mechanique Quantique*, Dunod, Paris, 1964.
- [12] J. Goldstone and R. Jackiw, in: *Gauge Theories and Modern Field Theory*, eds. R. Amowitt and P Nath, MIT. Cambridge, USA (1976) ; E. Tomboulis and G. Woo, Nucl. Phys. B **107** (1976) 221; N. Christ, A. Guth and E. Weinberg, Nucl. Phys. B **114** (1976) 61.

- [13] C.G. Callan, Phys. Rev. **D 25**
- [14] V.A. Rubakov, Nucl. Phys. **B 203** (1982) 311.
- [15] F. Wilczek, Phys. Rev. Lett. **48** (1982) 1144 and 1146.
- [16] K.I. Takada and H.Kuratsuji, Il Nuovo Cim. **106 A** (1993) 661.
- [17] J. Zinn-Justin, *Field Theory and Critical Phenomena*, Oxford Science Publ., 1990.
- [18] J. C. Le Guillou and F. A. Schaposnik, “On the electric charge of monopoles at finite temperature,” Phys. Lett. B **383** (1996) 339 [hep-th/9603102]
- [19] M. K. Prasad and C. M. Sommerfield, Phys. Rev. Lett. **35** (1975) 760.
- [20] S. R. Coleman, S. J. Parke, A. Neveu and C. M. Sommerfield, “Can One Dent a Dyon?,” Phys. Rev. D **15** (1977) 544.
- [21] R. S. Ward, “A Yang-Mills Higgs Monopole of Charge 2,” Commun. Math. Phys. **79** (1981) 317.
- [22] M. K. Prasad and P. Rossi, “Mirror Symmetry and Exact Multi-Monopole Solutions,” Phys. Rev. D **23** (1981) 1795.
- [23] M. F. Atiyah, R. S. Ward, Comm. Math. Phys. **55** (1977) 117.
- [24] N. J. Hitchin, “On the construction of monopoles”. Comm. Math. Phys **83** 1982; ibid 579 **89** (1983). 190.
- [25] C. H. Taubes. “Stability in Yang-Mills theories”, Comm. Math. Phys. 91 (1983) 235.
- [26] E. Weinberg, “Classical Solutions in Quantum Field Theory: Solitons and Instantons in High-Energy Physics”, Cambridge University Press, Cambridge, UK, 2012.
- [27] R. Jackiw and C. Rebbi, “Spin from Isospin in a Gauge Theory” Phys. Rev. Lett. **36** (1976) 1116.
- [28] P. Hasenfratz, G. ’t Hooft “Fermion-Boson Puzzle in a Gauge Theory” Phys. Rev. Lett. **36** (1976) 1119.
- [29] A. A. Goldhaber, "Connection of Spin and Statistics for Charge-Monopole Composites" Phys. Rev. Lett. **36** (1976) 1122.

Chapter 5

Skyrmions

5.1 Skyrmions in $3 + 1$ space-time dimensions

The Skyrme model [1] was proposed by Tony Hilton Royle Skyrme in 1962 as a nonlinear theory of pions in $3 + 1$ space-time dimensions, with the “Skyrme field”, $U(t, \vec{x})$, being an $SU(2)$ -valued scalar. More recently, it was regarded as an approximate, low energy effective theory of QCD, becoming exact as the number of quark colours becomes large [2]. Remarkably, and this was Skyrme’s main motivation for constructing and studying this model, it has topological soliton solutions that can be interpreted as baryons. These solitons are called Skyrmions.

The model is defined by the Lagrangian

$$L = \frac{F_\pi^2}{16} \text{tr} \left(\partial_i U \partial^i U^\dagger + \frac{1}{32e^2} [\partial_i U U^\dagger, \partial_j U U^\dagger] [\partial^i U U^\dagger, \partial^j U U^\dagger] \right) \quad (5.1.1)$$

Originally F_π was the pion decay constant (experiments indicate $F_\pi \approx 130$ MeV) and e an undetermined dimensionless parameter. These parameters can be scaled away by using energy and length units of $F_\pi/4e$ and $2/eF_\pi$ respectively, which we adopt from now on. In terms of these units the Skyrme Lagrangian can be written as

$$L = \text{tr} \left(-\frac{1}{2} R_\mu R^\mu + \frac{1}{16} [R_\mu, R_\nu] [R^\mu, R^\nu] \right) \quad (5.1.2)$$

with R_μ taking values in the Lie algebra of $SU(2)$,

$$R_\mu = (\partial_\mu U) U^\dagger \quad (5.1.3)$$

The Euler-Lagrange equations following from Lagrangian (5.1.1) read

$$\partial_\mu \left(R^\mu + \frac{1}{4} [R^\nu, [R_\nu, R^\mu]] \right) = 0 \quad (5.1.4)$$

Note that this equation takes the form of a current conservation equation

$$\partial_\mu \tilde{R}^\mu = 0 \quad (5.1.5)$$

with

$$\tilde{R}^\mu = R^\mu + \frac{1}{4} [R_\mu, [R_\nu, R^\mu]] \quad (5.1.6)$$

The boundary condition for the original fields U at fixed time are taken as

$$\lim_{|x| \rightarrow \infty} U(t, \vec{x}) = I_{2 \times 2} \quad (5.1.7)$$

so that the “vacuum”, the field of minimal energy will be $U_v(t, x) = I_{2 \times 2}$.

If one considers static configurations $U(t, \vec{x}) = U(\vec{x})$, the energy functional associated to Lagrangian (5.1.2) is

$$E = \frac{1}{12\pi^2} \text{tr} \int d^3x \left(-\frac{1}{2} R_i R^i + \frac{1}{16} [R_i, R_j] [R^i, R^j] \right) \quad (5.1.8)$$

where we have introduced the additional $12/\pi^2$ factor for later convenience. Static solutions of the Skyrme model can then be obtained by extremizing this energy.

At fixed time U is in principle a map from $\mathbb{R}^3$ into S^3 . However the imposed boundary condition (5.1.7) implies a one point compactification of space to $S^3 \equiv \mathbb{R}^3 \cup \{\infty\}$ while the $SU(2)$ gauge group can be also identified with S^3 so that topologically at fixed time U establishes an application between 3-spheres,

$$U : S^3 \rightarrow S^3 \quad (5.1.9)$$

Then, the relevant homotopy group classifying static field configurations is $\Pi_3(S^3) = \mathbb{Z}$. It is this topological charge that Skyrme identified with the baryon number.

The corresponding integers give the winding number (degree) of $U(\vec{x})$ and can be determined through the formula

$$\begin{aligned} \omega[U] \equiv B &= \frac{1}{24\pi^2} \text{tr} \int \varepsilon_{ijk} R_i R_j R_k \\ &= \frac{1}{24\pi^2} \text{tr} \int \varepsilon_{ijk} (U \partial_i U^\dagger) (U \partial_j U^\dagger) (U \partial_k U^\dagger) = n \in \mathbb{Z} \quad i = 1, 2, 3 \end{aligned} \quad (5.1.10)$$

A Bogomolny bound for the energy (5.1.8) can be found, $E > |B|$, but in contrast with monopoles and vortices this bound cannot be saturated for any field configuration with non-zero B [3]. The minimal energy solution for each B is what we call a Skyrmion, and their energy E is identified with their mass, m_B . (More loosely, local minima and saddle points of E with nearby energies are also sometimes called Skyrmions.)

5.2 The hedgehog

A spherically symmetric $B = 1$ skyrmion was presented in Skyrme paper. It takes the form

$$U_n = \exp(if(r)\vec{\sigma}.\vec{x}) = \cos f(r) + i\vec{\sigma}.\vec{x} \sin f(r) \quad (5.2.11)$$

The name hedgehog derives from the fact that the pion fields of this configuration point radially outward from the origin at all points in space,

For finite energy U_n should approach to a constant at infinity. Then $f(\infty) = n\pi, n = 0, 1, 2, \dots$. We choose $n = 0$.

If we choose the same value also at the origin ($n = 0$, the origin is mapped to a single point on the sphere) then $f(r)$ returns to 0 and the covering of S^3 may be contracted to a point and $U_{n=0} = I$. But if at the origin we choose $f(0) = \pi$ we get a non-contractible single and complete cover of S^3 , that is, we have a skyrmion with $B = 1$.

The value of B for Skyrme ansatz is confirmed by substituting the ansatz in the expression (5.1)

$$B = -\frac{2}{\pi} \int_0^\infty f'(r) \sin^2 f(r) dr = \frac{1}{\pi} f(0) = 1 \quad (5.2.12)$$

skyrmions with baryon numbers greater than 1 are not spherical like the basic $B = 1$ skyrmion. For example the $B = 2$ skyrmion is toroidal, the $B = 3$ skyrmion tetrahedral, etc.

Substituting Skyrme ansatz in the static equation of motion for the model one gets

$$(r^2 + 2 \sin^2 f) f'' + 2r f' + \sin 2f \left(f^2 - 1 - \frac{\sin^2 r}{r^2} \right) = 0 \quad (5.2.13)$$

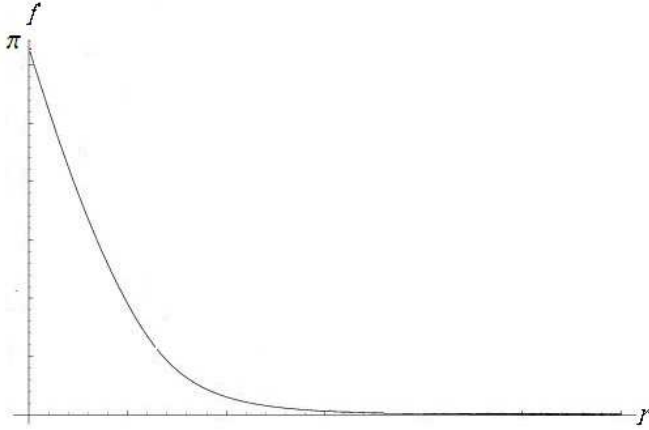


Figure 5.1

The profile function $f(r)$ for the $B = 1$ skyrmion

The solution to this equation cannot be found in closed form but can be solved numerically with high precision.

The energy of the solution is $E = 1.232$, 0.23 above the Bogomolny bound. As we see in Figure 1, the skyrmion is located at the origin but it can be centered at any point in space due to translation invariance of the theory. This together with rotational invariance makes that the moduli space of charge 1 skyrmion is six-dimensional. In fact, three moduli are lost (i.e. there are no additional collective coordinates) since $SU(2)$ gauge group rotation are, as in the monopole case, equivalent to spatial rotations.

5.3 $B > 1$ skyrmions

A construction of skyrmions using rational maps was developed by Houghton, Manton and Sutcliffe [4]. This map separates radial from angular coordinates introducing a complex coordinate

$$z = \tan\left(\frac{\theta}{2}\right) \exp(i\varphi) \quad (5.3.14)$$

with which one constructs a rational function of z ,

$$R(z) = \frac{p(z)}{q(z)} \quad (5.3.15)$$

where p and q are polynomials with no common root. One can think of R as a smooth map from concentric spheres S^2 in space to the target S^2 of latitude spheres on S^3

$$R(z) : S^2 \rightarrow S^2 \quad (5.3.16)$$

One treats each S^2 as a Riemann sphere, the first one having coordinate z . The point z corresponds to the cartesian unit vector

$$\vec{n}_z = \frac{1}{1 + |z|^2} (z + z^*, i(z^* - z), 1 - |z|^2) \quad (5.3.17)$$

and conversely

$$z_{\vec{n}} = \frac{n_1 + in_2}{1 - n_3} \quad (5.3.18)$$

Similarly, an image point R can be written as a unit vector

$$\vec{n}_{R_z} = \frac{1}{1 + |R|^2} (R + R^*, i(R^* - R), 1 - |R|^2) \quad (5.3.19)$$

A rational map (5.3.15) has degree N if at least one of each polynomials has degree and they have no common roots.

Rational maps were introduced into the theory of monopoles by Donaldson [?]. who showed that there is a one-to-one correspondence between maps of degree N (with the basing condition $R(z) \rightarrow 0$ as $z \rightarrow \infty$) and N -monopoles. Donaldson's work, following Hitchin [?].

Then, the rational map ansatz for the skyrmion is just a generalization of the $B = 1$ ansatz (5.2.11),

$$U(r, z) = \exp(i f(r) \vec{n}_{R_z} \cdot \vec{\sigma}) = \cos f(r) + i \sin f(r) \vec{n}_{R_z} \cdot \vec{\sigma} \quad (5.3.20)$$

Here the radial function $f(r)$ satisfies $f(0) = \pi$ and $f(\infty) = 0$ as in the hedgehog case. In the $N = 1$ case the basic map is $R(z) = z$ and this formula reduces to the Skyrme hedgehog field. Inserting this element U in the formula giving B one finds that $B = N$. As examples, rational functions $R(z)$ leading to $B = 2, 3$ are

$$R^{(2)}(z) = \frac{z^2 - a}{-az^2 + 1}, \quad R^{(3)}(z) = \frac{\sqrt{3}az^2 - 1}{z(z^2 - \sqrt{3}a)} \quad (5.3.21)$$

An attractive feature of the ansatz (5.3.20) is that it leads to a simple energy expression which can be minimized with respect to the rational map R and the profile function f

to obtain close approximations to several known skyrmion solutions. Starting with these approximations is an efficient method to find new exact solutions.

The solutions for negative B are obtained from those with positive B by reversing the sign of f , and the energy is unchanged (for $B = 0$ the solution is trivial). The solution for $B = 2$ has energy 3.668 which is significantly greater than the Bogomolny bound 2.

5.4 Skyrmions in diverse dimensions

In addition to the $d = 3 + 1$ space-time dimensional case that we have discussed above, skyrmions exist in diverse dimensions. Those constructed by Belavin and Polyakov in $d = 2$ dimensional space [5] to describe ferromagnetic states are also known as lumps (grumos) or sigma model instantons. They are solutions to the equations of motion of an $O(3)$ non-linear sigma model and they are characterized by $\pi_2(S^2)$.

In $2 + 1$ dimensional space-time there are the so called baby-skyrmions, solutions in an $O(3)$ sigma model with quartic derivative term and a potential term [6].

In general the $O(N + 1)$ sigma model admits skyrmions in N spatial dimensions characterized by $\pi_N(S^N) \simeq \mathbb{Z}$, at least for $N = 1, 2, 3$.

5.4.1 Skyrmions in the $O(3)$ nonlinear σ model [5],[10]

The model is described by the $(2 + 1)$ space-time Lagrangian

$$L = \frac{1}{2}(\partial_\mu n^a)(\partial^\mu n^a) \quad \mu = 0, 1, 2 \quad (5.4.22)$$

where n^a is a unit vector

$$n^a n^a = 1, \quad a = 1, 2, 3 \quad (5.4.23)$$

The energy for static configurations is given by

$$E = \int d^2x \frac{1}{2}(\partial_i n^a)^2 \quad i = 1, 2 \quad (5.4.24)$$

It is clear that in order to have finite energy we need

$$r|\vec{\partial}\vec{n}(r, \varphi)| \rightarrow 0 \quad \text{as } r \rightarrow \infty \Rightarrow \lim_{r \rightarrow \infty} n(r, \varphi) = \vec{n}^{(0)} \quad (5.4.25)$$

where we have introduced polar coordinates (r, φ) and $\vec{n}^{(0)}$ a constant vector (for example $\vec{n} = (0, 0, 1)$).

Since $\vec{n}(\vec{x})$ takes the same value at all points at infinity the physical coordinate plane $\mathbb{R}^2$ is essentially compacted into a spherical surface that we denote S_∞^2 (i.e, the circle at infinity is reduced to the north pole, $\mathbb{R}^2 \cup \{\infty\} \cong S_\infty^2$). Concerning the field “internal” space, the condition (5.4.23) is also a spherical surface of radius 1 (S^2) so that at fixed time n^a establishes the mapping

$$n^a : S_\infty^2 \rightarrow S^2 \quad (5.4.26)$$

As we have seen, these mappings are classified into homotopy sectors characterized by integers according to

$$\pi_2(S^2) = \mathbb{Z} \quad (5.4.27)$$

The existence of solitons in the (2+1)-dimensional O(3) nonlinear σ model, as first discussed by Belavin and Polyakov [5], is implied by this homotopy group.

The topological conserved current that one can construct from fields $n^a(\vec{x}, t)$ is

$$j^\mu = \frac{1}{8\pi} \varepsilon^{\mu\nu\alpha} \varepsilon^{abc} n^a \partial_\nu n^b \partial_\alpha n^c \quad (5.4.28)$$

from which the topological charge associated to (5.4.27) can be calculated integrating j^0 over the surface S_∞^2

$$B = \frac{1}{8\pi} \int dS_\infty \varepsilon^{ij} \varepsilon^{abc} n^a \partial_i n^b \partial_j n^c = \frac{1}{8\pi} \int dS_\infty \varepsilon^{ij} \vec{n} \cdot (\partial_i \vec{n} \wedge \partial_j \vec{n}). \quad (5.4.29)$$

Now, if one writes the unit vector $\vec{n}$ in the form

$$\vec{n} = (\sin \eta \cos \chi, \sin \eta \sin \chi, \cos \eta) \quad (5.4.30)$$

one can see that

$$\begin{aligned} \varepsilon_{ij} \vec{n} \cdot (\partial_i \vec{n} \wedge \partial_j \vec{n}) \sin \theta d\theta d\phi &= 2\vec{n} \cdot (\partial_\eta \vec{n} \wedge \partial_\chi \vec{n}) \sin \eta d\eta d\chi \\ &= 2\vec{n} \cdot \vec{n} \sin \eta d\eta d\chi = 2dS^2[\eta, \chi] \end{aligned} \quad (5.4.31)$$

so that

$$B = \frac{1}{4\pi} \int dS^2[\eta, \chi] = n \quad (5.4.32)$$

with n the number of times the field $\vec{n}$ sphere is traversed as we span the coordinate space $\mathbb{R}^2$ compacted to S^2 .

In order to find solutions in any given sector we employ the Belavin-Polyakov trick [5] which starts from the identity

$$\int d^2x (\partial_i \vec{n} \pm \varepsilon_{ij} \vec{n} \wedge \partial_j \vec{n}) \cdot (\partial_i \vec{n} \pm \varepsilon_{ik} \vec{n} \wedge \partial_k \vec{n}) \geq 0 \quad (5.4.33)$$

Now, using that

$$\vec{n} \cdot \vec{n} = 1 \Rightarrow \vec{n} \cdot \partial_i \vec{n} = 0 \quad (5.4.34)$$

it is easy to show that

$$\varepsilon_{ij}\varepsilon_{ik}(\vec{n} \wedge \partial_j \vec{n})(\vec{n} \wedge \partial_k \vec{n}) = (\partial_j \vec{n}) \cdot (\partial_k \vec{n}) \quad (5.4.35)$$

so that inequality (5.4.33) can be written as

$$2(\partial_i \vec{n})(\partial_i \vec{n}) \mp 2\varepsilon_{ij}\vec{n}(\partial_i \vec{n}) \wedge (\partial_j \vec{n}) \geq 0 \quad (5.4.36)$$

or, integrating out over x^1, x^2

$$E \geq 4\pi|B| \quad (5.4.37)$$

The inequality sets the bound for the energy of any static configuration in a given B -sector. The extrema of the energy correspond to minima of the energy and since configurations in each sector cannot be deformed continuously to configurations in any other sector, extremisation can be done in each sector separately. Now, in a given sector the bound is saturated when the following first order (Bogomolny) equation holds

$$\partial_i \vec{n} = \pm \varepsilon_{ij} \vec{n} \wedge \partial_j \vec{n} \quad (5.4.38)$$

One can verify that any solution of this equations satisfies the second order equations of motion for Lagrangian (5.4.22) subject to the constraint (5.4.23).

Analogously to the original Skyrme model discussed previously the $B = 1$ skyrmion solutions in the non-linear sigma model can be written in the form

$$\vec{n} = (\cos \varphi \sin f(r), \sin \varphi \sin f(r), \cos f(r)) = (\tilde{x} \sin f(r), \cos f(r)) \quad (5.4.39)$$

where $\tilde{x} = \vec{x}/r$ and $f(0) = \pi$, $f(\infty) = 0$.

For general $B = n$ topological charge note that eq. (5.4.38) can be simplified in terms of new variables. To this end, let us write such equation in the form

$$\partial_1 \phi = \mp i(\phi \partial_2 n_3 - n^3 \partial_2 \phi) \quad \text{and} \quad \partial_2 \phi = \pm i(\phi \partial_1 n_3 - n^3 \partial_1 \phi) \quad (5.4.40)$$

where

$$\phi = n_1 + i n_2 \quad (5.4.41)$$

Now, defining

$$\omega = \omega_1 + i \omega_2 = \frac{2\phi}{1 - n_3} \quad (5.4.42)$$

one can rewrite eq.(5.4.40) as

$$\partial_1 \omega = \mp i \partial_2 \omega \quad (5.4.43)$$

Taking the real and imaginary part of these equations one gets

$$\frac{\partial \omega_1}{\partial x_1} = \pm \frac{\partial \omega_2}{\partial x_2} \quad \text{and} \quad \frac{\partial \omega_1}{\partial x_2} = \pm \frac{\partial \omega_2}{\partial x_1} \quad (5.4.44)$$

Now, introducing a complex variable $z = x_1 + ix_2$ and taking the upper signs the equations we recognize the Cauchy-Riemann equations for ω as an analytic function of z^* while if we take the lower signs we get tohse for an analytic function of z . Poles are permitted (cuts are not if one demands single-valuedness). That ω diverges at the poles is not a problem because $\omega \rightarrow \infty$ corresponds to the north pole of S_∞^2 , i.e. to $n_3 = 1$. Indeed, $n_3 = 1$ corresponds to the case in which $f = 0$ and this happens at $r \rightarrow \infty$

In terms of ω the energy can be very simply written as

$$E = \int d^2x \frac{|d\omega/dx|^2}{(1 + |\omega|^2/4)^2} \quad (5.4.45)$$

and from this, using the explicit form of ω in terms of polar coordinates one gets

$$|B| = \frac{E}{4\pi} \quad (5.4.46)$$

A prototype solution for arbitrary positive topological charge n is

$$\omega(z) = \left(\frac{z - z_0}{\lambda} \right)^n \quad (5.4.47)$$

with λ any real number and z_0 any complex number. Parameters λ and z_0 correspond to the size and position of the solution and are related to scale and translation invariance of the sigma model Lagrangian.

5.4.2 The spin of Skyrmions in the $d = 3$ $O(3)$ non-linear sigma model

One can determine the angular momentum of the skyrmion following different routes [10]-[14]. We present below an heuristic derivation of the angular momentum.

First it should be noted that since the j_μ (5.4.28) is a conserved current it can be written as

$$j_\mu = \frac{1}{4\pi} \varepsilon_{\mu\nu\alpha} \partial_\nu A_\alpha \quad (5.4.48)$$

where

$$A_\mu[\vec{n}] = \nabla^{-2} \epsilon^{\mu\nu\lambda} \partial_\nu j_\lambda[\vec{n}] , \quad (5.4.49)$$

a non-local expression. Then one can add to the action a Chern-Simons (Hopf) term

$$S_{tot} = S + \theta S_{CS} \quad (5.4.50)$$

where

$$S_{CS} = \frac{1}{4\pi} \int d^3x \epsilon^{\mu\nu\alpha} A_\mu[\vec{n}] F_{\nu\alpha}[\vec{n}] = \int d^3x A_\mu j^\mu \quad (5.4.51)$$

is a Chern-Simons (Hopf) term. First note that written in terms of $F_{\mu\nu}$ and A_μ the action is explicitly gauge invariant so that θ is not quantized (as it happens in the non-Abelian term). Second, note the non-locality of S_{CS} in $\vec{n}$ through the relation (5.4.51).

The fact that the CS-Hopf density is locally a total divergence is very hard to be seen from the original nonlocal form (5.4.51). However, this property is needed for the action S_{CS} to be a homotopic invariant which because of this property does not modifying the equations of motion. Let us attack this problem introducing complex doublets Z

$$Z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \quad (5.4.52)$$

with complex variables z_i satisfying

$$z_1 z_1^* + z_2 z_2^* = 1 \quad (5.4.53)$$

and related to vectors n^a in the form

$$n^a = Z^{*T}(x) \sigma^a Z(x) \quad (5.4.54)$$

In terms of the Z fields, the CS-Hopf term can be written in the form

$$S_{CS} = \frac{\theta}{4\pi} \int d^3x \rho_{CSH} \quad (5.4.55)$$

with

$$\rho_{CSH} = \epsilon^{\mu\nu\lambda} (Z^T \partial_\mu Z) (\partial_\nu Z^{*T} \partial_\lambda Z) \quad (5.4.56)$$

A last change of variables is still necessary to show that the CS-Hopf density is a total derivative. One writes

$$z_1 = y_1 + i y_2 , \quad z_2 = y_3 + i y_4 \quad (5.4.57)$$

and after some work one gets from (5.4.55)

$$\rho = 2\varepsilon^{\mu\nu\lambda}\partial_\lambda (f(y)\partial_\mu y_1\partial_\nu y_2) \quad (5.4.58)$$

with

$$f(y) = \pm \sin^{-1} \left(y_3 / \sqrt{1 - y_1^2 - y_2^2} \right) \quad (5.4.59)$$

What does it happens when one analyzes the static skyrmions with a CS-Hopf term added to the non-linear sigma model as in (5.4.50). To begin with, the S_{CS} action is a total derivative and hence the skyrmion solutions (5.4.39) are still solutions of the action (5.4.50). Let us consider the $B = 1$ solution. In view of the form of the solution (5.4.39) a spatial rotation is equivalent to an internal $O(3)$ rotation. Let us then, instead of considering a spatial rotation, to make a time dependent $O(3)$ rotation in the $[0, T]$ interval

$$n^1(\vec{x}) \pm in^2(\vec{x}) \rightarrow \exp(\pm i\alpha(t))(n^1(\vec{x}) \pm in^2(\vec{x})) , \quad n^3(\vec{x}) \rightarrow n^3(\vec{x}) , \quad \alpha(T) - \alpha(0) = 2\pi \quad (5.4.60)$$

This rotation defines a map $S^2 \times [0, 1] \rightarrow S^2$. From eq.(5.4.28) we have that

$$j_i = \frac{1}{8\pi} \frac{d\alpha}{dt} \varepsilon_{ij} \partial_j n^3 , \quad j_0 = \varepsilon_{ij} \partial_i A_j \quad (5.4.61)$$

from the fact that $j_0 = \varepsilon_{ij} \partial_i A_j$ one infers that

$$A_j = \varepsilon_{jk} x_k \frac{g(r)}{r^2} , \quad g(r) = \int_0^r dr' r' j_0(r') \quad (5.4.62)$$

and

$$A_0 = \frac{d\alpha}{dt} n^3 \quad (5.4.63)$$

Inserting all this in the S_{CS} term one gets

$$S_{CS} \rightarrow S_{CS} = g(\infty) n_3(\infty) (\alpha(T) - \alpha(0)) / 2\pi = 1 \quad (5.4.64)$$

Now, if one considers a rotation for $\exp(iS)$, one can write

$$\exp(2\pi \hat{J}) \exp(iS) \rightarrow \exp(i\theta) \quad (5.4.65)$$

from which one infers

$$J = \frac{\theta}{2\pi} \quad (5.4.66)$$

A more formal derivation can be found in [11] where the canonical quantization of the theory is achieved considering it as a gauge theory. One then calculates the energy-momentum tensor $T^{\mu\nu}$ and from it the angular momentum

$$J = \varepsilon_{ij} \int d^2x x^i T^{0j} \quad (5.4.67)$$

They finally find

$$J = \int d^2x \varepsilon_{ij} \pi^a(x) \partial_j n^a(x) + \frac{\theta}{2\pi} B^2 \quad (5.4.68)$$

with $\pi^a(x)$ the canonical momentum conjugate to n^a . The first term is the usual orbital angular momentum while the second one corresponds to the fractional spin which, for $B = 1$ gives (5.4.66).

Since the $O(3)$ nonlinear sigma model in $2 + 1$ dimensions describes the 2-dimensional isotropic (Heisenberg) ferromagnet the main application of the results discussed corresponds to the domain of condensed-matter systems.

5.4.3 Baby Skyrme models [14]-[15]

A baby Skyrme model is a modified version of the Skyrme model in $d = 2 + 1$ dimensions, i.e, in one spatial dimension less than the original Skyrme proposal. The reduction from 3 to 2 spatial dimensions greatly simplifies the analysis providing a toy model to explore Skyrme-like properties. Also, in recent years people have used the baby Skyrme model to model aspects of the Quantum Hall Effect and other phenomena in planar Physics.

The baby Skyrme model is constructed by adding to the $O(3)$ model discussed in the precedent section a second term, fourth order in derivatives, the analogue of the Skyrme term in the usual Skyrme model. There is also a third term which does not contain any derivatives and is often referred to simply as potential.

The Lagrangian of the baby Skyrme model can be written in the form

$$L = \partial_\mu \vec{n} \cdot \partial^\mu \vec{n} - \theta_S \left[(\partial_\mu \vec{n} \cdot \partial^\mu \vec{n})^2 - \frac{\kappa^2}{4} (\partial_\mu \vec{n} \wedge \partial_\nu \vec{n}) (\partial^\mu \vec{n} \wedge \partial^\nu \vec{n}) \right] - \mu^2 (1 - \vec{n}_0 \cdot \vec{n}) \quad (5.4.69)$$

Here $\vec{n}^2 = 1$, $\vec{n}_0 = (0, 0, 1)$, μ (with dimensions of inverse mass) is a free parameter. In three spatial dimensions the Skyrme term is necessary for the existence of soliton solutions but the inclusion of a potential is optional from the mathematical point of view. Physically, however, a potential of a certain form is required to give the pions a mass. By contrast, in two dimensions a potential must be included in the above Lagrangian since otherwise non-trivial soliton solutions do not exist.

Let us define kinetic and potential energies as

$$T = \frac{1}{2} \int d^2x (\partial_t \vec{n} \cdot \partial_t \vec{n} + (\partial_t \vec{n} \wedge \partial_i \vec{n}) \cdot (\partial_t \vec{n} \wedge \partial_i \vec{n})) \quad (5.4.70)$$

$$V = \int d^2x \left(\frac{1}{2} \partial_i \vec{n} \cdot \partial_i \vec{n} + \frac{1}{4} (\partial_i \vec{n} \wedge \partial_j \vec{n}) \cdot (\partial_i \vec{n} \wedge \partial_j \vec{n}) + \mu^2 (1 - \vec{n}_0 \cdot \vec{n}) \right) \quad (5.4.71)$$

Finite energy requires that

$$\lim_{|\vec{x}| \rightarrow \infty} \vec{n}(t, \vec{x}) = \vec{n}_0, \quad \forall t \quad (5.4.72)$$

As a result we can formally compactify physical space to a 2-sphere S^2 space, as we did for the $O(3)$ sigma model, and regard the fields $\vec{n}$ at fixed time as a map

$$\vec{n} : S_\infty^2 \rightarrow S^2 \quad (5.4.73)$$

with the topological charge given by (5.4.29).

It is interesting to note that the same steps leading from the inequality (5.4.33) to the energy bound (5.4.37) for the $O(3)$ sigma model can be followed here to give a bound for the potential,

$$V[\vec{n}] \geq 4\pi |B| \quad (5.4.74)$$

We pointed that the usual $O(3)$ σ -model Lagrangian, where both the Skyrme term and the potential are omitted, is scale invariant. In particular, this means that a soliton solution of the σ -model can have arbitrary size. The inclusion of the Skyrme term breaks the scale invariance and it follows from a simple scaling argument that the potential energy contributed by the Skyrme term can be lowered by increasing the scale of a configuration. Thus, in order to obtain stable solutions it is necessary to include a potential which, on its own, would favour small scales. It follows that soliton solutions in a Lagrangian with both Skyrme term and potential have a definite size.

The Euler-Lagrange equations of motion read

$$\partial^\alpha (\vec{n} \wedge \partial_\alpha \vec{n} - \partial_\beta \vec{n} (\partial^\beta \vec{n} \cdot (\vec{n} \wedge \partial_\alpha \vec{n}))) = \mu^2 \vec{n} \wedge \vec{n}_0 \quad (5.4.75)$$

A trivial $B = 0$ solution is just $\vec{n} = \vec{n}_0$.

A conservation law can be read off immediately from the equation of motion (5.4.75). Indeed, taking the scalar product with $\vec{n}_0$ on both sides of the equation we find that the current

$$\partial_\alpha j^\alpha = 0 \quad (5.4.76)$$

with

$$j_\alpha = \vec{n}_0 \cdot (\vec{n} \wedge \partial_\alpha \vec{n}) - (\vec{n}_0 \cdot \partial_\beta \vec{n}) \cdot (\partial^\beta \vec{n} \cdot (\vec{n} \wedge \partial_\alpha \vec{n})) \quad (5.4.77)$$

The symmetry that leads to this conservation law is $SO(2)$ rotations of the field $\vec{n}$ about $\vec{n}_0$, which can be written in terms of an angle χ as

$$(n_1, n_2, n_3) \rightarrow (n_1 \cos \chi + n_2 \sin \chi, -n_1 \sin \chi + n_2 \cos \chi, n_3) \quad (5.4.78)$$

If one calls such a rotation an “isorotation”, then the conserved quantity can be called (with reference to the original Skyrme model) isospin,

$$I = \int j_0 d^2x \quad (5.4.79)$$

Time-independent solutions of the equations of motion (5.4.75), which are stationary points of the potential energy functional V , were studied in detail in [14]. An important class of static solutions of the equations of motion consists of fields which are invariant under the group of simultaneous spatial rotations by some angle $\alpha \in [0, 2\pi)$ and iso-rotations by $-n\alpha$, where n is a non-zero integer. Such fields are of the form

$$\vec{n} = \begin{pmatrix} \sin f(r) \sin(n\varphi - \chi) \\ \sin f(r) \cos(n\varphi - \chi) \\ \cos f(r) \end{pmatrix} \quad (5.4.80)$$

Here $f(r)$ is a function satisfying boundary conditions to be discussed below, χ is an arbitrary angle which can be eliminated by isorotations and so will be chosen to be zero and in this case vectors $\vec{n}$ of this form are the analogue of the hedgehog discussed before.

The function f has to satisfy

$$f(0) = m\pi, \quad m \in \mathbb{Z} \quad (5.4.81)$$

for the field to be regular at the origin. Concerning boundary condition we set

$$\lim_{r \rightarrow \infty} f(r) = 0 \quad (5.4.82)$$

The general m case is discussed in [14]. Here we shall consider $m = 1$, leaving arbitrary n which can be seen to correspond to the topological charge B ,

$$B[\vec{n}] = n \quad (5.4.83)$$

The field (5.4.80) will be a stationary point of the energy functional (which is just V since $T = 0$ for static solutions) if $f(r)$ satisfies the equation

$$\left(r + \frac{n^2 \sin^2 f}{r}\right) f'' + \left(1 - \frac{n^2 \sin^2 f}{r^2}\right) f' - \frac{n^2 \sin f \cos f}{r} - r\mu^2 \sin f = 0 \quad (5.4.84)$$

It was shown in [14] that the hedgehog fields with $n = 1$ and $n = 2$ and f functions satisfying the equation above for those values of n are the absolute minima of V amongst all field of topological charge $B = 1$ and $B = 2$ respectively. We write $\vec{n}^{(1)}$ and $\vec{n}^{(2)}$ for those fields in standard iso-orientation, i.e. with $\chi = 0$ in (5.4.80), and denote their profile functions by $f^{(1)}$ and $f^{(2)}$. Translations in physical space and iso-rotations act nontrivially on $\vec{n}^{(1)}$ and $\vec{n}^{(2)}$, so there is a three-dimensional family of minima of the energy functional for both $n = 1, 2$. We call any field obtained by translating and iso-rotating the field $\vec{n}^{(1)}$ a baby Skymion. Baby Skymions are the basic solitons of our model; they have three degrees of freedom: two translational and one rotational. For the total energy, or mass, of a baby Skymion one has $m = 1.564 \times 4\pi$; the energy density is rotationally symmetric and peaked at the baby Skymion's centre (where $\vec{n}^{(1)} = (0, 0, -1)$). The field $\vec{n}^{(2)}$ (and all those obtained by translating and rotating it) may be thought of as a bound state of two baby Skymions and is described in detail in [14]. The energy density is again rotationally symmetric but peaked at a distance $r \approx 1.8$ from the centre. The mass is $2.936 \times 4\pi$.

The profile function $f^{(1)}$ take, for large r the form

$$f^{(1)}(r) \sim \frac{p\mu}{2\pi} K_1(\mu r) \quad (5.4.85)$$

where p is a constant the asymptotic form of the modified Bessel function K_1 is

$$K_1(\mu r) = \sqrt{\frac{\pi}{2\mu r}} \exp(-\mu r) \left(1 + \mathcal{O}\left(\frac{1}{r}\right)\right) \quad (5.4.86)$$

This shows that the (potential) energy distribution of the field $\vec{n}^{(l)}$ is exponentially localised.

Bibliography

- [1] T. H. R. Skyrme, “A Unified Field Theory of Mesons and Baryons,” Nucl. Phys. **31** (1962) 556.
- [2] E. Witten, “Global Aspects of Current Algebra,” Nucl. Phys. B **223** (1983) 422; “Current Algebra, Baryons, and Quark Confinement,” Nucl. Phys. B **223** (1983) 433.
- [3] R. A. Battye, N. S. Manton, P. M. Sutcliffe, “Skyrmions and Nuclei”, Ed. G. E. Brown and M. Rho, World Sci., Singapore, 2009.
- [4] C. J. Houghton, N. S. Manton and P. M. Sutcliffe, “Rational maps, monopoles and skyrmions”, Nucl. Phys. B **510** (1998) 507.
- [5] A. M. Polyakov and A. A. Belavin, Pis'ma Zh. Eksp. Teor. Fiz. **22** (1975) 503 [JETP Lett. **22** (1975), 245.
- [6] B. M. A. Piette, B. J. Schroers and W. J. Zakrzewski, Z. Phys. C **65** (1995) 165; Nucl. Phys. B **439** (1995) 205.
- [7] A. Jackson, Nucl. Phys. A **493** (1989) 365.
- [8] P. M. Sutcliffe, Phys. Lett. B **283** (1992) 85; G. N. Stratopoulos and W. J. Zakrzewski, Z. Phys. C **59** (1993) 307.
- [9] M. F. Atiyah and N. S. Manton, Phys. Lett. B **222** (1989) 438; Commun. Math. Phys. **153** (1993) 391.
- [10] F. Wilczek and A. Zee, “Linking Numbers, Spin, and Statistics of Solitons,” Phys. Rev. Lett. **51** (1983) 2250.
- [11] M. J. Bowick, D. Karabali and L. C. R. Wijewardhana, “Fractional Spin via Canonical Quantization of the O(3) Nonlinear Sigma Model,” Nucl. Phys. B **271** (1986) 417.

- [12] A. M. Polyakov, “Fermi-Bose Transmutations Induced by Gauge Fields,” *Mod. Phys. Lett. A* **3** (1988) 325.
- [13] J. R. Wen, “A Note on spin of Skyrmions in the (2+1)-dimensions $O(3)$ nonlinear sigma model,” *Commun. Theor. Phys.* **16** (1991) 111.
- [14] B. M. A. G. Piette, B. J. Schroers and W. J. Zakrzewski, “Multi - solitons in a two-dimensional Skyrme model,” *Z. Phys. C* **65** (1995) 165 [hep-th/9406160].
- [15] B. M. A. G. Piette, B. J. Schroers and W. J. Zakrzewski, “Dynamics of baby skyrmions,” *Nucl. Phys. B* **439** (1995) 205 [hep-ph/9410256].

Chapter 6

Instantons

Instantons is the name given by 't Hooft (see ref. [1]) to the localized finite action solutions of the classical equations of motion of Yang-Mills theory in Euclidean space that were found by Belavin, Polyakov, Schwartz and Tyupkin [2]. These authors originally called their solution a "pseudoparticle". Today the name instanton applies to the classical solution to the equations of motion with localized finite Euclidean action of any non-linear field theory¹.

6.1 $SU(2)$ Euclidean Yang-Mills configurations and topology

Let us start by writing the $SU(2)$ Yang-Mills Euclidean action

$$S = -\frac{1}{2g^2} \int d^4x \text{tr} F_{\mu\nu} F_{\mu\nu} \quad (6.1.1)$$

Here g is the coupling constant (factored out with an appropriate redefinition of the gauge field $A_\mu(x)$) taking values in the Lie algebra of $SU(2)$,

$$A_\mu(x) = \frac{\sigma^a}{2i} A_\mu^a, \quad a = 1, 2, 3. \quad (6.1.2)$$

with σ^a the Pauli matrices and $\mu = 0, 1, 2, 3$,

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (6.1.3)$$

¹Note that any soliton (localized static finite energy solution) in $d+1$ Minkowski space-time can be seen as an instanton (localized finite action solution) in d Euclidean space-time.

Defined as in (6.1.2) $A_\mu(x)$ are anti-Hermitian matrices. Correspondingly, the field strength is a matrix-valued tensor

$$F_{\mu\nu} = \frac{\sigma^a}{2i} F_{\mu\nu}^a \quad (6.1.4)$$

which in terms of A_μ reads

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \quad (6.1.5)$$

which, using the Lie algebra

$$\left[\frac{\sigma^a}{2}, \frac{\sigma^b}{2} \right] = i\varepsilon^{abc} \frac{\sigma^c}{2} \quad (6.1.6)$$

becomes in components

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + \varepsilon^{abc} A_\mu^b A_\nu^c \quad (6.1.7)$$

Under an $SU(2)$ (local) gauge transformation A_μ changes as a connection

$$A_\mu(x) \rightarrow A^U_\mu(x) = U(x)A_\mu U^{-1}(x) + U(x)\partial_\mu U^{-1}(x) \quad (6.1.8)$$

For infinitesimal gauge transformations

$$U(x) \approx 1 + i\epsilon(x) \quad (6.1.9)$$

the change reads

$$\delta A_\mu = A_\mu^U - A_\mu = -iD_\mu[A]\epsilon(x) \quad (6.1.10)$$

with $D_\mu[A]$ the covariant derivative

$$D_\mu[A] = \partial_\mu + [A_\mu, \] \quad (6.1.11)$$

Concerning the field strength, it changes covariantly

$$F_{\mu\nu}(x) = U(x)F_{\mu\nu}(x)U^{-1}(x) \quad (6.1.12)$$

The equation of motion resulting from the Euclidean action (6.1.1) reads

$$D_\mu[A]F_{\mu\nu} = 0 \quad (6.1.13)$$

In order to determine the appropriate boundary conditions let us note that finite action configurations require that $F_{\mu\nu}$ must vanish on the boundary of Euclidean four-space, i.e.,

$$\lim_{r \rightarrow \infty} F_{\mu\nu}(x) = 0, \quad r = |x| = \sqrt{x_0^2 + x_1^2 + x_2^2 + x_3^2} \quad (6.1.14)$$

this implying that at infinity A_μ must be a pure gauge

$$\lim_{r \rightarrow \infty} A_\mu(x) = U(x) \partial_\mu U^{-1}(x) \quad (6.1.15)$$

(In fact $F_{\mu\nu}$ must fall to zero faster than $1/r^2$ as $r \rightarrow \infty$.)

The surface at infinity S_∞^3 may be parametrized by three parameters α_i , $i = 1, 2, 3$ (e.g. the 3 polar angles in four dimensional space). Then, working in a gauge in which $A_r = 0$ one can think of $U(x)$ in (6.1.15) at infinity as establishing an application between the spatial sphere at infinity, S_∞^3 , and the gauge group manifold which can be identified with S^3 ,

$$U(x)|_{r \rightarrow \infty} : S_\infty^3 \rightarrow S^3 \quad (6.1.16)$$

Then the possible U elements can be classified according to the homotopy classes to which they belong, characterized by integer numbers

$$\pi_3(S^3) = \mathbb{Z} \quad (6.1.17)$$

That is, the finite action gauge field configurations can be classified by integers which can be calculated by the so called Pontryagin (or Chern-Pontryagin) index Q which can be written in terms of the field-strength (curvature) $F_{\mu\nu}$ and its dual $*F_{\mu\nu}$

$$Q = -\frac{1}{16\pi^2} \text{tr} \int d^4x F_{\mu\nu} *F_{\mu\nu} = n \in \mathbb{Z} \quad (6.1.18)$$

$$*F_{\mu\nu} = \frac{1}{2} \varepsilon_{\mu\nu\alpha\beta} F_{\alpha\beta} \quad (6.1.19)$$

One can easily prove **Problem** that

$$F_{\mu\nu} *F_{\mu\nu} = 2\partial_\mu j_\mu \quad (6.1.20)$$

where

$$j_\mu = -2\text{tr} \left(\varepsilon_{\mu\nu\alpha\beta} \partial_\nu \partial_\alpha A_\beta + \frac{2}{3} A_\nu A_\alpha A_\beta \right) \quad (6.1.21)$$

Note that j_0 can be identified with the Chern-Simons Lagrangian after reducing Euclidean space-time from $d = 4$ to $d = 3$ dimensions.

Using eq.(6.1.20) one can write the Pontryagin index as an integral over S_∞^3

$$Q = \int_{S_\infty^3} ds_\mu j_\mu \quad (6.1.22)$$

which can be written (Problem: Prove the following formula using $F_{\mu\nu}|_{r \rightarrow \infty} = 0$)

$$Q = -\frac{1}{24\pi^2} \text{tr} \int_{S_\infty^3} ds_\mu \varepsilon_{\mu\nu\alpha\beta} U(\partial_\nu U)U^{-1}(\partial_\alpha U)U^{-1}(\partial_\beta U)U^{-1} \quad (6.1.23)$$

Now since the group manifold is also the 3-sphere, its measure $d\mu(U)$ can be also written in terms of three angles ξ ,

$$d\mu(U) = J d\xi_1 d\xi_2 d\xi_3 \quad (6.1.24)$$

The integral in (6.1.23) then counts how many times the group has been wrapped around the spatial S_∞^3 sphere and this number can be seen to be an integer (see a simple proof of this in [3].).

6.2 The BPST instanton solution

As stated in the previous section, a *chetvyorka* of russian physicists (Alexander Belavin, Alexander Polyakov, Albert Schwarz and Yu. S. Tyupkin) found the first solution to the Yang-Mills equations of motion in Euclidean space which they called pseudoparticle (the name instanton was afterwards imposed after 't Hooft) They followed a trick similar to that we described for the $O(3)$ nonlinear sigma model (see chapter 5, and the derivation after eq.(5.4.33)).

In the present case, one starts from the trivial inequality

$$-\text{tr} \int d^4x (F_{\mu\nu} \pm {}^*F_{\mu\nu})^2 \geq 0 \quad (6.2.25)$$

(The minus sign in the l.h.s. is due to the fact that with our conventions for generators $(\sigma^a/(2i))A_\mu^a$ is anti-hermitian.)

Eq.(6.2.25) can be written as

$$-\text{tr} \int d^4x F_{\mu\nu} F_{\mu\nu} \geq \mp \text{tr} \int d^4x F_{\mu\nu} {}^*F_{\mu\nu} \quad (6.2.26)$$

or

$$S \geq \frac{8\pi^2}{g^2} |Q| \quad (6.2.27)$$

where S is the Euclidean Yang-Mills action (6.1.1) and Q the Chern-Pontryagin index (6.1.18).

Obviously the field configuration for which the bound for the action is attained should correspond to a solution to the equations of motion. In view of eq.(6.2.25), such configuration should satisfy the first order equation

$$F_{\mu\nu} = \pm^* F_{\mu\nu} \quad (6.2.28)$$

thus self-dual and self-antidual configurations extremise the action and hence solve the Euclidean Yang-Mills second-order equations of motion (6.1.13).

Belavin et al [2] offered a very simple derivation of the $Q = 1$ (one-instanton) solution which they showed takes the form

$$A_\mu = \frac{|x|^2}{|x|^2 + \lambda^2} U_1(x) \frac{\partial U_1^{-1}(x)}{\partial x_\mu} \quad (6.2.29)$$

with $|x|^2 = x^{0^2} + \vec{x}^2$ and

$$U_1 = \frac{x_0 + i\vec{x} \cdot \vec{\sigma}}{|x|} \quad (6.2.30)$$

with $\vec{\sigma}$ the Pauli matrices. The scale λ arises from the scale invariance of the theory and determines the instanton size. Apart from this parameter, one can introduce a 4-vector a_μ associated to the instanton center by making the shift $x^\mu \rightarrow x^\mu - a^\mu$, associated with translation invariance of the Yang-Mills Lagrangian.

Belavin et al ansatz is related to the $O(4)$ symmetry that can be introduced by extending the original $SU(2)$ gauge group to $SU(2) \times SU(2) \approx O(4)$. Working with that group they considered the more general spherically symmetric ansatz in $d = 4$ Euclidean space time leading to the $Q = 1$ instanton solution.

In order to find instanton solutions with Pontryagin index $Q = n > 1$ 't Hooft introduced an η tensor that maps antisymmetric representations of $SO(4)$ onto vectors of one of its own subgroups of $SO(3)$ [4]-[1]

$$\eta_{a\mu\nu} = \begin{cases} \varepsilon_{a\mu\nu} & \text{if } a, \mu, \nu = 1, 2, 3; \\ -\delta_{a\nu} & \text{if } \mu = 4; \\ \delta_{a\mu} & \text{if } \nu = 4; \\ 0 & \text{if } \mu = \nu = 4. \end{cases} \quad (6.2.31)$$

in terms of this η tensor 't Hooft ansatz reads

$$A_\mu^a = i\eta_{a\mu\nu} \partial_\nu \log \phi(x) \quad (6.2.32)$$

Inserting this ansatz into the selfduality equations (6.2.28) and after some work one gets

$$\frac{1}{\phi(x)} \square \phi(x) = 0 \quad (6.2.33)$$

where $\square\phi$ is the $d = 4$ Laplacian $\square = \partial_\mu\partial_\mu$ (remember taht we are working in Euclidean space).

When ϕ is non-singular eq.(6.2.33) reduces to $\square\phi(x) = 0$ which, in order to have finite action configurations, permits only the trivial solution $\phi = \text{constant}$ so that $A_\mu = 0$. There are however singular $\phi(x)$ solutions leading to non-singular gauge fields. For example consider the case

$$\phi^{(1)} = \frac{1}{x^2} \quad (6.2.34)$$

which for $x \neq 0$ satisfies

$$\frac{1}{\phi^{(1)}}\square\phi^{(1)} = \frac{1}{\phi^{(1)}}\partial_\mu\left(-2\frac{x_\mu}{|x|^4}\right) = 0 \quad (6.2.35)$$

For $x = 0$ one can easily see using Gauss theorem that that

$$\int d^4x \square \frac{1}{|x|^2} = \int dS_\mu \partial_\mu \left(\frac{1}{|x|^2} \right) = -4\pi \quad (6.2.36)$$

showing that $1/|x|^2$ is the 4-dimensional Laplacian Green function. Hence one has

$$\frac{1}{\phi^{(1)}}\square\phi^{(1)} = -4\pi|x|^2\delta^{(4)} = 0 \quad (6.2.37)$$

The precedent discussion shows that $\phi^{(1)}$ solves the equation (6.2.33) and, more generally one has that

$$\phi^{(n)}(x) = 1 + \sum_{i=1}^n \frac{\lambda_i^2}{|x_\mu - a_\mu^i|^2} \quad (6.2.38)$$

this leading to an instanton solution with Chern-Pontryagin index $Q = n$.

In fact, when computing the n -instanton solution from $\phi^{(n)}$ in the form (6.2.38) one finds that the solution $A_\mu^{(n)}$ is singular at points $x_\mu = a_\mu^i$. However these singularities can be removed to infinity by singular gauge transformations, one for each singularity. Such transformations are of the form (6.2.30)

$$U^i(x) = \frac{(x_0 - a_0^i) + i\vec{\sigma} \cdot (\vec{x} - \vec{a}^i)}{|x - a^i|} \quad (6.2.39)$$

and the final form of the regular 't Hooft n -instanton solution reads

$$A_\mu^{(n)} = 2i\eta_{\mu\nu a} \frac{\sum_i^n \frac{\lambda_i^2(x_\mu - a_{i\mu})}{|x - a_i|^4}}{1 + \sum_i^n \frac{\lambda_i^2}{|x - a_i|^2}} \quad (6.2.40)$$

The associated field strength reads

$$F_{\mu\nu}^a = 2\eta_{a\mu\nu} \sum_{i=1}^n \frac{\lambda^{i^2}}{(|x - a^i|^2 + \lambda^{i^2})^2} \quad (6.2.41)$$

This solution corresponds to an instanton with n units of topological charges, each one located at points a_μ^i in $d = 4$ Euclidean space-time.

6.3 Witten cylindrically symmetric multi-instanton solution

Following a different approach Witten presented in ref. [5] a cylindrically symmetric n -instanton solution with arbitrary scale parameters and arbitrary separations, arranged along a line.

Witten looked for solutions of the self-dual equations (6.2.28) that are invariant under three-dimensional rotations combined with gauge transformations. He calls this a cylindrical symmetry, because it determines the dependence of the fields on the polar angles of three-dimensional space and leaves unknown only the dependence on the radius r of three-dimensional space and the Euclidean time x_0 .

The most general gauge field with cylindrical symmetry takes the form

$$\begin{aligned} A_j^a &= \frac{(\phi_2 + 1)}{r^2} \varepsilon_{jka} x_k + A_1 \frac{\phi_1}{r^3} (\delta_{ja} r^2 - x_j x_a) \\ A_0^j &= A_0 \frac{x^j}{r} \end{aligned} \quad (6.3.42)$$

Here $j, k = 1, 2, 3$ and ϕ_1, ϕ_2, A_1, A_0 are functions that depend on (r, t) to be defined below.

Inserting this ansatz into the Yang-Mills action one finds, after a lengthy calculation

$$S = 8\pi \int_{-\infty}^{\infty} dx_0 \int_0^{\infty} dr \left(\frac{1}{2} (D_\mu \phi_i)^2 + \frac{1}{8} r^2 (\partial_\mu A_\nu - \partial_\nu A_\mu)^2 + \frac{1}{r^2} (1 - \phi_1^2 - \phi_2^2)^2 \right) \quad (6.3.43)$$

Here we introduced indices $\mu, \nu = 0, 1$ and put $\partial_0 = \partial/\partial x_0$, $\partial_1 = \partial/\partial r$. Written in this form action (6.3.43) is nothing but the Abelian Higgs model action in a curved 2 dimensional space with

$$g_{\mu\nu} = r^2 \delta_{\mu\nu} \quad (6.3.44)$$

This metric has constant negative curvature and corresponds to the hyperbolic space metric.

With the Nielsen-Olesen-Bogomolny experience in flat space, it is not difficult to find first order self-dual equations which read in the present hyperbolic space case

$$\begin{aligned}\partial_0\phi_1 + A_0\phi_2 &= \partial_1\phi_2 - A_1\phi_1 \\ \partial_1\phi_1 + A_1\phi_2 &= -(\partial_0\phi_2 - A_0\phi_1)\end{aligned}\tag{6.3.45}$$

$$r^2(\partial_0A_1 - \partial_1A_0) = 1 - \phi_1^2 - \phi_2^2\tag{6.3.46}$$

In the Lorenz gauge, $\partial_\mu A_\mu = 0$, one can write the gauge field in the form

$$A_\mu = \varepsilon_{\mu\nu}\partial_\nu\psi\tag{6.3.47}$$

Then the first two equations become

$$\begin{aligned}(\partial_0 - (\partial_0\psi))\phi_1 &= (\partial_1 - (\partial_1\psi))\phi_2 \\ (\partial_1 - (\partial_1\psi))\phi_1 &= -(\partial_0 - (\partial_0\psi))\phi_2\end{aligned}\tag{6.3.48}$$

which, after writing

$$\phi_1 = \chi_1 \exp \psi, \quad \phi_2 = \chi_2 \exp \psi\tag{6.3.49}$$

become the Cauchy-Riemann equations for the function

$$f = \chi_1 - i\chi_2, \tag{6.3.50}$$

Indeed, inserting the change of variables (11.1.13) in eqs.(6.3.48) one gets

$$\begin{aligned}\partial_0\chi &= \partial_1\chi_2 \\ \partial_0\chi_1 &= \partial_0\chi_2\end{aligned}\tag{6.3.51}$$

Hence f is any analytic function of $z = r + it$.

Concerning the third equation in (6.3.45), it takes the form

$$-r^2\nabla^2\psi = 1 - f^*f \exp(2\psi)\tag{6.3.52}$$

There is a residual gauge invariance that we have not yet commented. Indeed, the gauge $\partial_\mu A_\mu = 0$ that we have chosen permits transformations $A_\mu \rightarrow A_\mu + \partial_\mu\lambda$ provided $\nabla^2\lambda = 0$. Associated with this invariance there is an invariance of function ψ introduced through eq.(6.3.47),

$$\psi \rightarrow \psi - \frac{1}{2} \log(h^*h)\tag{6.3.53}$$

provided $h(z)$ is analytic since in that case

$$\nabla^2(h^*h) = 0\tag{6.3.54}$$

This transformation should be accompanied by an appropriate change in f , which in turn was defined in terms of the scalars ϕ_1 and ϕ_2 according to (11.1.13).

Then, if we just change ψ as in eq.(6.3.53) ϕ_1, ϕ_2 would change as

$$\phi_1 = \chi_1 \exp \psi \rightarrow \chi_1 |h|^{-1} \exp \psi, \quad \phi_2 = \chi_2 \exp \psi \rightarrow \chi_2 |h|^{-1} \exp \psi \quad (6.3.55)$$

which means that f defined in (6.3.50) would change as

$$f \rightarrow |h|^{-1} f \quad (6.3.56)$$

Then we can compensate this change by imposing a complete transformation

$$\begin{aligned} \psi &\rightarrow \psi - \frac{1}{2} \log(h^* h) \\ f &\rightarrow h f \end{aligned} \quad (6.3.57)$$

so that the gauge invariance would manifest as an invariance of equation (6.3.52). To solve this equation Witten proposed the following ansatz,

$$\psi = \log r - \frac{1}{2} \log(f^* f) + \rho \quad (6.3.58)$$

where ρ is a new unknown function. Applying ∇^2 to this equation we get

$$\nabla^2 \rho = \exp(2\rho) \quad (6.3.59)$$

where we have ignored for the moment possible singularities of the analytic function f .

Eq. (6.3.59) is the well-known Liouville equation whose general solution is

$$\rho(z) = -\log \left(\frac{1 - g^* g}{2} \right) + \frac{1}{2} \log \left(\left| \frac{dg}{dz} \right|^2 \right) \quad (6.3.60)$$

with $g(z)$ any analytic function. Now, if and only if $(dg/dz)/z$ has neither zeroes nor poles in the right half plane the various singularities in (6.3.52) vanish and then its more general solution is (up to gauge transformations)

$$\psi = -\log \left(\frac{1 - g^* g}{2r} \right), \quad f = \frac{dg}{dz} \quad (6.3.61)$$

One has to impose $|g| = 1$ in order to have a non-singular ψ at $r = 0$ and also $|g| < 1$ for $r > 0$. The most general analytic function with these properties and smooth behavior for $z \rightarrow \infty$ is

$$g(z) = \prod_{i=1}^k \left(\frac{a_i - z}{a_i^* + z} \right) \quad (6.3.62)$$

where the a_i are an arbitrary set of complex numbers (some perhaps equal) constrained only to have $\text{Re } a_i > 0$. Eqs. (7.3.97)-(6.3.62) provide the most general solution of $F_{\mu\nu} = {}^*F_{\mu\nu}$ with cylindrical symmetry and finite action.

Let us now analyze the role of the zeroes of f in the right half plane which in view of (6.3.57) are gauge invariant quantities. If $k = 1$ f has no zeroes in the right half plane (r.h.p.) and it is easy to see that the solution is the vacuum. If $k = 2$ f has one zero in the r.h.p.. For general k f has $k - 1$ zeroes in the r.h.p.

One can then see that a k solution correspond to an instanton solution with topological charge $Q = k - 1$. One way to prove this is to first consider the case in which the zeroes of f are widely separated. Then one can extend the result to arbitrary positions since they can be reached continuously thus without affecting the topological charge. We shall not describe in detail this proof which can be found [5]. Instead, we shall present a proof based in the connection between the the topological charge of instantons and those of vortices associated to the reduced action (6.3.43).

We start from the topological charge expression written in terms of the fields A_1, A_0, ϕ_1, ϕ_2

$$Q = \frac{1}{2\pi} \int d^2x \left(\varepsilon_{\mu\nu} \varepsilon_{ij} D_\mu \phi_i D_\nu \phi_j + \frac{1}{2} \epsilon_{\mu\nu} F_{\mu\nu} \right) \quad (6.3.63)$$

where here $\mu, i = 1, 2$. The first term can be rewritten as a total divergence and converted to a boundary integral which vanishes because $D_i \phi_i = 0$ on the boundary (finite energy requirement). Concerning the second term, it is nothing but the usual topological charge Q_v of the vortex solution for the Abelian Higgs model so that it should be an integer. Now, as we have seen, the topological charge can be written in terms of the phase of the Higgs field $\phi = \phi_1 - i\phi_2$

$$Q_v = \frac{1}{2\pi i} \oint dx^\mu \phi * \partial_\mu \phi = \frac{1}{2\pi i} \oint ds \frac{1}{f} \frac{df}{ds} + \frac{1}{2\pi i} \oint ds \frac{d\psi}{ds} \quad (6.3.64)$$

where ds is the line element along the contour. Since ψ is a singled valued function the second integral vanishes while the first one picks contributions from each one of the zeroes of f so that

$$Q = Q_v = k - 1 \quad (6.3.65)$$

6.4 Collective coordinates, zero modes and instanton moduli space

6.4.1 Collective coordinates

The n instanton 't Hooft solution presented in eq.(6.2.40) exhibit a number of parameters that in the context of soliton quantization should be treated as collective coordinates.

The parameters associated to instanton configurations include

1. $4n$ coordinates $a_{i\mu}$ specifying the centers around which each one of the n topological densities are localized. They are associated to translation invariance of the theory.
2. n scale parameters λ_i measuring the size of each region in which the topological density is localized. They are associated to scale invariance of the theory
3. 3 global $SU(2)$ gauge transformations that originally were ignored on physical grounds

One should also take into account the fact that Yang-Mills theory exhibits a complete conformal invariance (not just the scale invariance manifested through the presence of λ_i . As Jackiw and Rebbi have proven [6] one has to take into account invariance under the whole conformal group which includes not only translations and dilatations but also the 6 $O(4)$ rotations and the 4 special conformal transformations

$$x_\mu \rightarrow \frac{x_\mu - c_\mu x^2}{1 - 2c_\mu x_\mu + x^2 c^2} \quad (6.4.66)$$

Looking at the solution obtained from 't Hooft ansatz permits to identified $5n + 3$ parameters. When quantizing Yang-Mills theory around an n -instanton those parameters will become collective coordinates. It should be stressed that the correct number of parameters is not the one revealed by the ansatz but, as proven by Atiyah and Ward [7] (see also [6]) the number of parameters is $8n - 3$. If global rotations are counted, the number becomes $8n$ for $SU(2)$ and $4nN$ for $SU(N)$.

6.4.2 Bosonic zero-modes

Classical solutions to the equations of motion of a given field theory play an important role in non-perturbative studies at the quantum level within the path-integral framework.

We shall start by considering a bosonic theory with action $S[\phi]$ with ϕ scalar fields, gauge fields, etc. having a classical solution (say an instanton solution) that we call $\phi_{cl}(x, a_i)$, thus satisfying

$$\frac{\delta S[\phi(x)]}{\delta \phi_{cl}(x; a)} = 0 \quad (6.4.67)$$

Here a represent the set of parameters associated to the invariances of the action (e.g. translation invariance, rotation invariance, gauge invariance, etc). For simplicity we shall discuss below the case in which just one parameter a exists.

Within the path-integral framework the quantum field theory is studied starting from the path-integral

$$Z = \int D\phi(x) \exp \left(-\frac{1}{\hbar} S[\phi] \right) \quad (6.4.68)$$

which we have written in Euclidean space where it can be more rigorously defined. The measure $D\phi$ should be appropriately defined in order to apply the Feynmann recipe of *integration over all fields* (with given boundary conditions). In view of the well-appreciated connection between Euclidean quantum field theory and statistical mechanics, Z is usually called the partition function and

$$Z[j] = \int D\phi(x) \exp \left(-\frac{1}{\hbar} S[\phi] \right) \exp \left(\int dx j(x) \phi(x) \right) \quad (6.4.69)$$

the associated generating functional with $j(x)$ an external source introduced in order to calculate correlation functions by functional differentiation,

$$\langle \phi(x_1) \phi(x_2) \dots \phi(x_n) \rangle \equiv \frac{1}{Z[j]} \frac{\delta^n}{\delta \phi(x_1) \delta \phi(x_2) \dots \delta \phi(x_n)} Z[j] \Big|_{j=0} \quad (6.4.70)$$

Given Z in eq.(6.4.68), classical solutions enter in the quantum game through a change of path-integral variables

$$\phi = \phi_{cl}(x, a) + \sqrt{\hbar} \varphi(x) \quad (6.4.71)$$

Written in this form it becomes clear that the new variable φ will take into account, when integrated, of quantum fluctuations while $\phi_{cl}(x, a)$ renders the dominating contribution from classical configurations in the $\hbar \rightarrow 0$ limit. Indeed, making an expansion in powers of $\hbar$ after changing variables in the action one gets

$$Z = \exp \left(-\frac{1}{\hbar} S[\phi_{cl}] \right) \int D\varphi(x) \exp \left(- \int dx dy \varphi(y) \frac{\delta^2 S[\phi]}{\delta \phi_{cl}(y; a) \delta \phi_{cl}(x; a)} \varphi(x) + \mathcal{O}(\hbar^{3/2}) \right) \quad (6.4.72)$$

Were the $\mathcal{O}(\hbar^{3/2})$ the term quadratic in φ in the exponential can be integrated in close form. The most convenient way to do this is to consider a Bessel Fourier expansion for φ ,

$$\varphi = \sum_n c_n \varphi_n(x) \quad (6.4.73)$$

with $\varphi_n(x)$ the eigenfunctions of the equation

$$\int dx \left(\frac{\delta^2 S[\phi]}{\delta \phi_{cl}(y; a) \phi_{cl}(x; a)} \right) \varphi_n(x) = \lambda_n \varphi_n(y) \quad (6.4.74)$$

and c_n being real numbers. One then defines the path integral measure as

$$D\varphi = \prod_n dc_n \quad (6.4.75)$$

this leading, after integrating over the c_n 's to

$$Z = \exp \left(-\frac{1}{\hbar} S[\phi_{cl}] \right) \det^p \left(\frac{\delta^2 S[\phi]}{\delta \phi_{cl}(y; a) \phi_{cl}(x; a)} \right) (1 + \dots) \quad (6.4.76)$$

Here the determinant is defined as the product of the eigenvalues associated to eq.(6.4.74),

$$\det \left(\frac{\delta^2 S[\phi]}{\delta \phi_{cl}(y; a) \phi_{cl}(x; a)} \right) = \prod_n \lambda_n \Big|_{Reg} \quad (6.4.77)$$

and it arises from the product of gaussian c_n -integrations. Here *Reg* indicates that in general the product of eigenvalues is unbounded and it should be appropriately regularized. The power p is $-1/2$ for real scalar fields (or for each vector component, tensor component).

Now, apart from the fact that in principle the product of eigenvalues grows without bound, there is a zero-eigenvalue problem. Indeed, if one differentiates eq.(6.4.67) with respect to the a parameter associated to some invariance of the theory one gets

$$\frac{d}{da} \frac{\delta S[\phi(x)]}{\delta \phi_{cl}(x; a)} = \int dy \frac{\delta^2 S[\phi(x)]}{\delta \phi_{cl}(y; a) \delta \phi_{cl}(x; a)} \frac{d\phi_{cl}(x; a)}{da} = 0 \quad (6.4.78)$$

Comparing this equation with eq.(6.4.74) we see that $\varphi_i^{(0)} = d\phi_{cl}(x; a)/da$ with $i = 1, 2, \dots, r$ and r the number of invariances are eigenfunction of the operator $\hat{S}^{(II)}$,

$$\hat{S}^{(II)} \equiv \int dy \frac{\delta^2 S[\phi(x)]}{\delta \phi_{cl}(y; a) \delta \phi_{cl}(x; a)} \quad (6.4.79)$$

with zero eigenvalue,

$$\hat{S}^{(II)} \frac{d\phi_{cl}(y; a)}{da} = 0 \quad (6.4.80)$$

There are in general as many zero-eigenvalues (zero-modes) as invariance parameters of the theory and they have to be treated appropriately in order to make sense from the path-integration. Otherwise the quadratic approximation would give a zero answer (once the large eigenvalue problem is under control).

Consider as an example the zero-mode associated to gauge invariance of the $SU(2)$ Yang-Mills theory. There are 3 parameters $\alpha^a(x)$ ($a = 1, 2, 3$) associated with the $SU(2)$ group so that

$$\varphi_\mu^{ab(0)} = \frac{\delta A_\mu^{a\,cl}}{\delta \alpha^b} = D_\mu^{ab}[A_{cl}] \delta(x - y) \quad (6.4.81)$$

where $D_\mu^{ab}[A_{cl}]$ is the covariant derivative in the classical solution background. One should remember that the eigenvalue equation is an integral eigenvalue equation (see eq.(6.4.74)) so its derivative makes sense after integration by parts. An alternative form to handle eq. (7.3.94) is to explicitly compute the second derivative of the action and then arrive a second order differential operator with zero-eigenvalue given by

$$\varphi_\mu^{a(0)} = D_\mu^{ab}[A_{cl}] \alpha^b \quad (6.4.82)$$

with $\alpha^b(x)$ arbitrary.

Interestingly enough, eliminating gauge zero-modes illuminates the way in which the natural gauge condition should be chosen. Indeed, the natural thing to do to avoid the integrating over the gauge zero mode is to fix the gauge so that all non-zero modes are orthogonal to it. Imposing this condition to any nonzero mode $\varphi_\nu^{a(n)}$ implies that

$$\int dx \varphi_\nu^{a(n)}(x) D_\mu^{ab}[A_{cl}] \alpha^b(x) = 0 \quad (6.4.83)$$

or, after integrating by parts

$$\int dx D_\mu^{ab}[A_{cl}] \varphi_\nu^{a(n)}(x) \alpha^b(x) = 0 \quad (6.4.84)$$

so that a gauge choice of the form

$$D_\mu[A_{cl}] A_\mu = 0 \quad (6.4.85)$$

and the corresponding one for each non-zero mode will guarantee their orthogonality with the zero-mode which can then be trivially eliminated using the Faddeev-Popov method. Gauge condition (6.4.85) is known as the "natural gauge" [8]-[9].

6.4.3 Fermion zero-modes: the case of instantons

The occurrence of zero-eigenvalue modes for the Dirac equation in a topologically non-trivial background field is by now a familiar and well-studied phenomenon. Examples exist for spinors in topologically non-trivial backgrounds like kinks (in $d = 1$ dimensions), vortices ($d = 2$), 't Hooft-Polyakov monopoles, Julia-Zee dyons ($d = 3$) and instantons ($d = 4$).

Here we shall discuss the zero-mode problem for the Dirac operator in an instanton background following Jackiw-Rebbi discussion of Euclidean Yang-Mills theory [10].

We are interested in Euclidean 4-space solutions to the Dirac equation

$$\not{D}[A]\psi \equiv i\gamma^\mu(\partial_\mu + A_\mu)\psi = 0 \quad (6.4.86)$$

where the $SU(2)$ gauge field A_μ corresponds to an instanton solution and ψ represents a 4-component spinor transforms according to some definite representation of $SU(2)$ with generators T^a ,

$$\delta\psi = i\epsilon^a T^a \psi \quad (6.4.87)$$

where

$$[T^a, T^b] = i\epsilon^{abc} T^c \quad (6.4.88)$$

and

$$iA_\mu = A_\mu^a T^a \quad (6.4.89)$$

The 4×4 Dirac matrices γ_μ satisfy Euclidean anticommutation relations,

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu} \quad (6.4.90)$$

(There is no distinction between upper and lower indices.) A convenient realization is

$$\vec{\gamma} = \begin{pmatrix} 0 & -i\vec{\sigma} \\ i\vec{\sigma} & 0 \end{pmatrix}, \quad \gamma_0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \quad (6.4.91)$$

with $\vec{\sigma}$ the Pauli matrices given in eq.(6.1.3). In this representation the γ_5 matrix takes the form

$$\gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \quad (6.4.92)$$

Now, γ_5 anticommutes with the Hermitian Dirac operator $\not{D}[A]$ and then if we consider the full eigenvalue spectrum

$$\not{D}[A]\psi_E = E\psi_E \quad (6.4.93)$$

We see that γ_5 takes eigenfunctions ψ_E with $E \neq 0$ into ψ_{-E} ,

$$\gamma_5\psi_E = \psi_{-E} \quad (6.4.94)$$

Concerning zero-eigenvalue modes, they can be chosen as eigenstates of γ_5 and in this way they have either positive or negative chirality. That is, given ψ_0 such that

$$\not{D}[A]\psi_0 = 0 \quad (6.4.95)$$

we can construct $\psi_0^\pm$

$$\psi_0^\pm = \frac{1}{2}(1 \pm \gamma_5)\psi_0 \quad (6.4.96)$$

such that

$$\gamma_5\psi_0^\pm = \pm\psi_0^\pm \quad (6.4.97)$$

The eigenvalue problem (6.4.93) may be written in a form which makes explicit the chiral structure,

$$\begin{pmatrix} 0 & L \\ L^\dagger & 0 \end{pmatrix} \begin{pmatrix} \psi_E^+ \\ \psi_E^- \end{pmatrix} = E \begin{pmatrix} \psi_E^+ \\ \psi_E^- \end{pmatrix} \quad (6.4.98)$$

where

$$L = i\alpha^\mu(\partial_\mu + A_\mu), \quad L^\dagger = i\bar{\alpha}^\mu(\partial_\mu + A_\mu) \quad (6.4.99)$$

and

$$\alpha^\mu = (-i\vec{\sigma}, 1), \quad \bar{\alpha}^\mu = (i\vec{\sigma}, 1) = (\alpha^\mu)^\dagger \quad (6.4.100)$$

The zero-eigenvalue modes satisfy

$$\bar{\alpha}_\mu(\partial_\mu + A_\mu)\psi^+ = 0 \quad (6.4.101)$$

$$\alpha_\mu(\partial_\mu + A_\mu)\psi^- = 0 \quad (6.4.102)$$

Here $\psi^\pm$ are two-component spinors, which also carry an isospin label appropriate to the representation (6.4.88).

The explicit form of zero modes for an isovector massless fermion field belonging to the adjoint representation of SU(2) in an instanton background can be found in [10]. We shall postpone the derivation and now discuss the index theorem.

The Atiyah-Singer index theorem

Within the path-integral quantization approach the existence of zero modes poses a problem: integrating out the Dirac fermion action gives as a result the Dirac fermion determinant, i.e., the product of Dirac operator eigenvalues. Apart from large eigenvalues which grow without bound requiring an appropriate regularization, zero eigenvalues made should also be handled with care. That is, a naive calculation would lead the fermionic path-integral to a meaningless $0 \times \infty$ answer!!!

Once the large eigenvalue problem is cured within some regularization approach, a natural procedure to handle the zero-mode problem starts by adding a mass term to the Lagrangian, so that the generating functional reads

$$Z_m[A] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(- \int d^4x \bar{\psi} (i\not{D} + \not{A} + m) \psi \right) \quad (6.4.103)$$

From here on, we shall consider the case of fermions in the fundamental representation but the procedure holds for any other representation. If we now call

$$D_m[A] = \not{D}[A] + m \quad (6.4.104)$$

one easily sees that the zero-modes of $\not{D}$ continue to be eigenmodes of D_m but now with associated non-zero eigenvalue m ,

$$D_m[A] \varphi_n^0 = m \varphi_n^0 \quad (6.4.105)$$

Of course, if one naively takes the $m \rightarrow 0$ limit after eliminating the large eigenvalue contribution one should obtain a trivial answer, since one can write for D_m ,

$$\det D_m[A] = m^N \prod_n \lambda'_n \quad (6.4.106)$$

where we indicate with λ'_n non-zero eigenvalues.

A natural question to pose is whether the operator D_m has no new zero eigenvalues even when it contains a diagonal term as the mass term produces. The answer is no: zero modes, in order to be included in the basis used to define the path-integral should be normalizable. Otherwise, they can be excluded and they do not appear in the product of eigenvalues defining the Dirac operator determinant. But, when this operator has diagonal elements (as it is the case with D_m), one can prove that no normalizable zero modes exist (We refer the interested persons to [11] and [12] for a thorough discussion of this point).

We now consider a particular case to show how one proceeds to take the $m \rightarrow 0$ limit. Suppose we want to calculate the Jacobian arising from gauge transformations of the fermionic variables in $Z_m[A]$. It is easy to see that it can be written as a ratio of determinants in the form

$$J_m[\delta\alpha] = \frac{\det D_m[A]}{\det D_m[A^{\delta\alpha}]} \quad (6.4.107)$$

where we have used the fact that the infinitesimal gauge transformation of fermions can be rewritten in terms of the gauge transformed gauge field $A_\mu^{\delta\alpha}$. Now, we can use (6.4.106)

in each determinant in the ratio (assuming some regularization for the product of non-zero eigenvalues),

$$J_m[\delta\alpha] = \frac{m^N \prod_n \lambda'_n}{m^N \prod_n \lambda'_n} \Big|_{reg'} \quad (6.4.108)$$

Here we have used that the non-zero eigenvalues λ'_n are gauge invariant and indicated with reg' the fact that the product of non-zero eigenvalues should be consistently regularized. Evidently, the numerator cancels the denominator so that finally we have $J_m = 1$. Then, taking the $m \rightarrow 0$ limit one can write

$$J \equiv \lim_{m \rightarrow 0} J_m = 1 \quad (6.4.109)$$

where J represents the gauge Jacobian for the *originally massless* theory.

This procedure will be the natural one each one faces the problem of Dirac operator zero modes: one adds a “small” mass, perform all calculations and at the end one takes the $m \rightarrow 0$ limit. Hence, it will be convenient to directly define a “primed” determinant just extracting from the beginning the zero eigenvalues

$$\det' \mathcal{D} \equiv \prod_n \lambda'_n \quad (6.4.110)$$

That this procedure is mathematically well-defined is proved in [12].

As it is well known the free massless Dirac Lagrangian is invariant under global quiral rotations

$$\psi \rightarrow \exp(\gamma_5 \delta\alpha) \psi, \quad \bar{\psi} \rightarrow \bar{\psi} \exp(\gamma_5 \delta\alpha) \quad (6.4.111)$$

with $\delta\alpha$ a constant infinitesimal transformation. When a fermion mass term is added, this symmetry is classically broken.

To see how the symmetry is broken the quantum level, one can apply the Noether method which consists in promoting the global transformation to a local one and take it as a change to new variables $\psi', \bar{\psi}'$ in the partition function,

$$\psi = \exp(\gamma_5 \delta\alpha(x)) \psi', \quad \bar{\psi} = \bar{\psi}' \exp(\gamma_5 \delta\alpha(x)) \quad (6.4.112)$$

After this change, Z_m reads

$$\begin{aligned} Z_m[A] = & J_{5m}[\delta\alpha] \int D\bar{\psi} D\psi \exp \left(- \int d^4x \bar{\psi} D_m[A] \psi \right) \\ & \exp(-2m \int d^4x \bar{\psi} \gamma_5 \psi \delta\alpha(x)) \exp \left(\int d^4x \delta\alpha(x) \partial_\mu j_5^\mu \right) \end{aligned} \quad (6.4.113)$$

(where we have renamed the new variables so as to ignore the primes). Concerning J_{5m} , it stands for the Jacobian associated to the chiral change of variables.

Now the generating functional $Z_m[A]$ cannot depend on $\delta\alpha(x)$, a parameter introduced by the change of variables. Then one has

$$0 = \frac{1}{Z} \frac{\delta Z_m}{\delta\alpha(x)} \Big|_{\alpha=0} = \frac{1}{J_5} \frac{\delta J_{5m}}{\delta\alpha(x)} \Big|_{\alpha=0} - 2m \langle \bar{\psi} \gamma_5 \psi \rangle + \langle \partial_\mu j_5^\mu \rangle \quad (6.4.114)$$

At this point we can take the $m \rightarrow 0$ limit. First, we note that the Jacobian does not depend on m . One can convince oneself of this by rememebring that the non trivial contribution to chiral Jacobians comes from the trace of a term containing γ_5 times the product of as many Dirac operators as the dimension of space-time is. Now, the mass term is proportional to the identity so that it reduces the number of γ_μ matrices in the product leading to a vanishing trace (see Problem). We then have

$$\langle \partial_\mu j_5^\mu \rangle = - \frac{\delta \log J_5}{\delta\alpha(x)} \Big|_{\alpha=0} + \lim_{m \rightarrow 0} 2m \langle \bar{\psi} \gamma_5 \psi \rangle \quad (6.4.115)$$

or, using the result for the Fujikawa Jacobian (see for exemple [9])

$$\langle \partial_\mu j_5^\mu \rangle = - \frac{1}{8\pi^2} \text{tr}^c(*F_{\mu\nu} F^{\mu\nu}) + \lim_{m \rightarrow 0} 2m \langle \bar{\psi} \gamma_5 \psi \rangle \quad (6.4.116)$$

We are left with the computation of $\langle \bar{\psi} \gamma_5 \psi \rangle$,

$$\langle \bar{\psi} \gamma_5 \psi \rangle = \frac{1}{Z} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(- \int d^4x \bar{\psi} D_m[A] \psi \right) \bar{\psi} \gamma_5 \psi \quad (6.4.117)$$

Concerning the normalization Z , one has, after expanding fermionic variables in Bessel-Fourier series in terms of the $\mathcal{D}[A]$ basis,

$$Z = \int \prod_n d\bar{c}_n dc_n \exp \left(- \sum_k \bar{c}_k c_k (\lambda_k + m) \right) = \prod_k (\lambda_k + m) \quad (6.4.118)$$

Concerning the numerator, which we call $\mathcal{N}$,

$$\mathcal{N} = \int \prod_n d\bar{c}_n dc_n \exp \left(- \sum_k \bar{c}_k c_k (\lambda_k + m) \right) \sum_{pq} \bar{c}_p c_q \varphi_p^\dagger(x) \gamma_5 \varphi_q(x) \quad (6.4.119)$$

In view of anticommuting integration rules, only those terms in the sum with $p = q$ will contribute

$$\mathcal{N} = \sum_p \varphi_p^\dagger(x) \gamma_5 \varphi_p(x) \int \prod_n d\bar{c}_n dc_n \exp \left(- \sum_k \bar{c}_k c_k (\lambda_k + m) \right) \bar{c}_p c_p \quad (6.4.120)$$

When computing Z , one expands the exponential and only the term containing once each eigenvalue contributes. Here, it is the term containing each but one, since there is also the product $\bar{c}_p c_p$ to integrate. And this for $p = 1, 2, \dots$. The answer is then

$$\mathcal{N} = \sum_p \varphi_p^\dagger(x) \gamma_5 \varphi_p(x) (\lambda_1 + m)(\lambda_2 + m) \dots (\lambda_{p-1} + m)(\lambda_{p+1} + m) \dots \quad (6.4.121)$$

Then, from (6.4.118) and (6.4.121) we have

$$\langle \bar{\psi} \gamma_5 \psi \rangle = \sum_p \varphi_p^\dagger(x) \gamma_5 \varphi_p(x) \frac{1}{\lambda_p + m} \quad (6.4.122)$$

Then, we have

$$\lim_{m \rightarrow 0} 2m \langle \bar{\psi} \gamma_5 \psi \rangle = \lim_{m \rightarrow 0} 2m \sum_p \varphi_p^\dagger(x) \gamma_5 \varphi_p(x) \frac{1}{\lambda_p + m} \quad (6.4.123)$$

Once the limit is taken, the m factor in the numerator eliminates all contributions from the sum except those corresponding to $\lambda_p = 0$ since in that case the m factor in the numerator is cancelled by that in the denominator. That is, only zero modes give a non-vanishing contribution and one ends with

$$\lim_{m \rightarrow 0} 2m \langle \bar{\psi} \gamma_5 \psi \rangle = 2 \sum_{p=1}^N \varphi_p^{0\dagger}(x) \gamma_5 \varphi_p^0(x) \quad (6.4.124)$$

We can then write (6.4.116) in the form

$$\langle \partial_\mu j_5^\mu \rangle = -\frac{1}{8\pi^2} \text{tr}^c(*F_{\mu\nu} F^{\mu\nu}) + 2 \sum_{p=1}^N \varphi_p^{0\dagger}(x) \gamma_5 \varphi_p^0(x) \quad (6.4.125)$$

Let us integrate this identity over all space. The l.h.s. can be seen to vanish using Stokes theorem,

$$\int_M d^4x \langle \partial_\mu j_5^\mu \rangle = \langle \int_{\partial M} dS_\mu j_5^\mu \rangle = 0 \quad (6.4.126)$$

Here we interchanged the expectation value with integration and also assumed that fields go to zero on the boundary. So we have

$$2 \sum_{p=1}^N \int d^4x \varphi_p^{0\dagger}(x) \gamma_5 \varphi_p^0(x) = \frac{1}{8\pi^2} \text{tr}^c \int d^4x (*F_{\mu\nu} F^{\mu\nu}) \quad (6.4.127)$$

The l.h.s. can be easily computed recalling that zero-modes can be written so as to be eigenfunctions of γ_5 with eigenvalues ± 1 ,

$$\varphi^{0\pm} = \frac{1 \pm \gamma_5}{2} \chi^0, \quad \gamma_5 \varphi^{0\pm} = \pm \varphi^{0\pm} \quad (6.4.128)$$

Then, the sum in (6.4.127) becomes

$$\sum_{p=1}^N \int d^4x \varphi_p^{0\dagger}(x) \gamma_5 \varphi_p^0(x) = n_+ - n_- \quad (6.4.129)$$

so that we end with the formula

$$\frac{1}{16\pi^2} \text{tr}^c \int d^4x (*F_{\mu\nu} F^{\mu\nu}) = n_+ - n_- \quad (6.4.130)$$

or, using the definition of the Chern-Pontryagin topological charge (eq.(6.1.18), with $e = 1$)

$$Q = -\frac{1}{16\pi^2} \text{tr} \int d^4x F_{\mu\nu} *F_{\mu\nu} = n \in \mathbb{Z} \quad (6.4.131)$$

we can write (6.4.130) in the form

$$n_+ - n_- = Q \quad (6.4.132)$$

We obtained this result, known as the index theorem (or the Atiyah-Singer theorem) for fermions in the fundamental representation. For any other representation the result can be obtained in analogous way leading to the formula

$$n_+ - n_- = \frac{2}{3} T(T+1)(2T+1)Q \quad (6.4.133)$$

where T is the fermion isospin ($T = 1/2$ in the fundamental, $T = 1$ in the adjoint, etc.)

Although the index theorem should be applied only to compact manifolds, it may, nevertheless, also be used in our Euclidean 4-space problem, since the conformal invariance of the theory and the assumption that the gauge fields decrease rapidly at infinity allow our problem to be mapped onto the surface of a 5-dimensional hypersphere which is compact.

The index theorem may be strengthened by showing that only n_- or n_+ is nonzero ("vanishing theorem") may sometimes hold. In particular, it can be always established when the gauge field is self-dual or anti-self-dual. To see this apply $L^\dagger$ to (6.4.101) and L to (6.4.102) to get

$$((\partial_\mu + A_\mu)^2 + 2i\bar{\sigma}_{\mu\nu} F^{\mu\nu}) \psi^+ = 0 \quad (6.4.134)$$

$$((\partial_\mu + A_\mu)^2 + 2i\sigma_{\mu\nu} F^{\mu\nu}) \psi^- = 0 \quad (6.4.135)$$

Here the spin matrices $\sigma_{\mu\nu}$ and $\bar{\sigma}_{\mu\nu}$ are defined as

$$\sigma_{\mu\nu} = \frac{1}{4i}(\bar{\alpha}_\mu\alpha_\nu - \bar{\alpha}_\nu\alpha_\mu) \quad (6.4.136)$$

$$\bar{\sigma}_{\mu\nu} = \frac{1}{4i}(\alpha_\mu\bar{\alpha}_\nu - \alpha_\nu\bar{\alpha}_\mu) \quad (6.4.137)$$

and obey the following duality properties

$$\begin{aligned} {}^*\sigma_{\mu\nu} &= \sigma_{\mu\nu} \\ {}^*\bar{\sigma}_{\mu\nu} &= -\bar{\sigma}_{\mu\nu} \end{aligned} \quad (6.4.138)$$

This duality properties assure that $\bar{\sigma}_{\mu\nu}F^{\mu\nu}$ ($\sigma_{\mu\nu}F^{\mu\nu}$) vanishes for self-dual (anti-self-dual) gauge fields. And since the remaining $(\partial_\mu + A_\mu)^2$ operator is positive definite the resulting differential equation has no normalizable solution.

All known Yang-Mills solutions with finite action are self-dual or anti-self-dual; hence in those instances there are precisely n zero-eigenvalue modes with chirality determined by the gauge field's duality properties.

Let us solve the zero-eigenvalue equation for the case of the $SU(2)$ gauge group with spinors in the adjoint for which the generators are

$$T_{nm}^a = i\varepsilon_{nam} \quad (6.4.139)$$

The N -instanton solution (6.2.32) can be rewritten in terms of $\bar{\sigma}^{\mu\nu}$ in the form

$$A_\mu = A_\mu^a \frac{\sigma^a}{2i} = i\bar{\sigma}^{\mu\nu}\partial_\nu \log \rho \quad (6.4.140)$$

where ρ is the gauge transformed to a regular solution obtained from ϕ^N in eq.(6.2.38),

$$\rho = \sum_{i=1} N + 1 \frac{\lambda_i^2}{(x - x_i)^2} \quad (6.4.141)$$

According to the index theory we expect to have $4N$ zero-eigenvalue modes of negative chirality. We then look for solutions to the equation

$$\alpha_\mu(\partial^\mu + \varepsilon^{abc}A_b^\mu)\psi_c^- = 0 \quad (6.4.142)$$

At this point one should use the following identity

$$\alpha_\mu A_a^\mu = -i\sigma^a\alpha^\nu\partial_\nu \log \rho \quad (6.4.143)$$

$$F_{\mu\nu} = \sigma_{\mu\nu}\rho\bar{\alpha}^\lambda\partial_\lambda\sigma^a\alpha^\sigma\partial_\sigma(1/\rho) \quad (6.4.144)$$

After some tedious manipulations left as a **Problem** (see [10]) one find for the zero modes. one obtains

$$\psi_a^- = \rho \bar{\alpha}^\mu \partial_\mu \frac{1}{\rho^2} \chi \quad (6.4.145)$$

where χ satisfies the equation

$$\alpha^\mu \partial_\mu \frac{1}{\rho} \bar{\alpha}^\nu \partial_\nu \chi = 0 \quad (6.4.146)$$

Writing

$$\chi = Mu \quad (6.4.147)$$

with u an arbitrary, constant, 2-component spinor and M a 2×2 matrix one has in fact to find $2N$ linearly independent solutions of the equation

$$\alpha^\mu \partial_\mu \frac{1}{\rho} \bar{\alpha}^\nu \partial_\nu M = 0 \quad (6.4.148)$$

which, become the $4N$ solutions we are searching by choosing either up or down spinors u .

The solutions for M are found to be

$$M_i^{(1)} = \frac{\lambda_i^2}{(x - x_i)^4} \alpha_\mu (x^\mu - x_i^\mu) \quad i = 1, 2, \dots, N+1 \quad (6.4.149)$$

$$M_i^{(2)} = \frac{\lambda_i^2}{(x - x_i)^2} + \sum_{j \neq i} \frac{\lambda_i^2 \lambda_j^2}{(x - x_i)^2} \left(\frac{\alpha^\mu (x_\mu - x_i^\mu) + \alpha^\mu (x_\mu - x_j^\mu)}{(x_\mu - x_i^\mu)^2 (x_\mu - x_j^\mu)^2} \right) \quad (6.4.150)$$

Although in principle these solutions implies $4N+4$ zero modes, this number can be reduced to the correct $4N$ by showing that there are 2 linear dependent solutions. One can see that the sum of all $M_i^{(2)}$ give a vanishing eigenmode and there is other linear combination leading to a zero eigenmode

$$\sum_i \lambda_i^2 M_i^{(1)} + \sum_i M_i^{(2)} \alpha_\mu x_i^\mu \quad (6.4.151)$$

Particularly interesting are two solutions , one obtained by summing al $M_i^{(1)}$'s and the otherone summing the terms $M_i^{(1)} \bar{\alpha}_\mu x_i^\mu$. The result is

$$\psi_{a1}^- = \frac{1}{2} \rho \bar{\alpha}^\mu \partial_\mu \sigma^a \alpha^\nu \partial_\nu (1/\rho) u = F_a^{\mu\nu} \sigma_{\mu\nu} u \quad (6.4.152)$$

$$\psi_{a2}^- = \left(\frac{1}{2} \bar{\alpha}^\mu \partial_\mu \sigma^a \alpha^\nu \partial_\nu (1/\rho) \right) \bar{\alpha}_\sigma x^\sigma u = F_a^{\mu\nu} \sigma_{\mu\nu} \bar{\alpha}_\sigma x^\sigma u \quad (6.4.153)$$

These solutions are connected to supersymmetry, as we shall see when discussing solitons and supersymmetry.

6.4.4 Moduli spaces

Let us now discuss very briefly the concept of moduli space, related to the above discussed collective coordinate issue.

In algebraic geometry, a moduli space is a space parametrizing the various classes of objects in an equivalence relation; interest is to provide these spaces of an additional structure. In differential geometry, the moduli space of a variety is the space of parameters defining the geometry modulo local and global diffeomorphisms.

Moduli spaces identify which objects are considered isomorphic and how these objects can modulate (vary) throughout the family.

Here we shall consider the moduli space of $SU(N)$ instantons as the space of all solutions to the selfduality equation $F_{\mu\nu} = *F_{\mu\nu}$ modulo gauge transformations in a given topological charge sector n . We shall call this space $\mathcal{I}_{k,N}$. The coordinates on $\mathcal{I}_{k,N}$ are given by the collective coordinates of the solution. For example, for the $k = 1$ $SU(2)$ instanton and including global gauge parameters we counted 8 collective coordinates so that $\dim \mathcal{I}_{1,2} = 8$. In the case of $SU(N)$ and general k one has

$$\dim \mathcal{I}_{k,N} = 4kN \tag{6.4.154}$$

Strictly speaking, the index theorem which tells us this formula doesn't count the number of collective coordinates, but rather related quantities known as zero modes.

A priori, it is not obvious that $\mathcal{I}_{k,N}$ is a manifold. In fact, it does turn out to be a smooth space apart from certain localized singularities corresponding to small instantons at the scale parameter $\lambda \rightarrow 0$. The moduli space $\mathcal{I}_{k,N}$ inherits a natural metric from the field theory, defined by the overlap of zero modes.

The zero-modes associated to the instanton solutions are those can be obtained by differentiation of a given solution with respect to a collective coordinate

Bibliography

- [1] G. 't Hooft, "Symmetry Breaking Through Bell-Jackiw Anomalies," Phys. Rev. Lett. **37** (1976) 8.
- [2] A. A. Belavin, A. M. Polyakov, A. S. Schwartz and Y. .S. Tyupkin, "Pseudoparticle Solutions of the Yang-Mills Equations," Phys. Lett. B **59** (1975) 85.
- [3] R. Rajaraman, *Solitons and Instantons*, Elsevier, Amsterdam, 1996.
- [4] G. 't Hooft, unpublished (1976).
- [5] E. Witten, "Some Exact Multi - Instanton Solutions of Classical Yang-Mills Theory," Phys. Rev. Lett. **38** (1977) 121.
- [6] R. Jackiw and C. Rebbi, "Degrees of Freedom in Pseudoparticle Systems," Phys. Lett. B **67** (1977) 189.
- [7] M. F. Atiyah and R. S. Ward, "Instantons and Algebraic Geometry," Commun. Math. Phys. **55** (1977) 117.
- [8] A. M. Polyakov, "Quark Confinement and Topology of Gauge Groups," Nucl. Phys. B **120** (1977) 429.
- [9] F. A. Schaposnik, "Path-Integral in Quantum Field Theory and Statistical Mechanics", https://docs.google.com/open?id=0B1Bztk_iEJigS3ItTG9MaTVUZkk
- [10] R. Jackiw and C. Rebbi, "Spinor analysis of Yang-Mills theory", Phys. Rev. D **77** (1977) 1052.
- [11] M. Atiyah and I.M. Singer, Bull. Amer. Math. Soc. **69** (1963) 422; M. Atiyah, Comm. Pure. Appl. Math. **20** (1967) 237.
- [12] R.E. Gamboa Saraví, M.A. Muschietti and J.E. Solomin, Comm. Math. Phys.

Chapter 7

Solitons and Supersymmetry

7.1 Bogomolny equations and supersymmetry algebra

We mentioned at the beginning of these lectures that in the case of Abelian and non-Abelian gauge theories coupled to Higgs scalars, particular relations between the gauge and scalar coupling constants lead to the existence of first order BPS equations whose static solutions also solve the more involved second order equations of motion.

As originally noted in [1], the coupling constants relations leading to the existence of BPS equations also allowed for the existence of supersymmetric extension of the original purely bosonic models. As pointed in that reference regarding the Abelian Higgs model and its vortex solutions, the existence of BPS first order equations requires in that case the following relation between the gauge coupling constant e and the quartic self-interaction coupling constant λ^1 ,

$$e^2 = 8\lambda \tag{7.1.1}$$

As we have seen, the resulting first-order set of Bogomolny equations allows to compute exact vortex solution (Note that this relation also corresponds to the limiting point separating type-I from type-II superconductivity).

Now, it is precisely when relation (7.1.1) holds that the Abelian Higgs Lagrangian can be extended to $\mathcal{N} = 2$ supersymmetric Lagrangian.

The case of the non-Abelian Higgs model and its monopole solutions is another example of this phenomenon but in this case the quartic coupling constant should vanish in order to have first order BPS equations [2]-[4]. Now, we know that topological stability is related to the non-triviality of the Higgs vacuum. Being the symmetry breaking potential absent, it

¹Note that the convention adopted in this lecture is different from that used in lecture 3

is through the boundary condition that one achieves in this case a topologically non trivial solution.

Observed in 1976, the origin of the connection between self-duality and supersymmetry began to be understood in 1978, after the work of Olive and Witten and [5], so that we shall start by reviewing this work in the simple 1 + 1 dimensional case they first discuss.

A 1 + 1 supersymmetric model

Consider the two 1 + 1 dimensional supersymmetric model with Lagrangian

$$L = -\frac{1}{2} ((\partial_\mu \phi)(\partial^\mu \phi) + \bar{\psi} \not{\partial} \psi + V^2[\phi] + V'[\phi] \bar{\psi} \psi) \quad (7.1.2)$$

where ψ is a Majorana² fermion and $V[\phi]$ an arbitrary function. The metric is $\text{diag } g_{\mu\nu} = (-1, 1)$, with $\mu = 0, 1$. The gamma matrices are chosen as follows

$$\gamma^0 = -i\sigma^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \gamma^1 = \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (7.1.3)$$

and the charge conjugation matrix C is

$$C = \sigma^2, \quad (\gamma^\mu)^\dagger C = -C \gamma^\mu \quad (7.1.4)$$

Here σ^i are the standard Pauli matrices. We write the Majorana spinor in the form

$$\psi = \begin{pmatrix} \psi^+ \\ \psi^- \end{pmatrix} \quad (7.1.5)$$

and then

$$\bar{\psi} = \psi^\dagger C = (\psi^- - \psi^+) \quad (7.1.6)$$

Note that the interaction between fermions is achieved with the derivative of the square root of the potential. Were this last a quartic symmetry breaking potential, the coupling would be of the form $\phi \bar{\psi} \psi$ and in this way a non trivial scalar vev would give mass to the fermion.

The equations of motion are,

$$\begin{aligned} \square \phi &= VV' + \frac{1}{2} V'' \bar{\psi} \psi \\ \not{\partial} \psi - V' \psi &= 0 \end{aligned} \quad (7.1.7)$$

²Majorana fermions are real fermions thus coinciding with their charge conjugate. They describe particles that coincide with its antiparticle.

The Lagrangian changes as a total divergence under the following supersymmetry transformations (with anticommuting parameter $\bar{\epsilon}, \epsilon$)

$$\begin{aligned}\delta\phi &= \bar{\epsilon}\psi \\ \delta\psi &= (\not{\partial}\phi - V[\phi])\epsilon \\ \delta\bar{\psi} &= -\bar{\epsilon}(\not{\partial}\phi + V[\phi])\end{aligned}\tag{7.1.8}$$

Using Noether theorem one finds that the conserved supersymmetry current takes the form

$$J^\mu = -(\not{\partial}\phi + V[\phi])\gamma^\mu\psi\tag{7.1.9}$$

so that the conserved supercharge is

$$Q = -\int dx (\not{\partial}\phi + V[\phi])\gamma^0\psi\tag{7.1.10}$$

Writing

$$Q = \begin{pmatrix} Q^+ \\ Q^- \end{pmatrix}\tag{7.1.11}$$

one finds from (7.1.10)

$$\begin{aligned}Q^+ &= \int dx (\partial_0\phi\psi^+ + (\partial_1\phi + V[\phi])\psi^-) \\ Q^- &= \int dx (\partial_0\phi\psi^- - (\partial_1\phi - V[\phi])\psi^+)\end{aligned}\tag{7.1.12}$$

Using the Majorana fermions equal-time anticommutator relations

$$\{\psi^\pm(x, t), \psi^\pm(y, t)\} = \delta(x - y)\tag{7.1.13}$$

one finds

$$\{Q^\pm, Q^\pm\} = 2(H \pm Z)\tag{7.1.14}$$

where H is the bosonic Hamiltonian,

$$H = \frac{1}{2}((\partial_0\phi)^2 + (\partial_1\phi)^2 + V[\phi])\tag{7.1.15}$$

and Z is defined as

$$Z = \int dx V[\phi] \frac{\partial\phi}{\partial x}\tag{7.1.16}$$

Note that defining $\bar{Q} = Q^\dagger \gamma^0 = Q^t \gamma^0$ we can write (7.1.14) in the form

$$\{\bar{Q}_\alpha, Q_\beta\} = -2\gamma_{\alpha\beta}^\mu P_\mu + \varepsilon_{\alpha\beta} Z \quad (7.1.17)$$

Problem: Obtain (7.1.14) and (7.1.17).

Now, Z can be written in the form

$$Z = \int dx V[\phi] \frac{\partial \phi}{\partial x} = \int dx \frac{\partial F[\phi]}{\partial x} \quad (7.1.18)$$

with

$$\frac{\partial F[\phi]}{\partial \phi} = V[\phi] \quad (7.1.19)$$

Then, taking as an example the usual symmetry breaking potential with two ground states $\phi = \pm a$,

$$V = -\lambda(\phi^2 - a^2) \quad (7.1.20)$$

one has

$$F = \lambda(a^2 \phi - \frac{1}{3} \phi^3) \quad (7.1.21)$$

and so

$$Z = \lambda \int_{-\infty}^{\infty} dx \frac{\partial}{\partial x} \left(a^2 \phi - \frac{1}{3} \phi^3 \right) = \lambda \left(a^2 \phi - \frac{1}{3} \phi^3 \right) \Big|_{-\infty}^{+\infty} \quad (7.1.22)$$

Let us now consider static configurations. The integral (7.1.22) vanishes for topologically trivial states, $\phi(x = -\infty) = \phi(x = \infty) = \pm a$. If instead one takes $\phi(x = -\infty) = \mp a$ and $\phi(x = \infty) = \pm a$ one has $Z = \pm 4\lambda a^3/3$. One associates the positive value for a kink state and the negative value for anti-kink state.

Appropriately normalized, the value of Z can be related to the kink topological charge defined before

$$T = \frac{1}{2a} \int_{-\infty}^{\infty} dx \frac{\partial \phi}{\partial x} = \phi(x = \infty) - \phi(x = -\infty) \quad (7.1.23)$$

The same approach can be followed in the case of the supersymmetric sine-Gordon model which corresponds to the case

$$F[\phi] = \cos \phi \quad (7.1.24)$$

This is so because the cosine potential in sine-Gordon Lagrangian can be written in terms of sine squared through their trigonometric relation.

One then has

$$Z = \int_{-\infty}^{\infty} dx \frac{\partial}{\partial x} (2 \cos \phi) = 2 \cos \phi \Big|_{-\infty}^{+\infty} \quad (7.1.25)$$

Imposing non trivial boundary conditions so that $\phi(+\infty) - \phi(-\infty) = n\pi$ we see that Z counts the soliton number (topological charge) modulo 2.

Eq.(7.1.14) implies, for a particle with rest mass $M = H$ that

$$(M \pm Z)^2 \geq 0 \quad (7.1.26)$$

or

$$M \geq |Z| \quad (7.1.27)$$

and the bound is saturated for those physical states $|phys\rangle$ such that

$$Q_+|phys\rangle = 0 \quad \text{or} \quad Q_-|phys\rangle = 0 \quad (7.1.28)$$

so that physical soliton or anti-soliton states such that (7.1.29) holds are annihilated by Q_+ or Q_- . Now, eq.(7.1.28) implies, according to (7.1.12) (in the static case),

$$(\partial_1 \phi + V[\phi])\psi_- = 0 \quad \text{or} \quad (\partial_1 \phi - V[\phi])\psi_+ = 0 \quad (7.1.29)$$

and then one has the following Bogomolny eqs.

$$\partial_1 \phi = -V[\phi] \quad \text{or} \quad \partial_1 \phi = V[\phi] \quad (7.1.30)$$

Let us note at this point that the energy for static configurations (the rest mass M) in the bosonic sector takes the form

$$M = \int dx \frac{1}{2} ((\partial_1 \phi)^2 + V^2[\phi]) \quad (7.1.31)$$

and can be trivially written a la Bogomolnyi in the form

$$M = \int dx \frac{1}{2} ((\partial_1 \phi) \pm V[\phi])^2 \mp Z \quad (7.1.32)$$

from which the bound (7.1.27) can be equally obtained.

We shall discuss below higher dimensional cases where this kind of connection between the supersymmetric algebra and soliton mass and topological charge. In contrast with the $1+1$ example above discussed in which there was just one supersymmetric charge, in higher dimension it will be necessary to have $N = 2$ extended supersymmetry in order to arrive to Bogomolny equations.

Of course, the solution to these BPS equations also solve the second order Euler-Lagrange eqs. Moreover, one can look at eq.(7.1.30) as the square root of the equation of motion for

static bosonic configurations. One can see this by integrating the equation for configurations such that $\partial\phi/\partial x$ and $V[\phi]$ vanish at infinity

$$\partial_1^2\phi = VV' \quad \rightarrow \quad (\partial_1\phi)^2 = V[\phi]^2 \quad (7.1.33)$$

Note that since the fermionic supersymmetry transformation law (7.1.8) can be written, for static configurations, as

$$\delta\psi = \begin{pmatrix} \partial_1\phi - V & 0 \\ 0 & -\partial_1\phi - V \end{pmatrix} \begin{pmatrix} \epsilon^+ \\ \epsilon^- \end{pmatrix} \quad (7.1.34)$$

putting this variation to zero is an alternative way to obtain the Bogomolny eqs without necessity of constructing the supercurrent.

One can easily see that there is a zero mode for the Dirac equation associated with each Bogomolny equation. These zero modes are

$$\chi_0 = (\not{\partial}\phi - V) \begin{pmatrix} \chi_0^+ \\ 0 \end{pmatrix} \quad \text{or} \quad \chi_0 = (\not{\partial}\phi + V) \begin{pmatrix} 0 \\ \chi_0^- \end{pmatrix} \quad (7.1.35)$$

where $\chi_0^\pm$ are constant components. One can understand this by noting that the kink (antikink) solution of the Bogomolny eqs.(7.1.30) solves, together with $\psi = 0$, the coupled Euler-Lagrange eqs. of the model. But a supersymmetric transformation of such solution will still be a solution and this means that $\delta\psi$ (the SUSY transform of $\psi = 0$) will solve the Dirac equation. Such $\delta\psi$ is precisely (167).

Note that the equality $M = Z$ when Bogomolny bound is saturated eliminates half of the SUSY generators (either Q_- or Q_+) which in turn is associated with a Dirac equation zero-mode.

Finally, let us stress that eqs.(7.1.17) has the form of an $N = 2$ extended supersymmetry algebra with central charge Z . In the present $1 + 1$ case there is not such extended supersymmetry which as we shall see will play a role in higher space dimensions.

7.2 Supersymmetry and BPS vortices

We are now ready to see how the vortex BPS equations for the Abelian Higgs model can be also derived by SUSY considerations. We shall follow here refs.[6]-[7].

One has in this case two possibilities: either one studies the $N = 2$ supersymmetric extension of the $d = 2 + 1$ -dimensional model looking at vortices as static solutions of eqs. of motion or one analyzes the 2-dimensional (Euclidean) model in which vortices can be seen as instanton solutions. Let us start by discussing the $d = 2 + 1$ approach.

The $2 + 1$ $N = 2$ supersymmetric model

The metric we shall use here satisfies $\text{diag} g_{\mu\nu} = (+ - -)$. A two component Majorana spinor will be written as

$$(\psi^\alpha) = \begin{pmatrix} \psi^+ \\ \psi^- \end{pmatrix} \quad (7.2.36)$$

Spinorial indices are raised and lowered using $C_{\alpha\beta} = i\varepsilon_{\alpha\beta}$ with $\varepsilon_{-+} = 1$,

$$\psi^\alpha = C^{\alpha\beta} \psi_\beta \quad (7.2.37)$$

Dirac matrices are chosen as

$$\gamma^1 = \sigma_1, \quad \gamma^2 = \sigma_2, \quad \gamma^0 = \sigma_3, \quad (7.2.38)$$

with σ^i the standard Pauli matrices. Superspace coordinates are written as $(x^\mu; \theta^\alpha)$ with θ^α two anticommuting spinor coordinates.

The Higgs field ϕ , together with a higgsino ψ and an auxiliary field F are accommodated in a complex scalar superfield,

$$\Phi(x; \theta) = \phi(x) + \bar{\theta}\psi(x) - \frac{1}{2}\bar{\theta}\theta F(x) \quad (7.2.39)$$

The gauge field A_μ together with a photino ρ_α are accommodated in a vector superfield which in the Wess-Zumino gauge reads

$$\Gamma_\alpha(x; \theta) = iA_\mu(x)(\gamma^\mu\theta)_\alpha - \bar{\theta}\theta\rho_\alpha(x) \quad (7.2.40)$$

Finally a real scalar superfield will be necessary in order to implement the Fayet-Iliopoulos gauge symmetry breaking mechanism,

$$S(x; \theta) = N(x) + \bar{\theta}\chi(x) - \frac{1}{2}\bar{\theta}\theta D(x) \quad (7.2.41)$$

where N is a scalar, χ is a Majorana fermion and D an auxiliary field.

Calling $\mathcal{D}_\alpha$ the (super) covariant derivative,

$$\mathcal{D}_\alpha = \frac{\partial}{\partial\theta^\alpha} + i(\gamma_\mu\theta)_\alpha\partial_\mu \quad (7.2.42)$$

the spinorial electromagnetic superfield W_α is given by

$$W_\alpha = \frac{1}{2}\mathcal{D}_\beta\mathcal{D}_\alpha\Gamma^\beta = \rho_\alpha(x) + \frac{i}{2}\varepsilon_{\mu\nu\sigma}F^{\nu\sigma}(x)(\gamma^\mu\theta)_\alpha - \frac{i}{2}\bar{\theta}\theta(\not{\partial}\rho)_\alpha(x) \quad (7.2.43)$$

With all this the $N = 1$ supersymmetric version of the Abelian Higgs model Lagrangian takes, in terms of superfields, the form

$$L_{N=1} = \int d^2\theta \left(\frac{1}{2} W^\alpha W_\alpha - \frac{1}{4} (D^\alpha + ie\Gamma^\alpha) \Phi^* (D_\alpha - ie\Gamma_\alpha) \Phi - \frac{1}{4} D^\alpha S D_\alpha S + \sqrt{2\lambda} S \Phi^* \Phi + \eta S \right) \quad (7.2.44)$$

where η is the real Fayet-Iliopoulos coefficient. Integration out over θ leads to the the $N = 1$ supersymmetric Lagrangian in component fields

$$\begin{aligned} L_{N=1} = & -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{1}{2} \partial^\mu N \partial_\mu N + \frac{1}{2} (D^\mu \phi)^* D_\mu \phi + \frac{1}{2} D^2 \\ & + \sqrt{2\lambda} D |\phi|^2 + \eta D + \frac{1}{2} |F|^2 + \sqrt{2\lambda} N (F^* \phi + F \phi^*) \\ & + \frac{i}{2} \bar{\rho} \not{\partial} \rho + \frac{i}{2} \bar{\chi} \not{\partial} \chi + \frac{i}{2} \bar{\psi} \not{D} \psi - \sqrt{2\lambda} N \bar{\psi} \psi \\ & + \frac{ie}{2} (\bar{\psi} \rho \phi - \bar{\rho} \psi \phi^*) - \sqrt{2\lambda} (\bar{\psi} \chi \phi + \bar{\chi} \psi \phi^*) \end{aligned} \quad (7.2.45)$$

Solving the equations of motion for the auxiliary fields one gets

$$D = \sqrt{2\lambda} |\phi|^2 + \xi, \quad F = -2\sqrt{2\lambda} N \phi \quad (7.2.46)$$

Here ξ is the Fayet-Iliopoulos parameter which will be written as

$$\xi = -\sqrt{2\lambda} \phi_0^2 \quad (7.2.47)$$

so that symmetry breaking takes place for

$$|\phi| = \phi_0 > 0 \quad (7.2.48)$$

With this, Lagrangian (7.2.45) takes the form

$$\begin{aligned} L_{N=1} = & -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{1}{2} \partial^\mu N \partial_\mu N + \frac{1}{2} (D^\mu \phi)^* D_\mu \phi - 4\lambda N^2 |\phi|^2 \\ & - \lambda (|\phi|^2 - \phi_0^2)^2 + \frac{i}{2} \bar{\rho} \not{\partial} \rho + \frac{i}{2} \bar{\chi} \not{\partial} \chi + \frac{i}{2} \bar{\psi} \not{D} \psi - \sqrt{2\lambda} N \bar{\psi} \psi \\ & + \frac{ie}{2} (\bar{\psi} \rho \phi - \bar{\rho} \psi \phi^*) - \sqrt{2\lambda} (\bar{\psi} \chi \phi + \bar{\chi} \psi \phi^*) \end{aligned} \quad (7.2.49)$$

This Lagrangian is invariant under the following infinitesimal supersymmetry transformations (with anticommuting parameters $\bar{\epsilon}, \epsilon$)

$$\delta\rho = -\frac{i}{2}\varepsilon_{\mu\nu\alpha}F^{\nu\alpha}\gamma^\mu\epsilon, \quad \delta A_\mu = -i\bar{\epsilon}\gamma_\mu\rho \quad (7.2.50)$$

$$\delta N = \bar{\epsilon}\chi, \quad \delta\chi = -\sqrt{2\lambda}(|\phi|^2 - \phi_0^2)\epsilon - i\gamma^\mu\epsilon\partial_\mu N \quad (7.2.51)$$

$$\delta\phi = \bar{\epsilon}\psi, \quad \delta\psi = -i\gamma^\mu\epsilon D_\mu\phi - \sqrt{8\lambda}N\phi\epsilon \quad (7.2.52)$$

with ϵ an anticommuting real (i.e. Majorana spinor) parameter.

It is possible to extend this $N = 1$ supersymmetry to an $N = 2$ supersymmetry. This can be achieved by considering SUSY transformations (7.2.50)-(7.2.52) where the parameter ϵ is replaced by a complex one, ϵ_c and real fermions χ and ρ are combined into a Dirac fermion Σ ,

$$\Sigma = \chi - i\rho \quad (7.2.53)$$

In terms of this new field the Lagrangian (7.2.49) becomes

$$\begin{aligned} L_{N=2} = & -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}\partial^\mu N\partial_\mu N + \frac{1}{2}(D^\mu\phi)^*D_\mu\phi - 4\lambda N^2|\phi|^2 \\ & -\lambda(|\phi|^2 - \phi_0^2)^2 + \frac{i}{2}\bar{\Sigma}\not{\partial}\Sigma + \frac{i}{2}\bar{\psi}\not{D}\psi - \sqrt{2\lambda}N\bar{\psi}\psi \\ & -\frac{e + \sqrt{8\lambda}}{4}(\bar{\Sigma}\phi + h.c.) + \frac{e - \sqrt{8\lambda}}{4}(\bar{\Sigma}^c\phi + h.c.) \end{aligned} \quad (7.2.54)$$

where Σ^c is the charge conjugate (the complex conjugate) of Σ .

Concerning the SUSY transformations (7.2.50)-(7.2.52), after the field redefinition they take the form, ,

$$\hat{\delta}\Sigma = -\left(\frac{1}{2}\varepsilon_{\mu\nu\alpha}F^{\nu\alpha}\gamma^\mu + \sqrt{2\lambda}(|\phi|^2 - \phi_0^2) + i\not{\partial}N\right)\epsilon^c \quad (7.2.55)$$

$$\hat{\delta}A_\mu = \frac{1}{2}(\bar{\epsilon}^c\gamma_\mu\Sigma + h.c.) , \quad \hat{\delta}N = \frac{1}{2}(\bar{\epsilon}^c\Sigma + h.c.) ; , \quad \hat{\delta}\phi = \bar{\epsilon}^c\psi \quad (7.2.56)$$

$$\hat{\delta}\psi = -\left(i\gamma^\mu D_\mu\phi + \sqrt{8\lambda}N\phi\right)\epsilon^c \quad (7.2.57)$$

Here ϵ^c is a complex parameter which can be written as

$$\epsilon^c = \epsilon \exp(i\alpha) \quad (7.2.58)$$

and so (7.2.57) can be interpreted as transformations (7.2.50)-(7.2.52) with real ϵ followed by a phase transformation for fermions,

$$\Sigma \rightarrow \exp(i\alpha)\Sigma, \quad \psi \rightarrow \exp(i\alpha)\psi \quad (7.2.59)$$

However the last term in (7.2.54) is not invariant under this phase rotation,

$$\bar{\psi}\Sigma^c\phi + h.c. \rightarrow \exp(2i\alpha)\bar{\psi}\Sigma^c\phi + h.c. \quad (7.2.60)$$

so one needs

$$\lambda = \frac{e^2}{8} \quad (7.2.61)$$

as necessary and sufficient condition for the $N = 2$ SUSY. This was a well-known result holding in general when, starting from an $N = 1$ supersymmetric gauge model, one attempts to impose a second SUSY (see for example [9]): conditions on coupling constants have to be imposed so as to accommodate different $N = 1$ multiplets into an $N = 2$ multiplet. Eq.(7.2.61) is an example of such conditions. These kind of conditions for the Abelian Higgs model were obtained, following different routes, in [6]-[8].

We shall now study the supersymmetry algebra in order to reobtain the results above in a way closer to the one used by Olive and Witten [5] for their 1+1 model.

The Noether current associated with supersymmetry invariance can be compactly written in the form

$$\mathcal{J}^\mu = \sum_{\Phi, \Psi} \left(\frac{\delta L}{\delta \nabla_\mu \Phi} \delta \Phi + \frac{\delta L}{\delta \nabla_\mu \Psi} \delta \Psi \right) - \Lambda^\mu \quad (7.2.62)$$

where Φ and Ψ represent generic bosonic and fermionic fields and is included to take into account the possible variation of the Lagrangian through a divergence term,

$$\delta L = \partial_\mu \Lambda^\mu \quad (7.2.63)$$

From this, the conserved supersymmetric charge Q can be defined as

$$\mathcal{Q} = \int d^2x \mathcal{J}^0 \quad (7.2.64)$$

Writing

$$\mathcal{Q} = \bar{\epsilon}^c Q + \bar{Q} \epsilon^c \quad (7.2.65)$$

in order to take rid of the infinitesimal Grassman parameter we find from Lagrangian (7.2.54)

$$\begin{aligned} Q = & \int d^2x \left(\left(-\frac{1}{2} \varepsilon^{\mu\nu\lambda} F_{\mu\nu} \gamma_\lambda + i \not{\partial} N - \frac{e}{2} (|\phi|^2 - v^2) \right) \gamma^0 \Sigma + \right. \\ & \left. \left(i(\not{D}\phi)^* - \frac{e}{2} N \phi^* \right) \gamma^0 \psi \right) \end{aligned} \quad (7.2.66)$$

Concerning the conjugate charge $\bar{Q}$ one has

$$\begin{aligned} \bar{Q} = & \int d^2x \bar{\Sigma} \gamma^0 \left(\left(-\frac{1}{2} \varepsilon^{\mu\nu\lambda} F_{\mu\nu} \gamma_\lambda - i \not{D} N - \frac{e}{2} (|\phi|^2 - v^2) \right) + \right. \\ & \left. \bar{\psi} \gamma^0 \left(-i \not{D} \phi - \frac{e}{2} N \phi \right) \right) \end{aligned} \quad (7.2.67)$$

If we restrict the model to the case $N = 0$ and, after computing the Poisson brackets we put all fermions to zero we end with the original Abelian Higgs model. Since we are interested in static configurations with $A_0 = 0$ we impose these conditions finding the following anticommutation relation among spinor supercharges,

$$\{Q_\alpha, \bar{Q}^\beta\} = 2 (\gamma_0)_\alpha^\beta P^0 + \delta_\alpha^\beta T \quad (7.2.68)$$

where

$$P^0 = E = \int d^2x \left(\frac{1}{4} F_{ij}^2 + \frac{1}{2} |D_i \phi|^2 + \frac{e^2}{8} (|\phi|^2 - \phi_0^2)^2 \right) \quad (7.2.69)$$

while the central charge T is given by

$$T = \int d^2x \left(\frac{e}{2} \varepsilon^{ij} F_{ij} (|\phi|^2 - \phi_0^2) + i \varepsilon^{ij} D_i \phi (D_j \phi)^* \right) \quad (7.2.70)$$

with $i, j = 1, 2$.

Problem: obtain eqs.(7.2.69)-(7.2.70) from the supersymmetry algebra.

One can easily see that T can be rewritten in the form

$$T = \int d^2x \partial_i \Omega^i \quad (7.2.71)$$

with

$$\Omega^i = \varepsilon^{ij} (e \phi_0^2 A_j + i \phi^* D_j \phi) \quad (7.2.72)$$

so that using Stokes theorem (and taking into account that $D_i \phi \rightarrow 0$ at infinity) one ends with

$$T = e \phi_0^2 \oint A_i dx^i = e \phi_0^2 \left(\frac{2\pi}{e} \right) n, \quad n \in \mathbb{Z} \quad (7.2.73)$$

where the integer n characterizes the homotopy class to which A_i belongs. To find the Bogomolny bound from the supersymmetry algebra, let us observe that, being the anticommutators (7.2.68) Hermitian one has

$$\{Q_\alpha, \bar{Q}^\beta\} \{Q^\alpha, \bar{Q}_\beta\} \geq 0 \quad (7.2.74)$$

Then, using (7.2.68) one has

$$4E^2 - T^2 \geq 0 \quad (7.2.75)$$

or

$$E \geq \frac{|T|}{2} \quad (7.2.76)$$

which is nothing but the bound (3.4.54) originally obtained by Bogomolny by completing squares. The bound is attained for states $|phys\rangle$ which are annihilated by the charges. To extract from this condition the Bogomolny eqs., it will be convenient to write

$$Q = \begin{pmatrix} Q_+ \\ Q_- \end{pmatrix}, \quad \bar{Q} = (\bar{Q}^+ \bar{Q}^-) \quad (7.2.77)$$

after a little work one finds

$$\begin{aligned} Q_+ &= -\frac{1}{2}\varepsilon_{ij}F^{ij}\Sigma_+ - \left(\frac{e}{2}(|\phi|^2 - v^2)\right)\Sigma_- - i(D_1\phi)^*\psi_+ + (D_2\phi)^*\psi_- \\ Q_- &= -\frac{1}{2}\varepsilon_{ij}F^{ij}\Sigma_- - \left(\frac{e}{2}(|\phi|^2 - v^2)\right)\Sigma_+ - i(D_1\phi)^*\psi_- + (D_2\phi)^*\psi_+ \end{aligned} \quad (7.2.78)$$

Hence, if one combines spinor components so that

$$Q_I = (Q_+ + iQ_-), \quad Q_{II} = (Q_+ - iQ_-) \quad (7.2.79)$$

one has

$$\begin{aligned} Q_I &= -\left(\frac{1}{2}\varepsilon_{ij}F^{ij} + \frac{e}{2}(|\phi|^2 - v^2)\right)\Sigma_I - (i(D_1\phi)^* - (D_2\phi)^*)\psi_I \\ Q_{II} &= -\left(\frac{1}{2}\varepsilon_{ij}F^{ij} - \frac{e}{2}(|\phi|^2 - v^2)\right)\Sigma_{II} - (i(D_1\phi)^* + (D_2\phi)^*)\psi_{II} \end{aligned} \quad (7.2.80)$$

We see that if the bosonic fields satisfy

$$\begin{aligned} \frac{1}{2}\varepsilon_{ij}F^{ij} + \frac{e}{2}(|\phi|^2 - v^2) &= 0 \\ i(D_1\phi)^* - (D_2\phi)^* &= 0 \end{aligned} \quad (7.2.81)$$

Q_I vanishes (while Q_{II} not). But if we put $\epsilon_{II} = 0$ (i.e., if we eliminate the charge $\bar{Q}$ contribution from the sum in (7.2.65)) we then have

$$Q|phys\rangle = 0 \quad (7.2.82)$$

for any configuration satisfying (7.2.81). This in turn will suffice to ensure that the bound (7.2.74) is attained.

Analogously, the supersymmetric charge will annihilate physical states if

$$\begin{aligned}\frac{1}{2}\varepsilon_{ij}F^{ij} - \frac{e}{2}(|\phi|^2 - v^2) &= 0 \\ i(D_1\phi)^* + (D_2\phi)^* &= 0\end{aligned}\tag{7.2.83}$$

together with $\epsilon_f^c = 0$. But (7.2.81) and (7.2.83) are nothing but Bogomolny equations so that we see that when the Bogomolny bound is attained half of the supersymmetry is lost. Again, a fermionic zero mode arises associated with this supersymmetry breaking and its form can be just inferred by SUSY transforming a trivial (zero) solution as in the Witten-Olive example we discussed.

7.3 Supersymmetry and uniqueness of Bogomolnyi equations

I will discuss here how the BPS structure of general gauge theories (depending on the Maxwell invariant $F^{\mu\nu}F_{\mu\nu}$ and the topological density $\tilde{F}^{\mu\nu}F_{\mu\nu}$) coupled to Higgs scalars is insensitive to the particular form of the gauge Lagrangian. Indeed, analyzing their supersymmetric extension, one can explicitly understand why this happens. Hence Maxwell, Born-Infeld or more complicated non-polynomial Lagrangians all satisfy the same Bogomolny equations and bounds which are dictated by the underlying supersymmetry algebra ([10]).

Due to the interest aroused by Born-Infeld theories in the context of the dynamics of D-branes (see [11] and references there) their Bogomolny equations for different cases were investigated [12]- [14] (early constructions were reported in [15]-[16]). Bogomolny equations were found to coincide with those arising in ordinary gauge theories. Concerning SUSY Born-Infeld theories, they were originally studied in [17]-[18].

One can quickly understand why BPS equations are not sensitive to the dynamics that one chooses for the gauge field by noting that they can be derived, in a SUSY framework, by imposing the vanishing of (half of) the supersymmetry variations of the gaugino and higgsino fields and these variations are formally the same for very different Lagrangians. The dynamics associated with the Lagrangian enters however through the equation of motion for the auxiliary field D (of the gauge field supermultiplet) which appears in the supersymmetric transformation law for the gaugino. It is then through D that the form of the Lagrangian may in principle determine the form of the BPS relations.

To see this in detail, let us take as an example the Abelian-Higgs model in $d = 3$ dimensions, for which Bogomolny equations were first derived and which we have already

discussed in Lecture 3. The arguments should hold, however, for other models like for example the $SO(3)$ gauge theory and in other dimensionalities of space-time.

We shall first consider $d = 4$ dimensional Minkowski space (with signature $(+, -, -, -)$) so that one could also use the results to analyze other $d = 4$ models and then proceed to dimensional reduction to $d = 3$. The gauge vector superfield V is written, in the Wess-Zumino gauge,

$$V = -\theta\sigma^\mu\bar{\theta}A_\mu + i\theta\theta\bar{\theta}\bar{\lambda} - i\bar{\theta}\bar{\theta}\theta\lambda + \frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}D \quad (7.3.84)$$

Here A_μ is a vector field, $\lambda = (\lambda_\alpha)$ and $\bar{\lambda} = (\bar{\lambda}^{\dot{\alpha}})$ are two-component spinors ($\alpha, \dot{\alpha} = 1, 2$) which can be combined to give a four-component Majorana fermion and D is an auxiliary field.

From V the chiral superfield W_α can be constructed,

$$W_\alpha(y, \theta, \bar{\theta}) = -i\lambda_\alpha + \theta_\alpha D - \frac{i}{2}(\sigma^\mu\bar{\sigma}^\nu\theta)_\alpha F_{\mu\nu} + \theta\theta(\sigma^\mu\partial_\mu\bar{\lambda})_\alpha \quad (7.3.85)$$

Here $\lambda, \bar{\lambda}, D$ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ are functions of the variable $y^\mu = x^\mu + i\theta\sigma^\mu\bar{\theta}$ where x^μ is the usual 4-vector position. The SUSY extension of (standard) gauge-invariant (Maxwell, Yang-Mills) theories are precisely constructed from W by considering W^2 and its hermitian conjugate $\bar{W}^2$.

Now, as stressed in [17], another superfield combination enters into play if one wishes to construct general gauge invariant SUSY Lagrangians. In particular, one needs to consider two superfields $\bar{X}$ and Y defined as

$$X = \frac{1}{8}(D^\alpha D_\alpha W^2 + \bar{D}_{\dot{\alpha}}\bar{D}^{\dot{\alpha}}\bar{W}^2) \quad (7.3.86)$$

$$Y = -\frac{i}{16}(D^\alpha D_\alpha W^2 - \bar{D}_{\dot{\alpha}}\bar{D}^{\dot{\alpha}}\bar{W}^2) \quad (7.3.87)$$

with covariant derivatives given by

$$D_\alpha = \frac{\partial}{\partial\theta^\alpha} + 2i(\sigma^\mu\bar{\theta})_\alpha \frac{\partial}{\partial y^\mu}, \quad \bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} \quad (7.3.88)$$

when acting on functions of $(y, \theta, \bar{\theta})$ and

$$D_\alpha = \frac{\partial}{\partial\theta^\alpha}, \quad \bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} - 2i(\theta\sigma^\mu)_{\dot{\alpha}} \frac{\partial}{\partial y^{\dagger\mu}} \quad (7.3.89)$$

on functions of $(y^\dagger, \theta, \bar{\theta})$. The only components of these superfields having purely bosonic terms are

$$W^2|_{\theta\theta} = D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} - i\frac{1}{2}F^{\mu\nu}\tilde{F}_{\mu\nu} \quad (7.3.90)$$

$$W^2|_{\theta\theta} = D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} + i\frac{1}{2}F^{\mu\nu}\tilde{F}_{\mu\nu} \quad (7.3.91)$$

$$\begin{aligned} X|_0 &= -(D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} - i\lambda\phi\bar{\lambda} - i\bar{\lambda}\bar{\phi}\lambda) \\ X|_{\theta\bar{\theta}} &= i\theta\sigma^p\bar{\theta}\partial_p(D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} - i\lambda\phi\bar{\lambda} - i\bar{\lambda}\bar{\phi}\lambda) \\ X|_{\theta\bar{\theta}\theta\bar{\theta}} &= \frac{1}{4}\theta\bar{\theta}\theta\bar{\theta}\square(D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} - i\lambda\phi\bar{\lambda} - i\bar{\lambda}\bar{\phi}\lambda) \end{aligned} \quad (7.3.92)$$

and

$$\begin{aligned} Y|_0 &= \frac{1}{2}(\frac{1}{2}F^{\mu\nu}\tilde{F}_{\mu\nu} + \lambda\phi\bar{\lambda} - \bar{\lambda}\bar{\phi}\lambda) \\ Y|_{\theta\bar{\theta}} &= -\frac{i}{2}\theta\sigma^p\bar{\theta}\partial_p(\frac{1}{2}F^{\mu\nu}\tilde{F}_{\mu\nu} + \lambda\phi\bar{\lambda} - \bar{\lambda}\bar{\phi}\lambda) \\ Y|_{\theta\bar{\theta}\theta\bar{\theta}} &= \frac{1}{8}\theta\bar{\theta}\theta\bar{\theta}\square(\frac{1}{2}F^{\mu\nu}\tilde{F}_{\mu\nu} + \lambda\phi\bar{\lambda} - \bar{\lambda}\bar{\phi}\lambda) \end{aligned} \quad (7.3.93)$$

with $\tilde{F}_{\mu\nu} = (1/2)\varepsilon_{\mu\nu\alpha\beta}F^{\alpha\beta}$.

A third superfield combination is necessary for constructing general gauge invariant SUSY Lagrangian. This combination is $W^2\bar{W}^2$ with its highest component taking the form

$$W^2\bar{W}^2|_{\theta\bar{\theta}\theta\bar{\theta}} = \theta\bar{\theta}\theta\bar{\theta}\left((D^2 - \frac{1}{2}F_{\mu\nu}F^{\mu\nu})^2 + (\frac{1}{2}\tilde{F}_{\mu\nu}F^{\mu\nu})^2\right) \quad (7.3.94)$$

Remark that in all three cases (7.3.92)-(7.3.94), all dependence on the curvature $F_{\mu\nu}$ and the auxiliary field D is through the combination

$$t = \frac{1}{\beta^2}\left(D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu}\right) \quad (7.3.95)$$

and this fact will have important consequences in our discussion. Here, in order to define a dimensionless variable t we have introduced a parameter β with the same dimensions as $F_{\mu\nu}$

(i.e. dimensions of a mass in $d = 4$). It corresponds to the *absolute field* in the Born-Infeld theory [5]-[?] as will become clear below.

As it happens for the last component of $W^2\bar{W}^2$, also the term W^2 ($\bar{W}^2$) depends on $F_{\mu\nu}$ and D through the combination (7.3.95). Indeed, the last component in W^2 ($\bar{W}^2$) contains the term $D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} + iF_{\mu\nu}\tilde{F}^{\mu\nu}$ ($D^2 - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} - iF_{\mu\nu}\tilde{F}^{\mu\nu}$) so that the sum of θ ($\bar{\theta}$) integrals leads to the well-known SUSY extension of the Maxwell theory.

We are ready to write a general $N = 1$ supersymmetric Lagrangian endowed with gauge-invariance in terms of X , Y and $W^2\bar{W}^2$

$$L^{d=4} = \frac{1}{4e^2} \int (W^2 d^2\theta + \bar{W}^2 d^2\bar{\theta}) + \frac{1}{e^2} \sum_{r,s,t=0}^{\infty} a_{rst} \int d^4\theta (W^2\bar{W}^2)^r X^s Y^t \quad (7.3.96)$$

with e the fundamental gauge coupling constant, which has been factorized in both terms for later convenience.

The second term accounts for the non-polynomial features of the general bosonic theory to be supersymmetrized. As explained in [17], supersymmetry imposes two constraints on coefficients a_{rst} . Their explicit form will not be relevant for our discussion. What one should retain is that expression (7.3.96) gives then the most general Lagrangian corresponding to the supersymmetric extension of a general bosonic Lagrangian depending on the two algebraic Maxwell invariants $F^{\mu\nu}F_{\mu\nu}$ and $\tilde{F}^{\mu\nu}F_{\mu\nu}$.

We shall focus now on a $d = 3$, $N = 2$ supersymmetric theory which can be obtained from Lagrangian (7.3.96) by dimensional reduction. The standard procedure for dimensional reduction, say in the x_3 spatial coordinate, implies identifying A_3 with a scalar field N . It can be shown that without including a Chern-Simons term, the bosonic part of the Lagrangian (7.3.96) can only yield electrically neutral configurations, so that as long as one looks for self-dual equations associated with (static) vortices, the A_0 field (as well as the N field) can be put to zero and so we will do from here on (the case $N \neq 0$ can be equally treated without additional complications). So far, without the addition of a Chern-Simons term, no electrically charged vortices exist and then the most general gauge field configurations are pure magnetic Nielsen-Olesen type soliton solutions. This implies that no $d = 3$ version of the $\tilde{F}_{\mu\nu}F^{\mu\nu}$ functional are available and that we can simply identify the field strength with the magnetic field B by

$$\frac{1}{2}F_{\mu\nu}F^{\mu\nu} = B^2 \quad (7.3.97)$$

with

$$B = \frac{1}{2}\varepsilon_{jk}F^{jk} \quad i, j = 1, 2 \quad (7.3.98)$$

Once the dimensional reduction is carried on, one ends with the $d = 3$ version of the SUSY

Lagrangian given in eq.(7.3.96). As it is well known, supersymmetry can be extended from $N = 1$ to $N = 2$ in this process.

From what we have seen, the gauge field dependent terms in the *bosonic* part of this $N = 2$ supersymmetric Lagrangian can be compactly written in the form

$$L_A[A_\mu, D] = \frac{1}{e^2} \sum_{n=0}^{\infty} c_n t^n \quad (7.3.99)$$

where t (defined in (7.3.95)) now reads

$$t = \frac{D^2 - B^2}{\beta^2} \quad (7.3.100)$$

and c_n are some coefficients which can be computed in terms of the a_{rst} 's.

Concerning the Higgs field sector, in $d = 4$ dimensions the coupling between the scalar Higgs field ϕ and the gauge field A_μ arises from the superfield interaction term

$$L_{A-\phi}^{d=4} = \Phi \exp(V) \Phi^* \quad (7.3.101)$$

where Φ is a chiral scalar superfield containing a Higgs field ϕ , a higgsino ψ and an auxiliary field F . One can easily see that the part of $L_{A-\phi}$ containing the auxiliary field D is [7]

$$L_{A-\phi}^{d=4}|_D = \frac{1}{2} D |\phi|^2 \quad (7.3.102)$$

On the other hand, gauge symmetry breaking can be achieved *à la* Fayet-Iliopoulos so that the complete D dependence of the supersymmetric Lagrangian arising from the Higgs coupling to A_μ and D is given by

$$L_D^{d=4}[A, \phi, D] \equiv L_{A-\phi}^{d=4}|_D + L_{FI}^{d=4} = \frac{1}{2} D (|\phi|^2 - \xi^2) \quad (7.3.103)$$

where ξ is a real constant. This Lagrangian remains unchanged after dimensional reduction so that we can write the D dependent terms of the $d = 3$ bosonic part of the Lagrangian as

$$L_D^{total}[A, \phi, D] = \frac{1}{e^2} \sum_{n=0}^{\infty} c_n \left(\frac{1}{\beta^2} (D^2 - B^2) \right)^n + \frac{1}{2} D (|\phi|^2 - \xi^2) \quad (7.3.104)$$

In $d = 3$ space-time, dimensions of parameters and fields are $[\beta] = m^2$, $[e] = m^{\frac{1}{2}}$, $[\xi] = m^{\frac{1}{2}}$, $[A_\mu] = m$, $[D] = m^2$ and $[\phi] = m^{\frac{1}{2}}$. Then, for dimensional reasons, one can infer that coefficients c'_n s can be written in the form

$$c_n = \beta^2 \lambda_n \quad (7.3.105)$$

where λ_n are dimensionless coefficients.

We can now obtain the equation of motion for D so as to eliminate the auxiliary field from the physical spectrum

$$\sum_{n=0}^{\infty} \frac{2n}{e^2} \lambda_n \left(\frac{1}{\beta^2} (D^2 - B^2) \right)^{n-1} D + \frac{1}{2} (|\phi|^2 - \xi^2) = 0 \quad (7.3.106)$$

One can easily see that the only nontrivial solution to eq.(7.3.106) takes the form

$$\begin{aligned} D &= -\frac{e^2}{4\lambda_1} (|\phi|^2 - \xi^2) \\ B &= \pm D \end{aligned} \quad (7.3.107)$$

These two equations can be readily combined into one which is nothing but the well-honored Bogomolny equation for the magnetic field of the Nielsen-Olesen vortices

$$B = \mp \frac{e^2}{4\lambda_1} (|\phi|^2 - \xi^2) \quad (7.3.108)$$

This shows that the Bogomolny gauge field equation for vortex configurations is independent of the particular form of the gauge field Lagrangian one chooses since we have proven formula (7.3.108) for the general supersymmetric Lagrangian (7.3.96)+(7.3.101). It should be noted that the choice of $\lambda_1 = -1/2$, $\lambda_n = 0$ for $n \neq 1$ corresponds to the usual value of the Maxwell term while the choice $\lambda_1 = -1/2$, $\lambda_2 = 1/8$, $\lambda_3 = 1/32$, $\dots$, gives a Dirac-Born-Infeld Lagrangian for the gauge field, which, once the auxiliary fields are put on shell takes the form

$$L_{DBI} = -\beta^2 \left(\sqrt{1 + \frac{1}{2\beta^2} F_{\mu\nu} F^{\mu\nu}} - 1 \right) \quad (7.3.109)$$

Let us now analyze the $N = 2$ supersymmetry transformations leaving invariant the three dimensional Born-Infeld SUSY theory. We shall not write the complete set of transformations but just those which are relevant for the discussion of Bogomolny equations, namely those for the higgsino and gaugino (which we call ψ and Σ):

$$\delta_\epsilon \psi = -i \not{D} \phi \epsilon = \begin{pmatrix} 0 & D_1 + iD_2 \\ D_1 - iD_2 & 0 \end{pmatrix} \begin{pmatrix} \epsilon_+ \\ \epsilon_- \end{pmatrix} \quad (7.3.110)$$

$$\delta_\epsilon \Sigma = \left(\frac{1}{2} \varepsilon_{\mu\nu\alpha} F^{\mu\nu} \gamma^\alpha + D \right) \epsilon = \begin{pmatrix} \frac{1}{2} \varepsilon_{ij} F^{ij} + D & 0 \\ 0 & \frac{1}{2} \varepsilon_{ij} F^{ij} - D \end{pmatrix} \begin{pmatrix} \epsilon_+ \\ \epsilon_- \end{pmatrix} \quad (7.3.111)$$

where we call ϵ the Dirac fermion transformation parameter (we have already made $N = A_0 = 0$ and considered the static case).

As it is well-known by now, making zero half of the SUSY variations associated with the higgsino and gaugino fields, one gets the Bogomolny equations. For instance, by demanding that those generated by ϵ_+ be zero, one gets the following self-dual equation from the higgsino's variation

$$\delta_{\epsilon_+} \psi = 0 \rightarrow D_1 \phi = i D_2 \phi \quad (7.3.112)$$

One should note that this transformation law just depends on the way the parallel displacement is defined in terms of the gauge connection and not on the explicit form of the gauge field action. One can then understand why eq.(7.3.112) is completely independent of the particular form the gauge field action is chosen, at least for minimally coupled gauge theories³. Regarding the equation derived from the gaugino transformation,

$$\delta_{\epsilon_+} \Sigma = \frac{1}{2} \varepsilon_{ij} F^{ij} + D = 0 \quad (7.3.113)$$

it could, in principle, depend on the particular Lagrangian chosen through the D term. However, as we have seen (eq.(7.3.107)), the solution to the equation of motion for D takes the same simple form for any gauge field Lagrangian since D always enters through the combination $D^2 - B^2$.

This feature can be also checked by analyzing the two supercharges which can be obtained following the usual Noether construction. As it has been shown in [14] for the Born-Infeld case, supercharges Q and $\bar{Q}$ can be always put in the form

$$\begin{aligned} \bar{Q} &= i \int d^2x \Sigma^\dagger \mathcal{H}[B, D] (\gamma^0 B + D) + \frac{i}{2} \int d^2x \psi^\dagger \not{D} \phi \\ Q &= -i \int d^2x (B + \gamma^0 D) \mathcal{H}[B, D] \Sigma - \frac{i}{2} \int d^2x \gamma^0 (\not{D} \phi)^\dagger \psi \end{aligned} \quad (7.3.114)$$

with $\mathcal{H}$ some real functional of D and B which can be computed order by order in $1/\beta^2$. Furthermore, eqs.(7.3.114) also hold when one considers not just SUSY Born-Infeld theory but the general Lagrangian, viz. eq.(7.3.96). Only the actual form of $\mathcal{H}$ will change, depending on the different sets of possible a 's coefficients. What one can easily see is that the following formula holds

$$\mathcal{H} = \mathcal{H}_{Maxwell} + \sum_{n=1}^{\infty} \frac{1}{\beta^{2n}} \mathcal{H}_n[B, D] \quad (7.3.115)$$

³For an analysis of Bogomolny equations in non-minimally coupled gauge theories, see ref.[?].

with

$$\mathcal{H}_{Maxwell} = 1 \quad (7.3.116)$$

$$\mathcal{H}_n[B, D]|_{B^2=D^2} = 0 \quad (7.3.117)$$

It is clear that condition $\bar{Q}|phys\rangle = 0$ is satisfied whenever $(B + \gamma^0 D)\epsilon = 0$ and $\not{D}\phi\epsilon = 0$, independently of the precise form the functional $\mathcal{H}$ takes. Choosing just the upper component of the transformation parameter, ϵ_+ , yields again the two Bogomolny equations (7.3.108), (7.3.112). Of course, this is to be expected since both $(B + \gamma^0 D)\epsilon$ and $\not{D}\phi\epsilon$, appearing in (7.3.114), provide the transformation laws of gaugino and higgsino respectively.

Concerning the supercharge algebra, when the Bogomolny equation $B = \pm D$ is used, only the Maxwell part of $\mathcal{H}$ survives, this showing again why the BPS structure is not sensitive to the particular form of the gauge field Lagrangian.

In conclusion, we have analyzed the most general Lagrangian corresponding to the supersymmetric extension of a general bosonic Lagrangian depending on the two algebraic Maxwell invariants $F^{\mu\nu}F_{\mu\nu}$ and $\tilde{F}^{\mu\nu}F_{\mu\nu}$. This general Lagrangian includes, for a particular choice of coefficients, the Born-Infeld supersymmetric Lagrangian, and also an infinite class of Lagrangians having causal propagation [17]. We have shown why the Bogomolny relations associated with the bosonic sector remain unchanged in spite of the actual form of the gauge field Lagrangian: Maxwell, Born-Infeld or more complicated non-polynomial Lagrangians all have the same BPS structure.

Bibliography

- [1] H. J. de Vega and F. A. Schaposnik, “A Classical Vortex Solution of the Abelian Higgs Model,” *Phys. Rev. D* **14** (1976) 1100.
- [2] E. B. Bogomolny, *Sov. J. Nucl. Phys.* **24** (1976) 449 [*Yad. Fiz.* **24** (1976) 861].
- [3] M. K. Prasad and C. M. Sommerfield, *Phys. Rev. Lett.* **35** (1975) 760.
- [4] S. R. Coleman, S. J. Parke, A. Neveu and C. M. Sommerfield, *Phys. Rev. D* **15** (1977) 544.
- [5] E. Witten and D. I. Olive, “Supersymmetry Algebras That Include Topological Charges,” *Phys. Lett. B* **78**, 97 (1978).
- [6] C. k. Lee, K. M. Lee and E. J. Weinberg, “Supersymmetry And Selfdual Chern-Simons Systems,” *Phys. Lett. B* **243** (1990) 105.
- [7] J. D. Edelstein, C. Núñez and F. Schaposnik, “Supersymmetry and Bogomolny equations in the Abelian Higgs model,” *Phys. Lett. B* **329** (1994) 39 [arXiv:hep-th/9311055].
- [8] A. Salam and J. Strathdee, “Supersymmetry, Parity And Fermion - Number Conservation,” *Nucl. Phys.* **B97**(1975)293.
- [9] M. F. Sohnius, “Introducing Supersymmetry,” *Phys. Rept.* **128**, 39 (1985).
- [10] H. R. Christiansen, C. Núñez and F. A. Schaposnik, “Uniqueness of Bogomolnyi equations and Born-Infeld like supersymmetric theories,” *Phys. Lett. B* **441**, 185 (1998) [arXiv:hep-th/9807197].
- [11] A. A. Tseytlin, “Born-Infeld action, supersymmetry and string theory,” in the *Yuri Golfand memorial volume*, ed. M. Shifman, World Scientific, 2000. In *Shifman, M.A. (ed.): The many faces of the superworld* 417-452. arXiv:hep-th/9908105.

- [12] G. W. Gibbons, “Born-Infeld particles and Dirichlet p-branes,” Nucl. Phys. B **514**, 603 (1998) [arXiv:hep-th/9709027].
- [13] J. P. Gauntlett, J. Gomis and P. K. Townsend, “BPS bounds for worldvolume branes,” JHEP **9801**, 003 (1998) [arXiv:hep-th/9711205].
- [14] S. Gonorazky, C. Núñez, F. A. Schaposnik and G. A. Silva, “Bogomolny bounds and the supersymmetric Born-Infeld theory,” Nucl. Phys. B **531**, 168 (1998) [arXiv:hep-th/9805054].
- [15] K. Shiraishi and S. Hirenzaki, Bogomolny Equations for Vortices in Born-Infeld Higgs Systems, International Journal of Modern Physics **A6** (1991) 2635.
- [16] A. Nakamura and K. Shiraishi, “Born-Infeld monopoles and instantons,” Hadronic J. **14**, 369 (1991).
- [17] S. Deser and R. Puzalowski, “Supersymmetric Nonpolynomial Vector Multiplets And Causal Propagation,” *J. Phys.* **A13** (1980) 2501.
- [18] S. Ceccotti and S. Ferrara, “Supersymmetric Born-Infeld Lagrangians,” *Phys. Lett.* **B187** (1987) 335.

Chapter 8

Monopoles and Duality

8.1 Maxwell-Dirac duality

Saying that a physical system exhibits “duality” implies that there are two complementary perspectives, formulations, or constructions of the theory. The first time we learn about duality as students in Physics is in the electromagnetism course, when we study Maxwell equations and their invariance under the duality transformation D

$$D : \quad \vec{E} \rightarrow \vec{B}, \quad \vec{B} \rightarrow -\vec{E} \quad (8.1.1)$$

which in terms of the field strengths $F_{\mu\nu}$ can be written as

$$D : \quad F^{\mu\nu} \rightarrow {}^*F^{\mu\nu}, \quad {}^*F^{\mu\nu} \rightarrow -F^{\mu\nu}$$

Note that D^2 takes $(\vec{E}, \vec{B}) \rightarrow (-\vec{B}, -\vec{E})$ which corresponds to charge conjugation satisfying $D^2 = -1$. This agrees with the fact that in Minkowski space $** = -1$.

Duality D can be generalized to duality rotations parametrized by an angle θ ,

$$D_\theta : \quad \begin{aligned} \vec{E} &\rightarrow \cos\theta \vec{E} + \sin\theta \vec{B} \\ \vec{B} &\rightarrow -\sin\theta \vec{E} + \cos\theta \vec{B} \end{aligned} \quad (8.1.2)$$

$$\vec{B} \rightarrow -\sin\theta \vec{E} + \cos\theta \vec{B} \quad (8.1.3)$$

Now, when we studied the Dirac monopole in Lecture 4, we discussed how the duality transformation (8.1.1) should be enhanced because of the presence of the magnetic current $k_\mu = (\kappa_0, \vec{k})$ introduced by Dirac who modified Maxwell equations according to

$$\vec{\nabla} \cdot \vec{E} = \rho \quad \vec{\nabla} \wedge \vec{B} - \dot{\vec{E}} = \vec{j} \quad (8.1.4)$$

$$\vec{\nabla} \cdot \vec{B} = \kappa \quad \vec{\nabla} \cdot \dot{\vec{E}} + \vec{B} = \vec{k} \quad (8.1.5)$$

or, in terms of $F_{\mu\nu}$,

$$\partial_\mu F^{\mu\nu} = j^\nu \quad \partial_\mu {}^*F^{\mu\nu} = k^\nu \quad (8.1.6)$$

The Dirac duality transformations that leaves invariant these equations now read

$$\begin{aligned} F^{\mu\nu} &\rightarrow {}^*F^{\mu\nu}, \quad {}^*F^{\mu\nu} \rightarrow -F^{\mu\nu} \\ D_{Dirac} : \\ j^\mu &\rightarrow k^\mu, \quad k^\mu \rightarrow -j^\mu \end{aligned} \quad (8.1.7)$$

8.2 Montonen-Olive and $SL(2, Z)$ duality

Let us now consider the case of non-Abelian gauge theories with symmetry breaking leading to the existence of monopoles and try to determine whether there exists some kind of duality of the type discussed above.

When we discussed in Lecture 4 monopole and dyon solutions in an $SO(3)$ gauge theory coupled to a Higgs scalar in the adjoint, we have seen that the dyon mass $\mathcal{M}$ in the center of mass frame is given by (see section 4.2)

$$\mathcal{M} = \int d^3x \left(\frac{1}{2} (E_i^a - \sin \alpha (D_i \phi)^a)^2 + \frac{1}{2} (B_i^a - \cos \alpha (D_i \phi)^a)^2 + V[\phi] \right) + v(Q_e \sin \alpha + Q_m \cos \alpha) \quad (8.2.8)$$

with Q_e and Q_m the already defined electric and magnetic dyon charges,

$$Q_m = \frac{1}{v} \int d^3x \vec{B}_i^a (D^i \phi)^a = n_m g, \quad n_m \in \mathbb{Z} \quad (8.2.9)$$

$$Q_e = \frac{1}{v} \int d^3x \vec{E}_i^a (D^i \phi)^a \quad (8.2.10)$$

Here g is the unit magnetic charge

$$g = \frac{4\pi}{e} \quad (8.2.11)$$

and $|\phi| = v$ corresponding to the minimum of the potential. In the case of a monopole solution with n_m units of magnetic charge we shall write

$$Q_m = n_m g, \quad (8.2.12)$$

Concerning the electric charge, we have seen when discussing the Witten effect, that takes place when a θ -term is added to the Yang-Mills-Higgs Lagrangian

$$L = -\frac{1}{4} \vec{F}^{\mu\nu} \cdot \vec{F}_{\mu\nu} + \frac{1}{2} \overrightarrow{D^\mu \phi} \cdot \overrightarrow{D_\mu \phi} + \theta \left(\frac{e^2}{32\pi^2} \right) \vec{F}_{\mu\nu} \cdot {}^* \vec{F}_{\mu\nu} - V[|\phi|] \quad (8.2.13)$$

that the electric charge, which should be quantized in the absence of the θ term ($Q_e = n_e e$, $n_e \in \mathbb{Z}$) changes to

$$Q_e = n_e e - \frac{e^2 g \theta}{8\pi^2} = n_e e - \frac{e \theta}{2\pi^2} n_m \quad (8.2.14)$$

where we have used

$$\frac{eg}{4\pi} = n_m \quad (8.2.15)$$

with n_m determining the total magnetic charge of the monopole.

We have found that a bound on the mass for a unit charged monopole was

$$M_{dyon} \geq v \sqrt{Q_e^2 + Q_m^2} \quad (8.2.16)$$

which, for the 't Hooft-Polyakov monopole with $Q_e = 0$ (no θ term) leads to

$$M_{mon} \geq v |g| \quad (8.2.17)$$

to be compared with the massive vector field W with mass $M_W = ev$.

The bound is not saturated unless the Higgs potential goes to zero (i.e., $\lambda \rightarrow 0$, the Prasad-Sommerfield limit) but still imposing the condition that at infinity the Higgs field takes a non-trivial vacuum expectation value,

$$\lim_{r \rightarrow \infty} |\vec{\phi}|^2 = v^2 \quad (8.2.18)$$

so that topologically non-trivial configurations could exist. Note that in this limit the Higgs scalar becomes massless.

The spectrum of the (Georgi-Glashow) model in the BPS limit ($\lambda = 0$ and $\phi(\infty) = a$) can be accommodated as follows:

particle	mass	elec. charge	mag.charge	spin/helicity
photon	0	0	0	± 1
Higgs	0	0	0	0
$W_{\pm}$ boson	ev	$\pm e$	0	1
$\pm$ BPS monopole	gv	0	$\pm g$	0

Table 8.1: the spectrum of the model

Note that the masses of all particles satisfy the Bogomolny bound (mass = $a \times$ charge with $a = e, g$). Also, it is important to stress that at weak coupling where an analysis taking account of monopoles would be valid at first order one has

$$M_W = ev \ll v, \quad M_{mon} = gv = \frac{4\pi}{e} v \gg v \quad (8.2.19)$$

so that the monopole mass is very large with respect to the $W_{\pm}$ mass.

Based on the classical spectrum shown in Table 1 and some other arguments Montonen and Olive proposed that an exact duality of the $SO(3)$ Yang-Mills-Higgs theory in the BPS limit [1] should hold taking the form

$$e \rightarrow g = \frac{4\pi}{e} \quad (8.2.20)$$

$$\pm \text{monopole} \leftrightarrow W_{\pm} \text{ boson} \quad (8.2.21)$$

However, as already noted in the original Montonen-Olive paper, there are some problems with this proposal:

1. Quantum corrections would be expected to generate a non-zero potential (Coleman-Weinberg effect) even if it is absent classically and should also modify the classical mass formula. Thus, there is no reason to think that the duality of the spectrum should be maintained by quantum corrections.
2. The $W_{\pm}$ bosons have spin one while the the model is rotationally invariant indicating that in principle monopoles have spin zero (In fact, we have seen that a careful analysis of their properties shows that because of their topological non-triviality the classical monopole solution behaves as spin 1/2 particle). Thus even if the mass spectrum is invariant under duality, there will not be an exact matching of states and quantum numbers.
3. The proposed duality symmetry seems impossible to test since rather than acting as a symmetry of a single theory it relates two different theories, one of which is necessarily at strong coupling where we have little control of the theory.

The first two problems can be solved by embedding the theory into $N = 4$ super Yang-Mills theory [2]. The third problem is still with us in that there are few concrete ways to test the proposal. However there are non-trivial tests and the first of these arose by first considering an extension of duality to a larger set of transformations.

Now, if the duality idea is correct one could expect an interesting extension when a theta term is introduced as described in section 4.2.1 We have seen that in that case, the Yang-Mills-Higgs Lagrangian of the model can be written in the BPS limit as

$$L = -\frac{1}{32\pi} \text{Im} \left(\frac{\theta}{2\pi} + \frac{4\pi i}{e^2} \right) \left(\vec{F}_{\mu\nu} + i^* \vec{F}_{\mu\nu} \right) \left(\vec{F}_{\mu\nu} + i^* \vec{F}_{\mu\nu} \right) + \frac{1}{2} \overrightarrow{D^{\mu}\phi} \cdot \overrightarrow{D_{\mu}\phi} \quad (8.2.22)$$

so that just a single complex parameter appears

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{e^2} \quad (8.2.23)$$

We have seen that physics is periodic in θ with period 2π , this implying that the transformation

$$T : \quad \tau \rightarrow \tau + 1 \quad (8.2.24)$$

should leave physics invariant up to a relabeling of states. Moreover, at $\theta = 0$ the duality transformation $e \rightarrow g = 4\pi/e$ can be written in terms of τ as

$$\tau = \frac{4\pi i}{e^2} \rightarrow \frac{4\pi i}{g^2} = \frac{4\pi i e^2}{(4\pi)^2} = -\frac{e^2}{4\pi i} = -\frac{1}{\tau} \quad (8.2.25)$$

that is

$$S : \quad \tau \rightarrow -\frac{1}{\tau} \quad (8.2.26)$$

It is important to note that, according to eq. (8.2.25) the transformation S is equivalent to change from a strong regime to a weak one since it corresponds to $g \rightarrow (4\pi)/(e^2)$.

It seems reasonable to suspect that also at arbitrary θ the full duality group is generated by transformations of the form (8.2.24)-(8.2.26). Now, it can be proved that these two transformations generate the group $SL(2, \mathbb{Z})$ of projective transformations (or modular group),

$$\tau \rightarrow \frac{a\tau + b}{c\tau + d}, \quad a, b, c, d \in \mathbb{Z}, \quad ad - bc = 1 \quad (8.2.27)$$

Note that since $e^2 > 0$, τ takes values on the upper half plane $\text{Im } \tau > 0$. Furthermore, one can see that one can use $SL(2, \mathbb{Z})$ transformations to map any τ in the upper half plane into the fundamental region defined by $-1/2 \leq \text{Re } \tau \leq 1/2$ and $|\tau| > 1$. In order for (8.2.27) to be a symmetry there must be in addition a relabeling of states.

Let us now consider a general dyon solution with n_m units of magnetic charge ($n_m = eg/(4\pi)$). As we have seen, the Bogomoln'y bound (eq. (8.2.16)) is

$$M_{dyon} \geq v \sqrt{Q_e^2 + Q_m^2} \quad (8.2.28)$$

On the other hand, if a theta term is present the total charge of the system is (Witten effect)

$$Q_e = n_e e - \frac{eg\theta}{8\pi^2} = n_e e - \frac{e\theta}{2\pi} n_m \quad (8.2.29)$$

with n_m an arbitrary integer.

The transformation $\tau \rightarrow \tau + 1$ is equivalent to $\theta \rightarrow \theta + 2\pi$ (see (8.2.24)) so that from Witten effect formula such change in θ makes $Q \rightarrow Q - n_m e$. That is, for $n_m = 1$ it shifts the electric charge in -1 unit and we know from the earlier discussion that the transformation (8.2.26) requires an exchange of electric and magnetic quantum numbers.

Putting these two facts together we deduce that the action of a generic $SL(2, Z)$ transformation on the quantum numbers should be, for a state with n_e units of electric charge and n_m units of magnetic charge

$$\begin{pmatrix} n_e \\ n_m \end{pmatrix} \rightarrow \begin{pmatrix} a & -b \\ c & -d \end{pmatrix} \begin{pmatrix} n_e \\ n_m \end{pmatrix} \quad (8.2.30)$$

We see that the duality group $SL(2, Z)$ includes a Z_2 corresponding to the interchange of electric and magnetic charges along with the interchange of strong and weak coupling. With all this, the BPS dyon mass (8.2.28) can be written in the form

$$M_{dyon}^2 = 4\pi v^2 = (n_e n_m) \frac{1}{\text{Im } \tau} \begin{pmatrix} 1 & -\text{Re } \tau \\ -\text{Re } \tau & \tau^2 \end{pmatrix} \begin{pmatrix} n_e \\ n_m \end{pmatrix} \quad (8.2.31)$$

or

$$M_{dyon} = v |n_e + n_m \tau| \quad (8.2.32)$$

Prove that the mass formula in this form is invariant under $SL(2, Z)$ transformations.

Given the existence of a state with quantum numbers $(n_e = 1, n_m = 0)$, the $SL(2, Z)$ symmetry of the theory requires the existence of other states, whose quantum numbers are related to those in the former state by an $SL(2, Z)$ transformation given in eq.(8.2.30). The charge quantum numbers carried by these states are given by

$$\begin{pmatrix} n'_e \\ n'_m \end{pmatrix} = \begin{pmatrix} a & -b \\ c & -d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} \quad (8.2.33)$$

If we choose $a = 0, c = 1$ the transformed state corresponds to the usual BPS monopole solution in the theory. States with $a \neq 0$ and $c = 1$ can be identified with the BPS dyon states in the theory.

Note that a and c cannot be arbitrary integers, but must be such that for some integers b and d , $ad - bc = 1$. One can show that integers b and d satisfy this requirement if and only if a and c are relatively prime [3].

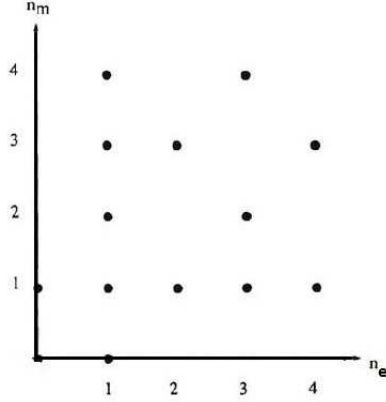


Figure 8.1

The lattice of BPS states in $N = 4$ super-Yang-Mills theory with gauge group $SU(2)$.

Since BPS states are precisely absolutely stable they cannot decay as we vary τ and hence we deduce, because of duality, that they must also exist at weak coupling where we can search for them using semiclassical techniques. It is then argued that they can be translated into the existence of certain geometric structures on the moduli space.

Note from the figure that the photon BPS multiplet lies on its own $SL(2, Z)$ orbit. There are extra states that could in principle exist but they are not stable: for example, the mass of a potential BPS state $(2, 2)$ is only marginally stable into the decay of two $(1, 1)$ states.

The extension of electromagnetic duality to $SL(2, Z)$ that we have discussed is usually called S-duality. Although this extension was first discovered in the context of lattice models [4] it was discussed as a symmetry of $N = 4$ Yang-Mills theory first in the low-energy limit of toroidal compactifications of string theory [5]. In that context the variable τ becomes a dynamical field usually denoted by S and the $SL(2, Z)$ transformations of S were called S -duality to distinguish it from other $SL(2, Z)$ transformations in string theory which are (superficially) unrelated.

8.3 Monopoles $N = 4$ Supersymmetric Gauge Theories

$N = 4$ super-Yang-Mills theory and some special theories with $N = 2$ supersymmetry provide the most natural setting for Montonen and Olive's original idea [1] since they have vanishing β -functions and hence the quantum corrections are under precise control. In many theories with $N = 2$ and $N = 1$ supersymmetry duality plays an important role in elucidating the

infrared dynamics. In these models one can study strong coupling phenomenon such as confinement and chiral symmetry breaking in an exact context [6]-[7].

8.3.1 $N = 4$, $G = SU(2)$ and S-Duality

$N = 4$ super-Yang-Mills theory has the maximal amount of supersymmetry corresponding to particles with spins less than or equal to one. It has vanishing beta-function and is conjectured to exhibit S-duality. The bosonic fields in the supermultiplet consist of a gauge field A_μ and six Higgs fields ϕ^I ($I = 1, \dots, 6$) transforming as a **6** of $\text{Spin}(6)$ ¹, all taking values in the adjoint representation of $SU(2)$. In the fermionic sector there are four Weyl fermions in the adjoint transforming as a **4** of $\text{Spin}(6)$.

The bosonic action is given by

$$S = -\frac{1}{16\pi} \text{Im} \int \tau \text{Tr} (F_{\mu\nu} F^{\mu\nu} + i^* F_{\mu\nu} F^{\mu\nu}) - \frac{1}{2e^2} \int d^4x \text{Tr} (D_\mu \phi^I D^\mu \phi^I + V[\phi^I]) \quad (8.3.34)$$

where the symmetry breaking potential takes the form

$$V[\phi^I] = \sum_{1 \leq I < J \leq 6} \text{Tr}[\phi^I, \phi^J]^2 \quad (8.3.35)$$

We have taken $\text{Tr}[T^a, T^b] = \delta^{ab}$.

The classical vacua of the theory are given by $V[\phi^I] = 0$ or equivalently $[\phi^I, \phi^J] = 0$ for all I, J . In this theory, there are no quantum corrections to the moduli space of vacua. For simplicity we will restrict our attention in the following to a single direction in the moduli space of vacua:

$$\phi^2 = \dots = \phi^6 = 0, \quad \phi^1 = \Phi, \quad \text{Tr} \Phi^2 = v^2 \quad (8.3.36)$$

which clearly satisfies $V[\phi^I] = 0$. Classical BPS monopole solutions are then obtained by solving the Bogomol'nyi equations we considered before.

Since the BPS mass formula (8.2.32) can be derived from supersymmetry, the quantum theory should also be invariant under S-duality. To see the invariance one should note that $v \rightarrow v' = v|c\tau r + d|$ under a $SL(2, Z)$ transformation because we rescaled the Higgs field by a factor of the coupling constant e .

There are more sophisticated tests of S-duality. We begin by noting that the perturbative spectrum can be determined at weak coupling and consists of a neutral massless photon

¹The Spin group $\text{Spin}(n)$ is the double cover of the special orthogonal group $SO(n) = SO(n, R)$. As a Lie group $\text{Spin}(n)$ therefore shares its dimension $(n(n-1)/2)$ with the special orthogonal group and its Lie algebra. For $n > 2$, $\text{Spin}(n)$ is simply connected and so coincides with the universal cover of $SO(n)$.

multiplet $(0, 0)$ and massive $W_{\pm}$ -boson BPS multiplets with charge $(0, \pm 1)$. S-duality maps the W-boson multiplets to BPS states (k, l) , with k and l relatively prime integers, typically at strong coupling. But since these are precisely the absolutely stable BPS states they cannot decay as we vary τ and we deduce that they must also exist at weak coupling where we can search for them using semiclassical techniques. We will argue that they can be translated into the existence of certain geometric structures on the moduli space. Note that the photon BPS multiplet lies on its own $SL(2, Z)$ orbit. The lattice of BPS states predicted by S-duality is drawn in Figure 1.

One can translate these predictions of the spectrum of BPS states with relatively prime charges (k, l) into statements about the moduli space of monopoles. The semiclassical analysis begins with the moduli space $\mathcal{M}_k$ of BPS monopole solutions.

As we learn when studying semiclassical quantization of solitons, there are zero modes associated to invariances of the Yang-Mills-Higgs theory, namely invariance under space translations and the surviving $U(1)$ gauge invariance. Each one of these zero modes should be treated separately from the non-zero modes and, instead of the integration of the Bessel-Fourier coefficient of the zero-mode one introduces an integration over a collective coordinate associated, in this case the position (3 collective coordinates) and the parameter of the gauge rotation (1 collective coordinate). These coordinates define the moduli space $\mathcal{M}$ of solutions to Bogomolny equations.

In the case of a monopole with magnetic charge 1 the number of collective coordinates is 4 and hence $\dim \mathcal{M}_1 = 4$. One can show that for BPS solutions with k units of charges $\dim \mathcal{M}_k = 4k$ [8]-[10].

In the $N = 4$ supersymmetry context there are also fermionic zero modes. These arise from solving the Dirac equation for the fermion fields in the presence of a given monopole solution. There are four Weyl or two Dirac spinors in the adjoint of $SU(2)$ and the index theorem shows that there are $4k$ -fermionic c-number zero modes that require the introduction of $4k$ complex Grassmann odd fermionic collective coordinates ψ^{α} . This means the low-energy ansatz for the fermions will include terms of the schematic form

$$\chi(x, t) = \psi(t)\chi^{cl}(x, Z(t)) \quad (8.3.37)$$

where χ^{cl} is a c-number fermion zero mode for the monopole solution specified by the moduli Z and $\psi(t)$ the generic associated collective coordinate.

In general low-energy effective theories are constructed just from considering the zero-mode contribution to the path-integral, ignoring all non-zero mode contributions. In mathematical terms, this corresponds to assume that the low-energy dynamics of k monopoles is approximated by assuming that it is geodesic motion on M_k , the moduli space of static k -monopole solutions., This is a finite-dimensional subspace of the configuration space that

possesses a natural metric coming from the kinetic energy term in the lagrangian. The metric on the moduli space for two monopoles was constructed by Atiyah and Hitchin [10]

We in the previous letter that that BPS states preserve half of the supersymmetry. This manifests itself in the present case in the fact that half of the supersymmetry generators leave the classical BPS monopole solution invariant. It can be shown that the bosonic and fermionic zero modes form a multiplet of the unbroken supersymmetries. This is essential in obtaining a supersymmetric low-energy ansatz for the fields. The ansatz for the low-energy fields is technically quite involved and has been carried out in [9]. The result of substituting the ansatz into the space-time Lagrangian leaving just the contribution of the zero-modes leads to the following supersymmetric quantum mechanics action:

$$S = \frac{1}{2} \int dt \left(\mathcal{G}_{\alpha\beta} \left(\dot{Z}^\alpha \dot{Z}^\beta + i \bar{\psi}^\alpha \gamma^0 D_t \psi^\beta \right) + \frac{1}{6} R_{\alpha\beta\gamma\delta} \bar{\psi}^\alpha \psi^\beta \bar{\psi}^\gamma \psi^\delta \right) - \frac{4\pi v}{e^2} k \quad (8.3.38)$$

Here

- We take $\{Z^\alpha\}$ as the natural set of coordinates of the moduli (with $\alpha = 1, 2, \dots, \dim M_k$)
- $\lambda^{(\alpha)}$ are the zero modes
- The complex ψ^α has been traded for a real two component Majorana spinor ψ_i^α .
- $\mathcal{G}_{\alpha\beta}$ is the metric on M_k ,

$$\mathcal{G}_{\alpha\beta} = \int d^3x \lambda_{(\alpha} \lambda_{\beta)} \quad (8.3.39)$$

- There is a Christoffel connection

$$\Gamma_{\alpha\beta,\gamma} = \int d^3x \lambda_{(\alpha} s_\beta \lambda_{\gamma)} \quad (8.3.40)$$

where s_β is a covariant derivative taking the gauge parameters as connections on M ,

$$s_\beta = \partial_\beta + [e_\beta, \] \quad (8.3.41)$$

where $\partial_\beta = \partial/\partial Z^\beta$.

- The covariant derivative D_t is defined as

$$D_t \psi^\alpha = \dot{\psi}^\alpha + \Gamma_\beta^\alpha \gamma \dot{Z}^\beta \quad (8.3.42)$$

Bibliography

- [1] C. Montonen and D. Olive, Phys. Lett. **72B** (1977) 117.
- [2] H. Osborn, Phys. Lett. **83B** (1979) 321.
- [3] A. Sen, Phys. Lett. **329B** (1994) 217.
- [4] J. Cardy and E. Rabinovici, Nucl. Phys. **B205** (1982) 1;
J. Cardy, Nucl. Phys. **B205** (1982) 17 ;
A. Shapere and F. Wilczek, Nucl. Phys. **B320** (1989) 669.
- [5] A. Font, L. Ibáñez, D. Lust and F. Quevedo, Phys. Lett. **B249** (1990) 35;
S. J. Rey, Phys. Rev. **D43** (1991) 526.
- [6] N. Seiberg and E. Witten, Nucl. Phys. **B426** (1994) 19.
- [7] N. Seiberg and E. Witten, Nucl. Phys. **B431** (1994) 484.
- [8] E. Weinberg, Phys. Rev. **20** (1979) 936.
- [9] J. P. Gauntlett, Nucl. Phys. **B411** (1994) 443.
- [10] M. F. Atiyah and N. J. Hitchin, *The Geometry and Dynamics of Magnetic Monopoles*,
Princeton Univ. Press, Princeton, NJ, 1988.

Chapter 9

Monopoles and confinement/Vortices and screening

Bibliography

- [1] Steven Weinberg, *Dreams of a Final Theory*, Ed. Random House, New York, 1992.

Chapter 10

Solitons in condensed matter

Bibliography

- [1] Steven Weinberg, *Dreams of a Final Theory*, Ed. Random House, New York, 1992.

Chapter 11

Solitons and Branes

The non-relativistic string

Consider for simplicity the 2-dimensional plane (x, y) and a string with endpoints fixed at $(0, 0)$ and $(a, 0)$. x is the longitudinal direction and y the transverse direction. In the static configuration the string is stretched along the x axis between 0 and a .

Transverse oscillations are described by the y coordinate while the x coordinate of any point of the string does not change in time. The equation describing the oscillations is

$$\frac{\partial^2 y(t, x)}{\partial^2 x} - \frac{\mu}{T} \frac{\partial^2 y(t, x)}{\partial^2 t} = 0 \quad (11.0.1)$$

where μ is the string mass per unit length and T its tension.

Dirichlet boundary conditions to solve this equation are:

$$y(t, x = 0) = y(t, x = a) = 0 \quad (11.0.2)$$

The corresponding string shape at a given time for the case of transverse oscillations is shown in figure 1.1

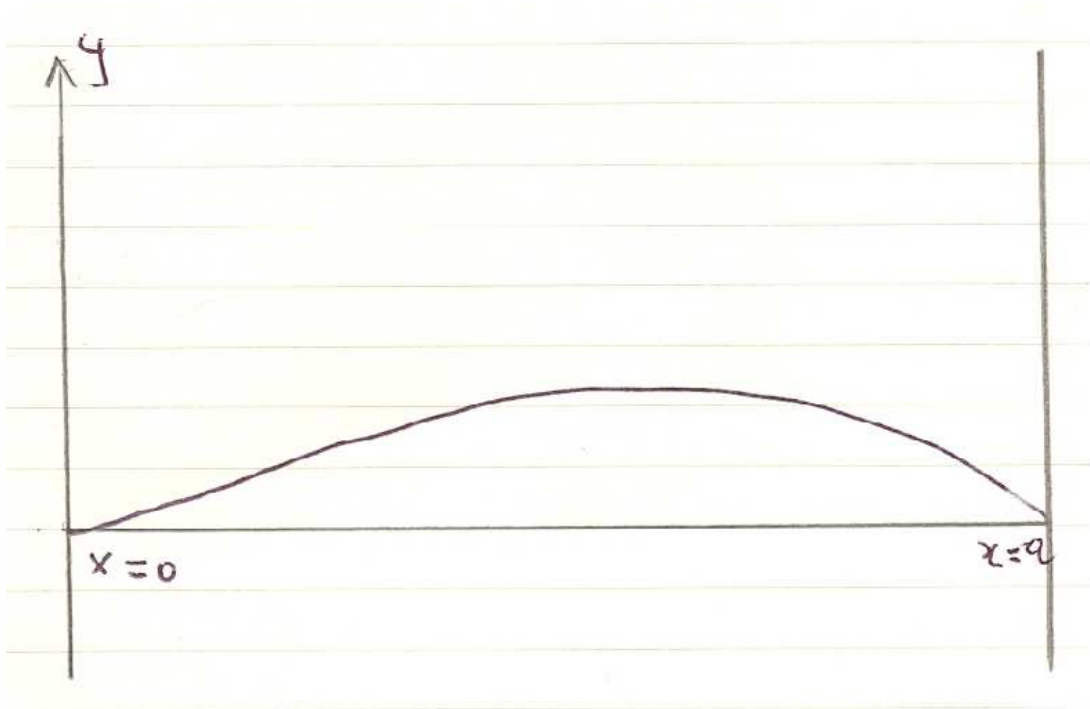


Figure 10.1

Alternatively, one can chose Neumann boundary conditions

$$\frac{\partial y}{\partial x}(t, x = 0) = \frac{\partial y}{\partial x}(t, x = a) = 0 \quad (11.0.3)$$

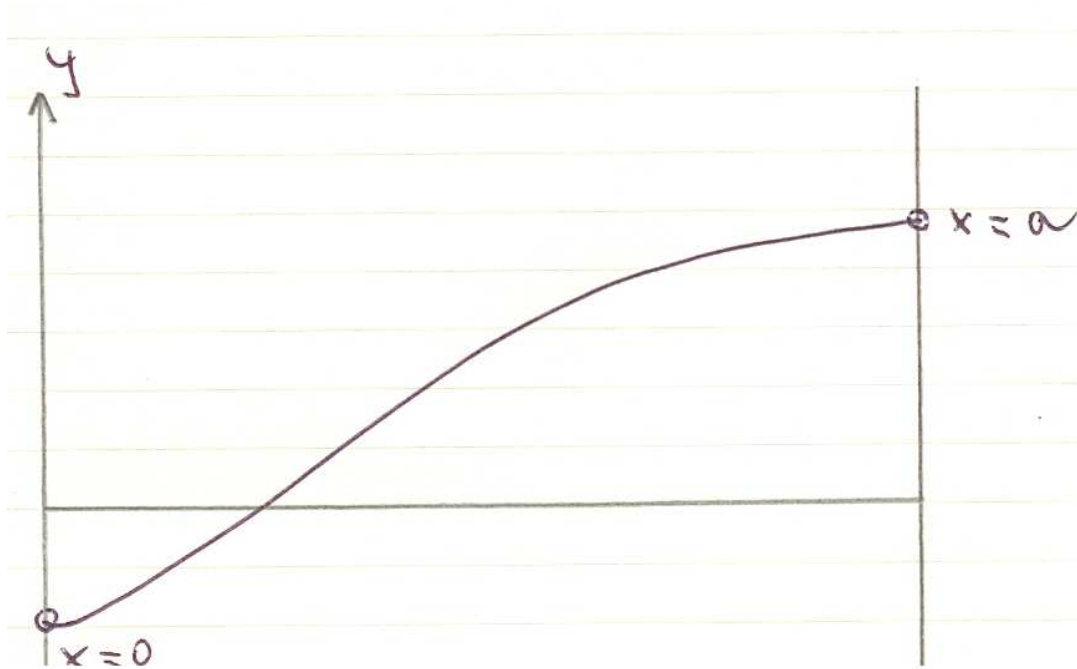


Figure 10.2

which correspond to a string with two massless loops attached at its ends which can slide along frictionless, as in figure 1.2

11.1 The relativistic strings

A one dimensional string sweeps out a two-dimensional world-sheet. One then needs 2 parameters to describe such worldsheet, σ and τ . In the case of relativistic strings the surface is described in terms of mapping functions

$$X^\mu(\sigma, \tau), \quad \mu = 0, 1, 2, \dots \quad (11.1.4)$$

which are written in capital letters to distinguish them from space-time coordinates x^μ . With this notation we can ignore the σ, τ dependence by just writing X^μ . For open strings one lets $0 \leq \sigma \leq \pi$ while for closed ones it is convenient to use the interval $0 \leq \sigma \leq 2\pi$. Concerning τ it may extend from $-\infty$ to $+\infty$.

The Nambu-Goto string action is defined in terms of the total proper area of the world-sheet. Using latin indices a, b for parameters σ, τ and defining an induced metric in the world-sheet

$$h_{ab} = \partial_a X^\mu \partial_b X_\mu = \eta_{\mu\nu} \partial_a X^\mu \partial_b X^\nu \quad (11.1.5)$$

$$\eta_{\mu\nu} = (-1, 1, 1, \dots, 1) \quad (11.1.6)$$

one can see that the area (and hence the Nambu-Goto action) can be written in terms of the determinant of the metric

$$S_{NG} = -T \int d\tau d\sigma \sqrt{-\det h} = -T \int d\tau d\sigma \sqrt{(\dot{X} \cdot X')^2 - (\dot{X})^2 (X')^2} \quad (11.1.7)$$

Here T is the total rest mass per unit length which coincides with the string tension and is related to the “Regge slope” α'

$$T = \frac{1}{2\pi\alpha'} \quad (11.1.8)$$

One can easily see that the integral in the r.h.s. gives the area of the world-sheet. surface. The minus sign in front of the integral has to do with the fact that the Lagrangian associated to S_{NG} becomes, for a stretched string with endpoints $x^1 = 0$ and $x^1 = a$ the constant $-Ta$. So it is a pure potential energy Lagrangian $L = -V$. So the potential energy is Ta which measures the energy to create a string with tension T . The rest mass per unit length satisfies $m_0 c^2 = V/a = T$.

Equations of motion

Starting from action (11.1.7) one can obtain the equations of motion for the relativistic bosonic string. Writing the action as the integral of a Lagrangian density $\mathcal{L}$ we get from its variation

$$\begin{aligned} \delta S &= \int d\tau d\sigma \left(\frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} \delta \dot{X}^\mu + \frac{\partial \mathcal{L}}{\partial X^{\mu'}} \delta X^{\mu'} \right) \\ &= - \int d\tau d\sigma \left(\frac{\partial}{\partial \tau} \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} - \frac{\partial}{\partial \sigma} \frac{\partial \mathcal{L}}{\partial X^{\mu'}} \right) \delta X^\mu + \int d\tau \left(\frac{\partial \mathcal{L}}{\partial X^{\mu'}} \delta X^{\mu'} \right) \Big|_0^\pi \end{aligned} \quad (11.1.9)$$

(we have assumed that $\delta X^\mu(\tau_f, \sigma) = \delta X^\mu(\tau_i, \sigma) = 0$). The resulting equations of motion can be written in terms of conjugate momenta

$$\mathcal{P}_\mu^\tau \equiv \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} = -T \frac{(\dot{X} \cdot X') X_\mu' - (X')^2 \dot{X}_\mu}{\sqrt{(\dot{X} \cdot X')^2 - (\dot{X})^2 (X')^2}} \quad (11.1.10)$$

$$\mathcal{P}_\mu^\sigma \equiv \frac{\partial \mathcal{L}}{\partial X^{\mu'}} - T \frac{(\dot{X} \cdot X') \dot{X}_\mu - (X')^2 X_\mu'}{\sqrt{(\dot{X} \cdot X')^2 - (\dot{X})^2 (X')^2}} \quad (11.1.11)$$

Here $\mathcal{P}_\mu^\sigma$ is the momentum running along the string while $\mathcal{P}_\mu^\tau$ is the momentum running transverse to it. In terms of these momenta the eqs. of motion take the form

$$\frac{\partial \mathcal{P}_\mu^\tau}{\partial \tau} + \frac{\partial \mathcal{P}_\mu^\sigma}{\partial \sigma} = 0 \quad (11.1.12)$$

with

$$\mathcal{P}_\mu^\sigma = 0 \quad \text{at} \quad \sigma = 0, \pi \quad (11.1.13)$$

In the case of the non-relativistic string, figure 1.1. represents a string attached to 2 points while figure 1.2 represents a string in which endpoints can slide up and down but cannot move horizontally. The objects in which endpoints must lie are characterized by their dimensionality or, more precisely, by the number of the spatial dimensions that they have. These objects are called D-branes. For figure 1.1 they are D0-branes. For figure 1.2 they are called D1-branes.

Boundary conditions

- For a closed string boundary conditions imply

$$X^\mu(\tau, \sigma + 2\pi) = X^\mu(\tau, \sigma) \quad (11.1.14)$$

- For an open string one can impose

1. Neumann boundary conditions (translation invariance is respected)

$$\partial_\sigma X^\mu(\tau, 0) = \partial_\sigma X^\mu(\tau, \pi) = 0 \quad (11.1.15)$$

2. Dirichlet boundary conditions (translation invariance is violated)

$$X^\mu(\tau, 0) = X^\mu(\tau, \pi) = 0 \quad (11.1.16)$$

Let us write the mapping functions X^μ in the form $X^\mu = (X^+, X^-, X^I)$ with $I = 1, \dots, D-2$ with

$$X^\pm = \frac{1}{\sqrt{2}}(X^0 \pm X^1) \quad (11.1.17)$$

The light-cone gauge implies fixing $X^+ = d\tau$ with d some constant. Then one can see that there is no dynamics in X^- and all the dynamics is reduced to the transverse coordinates X^I . One can also choose the coordinates so that the momentum density $\mathcal{P}^+$ is constant on the string.

A Fourier expansion of the string coordinates taking Neumann conditions

$$\partial_\sigma X^I(\tau, 0) = \partial_\sigma X^I(\tau, \pi) = 0 \quad (11.1.18)$$

takes the form

$$X^I = x^I + p^I \tau + i \sum_{n \neq 0} \frac{1}{n} \alpha_n^I \exp(-in\tau) \cos(n\sigma) \quad (11.1.19)$$

with $\alpha_n^I = \sqrt{n} \hat{a}_n^I$, $\alpha_{-n}^I = \sqrt{n} \hat{a}_n^{I\dagger}$ and $\hat{a}_n$ ($\hat{a}_n^\dagger$) ordinary annihilation (creation) operators.

The Hamiltonian takes the form

$$H = \bar{p}^2 + N + o \quad (11.1.20)$$

with o a normal ordering constant and N is given by

$$N = \sum_{n=1}^{\infty} \alpha_{-n}^I \alpha_n^I \quad (11.1.21)$$

Concerning Dirichlet directions one has, instead of (11.1.19)

$$X^I = c^I + \sum_{n \neq 0} \frac{1}{n} \exp(-in\tau) \sin(n\sigma) \quad (11.1.22)$$

There is no momenta associated with the Dirichlet directions since the string endpoints are fixed to the constants c^I which should be seen as collective coordinates.

11.2 Branes

Branes were originally introduced in the 1980s-1990s in the context of string theories and since then its leading role in particle physics non-perturbative calculations is continually growing.

According to the Oxford dictionary

brane

Definition of brane

noun

Physics an extended object analogous to the strings of string theory but having any number of dimensions rather than one dimension.

Origin:

1980s: short for membrane

The term “brane” is introduced in the first paragraph of the paper by Achúcarro et al [3]: *The covariant space-time-supersymmetric action of Green and Schwarz [1] can be generated [2,3] to an action for a p -dimensional extended object, a “ p -brane”, moving in a d -dimensional, N -extended superspace, at least for certain values of (p, d, N) .*

Taxonomy

- **p-brane:** A p -dimensional extended static object.

A **0-brane** is a point-like particle (zero spatial dimensions); a **1-brane** is a string: it has one spatial dimension; a **2-brane** is a two-dimensional object (a membrane, from which the apheresis “brane” has been proposed). In general a p -brane sweeps out a $p + 1$ dimensional world-volume as it propagates through space-time.

A p brane can be identified with a soliton in the $p + 1$ dimensional world-volume.

- **M2-brane:** In the context of M-theory there is an extra-dimension so that ordinary strings become 2-dimensional membranes. Its magnetic dual is called an **M5-brane**.
- **D-brane:** a class of branes upon which open strings can end with **Dirichlet** boundary conditions. They were introduced by Dai, Leigh and Polchinski [1], and independently by Horava [2] in 1989.
- **Dp -brane:** Let us call D the number of space-time dimensions, There are hence $d = D - 1$ space coordinates.

If one admits that $D - p - 1$ boundary conditions are of the Dirichlet type while the remaining $p + 1$ ones are Neuman type, the system is said to describe a Dp -brane.

To see this more clearly, consider simple Dp branes that are p -dimensional hyperplanes in d -dimensional space. To specify the position of an hyperplane we need $d - p$ linear conditions. When $d = 3$ a 2 brane is a plane and we need $d - p = 3 - 2 = 1$ conditions (for example $z = 0$ specifies the plane. A 1-brane (i.e, a string) in $d = 3$ requires $d - 1 = 2$ conditions: say

$x = 0$ and $y = 0$. One needs as many conditions as the number of coordinates perpendicular to the brane.

Consider now a D p -brane in d -dimensional space (plus 1 time). Let us split the d spatial coordinates in two groups:

$$\begin{array}{ll} x^0, x^1, \dots, x^p & \text{time + Dp tangential coordinates} \\ x^{p+1}, x^{p+2}, \dots, x^d & \text{Dp normal coordinates} \end{array} \quad (11.2.23)$$

To localize the D p brane we fix the coordinates normal to the brane (x^a) to certain constant values c^a

$$x^a = c^a, \quad a = p+1, p+2, \dots, d \quad (11.2.24)$$

They are $d - p$ conditions.

We now split the string coordinates $X^\mu(\sigma, \tau)$ in exactly the same fashion:

$$\begin{array}{ll} X^0, X^1, \dots, X^p & \text{time + Dp tangential coordinates} \\ X^{p+1}, X^{p+2}, \dots, X^d & \text{Dp normal coordinates} \end{array} \quad (11.2.25)$$

The ends of the string should lie on the D p brane so we need 2 times $d - p$ Dirichlet conditions:

$$X^a(\tau, 0) = X^a(\tau, \pi) = 0, \quad a = p+1, \dots, d \quad (11.2.26)$$

That is, the “ a -directions” are Dirichlet directions

The endpoints of the string can move freely along the direction tangential to the brane (i.e., remaining on the p dimensional brane

$$(X^m)'(\tau, 0) = (X^m)'(\tau, \pi), \quad m = 0, 1, \dots, p \quad (11.2.27)$$

so we have in this case Neumann conditions for the p spatial coordinates of the string.

That is, the “ μ -directions” are Neumann directions

Note that endpoints of the string ending on D-branes maybe on the same or on different branes, as shown in figure 11.3.

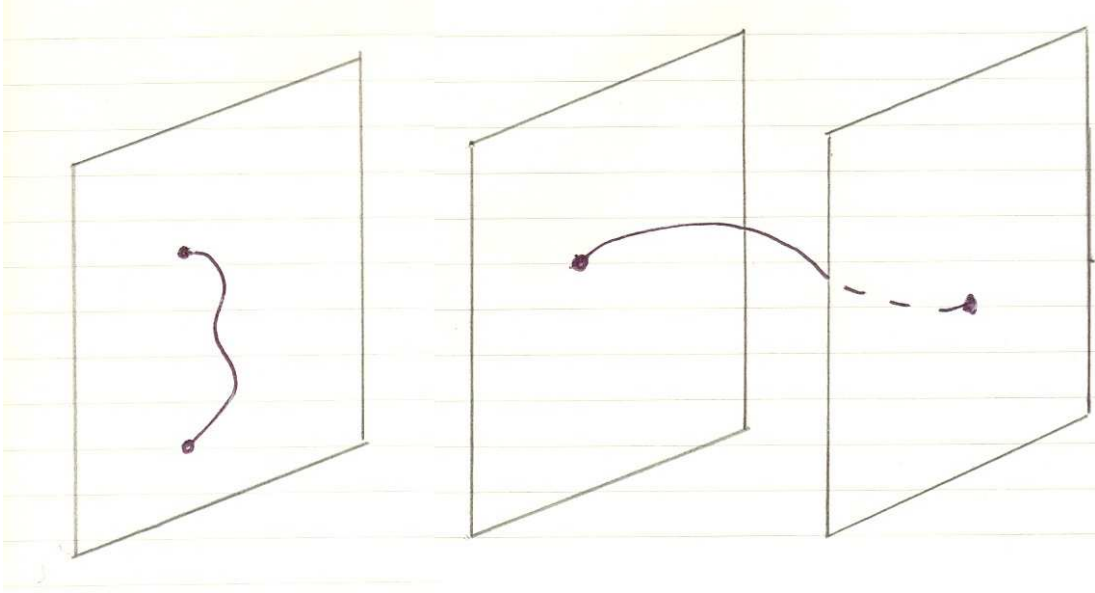


Figure 10.3

11.3 Branes become dynamical

Dai, Leigh and Polchinski have shown in [1] that the locus of the Dirichlet boundary conditions is dynamical, and in fact they coined the term Dirichlet-brane (D-brane) for the resulting object. Afterwards Leigh showed in [4] that D-brane dynamics is governed by the Dirac-Born-Infeld action. In this section we shall explain how D-branes become dynamical.

As they were introduced above, D-branes were flat fixed surfaces determined by the constants c^a introduced in (11.2.24). Now, open strings are attached to the D-branes and their oscillation modes are associated to fields which set up the “spectrum of string theory”. Among the massless boson particles appearing as oscillation modes of the string there is a gauge field $A_\mu(x)$,

$$A_\mu(x^0, x^1, \dots, x^p), \quad \mu = 0, 1, \dots, p \quad (11.3.28)$$

There appear also scalar fields Φ^m which are also function of time and spatial coordinates on the brane with $m = p + 1, \dots, d$

$$\Phi^m(x^0, x^1, \dots, x^p), \quad m = p + 1, \dots, d \quad (11.3.29)$$

The field theory associated to the string oscillations only depends on the brane coordinates is due to the fact that the center of mass of the string cannot oscillate in directions transverse to the brane since the string has on it fixed boundary conditions. The proof that these massless fields are related to oscillations of the string can be found in [7], section 15.2, pp 333-338. We shall sketch the proof below.

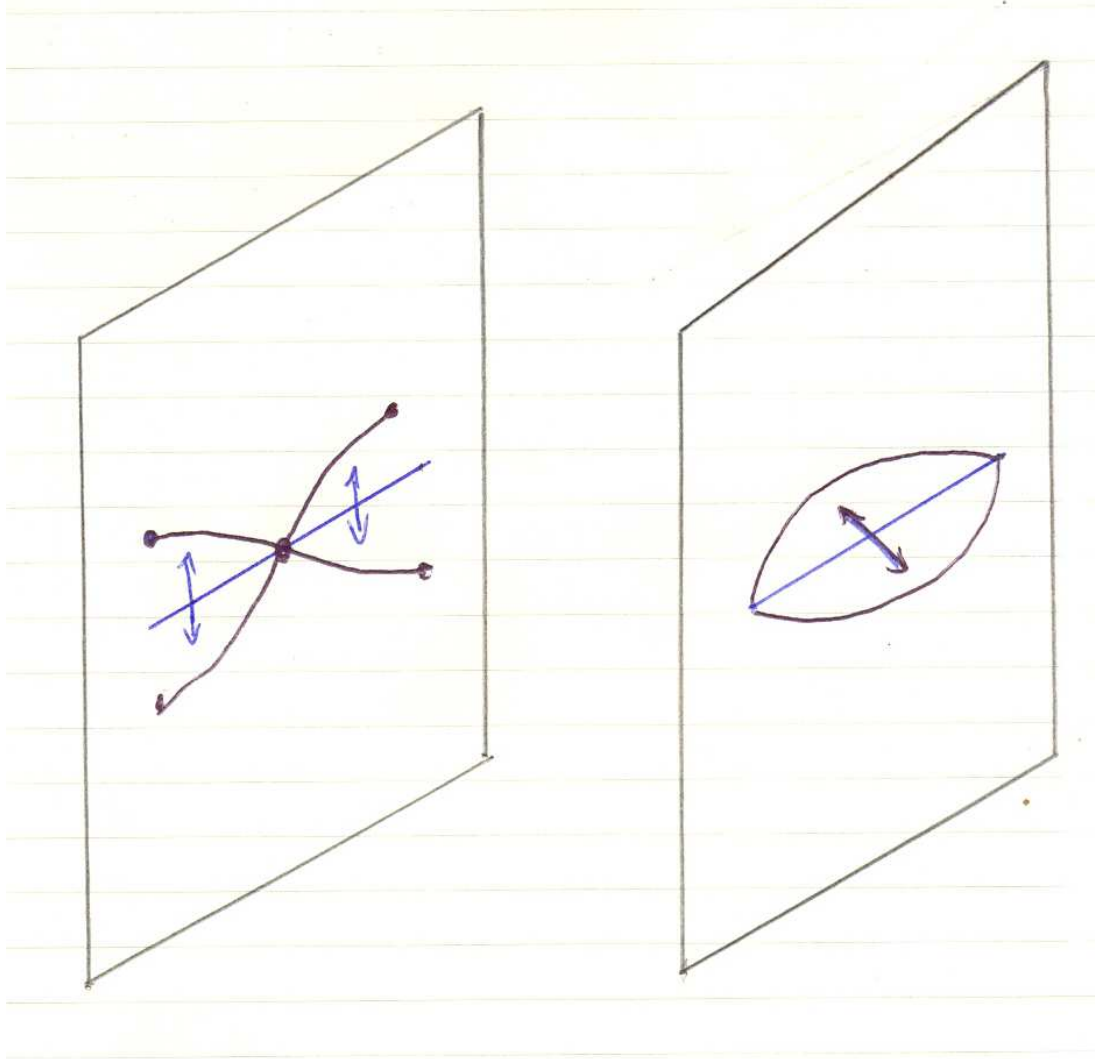


Figure 10.4

We show in figure 11.4 the two kind of oscillations: those in the left are oscillations on the plane of the free boundary conditions and do not depend on the fixed boundary conditions

of the D-brane. They are associated with the emergence of the gauge field $A_\mu(x^\mu)$. The oscillations transverse to the D-brane are subject to the fixed boundary condition. From these oscillations emerge the scalar field $\Phi^m(x^\mu)$.

Let us stress at this point that one can show that endpoints move on the brane with the speed of light and transverse to the string. The first property can be proven from eq.(11.1.13) (see [7], pp q124-125).

We summarize the discussion above by stressing two fundamental facts:

- **A Dp-brane has a massless gauge field living on its world-volume.**
- **A Dp-brane has a massless scalar for each one of its normal directions.**

Two parallel branes of the same dimensionality have the same set of longitudinal coordinates and the same set of normal coordinates. What distinguish one from the other are the values of the constants c 's (see eqs.(11.2.24)) fixing their position. Suppose that the first brane is located at $x^a = c_1^a$ and the second one at $x^a = c_2^a$. When $c_1^a = c_2^a$ the two branes coincide in space, they are on top of each other.

We represent in figure 10.5 the case two parallel D2-branes with longitudinal coordinates x^1 and x^2 and normal coordinate x^3 . Branes are located at $(0, 0, c_1^3)$ and $(0, 0, c_2^3)$. The figure shows the four different string configurations with endpoints on the branes: (i) both ends on brane 1, configuration [11] (ii) both ends on brane 2, [22] (iii) a *stretched string* going from brane 1 to brane 2, [12] and (iv) a stretched string going from brane 2 to brane 1, [21]. The last two are differentiated because the orientation of the string matters (it has to do with the sign charge of the string).

Concerning strings starting on one of the branes and ending on the other, the mass-square operator $M^2 = -p^2$ one gets ([7], pp 341) a new term with respect to the string ending in the same brane result

$$M^2 = \left(\frac{c_2^a - c_1^a}{2\pi\alpha'} \right)^2 - \frac{1}{\alpha'} + \frac{1}{\alpha'} \sum_{n>0} n N_n^\perp \quad (11.3.30)$$

since $1/2\pi\alpha' = T$ is just the string tension which coincides with the energy of the classical string, this new term is the square of the energy of the classical string stretched between the two branes. This constant valued term vanishes when the two branes coincide. Concerning N it is defined as

$$N_n^\perp = \sum_{n=1}^{\infty} \sum_{i=2}^p n \hat{a}_n^{i\dagger} \hat{a}_n^i + \sum_{m=1}^{\infty} \sum_{a=p+1}^d m \hat{a}_m^{a\dagger} \hat{a}_m^a \quad (11.3.31)$$

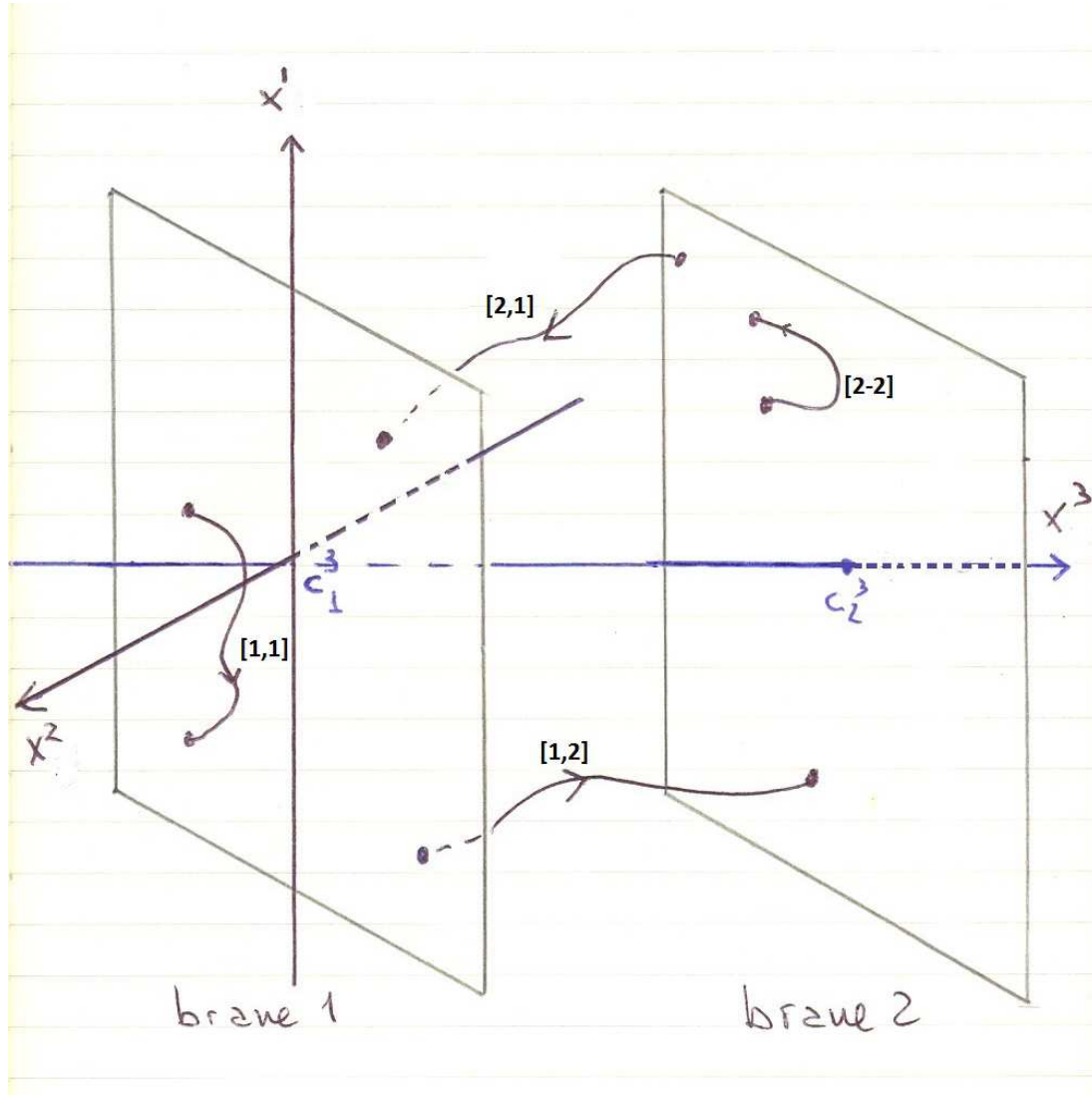


Figure 10.5

with $\hat{a}_n^i$ and $\hat{a}_n^{i\dagger}$ are annihilation and creation operators with components along the directions on the brane and $\hat{a}_m^a$ and $\hat{a}_m^{a\dagger}$ those along the normal directions. The ground state corresponding to the configuration [12] has a squared mass

$$M^2 = -\frac{1}{\alpha'} + \left(\frac{c_2^a - c_1^a}{2\pi\alpha'} \right)^2 + \frac{1}{\alpha'} N^\perp \quad (11.3.32)$$

The ground state of a quantum string is $|p^+, \vec{p}\rangle$ with $\vec{p} = (p^1, p^2, \dots, p^p)$ since, as we have seen the string carries no momentum in the a directions which are Dirichlet directions. Excited states of a quantum string are constructed with oscillators along the brane

$$\hat{a}_n^{i\dagger}, \quad i = 1, 2, \dots, p \quad (11.3.33)$$

and normal to the brane

$$\hat{a}_m^{a\dagger}, \quad a = p, p+1, \dots, d \quad (11.3.34)$$

Concerning the configuration represented in figure 10.5, there are four types of ground states:

$$|p^+, \vec{p}; [1, 1]\rangle, \quad |p^+, \vec{p}; [1, 2]\rangle, \quad |p^+, \vec{p}; [2, 1]\rangle, \quad |p^+, \vec{p}; [2, 2]\rangle \quad (11.3.35)$$

where k, l ($k, l = 1, 2$) indicates the brane on which the string starts and that on which it ends. Acting with powers $\lambda_{n,i}$ and $\lambda_{m,a}$ of operators (11.3.33) and (11.3.34) on these four ground states we build additional states. Now, where do the fields associated with oscillations of the string $[1, 2]$ live? We cannot say that the string lives on any single one of them so they are declared to live on some $p+1$ dimensional space not necessarily identified with any one of the two branes. It is here that noncommutative geometry can enter into play but we shall not discuss this...

The case $c_1^a = c_2^a$ leads to a tachyon state. Moreover, there is a critical separation

$$|c_2^a - c_1^a| = 2\pi\sqrt{\alpha'} = d_{crit} \quad (11.3.36)$$

which represents a massless scalar while for $|c_2^a - c_1^a| > d_{crit}$ the states correspond to massive scalar fields.

Now a simple state with one oscillator along the normal directions acting on the vacuum takes, for the $[1, 2]$ case the form

$$a_1^{a\dagger} |p^+, \vec{p}; [12]\rangle, \quad a = p+1, \dots, d, \quad M^2 = \left(\frac{c_2^a - c_1^a}{2\pi\alpha'} \right)^2 \quad (11.3.37)$$

Since the index a is not a Lorentz index for the $p+1$ dimensional state these states are Lorentz scalars. So one has $d-p$ Lorentz scalars (viewed from the brane). If instead we consider oscillations in the directions along the brane

$$a_1^{i\dagger} |p^+, \vec{p}; [12]\rangle, \quad i = 2, \dots, p, \quad M^2 = \left(\frac{c_2^a - c_1^a}{2\pi\alpha'} \right)^2 \quad (11.3.38)$$

they can be viewed from the brane as $p-1$ dimensional vector fields with the same mass as the scalars. However, to be massive gauge fields one needs p dimensional vectors (because of

lack of gauge invariance) so one needs to add one of the scalars as the additional component needed to have an actual gauge field. It is natural to guess that the scalar will be the combination

$$\sum_a (c_2^a - c_1^a) |p^+, \vec{p}; [1, 2]\rangle \quad (11.3.39)$$

When the separation of the branes goes to zero the masses above go to zero and one ends with a massless gauge field for each one of the four $[k, l]$ configurations, which means that one ends with a $U(2)$ gauge group. We then conclude that

N coincident D-branes carry $U(N)$ massless gauge fields

Solitons and branes

We have seen that kinks are 1+1 solitons (i.e. static localized solutions depending on just one coordinate x with finite energy). Now, if one considers a kink solution in 3 + 1 dimensional space (with say x, y and z spatial coordinates) the energy will obviously be infinite but one can consider the energy per unit area as the finite relevant quantity. The kink can be thought off as a wall located at its center, where the energy is concentrated. In this context it is dubbed as a domain wall.

Concerning vortices, they have finite energy if one works in a 2 + 1 dimensional space. In 3 + 1 they can be seen as filaments (strings) with finite energy per unit length.

In 3 + 1 dimensions the world volume of the kink is then 2 + 1 dimensional (world sheet) while that of the vortex is 1 + 1. For a monopole it is just 0 + 1,

How to live on solitons? [9]

Take the anharmonic oscillator equation

$$\partial_1^2 \phi(x^1) - m^2 \phi(x^1) - \lambda \phi^4(x^1) = 0 \quad (11.3.40)$$

It has a kink solution of the form

$$\phi_{kink}(x^1) = \frac{|m|}{\sqrt{\lambda}} \tanh \left(\frac{|m|}{\sqrt{2}} (x^1 - X^1) \right) \quad (11.3.41)$$

where X^1 is a constant (the center of the kink, see figure 10.6).

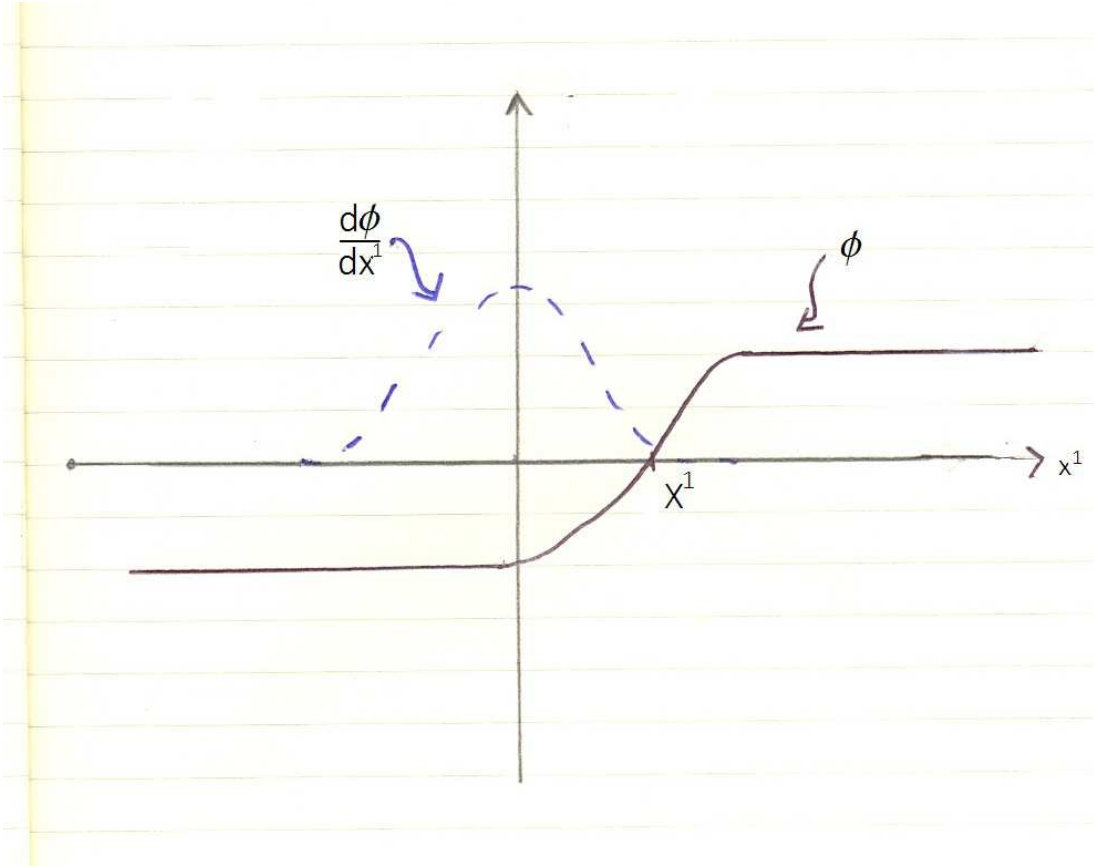


Figure 10.6

Let us now suppose that x^1 is just one of the $D - 1$ spatial dimensions of a D -dimensional space-time. Then, suppose that X^1 now depends on the rest of the spatial variables,

$$X^1 = X^1(x^0, x^2, \dots, x^{D-1}) \quad (11.3.42)$$

and satisfies the “wave-equation”

$$(-\partial_0^2 + \partial_2^2 + \dots + \partial_{D-1}^2) X^1 = 0 \quad (11.3.43)$$

thus coming from the action

$$S_{X^1} = \int dx^0 dx^2, \dots, dx^{D-1} \left(-(\partial_0 X^1)^2 + (\partial_2 X^1)^2 + \dots + (\partial_{D-1} X^1)^2 \right) \quad (11.3.44)$$

This action can be seen as the action for a massless field X^1 living in a $D - 1$ -dimensional space-time, The field theory associated with X^1 is called the effective field theory of the kink. The physical meaning of the field $X^1(x^0, x^2, \dots)$ is the following. The worldvolume of the kink solution is spanned by $x^0, x^2, \dots$ and X^1 is given as a function of these variables. The kink position in the direction x^1 is provided by the value of X^1 (see Fig. 10.7). In other words, the shape in the higher dimension of the kink is given by the field function X^1 . Since the field X^1 does not depend on x^1 and is a field in $(D - 1)$ -dimensional space-time, one can think that it just “lives” on the kink. Note in figure 10.6 that the x^1 -derivative of the kink is centered in $x^1 = 0$. Now, making a typical expansion of the kink solution (11.3.41) in powers of the collective coordinate X^1 ,

$$\phi_{kink} = \phi_{kink}|_{X^1=0} - X^1(x^0, x^2, \dots) \partial_1 \phi_{kink(x^1)}|_{X^1=0} + \dots \quad (11.3.45)$$

we see that the second term produces (because of the factor $\partial_1 \phi_{kink(x^1)}|_{X^1=0}$) the bent in the direction x^1 shown in figure 10.7,

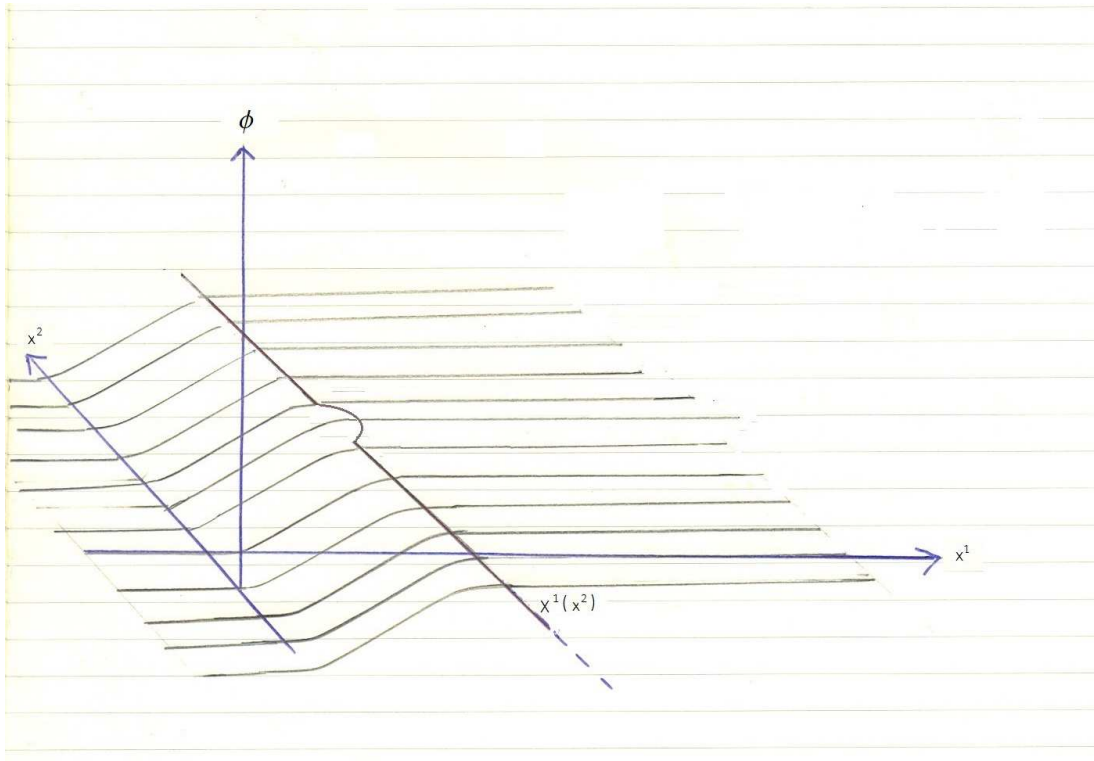


Figure 10.7

Consider now a vortex in 2 spatial dimensions. The collective coordinate appearing in the solution (locating the vortex center) is now a 2-component vector X^μ ($\mu = 1, 2$). Then, the number of massless fields associated to the vortex is 2. Associated with these 2 collective coordinates there are 2 zero modes which can be seen as massless Goldstone bosons.

There are monopoles in branes

We have seen that when two D3-branes coalesce gauge fields living on these branes become non-Abelian (with gauge group $U(2)$). If one now separates the branes from each other while keeping them parallel the two strings [12] and [21] massless states do not appear as before since now in the mass formula we have written $c_2^a \neq c_1^a$ (see eq.(11.3.38))

$$M^2 = \left(\frac{c_2^a - c_1^a}{2\pi\alpha'} \right)^2 \quad (11.3.46)$$

Then $A_\mu^{[12]}$ and $A_\mu^{[21]}$ have obtained masses which are in fact identical. Then, the combination $A_\mu^{[12]} - A_\mu^{[21]}$ remains massless so there is a remaining $U(1)$ symmetry. Separation of these branes can then be seen as a symmetry breaking mechanism $U(2) \rightarrow U(1)$

Bibliography

- [1] J. Dai, R. G. Leigh and J. Polchinski, “New Connections Between String Theories,” *Mod. Phys. Lett. A* **4** (1989) 2073.
- [2] P. Horava, “Strings on World Sheet Orbifolds,” *Nucl. Phys. B* **327** (1989) 461.
- [3] A. Achucarro, J. M. Evans, P. K. Townsend and D. L. Wiltshire, “Super p-Branes,” *Phys. Lett. B* **198** (1987) 441.
- [4] R. G. Leigh, “Dirac-Born-Infeld Action from Dirichlet Sigma Model,” *Mod. Phys. Lett. A* **4** (1989) 2767.
- [5] M. Blau “Supergravity solitons” Lectures given at the 1988 ICTP introductory school on string theory.
- [6] C. V. Johnson, “D-branes”, Cambridge University Press, Cambridge, UK, 2003.
- [7] , B. Zwiebach, “A first course on string theory”, Cambridge University Press, Cambridge, UK, 2009.
- [8] C. G. Callan, Jr., J. A. Harvey and A. Strominger, “Supersymmetric string solitons,” In *Trieste 1991, Proceedings, String theory and quantum gravity '91* 208-244 and Chicago Univ. - EFI 91-066 (91/11,rec.Feb.92) 42 p [hep-th/9112030].
- [9] K. Hashimoto, “D-brane: Superstrings and new perspective of our world,”

Chapter 12

Solitons in curved space

Bibliography

- [1] Steven Weinberg, *Dreams of a Final Theory*, Ed. Random House, New York, 1992.

Chapter 13

Solitons and holography

Bibliography

- [1] Steven Weinberg, *Dreams of a Final Theory*, Ed. Random House, New York, 1992.

Chapter 14

Appendix: Topological conservation laws and homotopy classes

A few notions of topology

Topology (from the Greek *τοπος*, “place”) was known in the past as *Analysis situs*, taken from the title of a Henri Poincaré paper where he presented the first systematic treatment of topology [1]. Today it is understood as the study of continuity and connectivity.

Let us start by defining a

- **topological space:** a set of points, along with a set of neighborhoods for each point, that satisfy a set of axioms relating points and neighborhoods.

We shall not discuss the above mentioned axioms but just state that manifolds and metric spaces are specializations of topological spaces with extra structures or constraints.

A fundamental concept in topology is that of topological equivalences. The basic one is

- **homeomorphism:** two spaces (think of geometrical figures) are homeomorphic if one can be deformed to the other by means of a continuous deformation (without cutting or gluing).

Another very important equivalence is called

- **homotopy equivalence:** Let X and Y be topological spaces and I the closed interval $[0, 1]$. Let f_0, f_1 be continuous maps from $X \rightarrow Y$. f_0 is homotopic to f_1 (written $f_0 \simeq f_1$) if there exist a continuous map $H : X \times I \rightarrow Y$ such that $H(x, 0) = f_0(x)$ and $H(x, 1) = f_1(x)$ for all $x \in X$. The map H is called a *homotopy* from f_0 to f_1 .

For $(x, t) \in X \times I$, one may regard t as measuring time. Then $f_t = F(x, t)$ is a one parameter family of maps $X \rightarrow Y$. At time $t = 0$ we have the map f_0 . At time $t = 1$ we have the map f_1 . As time increases from 0 to 1 the map f_0 is deformed continuously to f_1 .

One can show that homotopy is an equivalence relation.

In order to classify topological spaces one uses in algebraic topology *homotopy groups*. To begin with let us define the so-called fundamental group,

- **fundamental group Π_1** : Is is the first and simplest homotopy group. Intuitively, it gives information about the basic shape (in particular, holes) of a topological space.

In order to define mathematically the fundamental group let us start by considering loops

- A **loop** in space X is a continuous application f from the unit interval $[0, 1]$ to X such that $f[0] = f(1)$

$$f : [0, 1] \rightarrow X, \quad f[0] = f(1)$$

such that

One can see the loop as a continuous application from the unit circle S^1 on X

$$f : S^1 \rightarrow X, \quad f[0] = f(1)$$

taking into account that S^1 coincides with the quotient $[0, 1]/(0 \sim 1)$.

The set of all such loops with this method of combining and this equivalence (homotopy) between them is the fundamental group for that particular space.

Consider the set (family) of loops (continuous functions) with base point x_0 ,

$$\{f : [0, 1] \rightarrow X : f(0) = x_0 = f(1)\} \quad (14.0.1)$$

The fundamental group of X with base point x_0 is the set (14.0.1) modulo homotopy h (i.e, one is “counting” all homotopic maps as if it were just one. In practice, one choses one representative). This definition can be written in mathematical terms as

$$\{f : [0, 1] \rightarrow X : f(0) = x_0 = f(1)\}/h \quad (14.0.2)$$

The group multiplication is defined by

$$(f * g)(t) = \begin{cases} f(2t) & 0 \leq t \leq \frac{1}{2} \\ g(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases} \quad (14.0.3)$$

The product of two homotopy classes of loops $[f]$ and $[g]$ is then defined as $[f * g]$, and it can be shown that this product does not depend on the choice of representatives.

With the above product, the set of all homotopy classes of loops with base point x_0 forms the fundamental group of X at the point x_0 and is denoted

$$\pi_1(X, x_0), \quad (14.0.4)$$

The identity element is the constant map at the basepoint, and the inverse of a loop f is the loop g defined by $g(t) = f(1 - t)$. That is, g follows f backwards.

Example 1: $X = U(1)$, the group of complex number of modulus 1

Let us take $x_0 = 1$. We want to determine $\pi_1(U(1), 1)$ (to simplify notation we shall ignore the second argument that indicates the base point and just write $\pi_1(U(1))$)

For each integer m we introduce the angular variable φ ($0 \leq \varphi \leq 2\pi$) and define

$$f_m(t) = \exp(im2\pi t), \quad m \in \mathbb{Z} \quad (14.0.5)$$

If we identify $t = 0$ and $t = 1$ and introduce an angular variable $\varphi = 2\pi t$ we can see f_m as

$$f_m : S^1 \rightarrow U(1) \quad \text{or} \quad f_m : S^1 \rightarrow S^1 \quad (14.0.6)$$

The map f_m is a representative of a family of loops characterized by the integer m . Anyone of the loops in this family can be deformed into f_m . One has

$$\Pi_1(U(1)) = \mathbb{Z} \quad (14.0.7)$$

The homotopy group $\pi_1(U(1))$ is therefore an infinite cyclic group, and is isomorphic to the group $\mathbf{Z}$ of integers under addition: a homotopy class is identified with an integer by counting the number of times a mapping in the homotopy class wraps around the circle. This integer can also be thought of as the winding number of a loop around the origin in the plane.

We show in Figure A.1 three representatives of homotopy classes with $m = 1, 2, 3$. The interior circle represents the interval $[0, 1]/(0 \sim 1) \sim S^1$ while the arrowed circles represents loops defined by $\exp(i2\pi m)$ with $m = 1, 2, 3$ from left to right. The wrapping can be in either of two directions, and wrappings in opposite directions will cancel out after a deformation

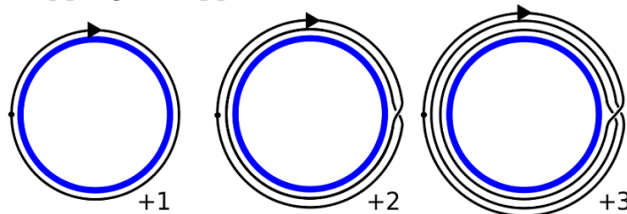


Figure A.1

In analogy with the fundamental group $\pi_1(X)$ one can define **the n -th homotopy group** $\pi_n(X)$ as follows

- In the n -sphere S^n one chooses a base point s_0 . For a space X with base point x_0 , one defines $\pi_n(X)$ to be the set of homotopy classes of maps

$$f : S^n \rightarrow X \quad (14.0.8)$$

that map the base point s_0 to the base point x_0 . In particular, the equivalence classes are given by homotopies that are constant on the basepoint of the sphere.

As for the case $n = 1$ the homotopy classes for $n > 1$ form a group with the product defined as in (14.0.3).

Some properties (given without proof)

- If $n > 1$ then $\Pi_n(X)$ is Abelian for any topological space X .

Homotopy groups for spheres S^N

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}
S^1	$\mathbf{Z}$	0	0	0	0	0	0	0	0	0	0
S^2	0	$\mathbf{Z}$	$\mathbf{Z}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_{12}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_3$	$\mathbf{Z}_{15}$	$\mathbf{Z}_2$
S^3	0	0	$\mathbf{Z}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_{12}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_3$	$\mathbf{Z}_{15}$	$\mathbf{Z}_2$
S^4	0	0	0	$\mathbf{Z}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z} \times \mathbf{Z}_{12}$	$\mathbf{Z}_2^2$	$\mathbf{Z}_2^2$	$\mathbf{Z}_{24} \times \mathbf{Z}_3$	$\mathbf{Z}_{15}$
S^5	0	0	0	0	$\mathbf{Z}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_{24}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_2$
S^6	0	0	0	0	0	$\mathbf{Z}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_{24}$	0	0
S^7	0	0	0	0	0	0	$\mathbf{Z}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_{24}$	0
S^8	0	0	0	0	0	0	0	$\mathbf{Z}$	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\mathbf{Z}_{24}$

Table A1: Homotopy groups for spheres

According to Table A1

$$\pi_1(S^2) = 0$$

One can see from Figure A2 how any continuous mapping from a circle to an ordinary sphere can be continuously deformed to a one-point mapping, and so its homotopy class is trivial. The homotopy group is therefore a trivial group, with only one element, the identity element, and so it can be identified with the subgroup of $\mathbf{Z}$ consisting only of the number zero. This group is often denoted by 0.

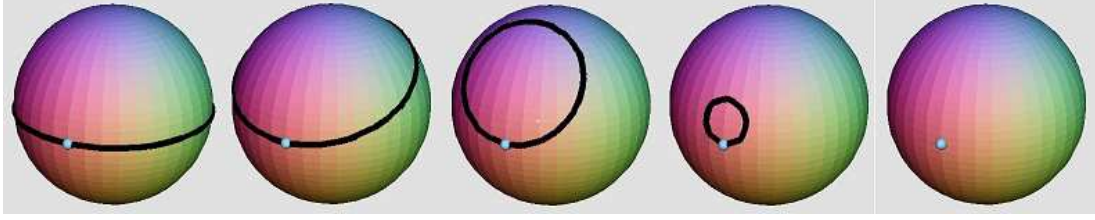


Figure A.2

This result generalizes to higher dimensions. All mappings from a lower-dimensional sphere into a sphere of higher dimension are similarly trivial

$$\Pi_i(S^n) = 0 \quad \text{if } i < n \quad (14.0.9)$$

Concerning Lie groups one can show

- $\Pi_2(SO(N)) = 0$ for $N \geq 2$
- $\Pi_3(SO(N)) = \mathbb{Z}$ for $N \geq 3, N \neq 4$
- $\Pi_2(SU(N)) = 0$ for $N \geq 2$
- $\Pi_3(SU(N)) = \mathbb{Z}$ for $N \geq 2$

Important result on homotopy groups

- If M and N are two topological spaces then for their direct product we have

$$\Pi_k(M \times N) = \Pi_k(M) \times \Pi_k(N) \quad (14.0.10)$$

- If M is a simply-connected topological space ($\Pi_0(M) = \Pi_1(M) = 0$) and group H acts on M then one can form topological space M/H identifying points of M which can be related by some element of H ($x \equiv hx$). We have

$$\Pi_1(M/H) = \Pi_0(H) \quad (14.0.11)$$

- If H is a discrete group $\Pi_0(H) = H$ and then $\Pi_1(M/H) = H$
- For $k > 1$ one has $\Pi_k(M/H) = \pi_k(M)$ if $\Pi_k(H) = \Pi_{k-1}(H) = 0$.

Homotopy groups for unitary groups $U(N)$ and $SU(N)$

- Bott periodicity theorem: for $k > 1$ and $N \geq \frac{k+1}{2}$

$$\Pi_k(U(N)) = \Pi_k(SU(N)) = \begin{cases} 0 & \text{if } k\text{-even} \\ \mathbb{Z} & \text{if } k\text{-odd} \end{cases} \quad (14.0.12)$$

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}
$U(1)$	$\mathbb{Z}$	0	0	0	0	0	0	0	0	0	0
$U(2)$	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$
$U(3)$	0	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_6$					
$U(4)$	0	0	$\mathbb{Z}$	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$				
$U(5)$	0	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$		

Table A2: Homotopy groups for unitary groups

In the previous and following tables we encircle the cells from which Bott periodicity theorem starts working.

Homotopy groups for ortogonal groups $O(N)$ and $SO(N)$

$O(N)$ is the group of orthogonal matrices of a given dimension N , where the group operation is given by matrix multiplication. An orthogonal matrix is a real matrix whose inverse equals its transpose.

- Bott periodicity theorem for orthogonal groups: for $N \geq k + 2$

$$\Pi_k(O(N)) = \Pi_k(SO(N)) = \begin{cases} 0 & \text{if } k = 2, 4, 5, 6 \pmod{8} \\ \mathbb{Z}_2 & \text{if } k = 0, 1 \pmod{8} \\ \mathbb{Z} & \text{if } k = 3, 7 \pmod{8} \end{cases} \quad (14.0.13)$$

- For homotopy groups of orthogonal groups

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}
$SO(2)$	$\mathbb{Z}$	0	0	0	0	0	0	0	0	0	0
$SO(3)$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$		
$SO(4)$	$\mathbb{Z}_2$	$\mathbb{Z}$	$\mathbb{Z}^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}^2$	$\mathbb{Z}_{12}^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_3^2$	
$SO(5)$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0		
$SO(6)$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	0	$\mathbb{Z}$	$\mathbb{Z}_{24}$	$\mathbb{Z}_2$	$\mathbb{Z}_{120}$	
$SO(N), N > 6$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	0				

Table A3: Homotopy groups for orthogonal groups

Homotopy groups for symplectic groups $\text{Sp}(n)$

Given z an element of a real vector space E , a symplectic form (or skew product) on E is a mapping $\omega : E \times E \rightarrow \mathbb{R}$ which is

- linear in each of its components
- antisymmetric, $\omega(z, z') = -\omega(z', z)$ for all $z, z' \in E$
- non-degenerate, $\omega(z, z') = 0$ for all $z \in E$ if and only if $z' = 0$.

The symplectic form can be written in matrix form as

$$\omega(z, z') = (z')^T J z = \langle J z, z' \rangle$$

where J is the standard symplectic matrix

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

Notice that $J^T = -J$ and $J^2 = -I$.

The matrix $S \in \text{Sp}(n)$ if and only if

$$S^T J S = J$$

- Bott periodicity theorem for symplectic groups: for $N \geq \frac{k-1}{4}$

$$\Pi_k(\text{Sp}(n)) = \begin{cases} 0 & \text{if } k = 0, 1, 2, 6 \pmod{8} \\ \mathbb{Z}_2 & \text{if } k = 4, 5 \pmod{8} \\ \mathbb{Z} & \text{if } k = 3, 7 \pmod{8} \end{cases} \quad (14.0.14)$$

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}
$\text{Sp}(1)$	0	$\mathbb{O}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$
$\text{Sp}(2)$	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{O}$	$\mathbb{Z}$	$\mathbb{O}$	$\mathbb{O}$	$\mathbb{Z}_{120}$	$\mathbb{Z}_2$
$\text{Sp}(3)$	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	$\mathbb{O}$	$\mathbb{Z}$
$\text{Sp}(4)$	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$
$\text{Sp}(5)$	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

Table A4: Homotopy groups for symplectic groups

Homotopy groups for Complex Projective space $\mathbb{C}P^n$

Complex projective space is a complex manifold that may be described by $n + 1$ complex coordinates as

$$Z = (Z_1, Z_2, \dots, Z_{n+1}) \in \mathbb{C}^{n+1}, \quad (Z_1, Z_2, \dots, Z_{n+1}) \neq (0, 0, \dots, 0)$$

with the tuples differing by an overall rescaling are identified:

$$(Z_1, Z_2, \dots, Z_{n+1}) \equiv (\lambda Z_1, \lambda Z_2, \dots, \lambda Z_{n+1}); \quad \lambda \in \mathbb{C}, \quad \lambda \neq 0.$$

- One has $CP_1 = S_2$ and generally $CP_n = S^{2n+1}/S^1$. Concerning homotopy groups

$$\begin{aligned} \Pi_1(CP^n) &= 0 \\ \Pi_2(CP^n) &= \mathbb{Z} \\ \Pi_k(CP^n) &= \Pi_k(S^{2n+1}) \quad \text{for } k \geq 3. \end{aligned} \tag{14.0.15}$$

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}
CP^1	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$
CP^2	0	$\mathbb{Z}$	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$
CP^3	0	$\mathbb{Z}$	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0
CP^4	0	$\mathbb{Z}$	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$

Table As: Homotopy groups for complex projective spaces

The connection between homotopy groups and homology groups.

The concept of homotopy is a mathematical formulation of the intuitive idea of a continuous transition between two geometrical configurations. The concept of homology gives a mathematical precision to the intuitive idea of a curve bounding an "area" or a surface bounding a "volume."

Both the homotopy group and the homology group are tools used in the study of topological properties of manifolds and are the bases of algebraic topology. They both apply algebraic methods such as groups and homomorphisms to study the topological properties of manifolds.

Homotopy groups are conceptually intuitive and have direct links with global properties and they are homotopy type invariants of manifolds. However, the calculation of homotopy groups can be very difficult while the calculation of homology groups is in general easier.

The homotopy class of a mapping determines its homology class, but not conversely, as shown by the example of the mappings of the sphere S^3 into S^2 which all belong to the same homology class although there is an infinite number of homotopy classes ($\pi_3(S^2) = \mathbf{Z}$).

The question arises: under what special conditions the homotopy classification of the mappings of P into X coincides with their homology classification. The classical result of Hopf states that this is the case if P is a polytope of dimension n and X the n -dimensional sphere S^n . Generalizations of this result will be described below.

tools

- As we have already seen, two topological spaces (spaces equipped with a definition of what is to be considered an open set) are regarded as being the “same”, or homeomorphic, if there is a one-to-one, onto, continuous map between them whose inverse is also continuous.
- Manifolds come with the additional structure of differentiability: we may therefore talk of “smooth” maps, meaning that their expression in coordinates is infinitely (C^∞) differentiable. We regard two manifolds as being the “same”, or diffeomorphic, if there is a one-to-one onto C^∞ map between them whose inverse is also C^∞ .
- In elementary geometry, a polytope is a geometric object with flat sides, which exists in any general number of dimensions. A polygon is a polytope in two dimensions, a polyhedron in three dimensions, and so on in higher dimensions (such as a polychoron in four dimensions). When referring to an n -dimensional generalization, the term n -polytope is used. For example, a polygon is a 2-polytope, a polyhedron is a 3-polytope, and a polychoron is a 4-polytope.

homotopy and homology

The first step toward connecting these two basic concepts of topology was taken by L. E. J. Brouwer in 1912 by demonstrating that two continuous mappings of a two-dimensional sphere into itself can be continuously deformed into each other if and only if they have the same degree (that is, if they are equivalent from the point of view of homology theory).

H. Hopf generalized Brouwer’s result to an arbitrary number of dimensions. Then, he undertook a systematic study of the problem of classifying the continuous mappings of a polytope P into a polytope X . Each mapping f induces homomorphisms of homology groups of P into the corresponding groups of X .

We need now to understand the basic idea of homology. Pictorially we can start from the following:

- Given a region Ω , we can find its boundary $\partial\Omega$. Inspection of a few simple cases will soon lead to the conclusion that the “boundary of a boundary” consists of nothing. In symbols, $\partial^2 = 0$. The statement “ $\partial^2 = 0$ ” is clearly analagous to “ $d^2 = 0$ ” and, pursuing the analogy, we can construct a vector space of “regions” and define two “regions” as being homologous if they differ by the boundary of another “region.”
- An element h of the Kernel of ∂ , $Ker(\partial)$ should satisfy $\partial h = 0$. An element of the Image of ∂ , $Im(\partial)$ is called a boundary.
- A chain complex is a sequence of Abelian groups C_n and a family of homomorphisms ∂_n appelés opérateurs de bord

$$\partial_n : C_n \rightarrow C_{n-1}$$

such that $\partial_i \partial_{i+1} = 0$. Elements of the kernel $Ker(\partial_i)$ are called cycles. Elements of the image $Im(\partial_{i+1})$ are called boundaries. Every boundary is a cycle. The chain can be represented as

$$\dots \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

where 0 denotes the trivial group and $C_k = 0$ for $k < 0$

- one defines n -th homology group of X as

$$H_n(X) = Ker(\partial_n)/Im(\partial_{n+1}) \quad (14.0.16)$$

- Evry cycle σ in H_n defines an element $[\sigma] \in H_n(X)$ called its homology class.
- Two mappings f and g are said to belong to the same **homology class** if they induce identical homomorphisms of homology groups (for all dimensions and all coefficient domains).
- The mappings f and g are said to belong to the same **homotopy class** if they can be embedded into a common one-parameter continuous family of mappings.

The homotopy class of a mapping determines its homology class, but not conversely, as shown by the example of the mappings of the sphere S^3 into S^2 which all belong to the same homology class although there is an infinite number of homotopy classes. The question arises: under what special conditions the homotopy classification of the mappings of P into Q coincides with their homology classification.

They are closely related to each other. There exist homomorphism mapping from homotopy group to homology group. There is a theorem:

- For any space X and positive integer k there exists a group homomorphism

$$h_*: \pi_k(X) \rightarrow H_k(X) \quad (14.0.17)$$

called the Hurewicz homomorphism from the k -th homotopy group to the k -th homology group (with integer coefficients).

Chern characters and classes. Topological charges

Characteristic classes serve to classify the structure of fiber bundles¹. Of particular importance are one type of these classes, namely Chern classes, which arise when the bundles are complex vector bundles equipped with connections (gauge fields)

The standard topological invariants of a vector bundle are its characteristic classes, namely the Chern classes. That is, Chern classes allow to associate a topological invariant quantity to each vector fiber bundle (= topological space -for example space, space-time, ...) + vector space (fiber).

Note that

- A topological invariant is an object that is invariant under homeomorphisms.
- An homotopic invariant is an object that is the same for homotopically equivalent topological spaces.
- As a rule, topological invariants are also homotopy invariants

Let us write the gauge field (connection) in the form

$$A = A_\mu dx^\mu \quad (14.0.18)$$

and the field strength (curvature) as

$$F = dA + A^2 \quad (14.0.19)$$

with

$$dA = \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu dx^\nu, \quad A^2 = \frac{1}{2}[A_\mu, A_\nu] dx^\mu dx^\nu \quad (14.0.20)$$

so that

$$F = \frac{1}{2}F_{\mu\nu} dx^\mu dx^\nu \quad (14.0.21)$$

1

The n -th Chern character is defined as

$$\text{ch}_n(F) = \text{tr} \left(\frac{1}{n!} \left(\frac{1}{2\pi} F \right)^n \right) \quad (14.0.22)$$

From this, one defines the total Chern character as

$$\text{ch}(F) = \text{ch}_0 + \text{ch}_1 + \text{ch}_2 + \dots = \exp \left(\left(\frac{1}{2\pi} F \right)^n \right) \quad (14.0.23)$$

For an Abelian $U(1)$ gauge field on S^2 , takes the form²

$$\text{ch}_1 = \frac{1}{2\pi} \int_S F = \frac{1}{2\pi} \oint_{S^1_\infty} d\chi = n \in \mathbf{Z} \quad (14.0.24)$$

so that ch_1 is the integer counting the winding χ of the $U(1)$ gauge group element $\exp(i\chi)$.

In general, Chern classes allow to associate a topological invariant quantity to each vector fiber bundle (= topological space -for example space, space-time, ...) + vector space (fiber)

²In the search of finite energy configurations one should consider that the field strength vanish at infinity this allowing to consider the compactification of $\mathbf{R}^2$ by identifying the points at infinity. Hence, at infinity gauge fields can onlo take the form of a “pure gauge” $\partial_\mu \chi$

Bibliography

- [1] H. Poincaré, *Analysis situs*, Journal de l'École Polytechnique. **2** (1895) 1.
- [2] A. Abanov, Graduate classes on *Topological terms in condensed matter physics*, Appendix 1, <http://felix.physics.sunysb.edu/~abanov/>
- [3] M. de Gosson, Introduction to symplectic mechanics: Lectures I-II-III, Lecture Notes at the University of São Paulo, May-June 2006.
<http://www.ime.usp.br/~piccione/Downloads/LecturesIME.pdf>
- [4]